
NSPL 0.5 μ m Analog CMOS 2P3M 5V Design Guide

Version 1.1

National Semiconductor Public Laboratory (NSPL)

- Electronics and Telecommunications Research Institute (ETRI)
- Seoul National University (SNU)
- Daegu Gyeongbuk Institute of Science & Technology (DGIST)



서울대학교
SEOUL NATIONAL UNIVERSITY



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1. Overview

1.1 Overview

The purpose of this guide is NSPL 0.5um Analog CMOS 2P3M 5V Design guide.

1.2 Software Version and Tools

- Virtuoso 6.1.8 version
- Calibre 2019.3_15.11

1.3 File Configuration

File : [NSPL_0p5um_Analog_CMOS_2P3M_5V.tar.gz](#)

- NSPL_0p5um_Analog_CMOS_2P3M_5V
 - NSPL_0p5um_Analog_CMOS_2P3M_5V.tf : tech file
 - NSPL_0p5um_Analog_CMOS_2P3M_5V.layermap : map file
 - display.drf : display file
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_pcell : Pcell library
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_std_sch : Standard sch library
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_std_lay : Standard lay library
 - ETRI_NSPL_GPIO : GPIO library
 - MPW_PAD_28pin : MPW library
 - Model_Parameter
 - 05cmos_model.lib , 05cmos_mosfet_corner.lib : Spice model parameter
 - 05cmos_model.scs , 05cmos_mosfet_corner.scs : Spectre model parameter
 - Calibre/DRC
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_DRC.cal : DRC rule file
 - Calibre/ANT
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_ANT.cal : DRC rule file
 - Calibre/ LVS
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_LVS.cal : LVS rule file
 - Calibre/LVL
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_LVL.cal : LVL rule file
 - Calibre/PEX
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_PEX.cal : PEX rule file
 - LPE_LIB : LPE Lib
 - Assura/DRC
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_DRC.rul : DRC rule file
 - Assura/LVS
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_LVS_compare.rul : LVS rule file 1
 - NSPL_0p5um_Analog_CMOS_2P3M_5V_LVS_extract.rul : LVS rule file 2
 - Assura/QRC
 - worst / typical / best : PEX rule file

1.4 Documents

- NSPL 0.5um Analog CMOS 2P3M 5V Design Rule : Design rule
- NSPL 0.5um Analog CMOS 2P3M 5V Design Guide : Design Guide

1.5 Document history

Rev	Date	From	Description
1.0	2023.08.21	Coressol	Added Items 1. Overview(version, file configuration, Documents) 2. Environment Files and About Use 3. Schematic Library 4. Parameter Cell Library 5. Standard Schematic Library 6. Standard Layout Library 7. Sealing 8. Calibre DRC 9. Calibre LVS 10. Calibre LVL
	2023.09.06	Coressol	Added Items 8. LVS Guide Modify Items 8.Calibre DRC → 9.Calibre DRC 9. Calibre LVS → 10.Calibre LVS 10. Calibre LVL → 11.Calibre LVL
1.1	2023.10.27	Coressol	Added Layers 2.2 Technology File : CHIPBOUNDARY Layer 2.3 Map File : CHIPBOUNDARY Layer Modify Layer Numbers 2.2 Technology File : ESDI(109→26) , ESD(26→52) Layer 2.3 Map File : ESDI(109→26) , ESD(26→52) Layer
			Title
			NSPL 0.5/μm Analog CMOS 2P3M 5V Design Guide
			Division
			Coressol

1.5 Document history

Rev	Date	From	Description
	2023.11.23	Coressol	Modify Items. - PPRES/NPRES → Merged P2RES - PPHRES/NPHRES → Merged P2HRES - Schematic Library : Added Cell “3.2 .Device_ALL” - Symbol Update : 3.5 p2res, 3.6 p2hres, 3.7 pipcap - Pcell Changed : 4.5 p2res, 4.6 p2hres - Std_Sch Library : Added Cell “5.1 .SAMPLE” Added Items. - 12. Assura DRC - 13. Assura LVS
	2025.06.18	Coressol	Modify Items. - 1.3 File Configuration : Data Update - 2.4 How to Setup Library : Data Update - 3. Schematic Library & 4. PCELL Layout Library : Merged Library - 3.1 Library Category : Data Update - 3.2 .Device_ALL : Data Update - 3.3 nmos4 (Schematic) : Changed Default Value - 3.4 pmos4 (Schematic) : Changed Default Value - 3.5 p2res (Schematic) : Device Structure Update - 3.6 p2hres (Schematic) : Device Structure Update - 3.7 pipcap (Schematic) : Device Structure Update - 3.8 npn2 (Layout) : Added Parameter - 3.9 npn5 (Layout) : Added Parameter - 3.10 npn10 (Layout) : Added Parameter - 3.11 pnp2 (Layout) : Added Parameter
			Title
			NSPL 0.5/μm Analog CMOS 2P3M 5V Design Guide
			Division
			Coressol

1.5 Document history

Rev	Date	From	Description
	2025.06.18	Coressol	Modify Items. - 3.12 pnp5 (Layout) : Added Parameter - 3.13 pnp10 (Layout) : Added Parameter - 3.14 ndiode (Layout) : Changed Parameter - 3.18 pdiode (Layout) : Changed Parameter - 3.22 cont (Layout) : Added Comment - 3.23 cont_bar (Layout) : Added Comment - 3.24 poly1cont (Layout) : Changed Default Value - 3.25 poly2cont (Layout) : Changed Default Value - 3.26 ntap (Lyaout) : Changed Default Value - 3.27 ntap_ring (Lyaout) : Changed Default Value - 3.28 ntap_ring_pdiode (Layout) : Changed Default Value - 3.29 ptap (Lyaout) : Changed Default Value - 3.30 ptap_ring (Lyaout) : Changed Default Value - 3.31 ptap_ring_ndiode (Layout) : Added Comment - 3.36 pad80 (Layout) : Added Parameter - 3.37 pad100 (Layout) : Added Parameter - 5. Standard Schematic Library → 4. Standard Schematic Library - 6. Standard Layout Library → 5. Standard Layout Library - 7. Sealring → 6. Sealring - 8. LVS Guide → 7. LVS Guide - 7.1 PIPCAP Layout/Circuit Structure : Data Update - 9. Calibre DRC → 8. Calibre DRC - 10. Calibre LVS → 9. Calibre LVS
			Title
			NSPL 0.5/μm Analog CMOS 2P3M 5V Design Guide
			Division
			Coressol

1.5 Document history

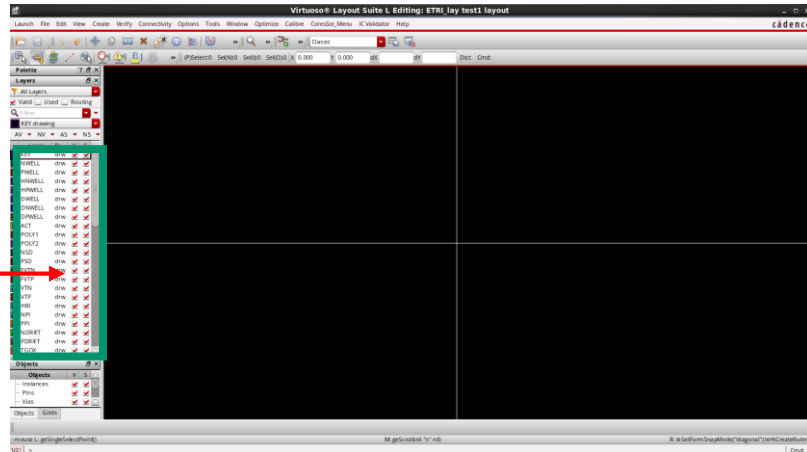
Rev	Date	From	Description
	2025.06.18	Coressol	Modify Items. - 11. Calibre LVL → 10. Calibre LVL - 12. Assura DRC → 11. Assura DRC - 13. Assura LVS → 12. Assura LVS
			Title NSPL 0.5μm Analog CMOS 2P3M 5V Design Guide
			Division Coressol

2. Environment Files and About Use

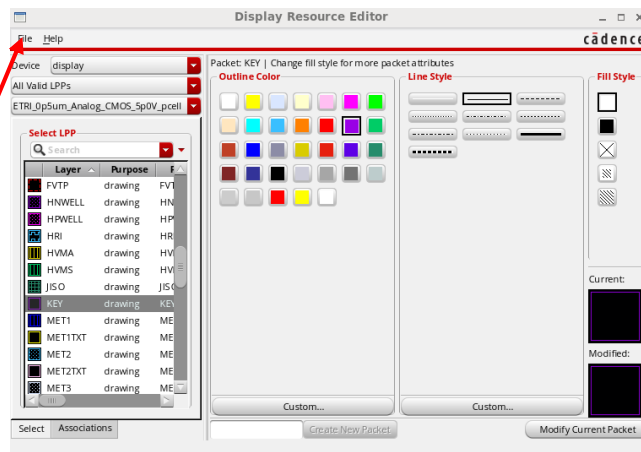
2.1 Display.drf File

- This file should exist in the location where virtuoso is run.
- This file the color of the layer is specified.
- If you want to change the color, you can do so in the following ways

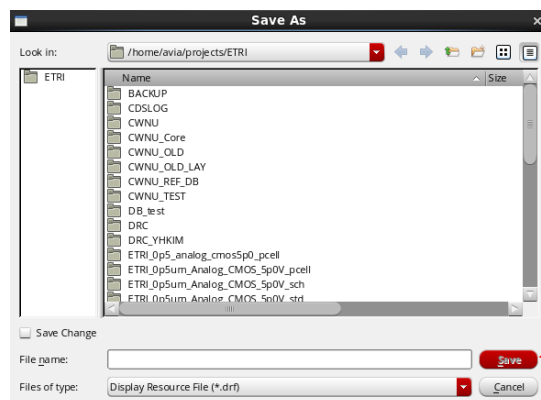
Shift +
left mouse
click.



Layer Edit
Save at file



Select file or
New file name



Save

2.2 Technology File

Layer Name	Layer No.	Function	Digitized Area
CHIPBOUNDARY	0	Define Chip Boundary	
KEY	1	Define Align key	C
NWELL	2	Define N-well for PMOS body	C
PWELL	3	Define P-well for NMOS body	C
ACT	11	Define Active area	D
POLY1	12	Define Poly-Si 1 for gate of transistor	D
POLY2	13	Define Poly-Si 2 for resistor and capacitor	D
NSD	14	Define N-type implant region for N-LDD and N+	C
PSD	15	Define N-type implant region for P-LDD and P+	C
NHRI	20	Define implant for N-type high resistance poly-Si resistor (poly)	C
PHRI	21	Define implant for P-type high resistance poly-Si resistor (poly)	C
ESDI	26	Define ESD Implnat	C
CONT	31	Define contact from MET1 to ACT, POLY1 and POLY2	C
MET1	32	Define Metal-1 for interconnection	D
VIA1	33	Define Via1 connecting MET1 and MET2	C
MET2	34	Define Metal-2 for interconnection	D
VIA2	35	Define Via2 connecting MET2 and MET3	C
MET3	36	Define Metal-3 for interconnection	D
PAD	37	Define open region of passivation for bonding pad	C
POLY1TXT	41	Poly1 Text	
MET1TXT	42	MET1 Text	
MET2TXT	44	MET2 Text	
MET3TXT	46	MET3 Text	
PADTXT	47	PAD Text	
TEXT	48	Text	
RES	51	Poly-Si Resistor for DRC & LVS	
ESD	52	Define ESD device region for DRC	
CAP	53	Poly-Si Capacitor for DRC & LVS	
BJT	56	BJT for DRC & LVS	
DIODE	57	Diode for DRC & LVS	
NODRC	58	DRC check blocking layer	
ESDD	59	Define ESD device drain electrode	
LVSD	60	Extract Poly-Si Resistor/PIP CAP Area	

2.3 Map File

Layer Name	Layer Purpose	Layer No.	Layer Data Type
CHIPBOUNDARY	drawing	0	0
KEY	drawing	1	0
NWELL	drawing	2	0
PWELL	drawing	3	0
ACT	drawing	11	0
POLY1	drawing	12	0
POLY2	drawing	13	0
NSD	drawing	14	0
PSD	drawing	15	0
NHRI	drawing	20	0
PHRI	drawing	21	0
ESDI	drawing	26	0
CONT	drawing	31	0
MET1	drawing	32	0
VIA1	drawing	33	0
MET2	drawing	34	0
VIA2	drawing	35	0
MET3	drawing	36	0
PAD	drawing	37	0
POLY1TXT	drawing	41	0
MET1TXT	drawing	42	0
MET2TXT	drawing	44	0
MET3TXT	drawing	46	0
PADTXT	drawing	47	0
TEXT	drawing	48	0
RES	drawing	51	0
ESD	drawing	52	0
CAP	drawing	53	0
BJT	drawing	56	0
DIODE	drawing	57	0
NODRC	drawing	58	0
ESDD	drawing	59	0
LVSD	drawing	60	0

2.4 How to Setup Library

- Bring the NSPL_0p5um_Analog_CMOS_2P3M_5V.tar.gz file to the location where you want to run virtuoso and release the component with a command like the following:

```
tar -zxvf NSPL_0p5um_Analog_CMOS_2P3M_5V.tar.gz
```

```
May 21 09:01 ../
May 21 09:01 Assura/
May 21 09:01 Calibre/
May 21 09:01 NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1.layermap*
May 21 09:01 NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1.tf*
May 21 14:08 NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_pcell/
May 21 09:01 NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_lay/
May 21 09:01 NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_sch/
May 21 09:01 display.drf*
```

- Define Library in cds.lib

```
NSPL_0p5um_Analog_CMOS_2P3M_5V_pcell
```

```
NSPL_0p5um_Analog_CMOS_2P3M_5V_sch
```

```
NSPL_0p5um_Analog_CMOS_2P3M_5V_std_lay
```

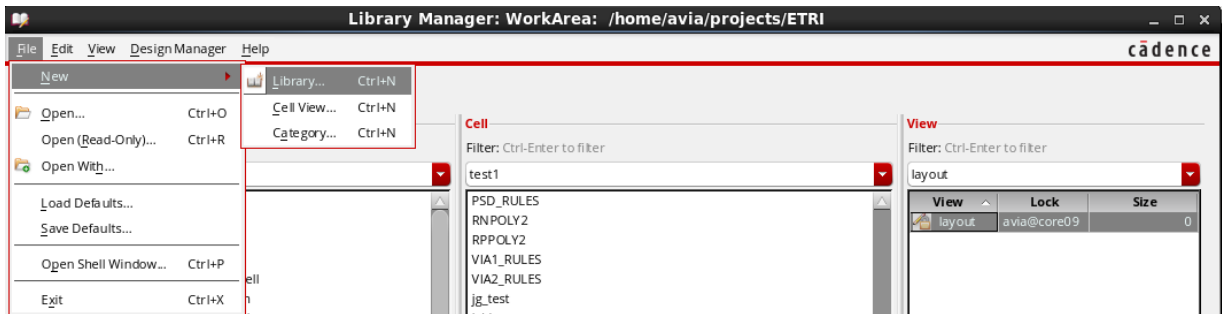
```
NSPL_0p5um_Analog_CMOS_2P3M_5V_std_sch
```

- DEFINE library_name library_path (following example – cds.lib)

```
DEFINE NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_pcell /home/projects/NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_pcell
DEFINE NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_lay /home/projects/NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_lay
DEFINE NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_sch /home/projects/NSPL_0p5um_Analog_CMOS_2P3M_5V_V1p1_std_sch
```

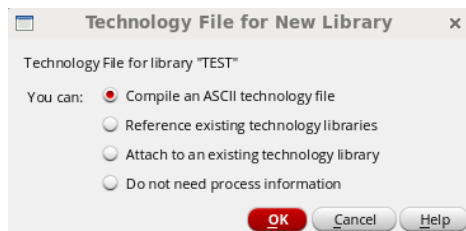
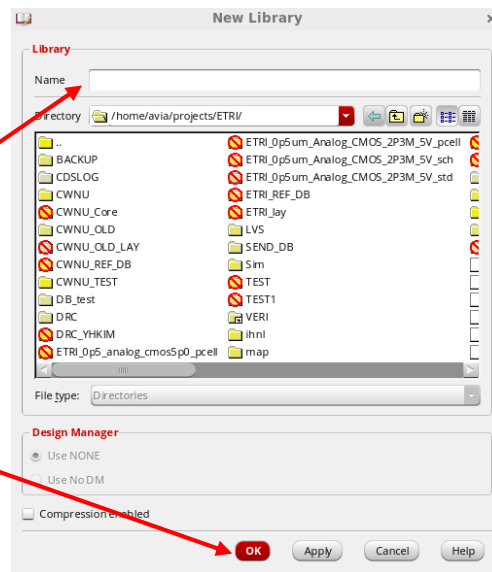
- Copy the files in the DRC Directory to the location where you want to run the DRC.
- Copy the files in the LVS Directory to the location where you want to run the LVS.
- Copy the files in the LVL Directory to the location where you want to run the LVL.
- display.drf file copy at virtuoso running.

2.5 How to Create a Library

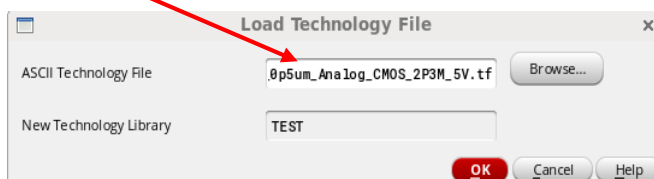


Library name

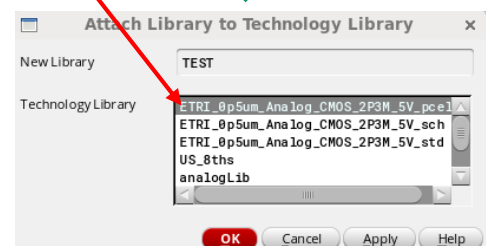
After write
Library name



Write Tech file with
location

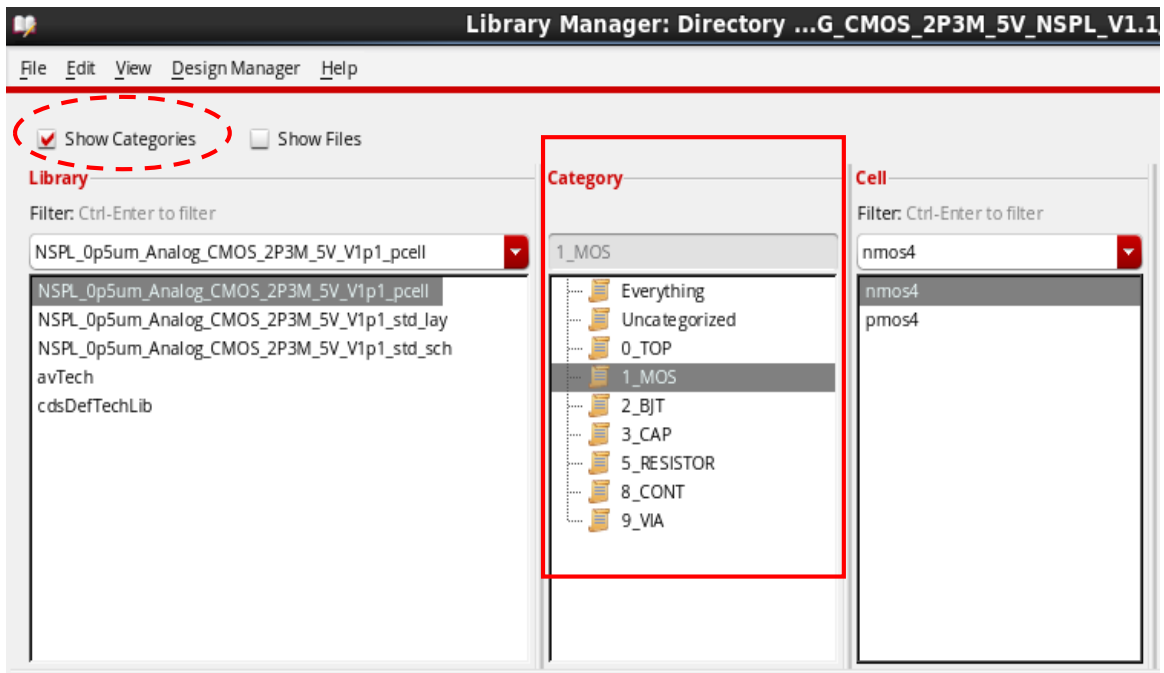


Select pcell or std
or sch library



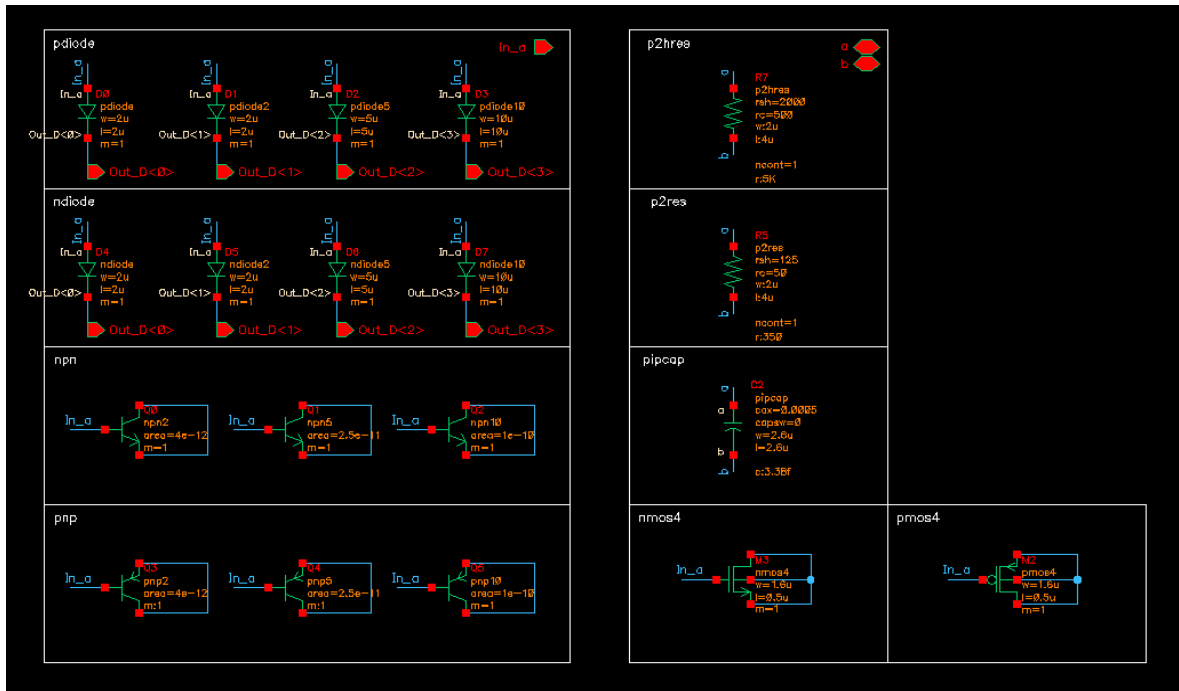
3. PCELL Library

3.1 Library Category

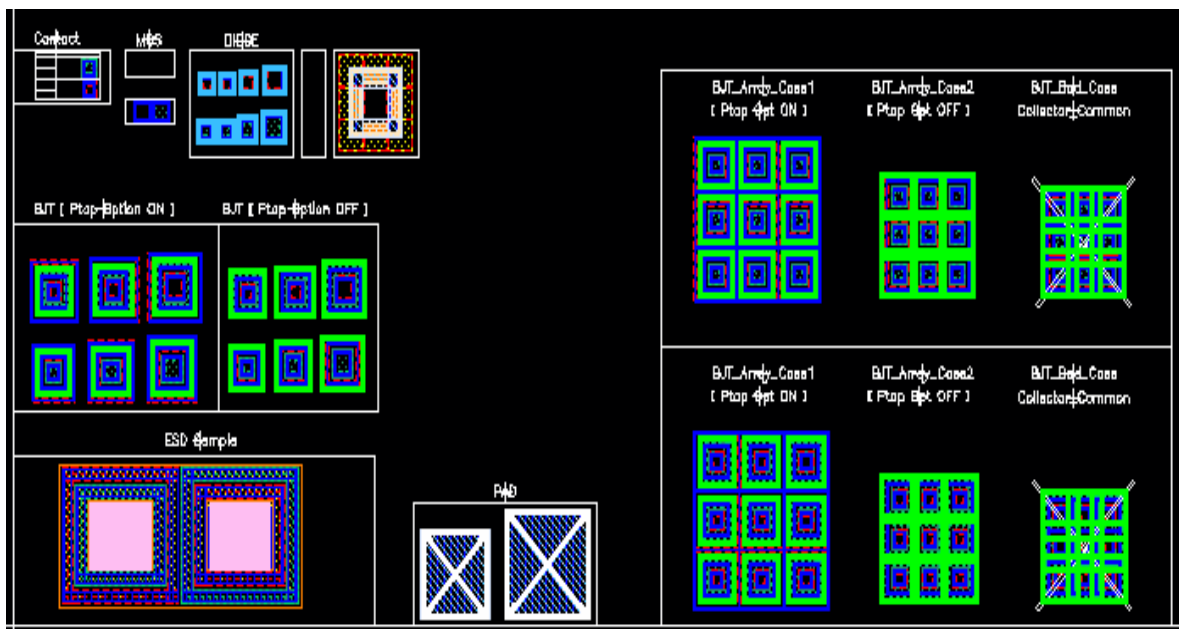


3.2 .Device_ALL

→ View : Schematic

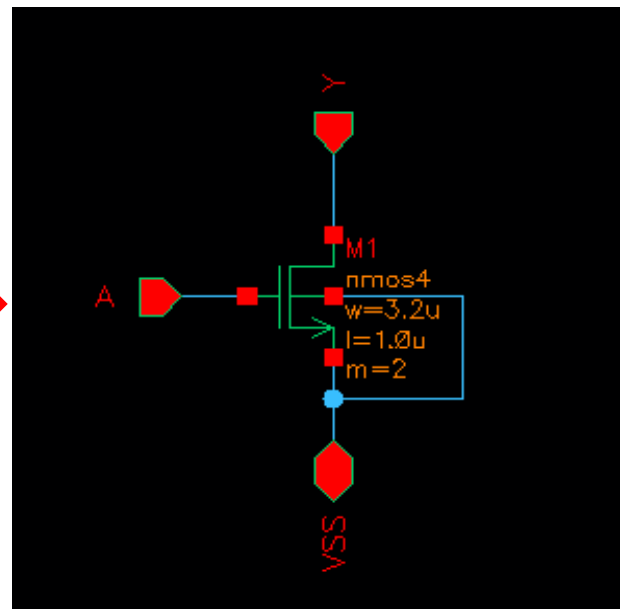
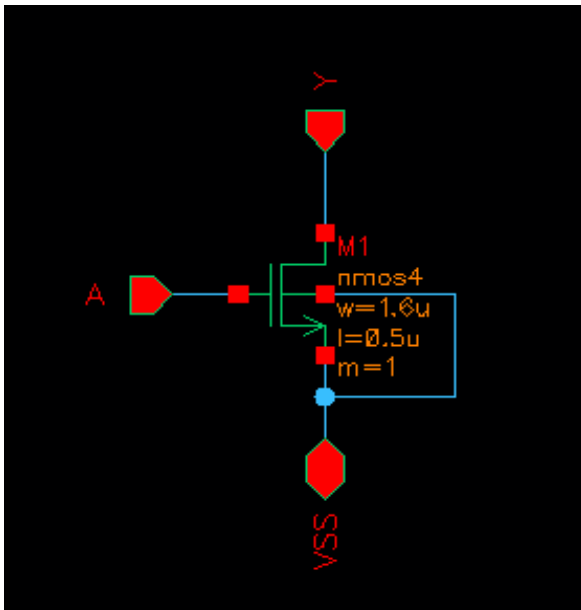


→ View : Layout



3.3 nmos4 (Schematic)

Parameter					
Width	1.6u → 3.2u	Length	0.5u → 1.0u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    nmos4
* View Name:    schematic
*****
.SUBCKT nmos4 A VSS Y
*.PININFO A:I Y:I VSS:B
MM1 Y A VSS VSS nmos4 W=1.6u L=0.5u m=1
.ENDS
```

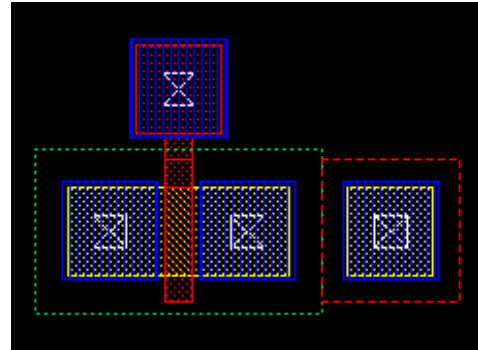


```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    nmos4
* View Name:    schematic
*****
.SUBCKT nmos4 A VSS Y
*.PININFO A:I Y:I VSS:B
MM1 Y A VSS VSS nmos4 W=3.2u L=1.0u m=2
.ENDS
```

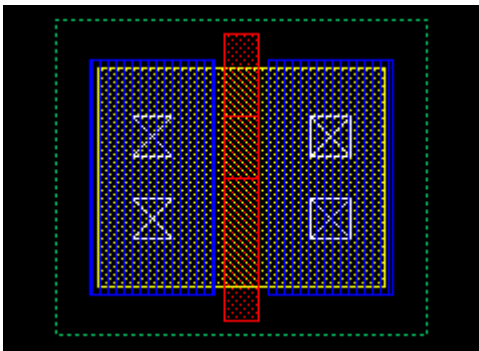
<cdl output>

3.3 nmos4 (Layout)

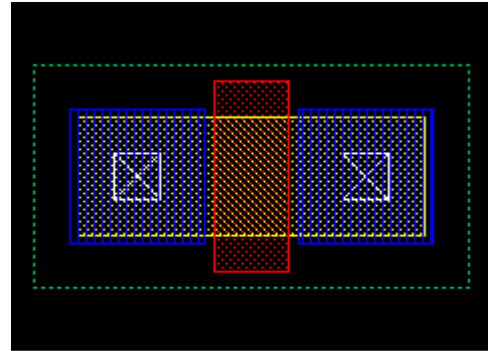
Parameter	Default	Change
width	1.6	<i>value</i>
length	0.5	<i>value</i>
multi	1	<i>value</i>
polyshift	0	<i>value</i>
polynumber	1	<i>value</i>
polycont	☒	☐
ptap	☒	☐



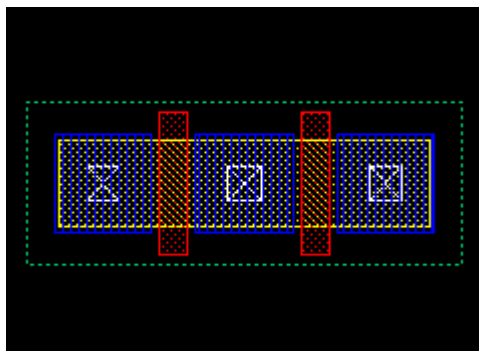
width	1.6	3.2
-------	-----	-----



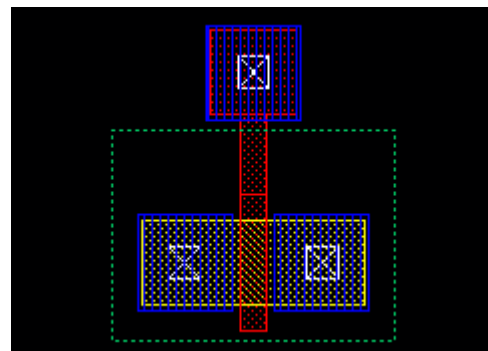
length	0.5	1.0
--------	-----	-----



multi	1	2
-------	---	---

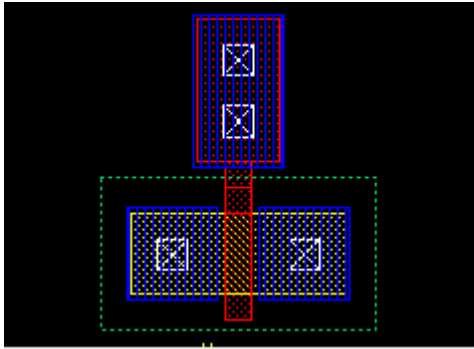


polyshift	0	1
-----------	---	---

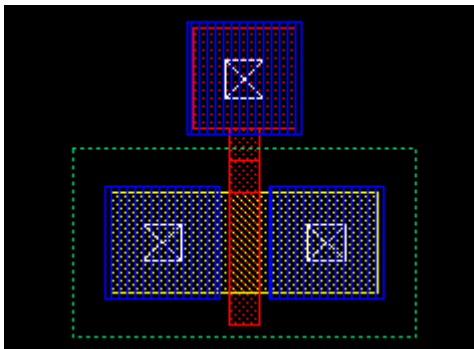


3.3 nmos4 (Layout)

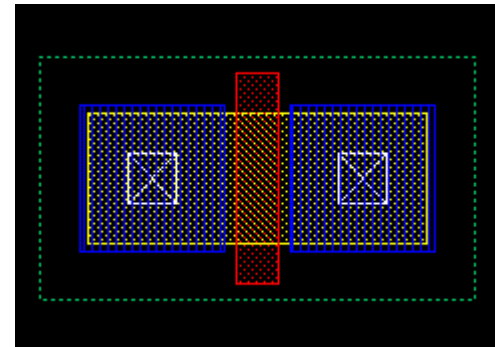
polynumber	1	2
------------	---	---



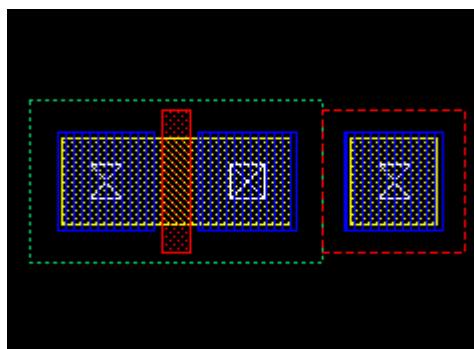
polycont	☒
----------	-------------------------------------



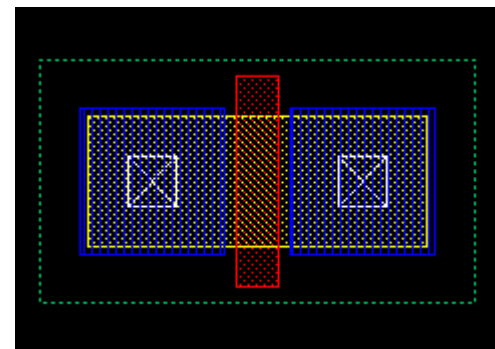
polycont	☐
----------	--------------------------



ptap	☒
------	-------------------------------------

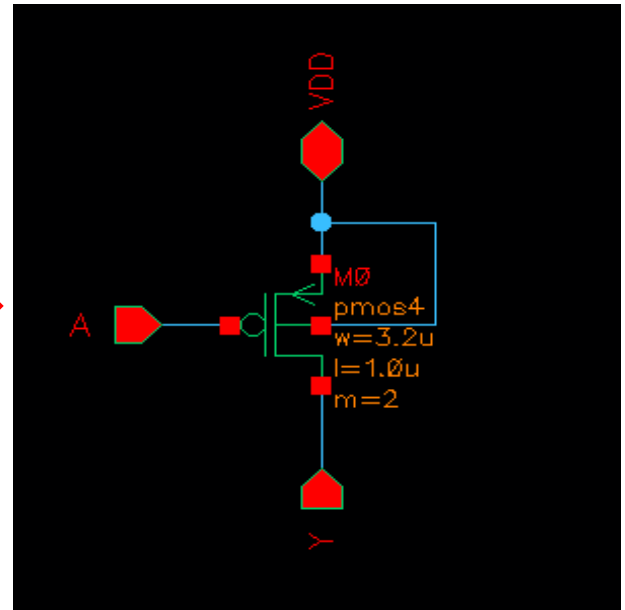
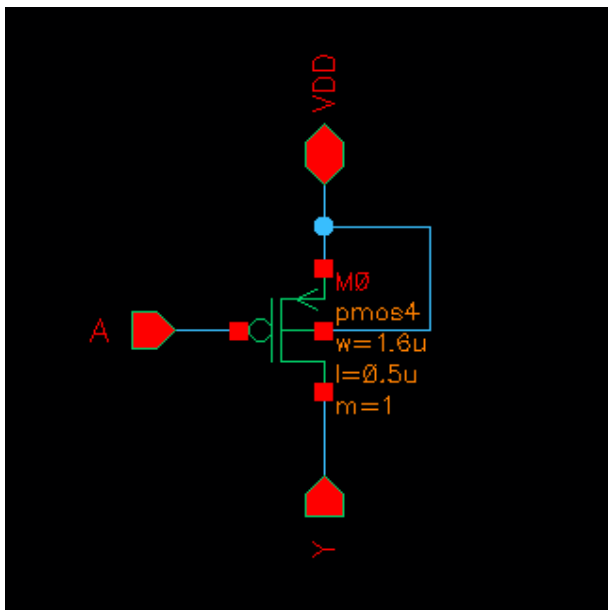


ptap	☐
------	--------------------------



3.4 pmos4 (Schematic)

Parameter					
Width	1.6u → 3.2u	Length	0.5u → 1.0u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    pmos4
* View Name:    schematic
*****
.SUBCKT pmos4 A VDD Y
*.PININFO A:I Y:I VDD:B
MM0 Y A VDD VDD pmos4 W=1.6u L=0.5u m=1
.ENDS
```

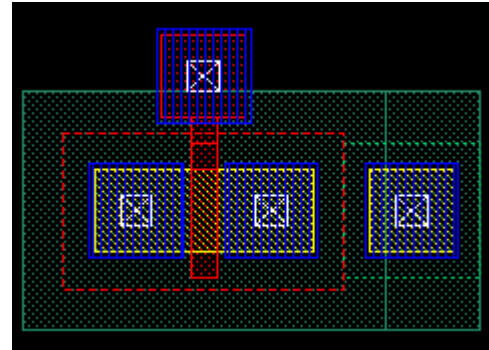


```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    pmos4
* View Name:    schematic
*****
.SUBCKT pmos4 A VDD Y
*.PININFO A:I Y:I VDD:B
MM0 Y A VDD VDD pmos4 W=3.2u L=1.0u m=2
.ENDS
```

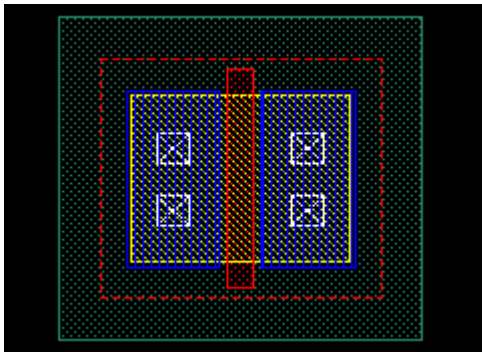
<cdl output>

3.4 pmos4 (Layout)

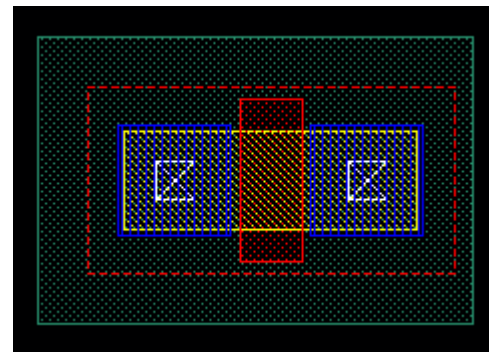
Parameter	Default	Change
width	1.6	<i>value</i>
length	0.5	<i>value</i>
multi	1	<i>value</i>
polyshift	0	<i>value</i>
polynumber	1	<i>value</i>
polycont	☒	☐
ntap	☒	☐



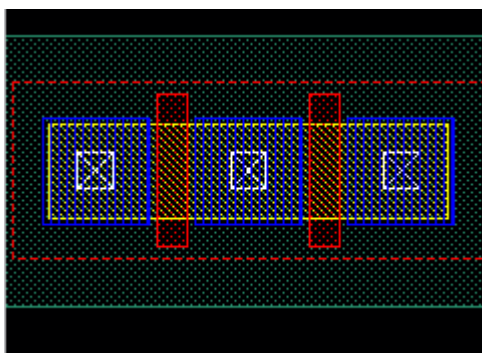
width	1.6	3.2
-------	-----	-----



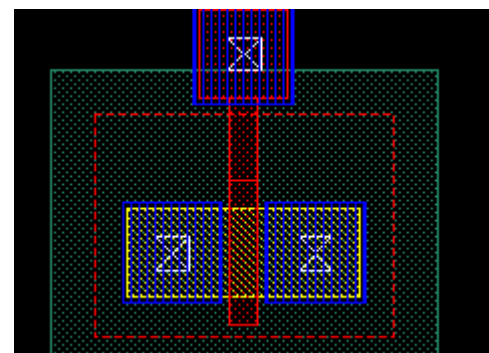
length	0.5	1.0
--------	-----	-----



multi	1	2
-------	---	---

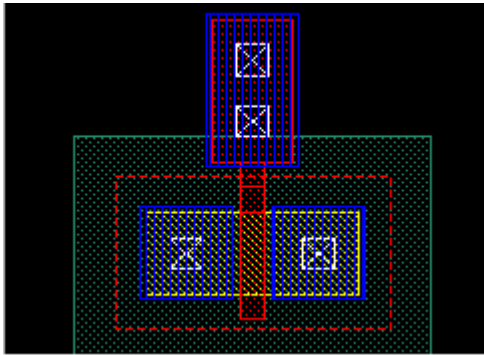


polyshift	0	1
-----------	---	---

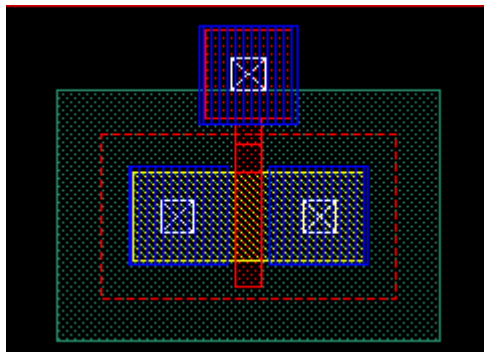


3.4 pmos4 (Layout)

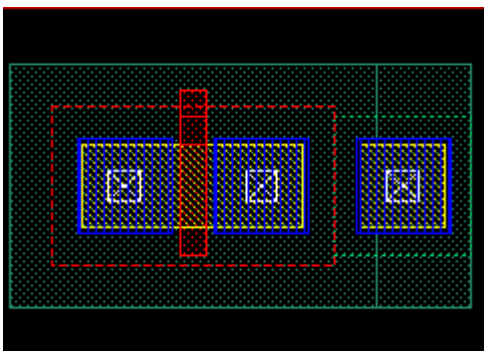
polynumber	1	2
------------	---	---



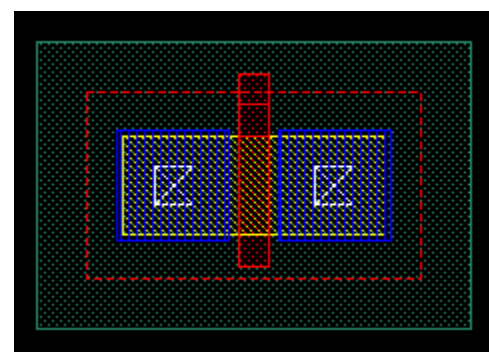
polycont	☒
----------	-------------------------------------



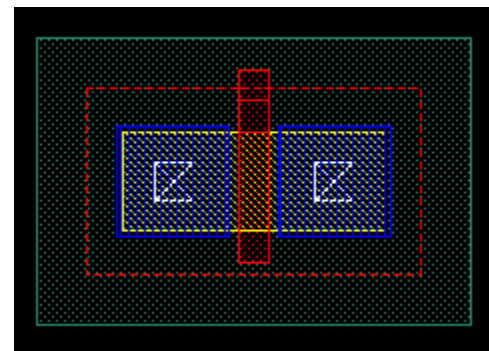
ntap	☒
------	-------------------------------------



polycont	☐
----------	--------------------------



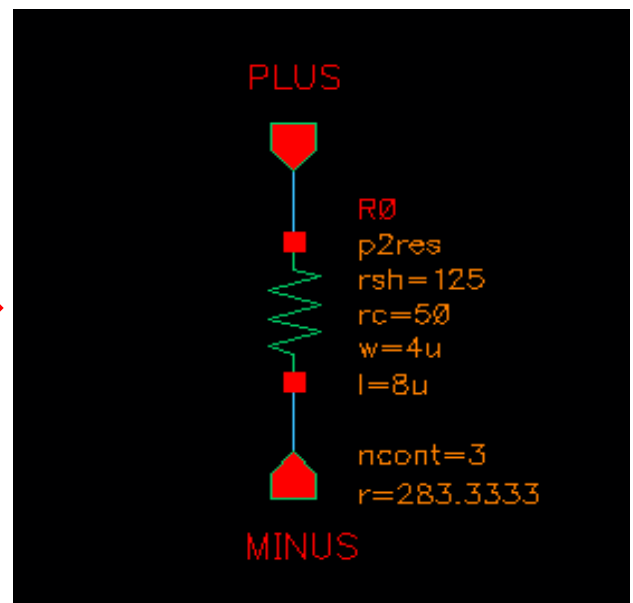
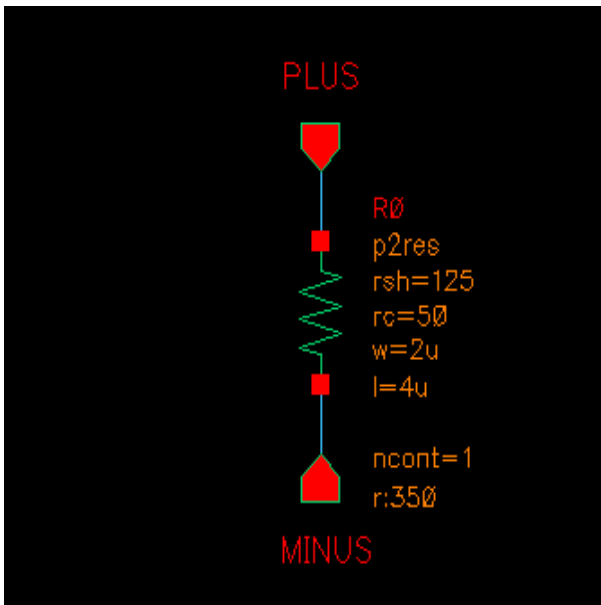
ntap	☐
------	--------------------------



3.5 p2res (Schematic)

* Process Parameter : rsh, rc

Parameter					
Width	2u → 4u	Length	4u → 8u	ncont	1 → 3



<Schematic parameter>

```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    p2res
* View Name:    schematic
*****

.SUBCKT p2res MINUS PLUS
*.PININFO MINUS:I PLUS:I
RR0 PLUS MINUS $[p2res] w=2u l=4u
.ENDS
```



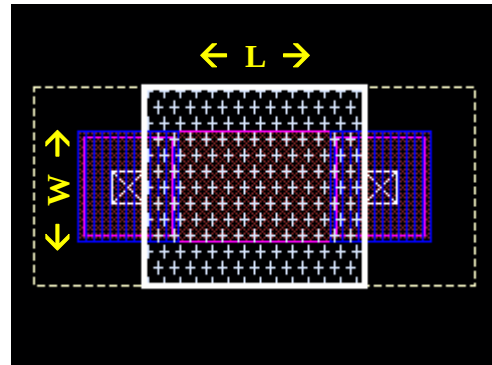
```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    p2res
* View Name:    schematic
*****

.SUBCKT p2res MINUS PLUS
*.PININFO MINUS:I PLUS:I
RR0 PLUS MINUS $[p2res] w=4u l=8u
.ENDS
```

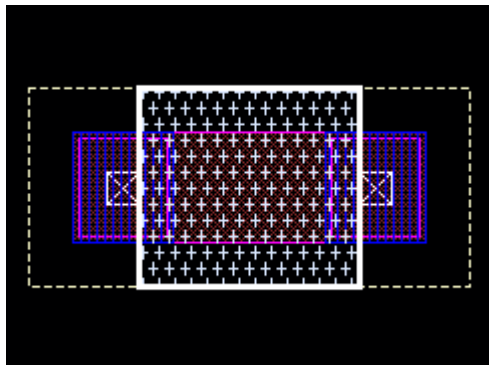
<cdl output>

3.5 p2res (Layout)

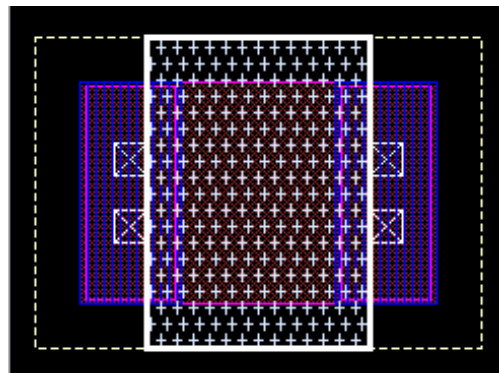
Parameter	Default	Change
W	2	value
L	4	value



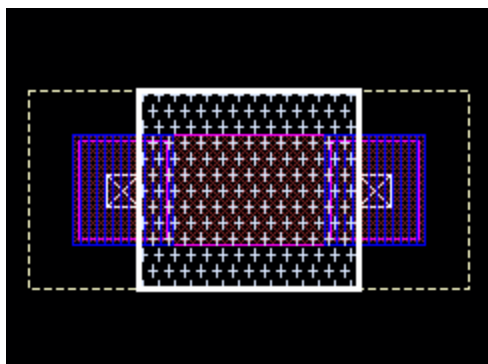
W	2
---	---



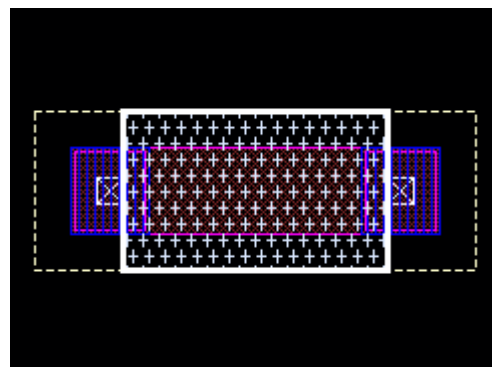
W	4
---	---



L	4
---	---



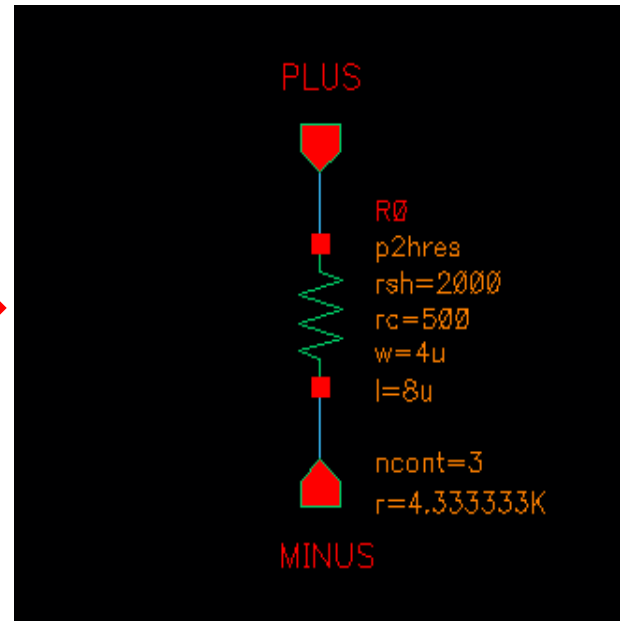
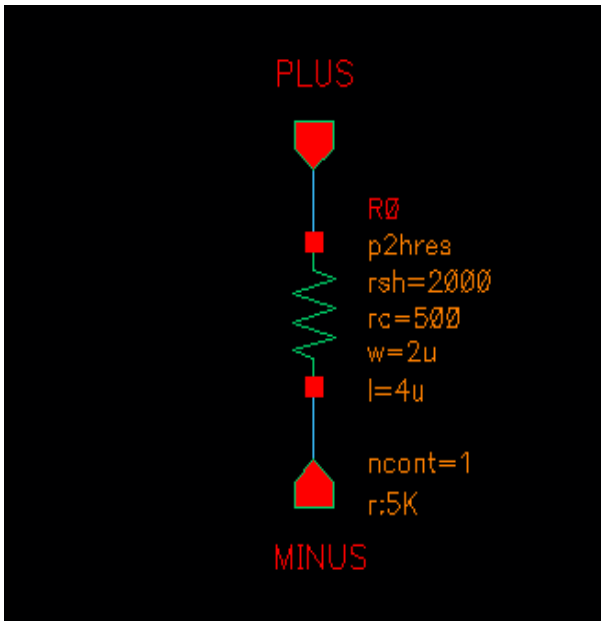
L	6
---	---



3.6 p2hres (Schematic)

* Process Parameter : rsh, rc

Parameter					
Width	2u → 4u	Length	4u → 8u	ncont	1 → 3



<Schematic parameter>

```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    p2hres
* View Name:    schematic
*****
.SUBCKT p2hres MINUS PLUS
*.PININFO MINUS:I PLUS:I
RR0 PLUS MINUS ${p2hres} w=2u l=4u
.ENDS
```

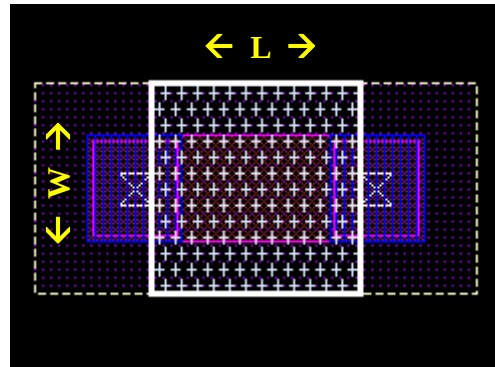


```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:    p2hres
* View Name:    schematic
*****
.SUBCKT p2hres MINUS PLUS
*.PININFO MINUS:I PLUS:I
RR0 PLUS MINUS ${p2hres} w=4u l=8u
.ENDS
```

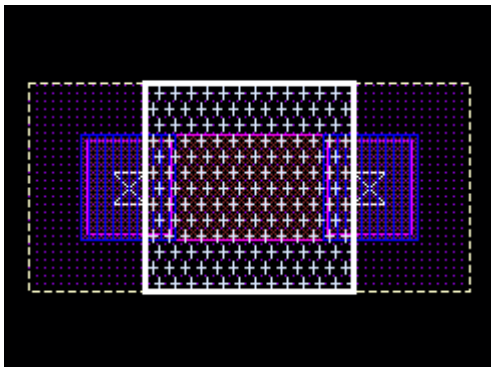
<cdl output>

3.6 p2hres (Layout)

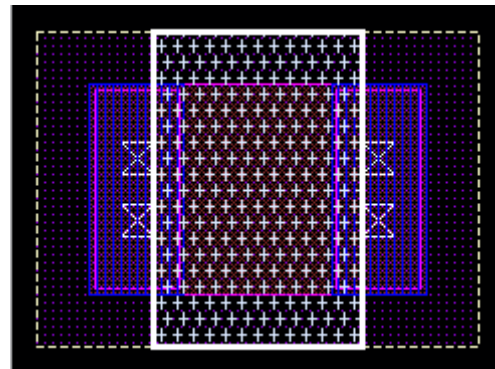
Parameter	Default	Change
W	2	value
L	4	value



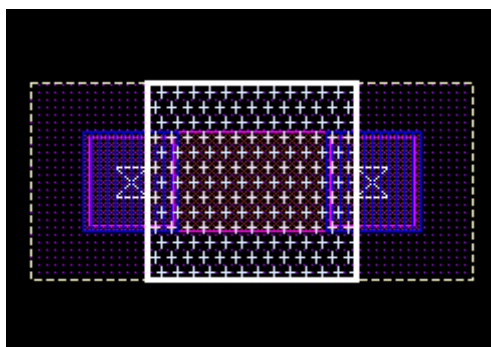
W	2
---	---



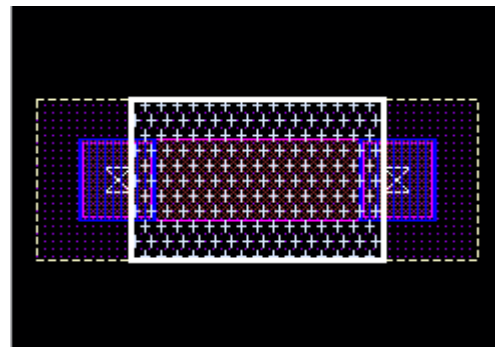
W	4
---	---



L	4
---	---



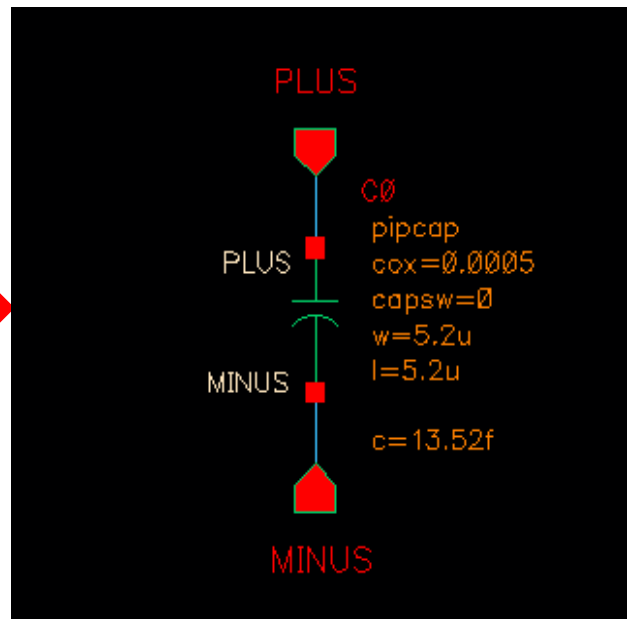
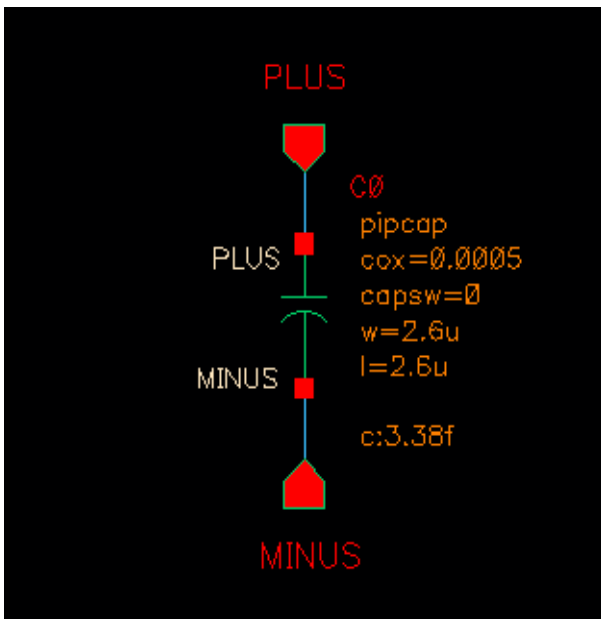
L	6
---	---



3.7 pipcap (Schematic)

* Process Parameter : cox, capsw

Parameter			
Width	2.6u → 5.2u	Length	2.6u → 5.2u



<Schematic parameter>

```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:   pipcap
* View Name:   schematic
*****

.SUBCKT pipcap MINUS PLUS
*.PININFO MINUS:I PLUS:I
CC0 PLUS MINUS ${pipcap} w=2.6u l=2.6u
.ENDS
```



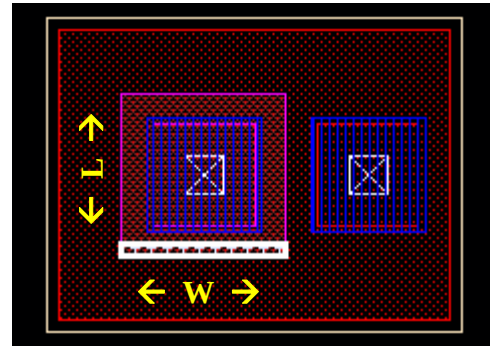
```
*****
* Library Name: ETRI_NSPL_lay
* Cell Name:   pipcap
* View Name:   schematic
*****

.SUBCKT pipcap MINUS PLUS
*.PININFO MINUS:I PLUS:I
CC0 PLUS MINUS ${pipcap} w=5.2u l=5.2u
.ENDS
```

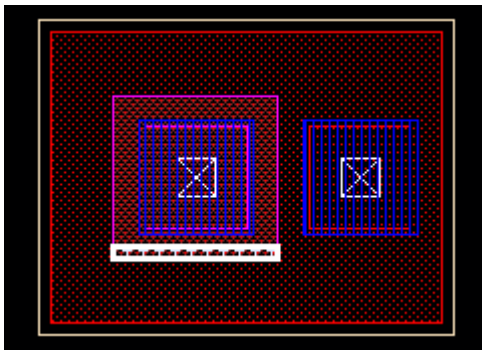
<cdl output>

3.7 pipcap (Layout)

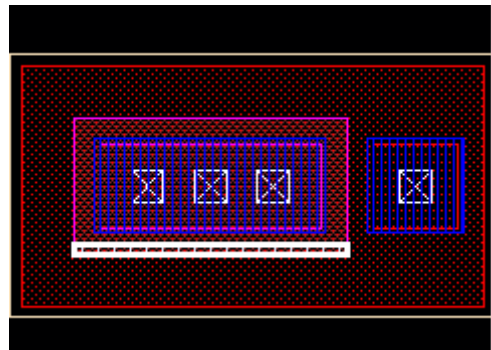
Parameter	Default	Change
W	2.6	value
L	2.6	value



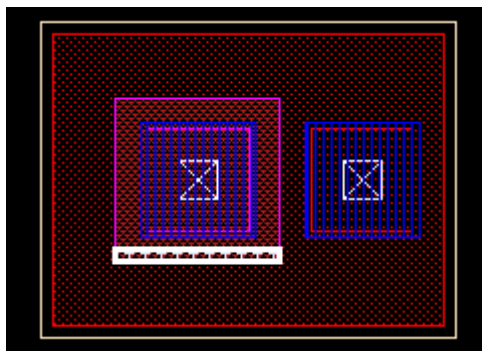
W	2.6
---	-----



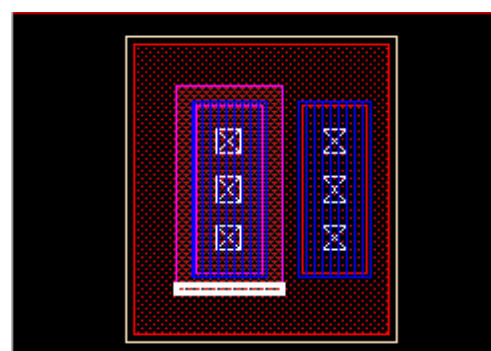
W	5.2
---	-----



L	2.6
---	-----

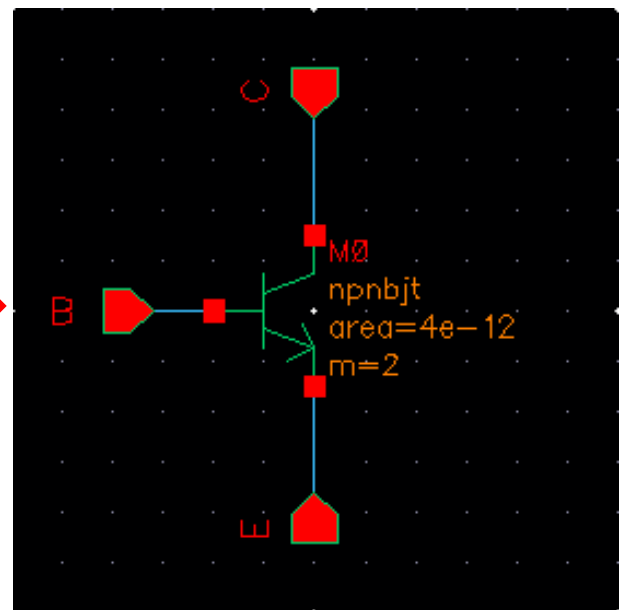
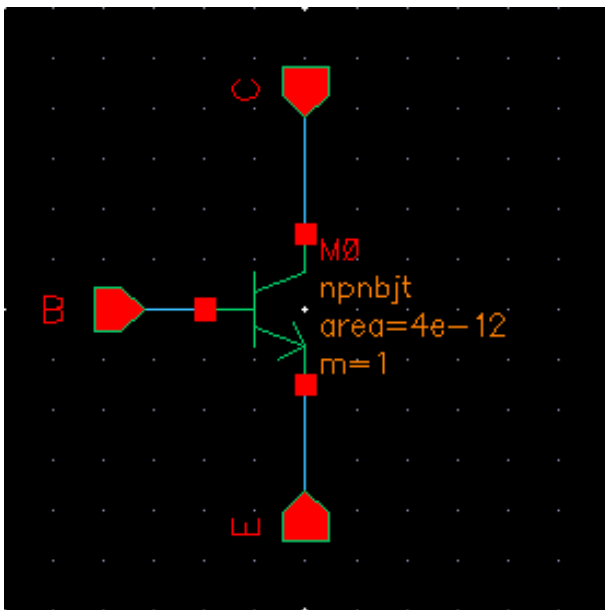


L	5.2
---	-----



3.8 npn2 (Schematic)

Parameter			
Area	4e-12	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    npn2
* View Name:    schematic
*****
.SUBCKT npn2 B C E
*.PININFO B:I C:I E:0
QM0 C B E npn2 m=1 area=4e-12
.ENDS
```



```
*****
* Library Name: ETRI_lay
* Cell Name:    npn2
* View Name:    schematic
*****
.SUBCKT npn2 B C E
*.PININFO B:I C:I E:0
QM0 C B E npn2 m=2 area=4e-12
.ENDS
```

<cdl output>

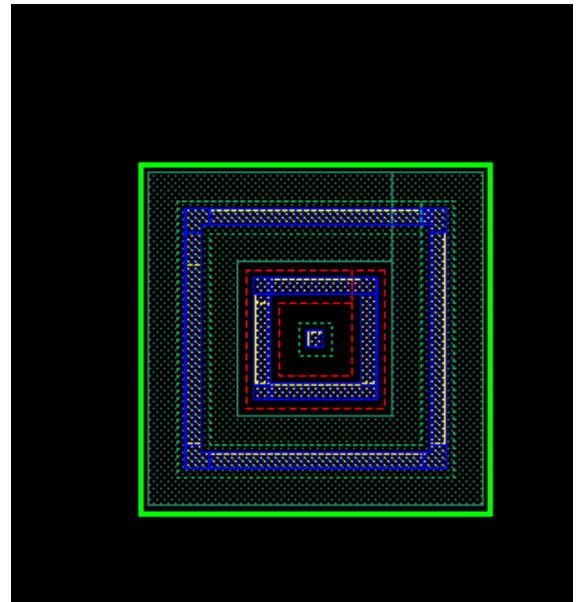
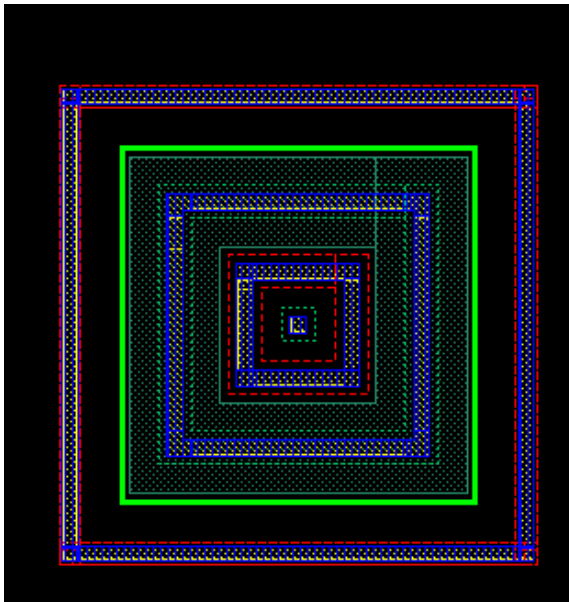
3.8 npn2 (Layout)

Parameter	Default	Change
PTAP_ON_OFF	☒	☐

BJT doesn't change PCELL
It must be used PCELL

PTAP_ON_OFF	☒
-------------	-------------------------------------

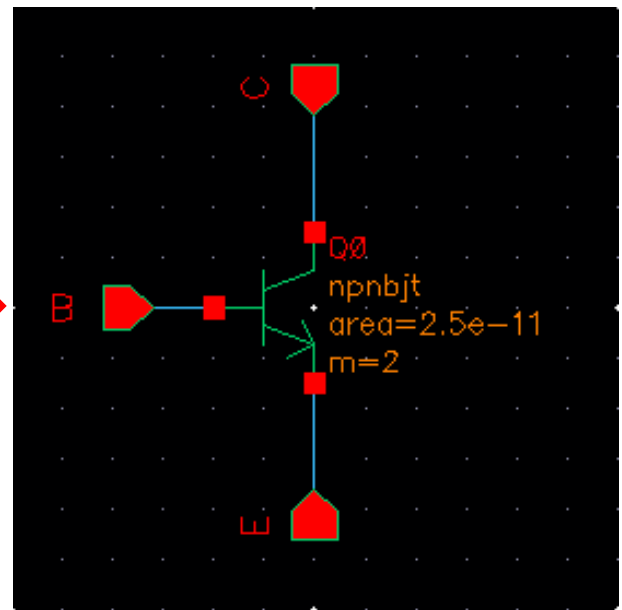
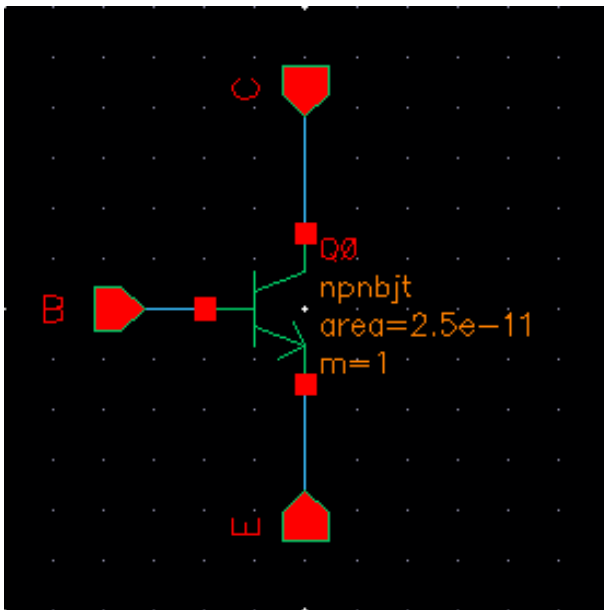
PTAP_ON_OFF	☐
-------------	--------------------------



**BJT layer on BJT device
can't be overlapped each other**

3.9 npn5 (Schematic)

Parameter			
Area	2.5e-11	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    npn5
* View Name:    schematic
*****
.SUBCKT npn5 B C E
*.PININFO B:I C:I E:0
QQ0 C B E npn5 m=1 area=2.5e-11
.ENDS
```



```
*****
* Library Name: ETRI_lay
* Cell Name:    npn5
* View Name:    schematic
*****
.SUBCKT npn5 B C E
*.PININFO B:I C:I E:0
QQ0 C B E npn5 m=2 area=2.5e-11
.ENDS
```

<cdl output>

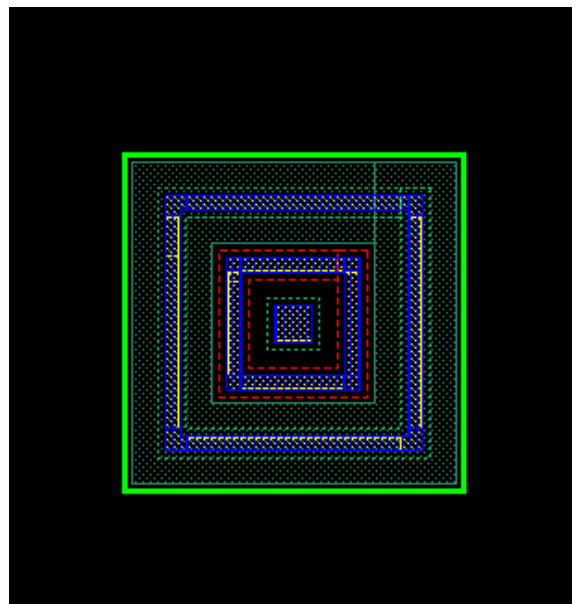
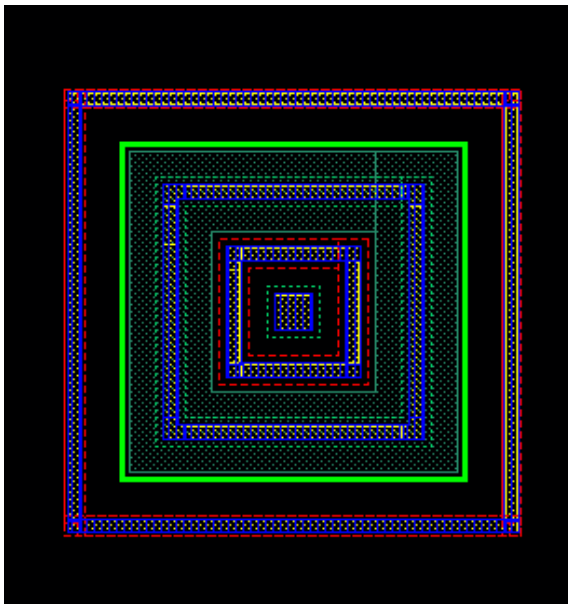
3.9 npn5 (Layout)

Parameter	Default	Change
PTAP_ON_OFF	☒	☐

BJT doesn't change PCELL
It must be used PCELL

PTAP_ON_OFF	☒
-------------	-------------------------------------

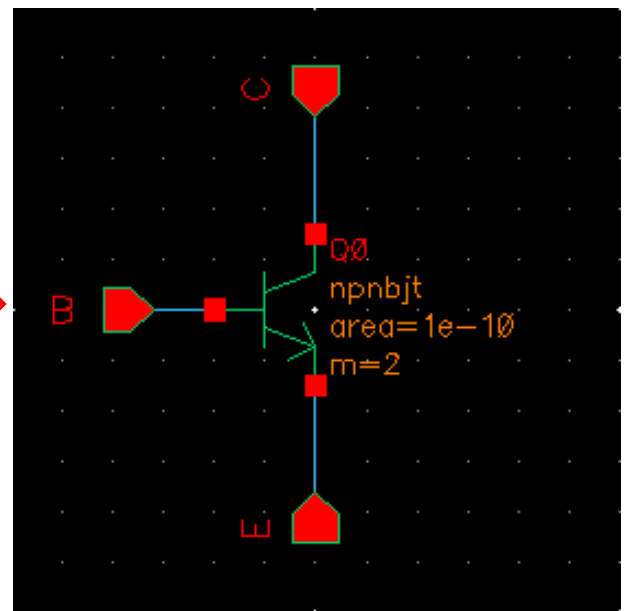
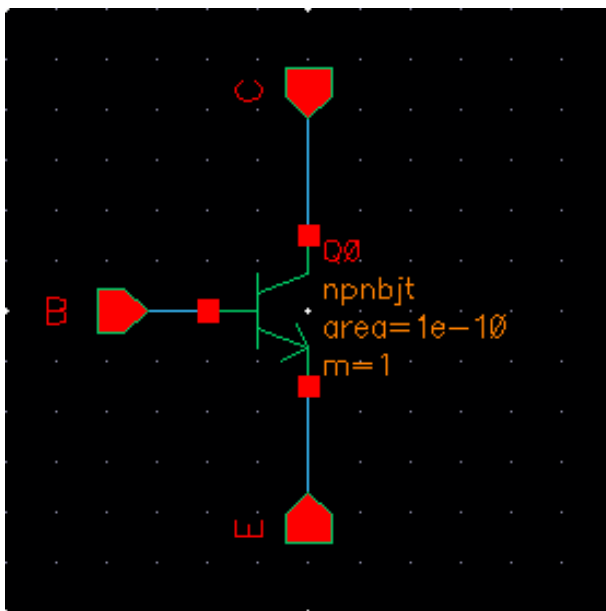
PTAP_ON_OFF	☐
-------------	--------------------------



**BJT layer on BJT device
can't be overlapped each other**

3.10 npn10 (Schematic)

Parameter			
Area	1e-10	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    npn10
* View Name:    schematic
*****
.SUBCKT npn10 B C E
*.PININFO B:I C:I E:0
QQ0 C B E npn10 m=1 area=1e-10
.ENDS
```



```
*****
* Library Name: ETRI_lay
* Cell Name:    npn10
* View Name:    schematic
*****
.SUBCKT npn10 B C E
*.PININFO B:I C:I E:0
QQ0 C B E npn10 m=2 area=1e-10
.ENDS
```

<cdl output>

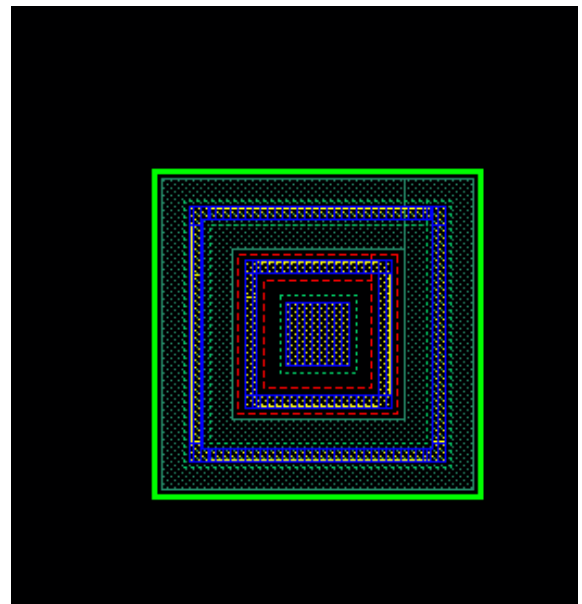
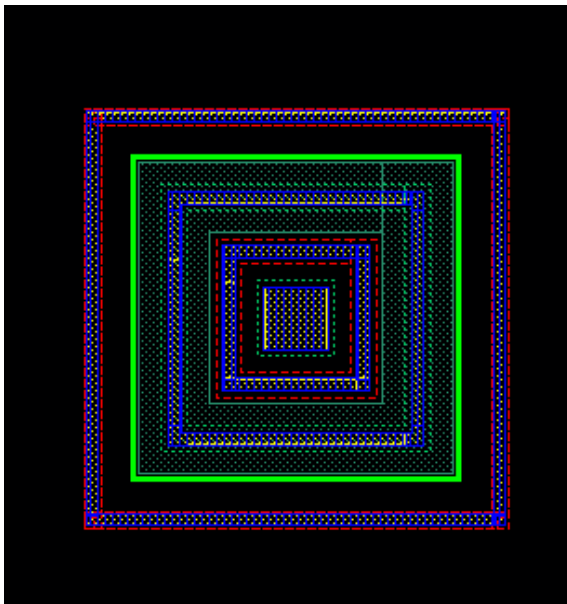
3.10 npn10 (Layout)

Parameter	Default	Change
PTAP_ON_OFF	☒	☐

BJT doesn't change PCELL
It must be used PCELL

PTAP_ON_OFF	☒
-------------	-------------------------------------

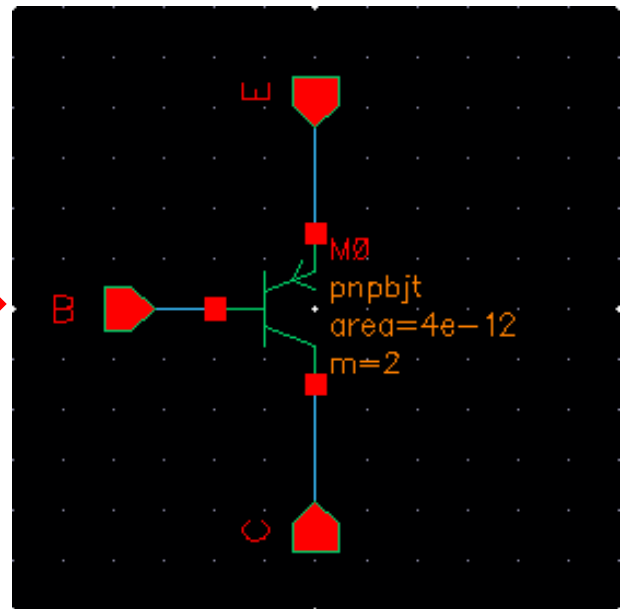
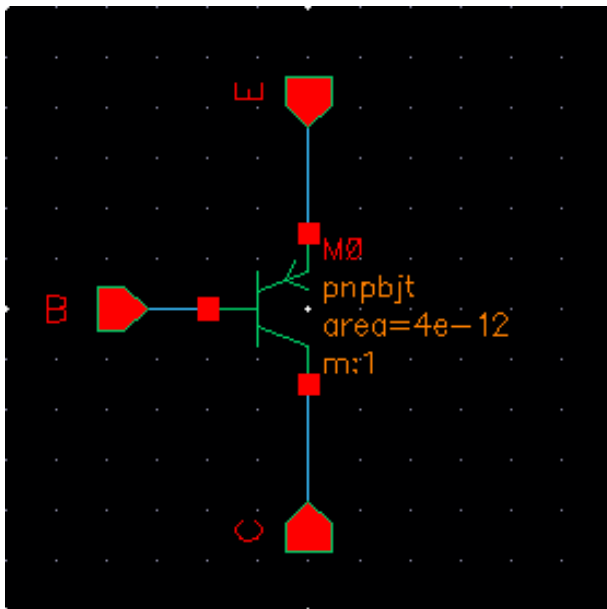
PTAP_ON_OFF	☐
-------------	--------------------------



**BJT layer on BJT device
can't be overlapped each other**

3.11 pnp2 (Schematic)

Parameter			
Area	4e-12	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    pnp2
* View Name:    schematic
*****
.SUBCKT pnp2 B C E
*.PININFO B:I C:I E:0
QM0 C B E pnpbjt m=1 area=4e-12
.ENDS
```

```
*****
* Library Name: ETRI_lay
* Cell Name:    pnp2
* View Name:    schematic
*****
.SUBCKT pnp2 B C E
*.PININFO B:I C:I E:0
QM0 C B E pnpbjt m=2 area=4e-12
.ENDS
```

<cdl output>

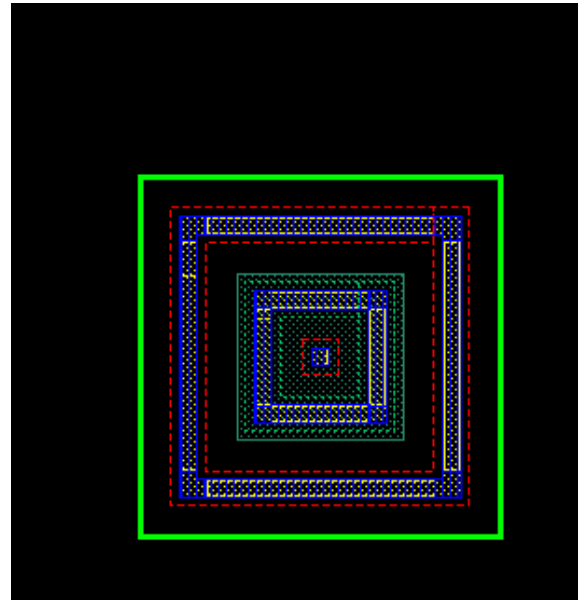
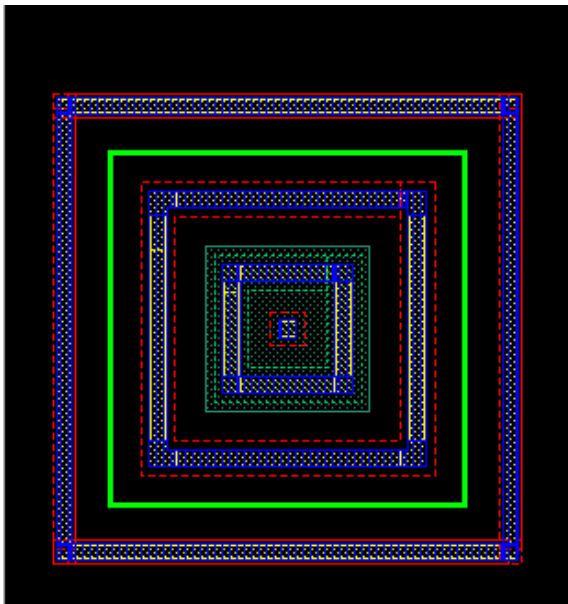
3.11 pnp2 (Layout)

Parameter	Default	Change
PTAP_ON_OFF	☒	☐

BJT doesn't change PCELL
It must be used PCELL

PTAP_ON_OFF	☒
-------------	-------------------------------------

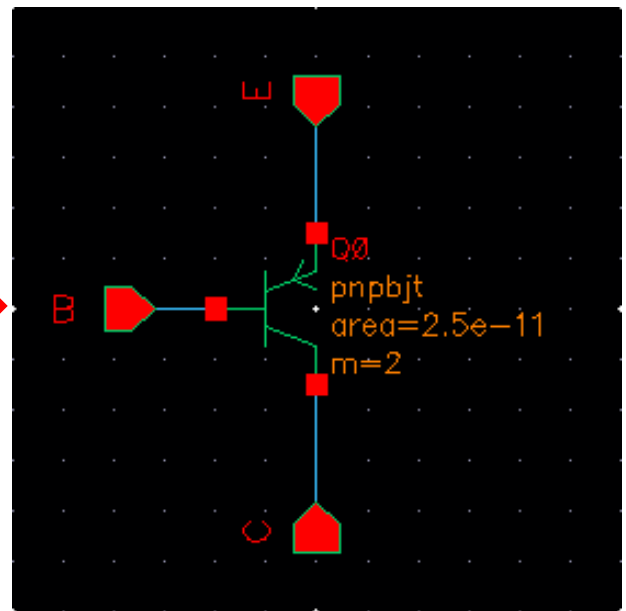
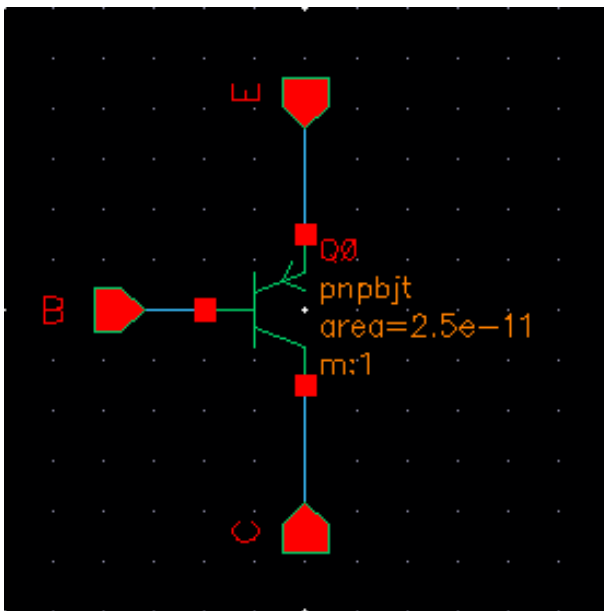
PTAP_ON_OFF	☐
-------------	--------------------------



**BJT layer on BJT device
can't be overlapped each other**

3.12 pnp5 (Schematic)

Parameter			
Area	2.5e-11	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    pnp5
* View Name:    schematic
*****
.SUBCKT pnp5 B C E
*.PININFO B:I C:I E:0
QQ0 C B E pnpbjt m=1 area=2.5e-11
.ENDS
```



```
*****
* Library Name: ETRI_lay
* Cell Name:    pnp5
* View Name:    schematic
*****
.SUBCKT pnp5 B C E
*.PININFO B:I C:I E:0
QQ0 C B E pnpbjt m=2 area=2.5e-11
.ENDS
```

<cdl output>

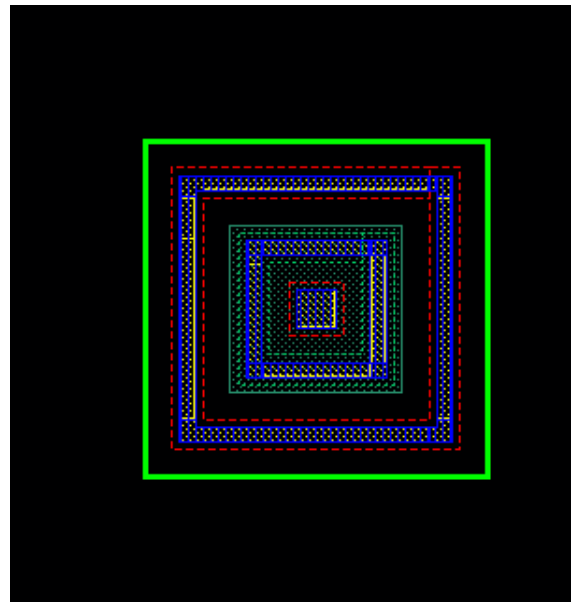
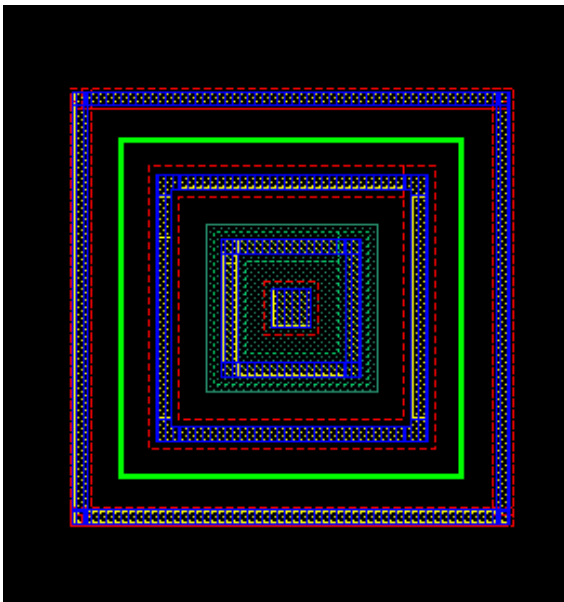
3.12 pnp5 (Layout)

Parameter	Default	Change
PTAP_ON_OFF	☒	☐

BJT doesn't change PCELL
It must be used PCELL

PTAP_ON_OFF	☒
-------------	-------------------------------------

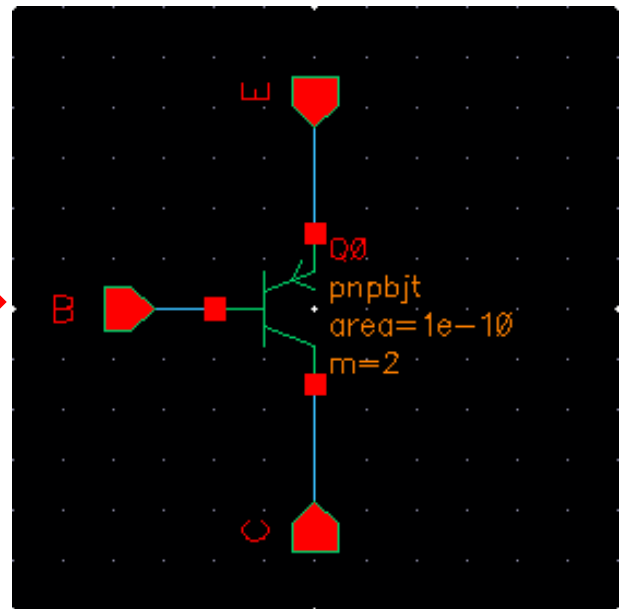
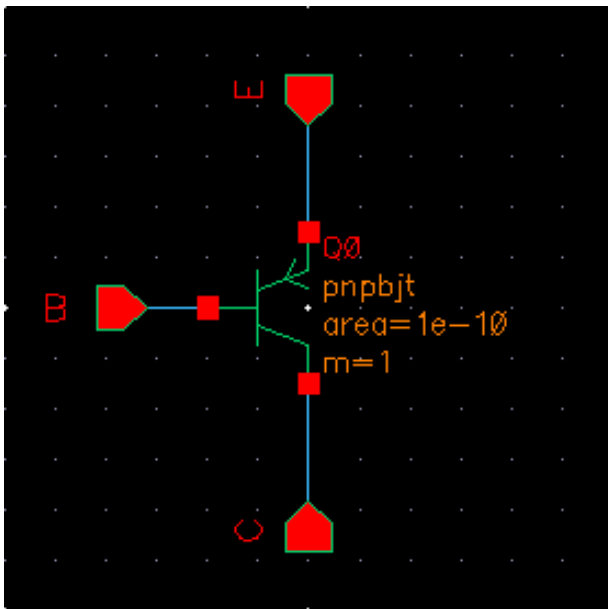
PTAP_ON_OFF	☐
-------------	--------------------------



**BJT layer on BJT device
can't be overlapped each other**

3.13 pnp10 (Schematic)

Parameter			
Area	1e-10	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    pnp10
* View Name:    schematic
*****
.SUBCKT pnp10 B C E
*.PININFO B:I C:I E:0
QQ0 C B E pnpbjt m=1 area=1e-10
.ENDS
```



```
*****
* Library Name: ETRI_lay
* Cell Name:    pnp10
* View Name:    schematic
*****
.SUBCKT pnp10 B C E
*.PININFO B:I C:I E:0
QQ0 C B E pnpbjt m=2 area=1e-10
.ENDS
```

<cdl output>

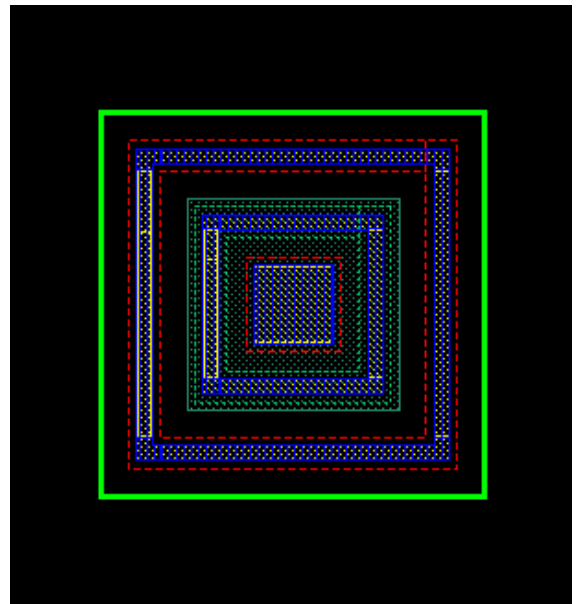
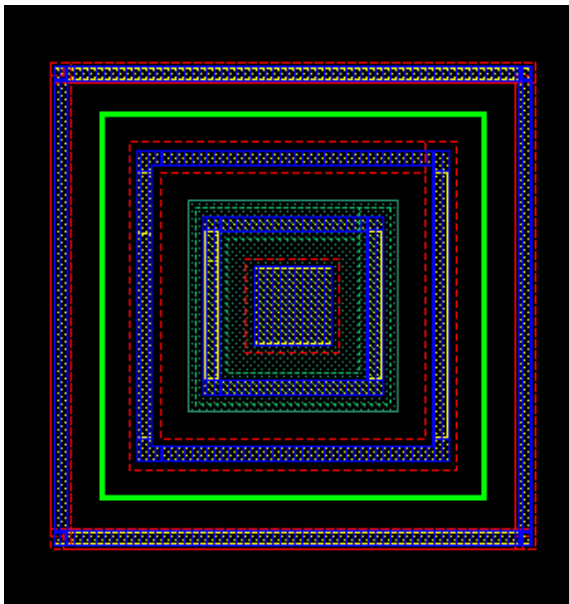
3.13 pnp10 (Layout)

Parameter	Default	Change
PTAP_ON_OFF	☒	☐

BJT doesn't change PCELL
It must be used PCELL

PTAP_ON_OFF	☒
-------------	-------------------------------------

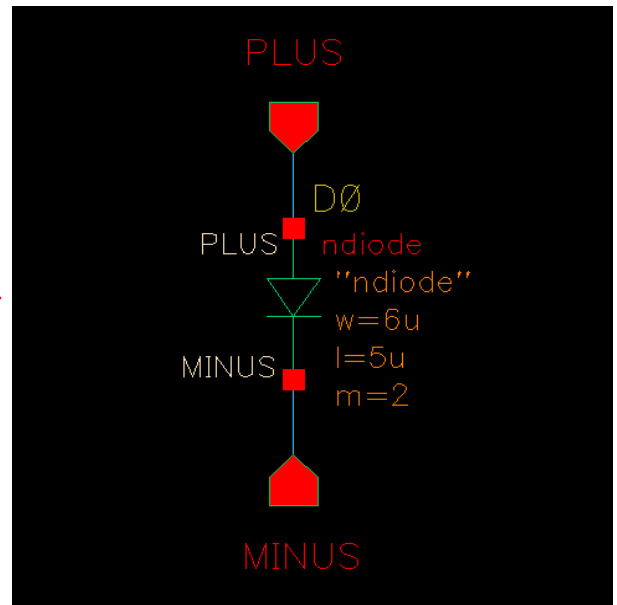
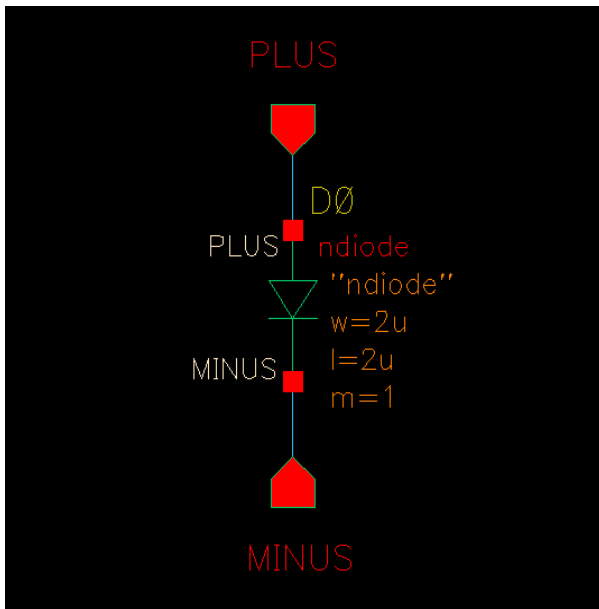
PTAP_ON_OFF	☐
-------------	--------------------------



**BJT layer on BJT device
can't be overlapped each other**

3.14 ndiode (Schematic)

Parameter					
Width	2.0u → 6u	Length	2.0u → 5u	multiplier	1 → 2



<Schematic parameter>

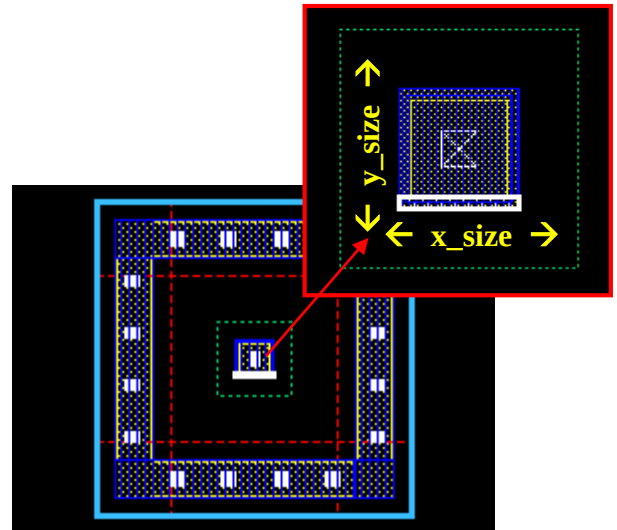
```
*****
* Library Name: ETRI lay
* Cell Name:   ndiode
* View Name:   schematic
*****
.SUBCKT ndiode MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS ndiode m=1 w=2u l=2u
.ENDS
```

```
*****
* Library Name: ETRI_lay
* Cell Name:   ndiode
* View Name:   schematic
*****
.SUBCKT ndiode MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS ndiode m=2 w=6u l=5u
.ENDS
```

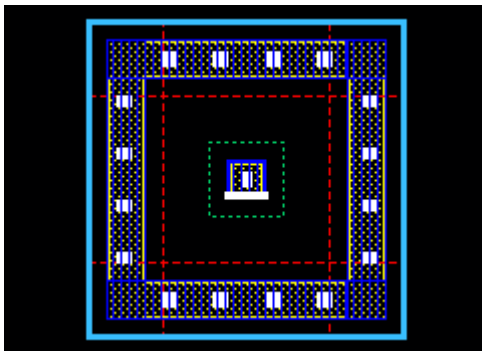
<cdl output>

3.14 ndiode (Layout)

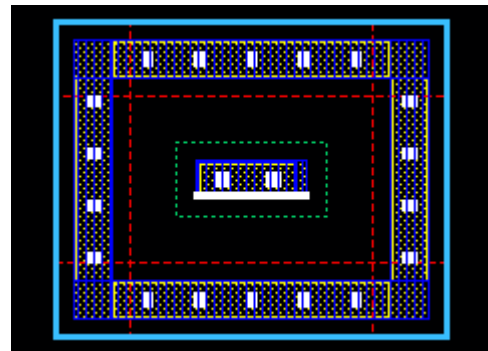
Parameter	Default	Change
x_size	2	value
y_size	2	value



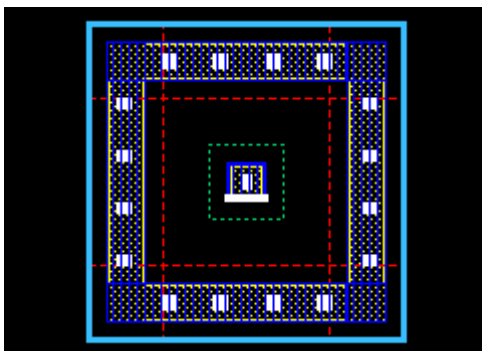
x_size	2
--------	---



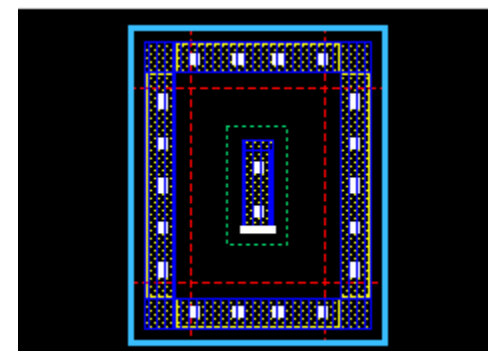
x_size	6
--------	---



y_size	2
--------	---

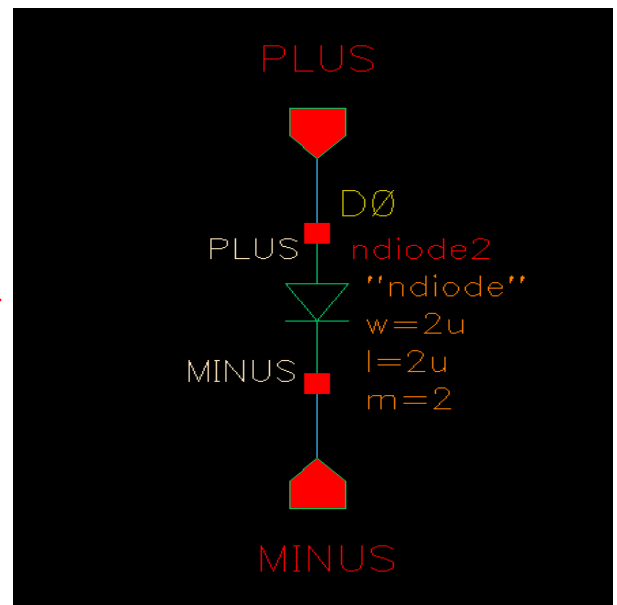
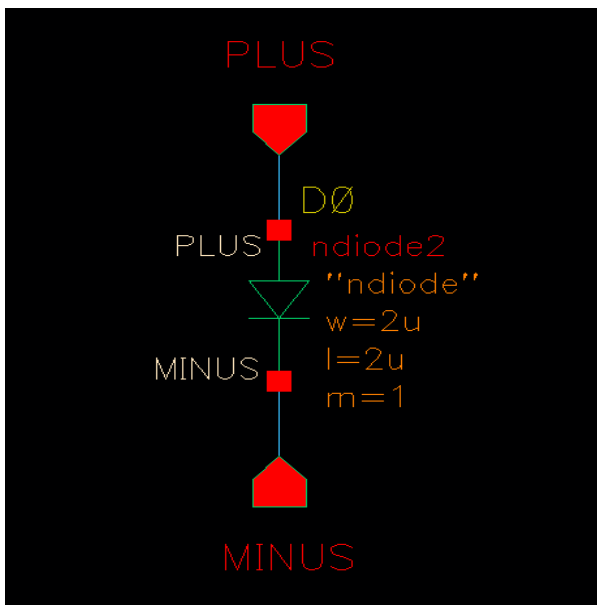


y_size	6
--------	---



3.15 ndiode2 (Schematic)

Parameter					
Width	2.0u	Length	2.0u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:   ndiode2
* View Name:   schematic
*****
.SUBCKT ndiode2 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS ndiode m=1 w=2u l=2u
.ENDS
```

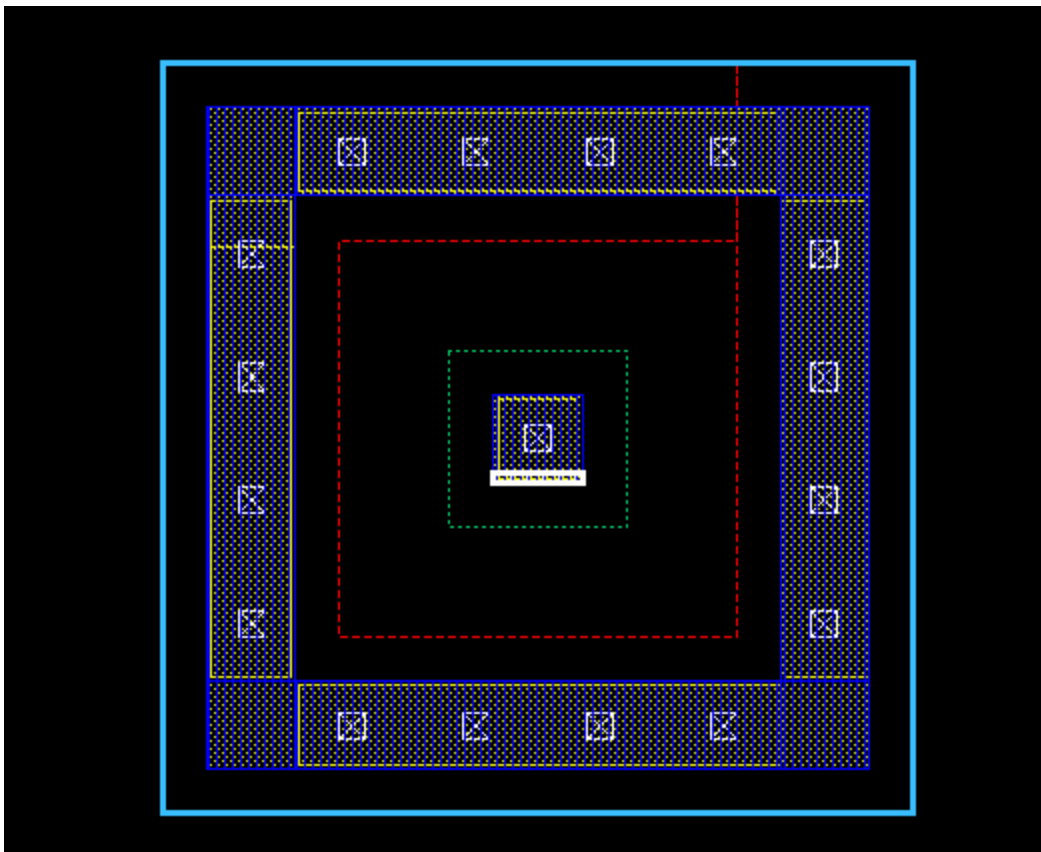


```
*****
* Library Name: ETRI_lay
* Cell Name:   ndiode2
* View Name:   schematic
*****
.SUBCKT ndiode2 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS ndiode m=2 w=2u l=2u
.ENDS
```

<cdl output>

3.15 ndiode2 (Layout)

Parameter	Default	Change
-	-	-

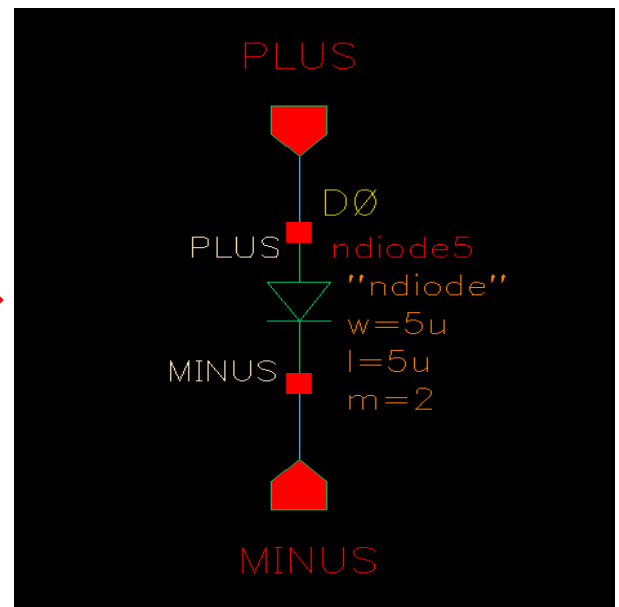
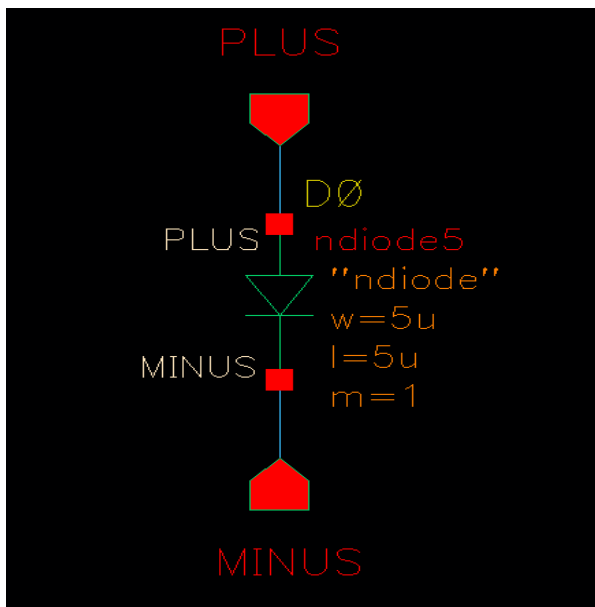


NDIODE Fixed Size

**→ doesn't provide parameter
modification**

3.16 ndiode5 (Schematic)

Parameter					
Width	5.0u	Length	5.0u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:    ndiode5
* View Name:    schematic
*****
.SUBCKT ndiode5 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DDØ PLUS MINUS ndiode m=1 w=5u l=5u
.ENDS
```

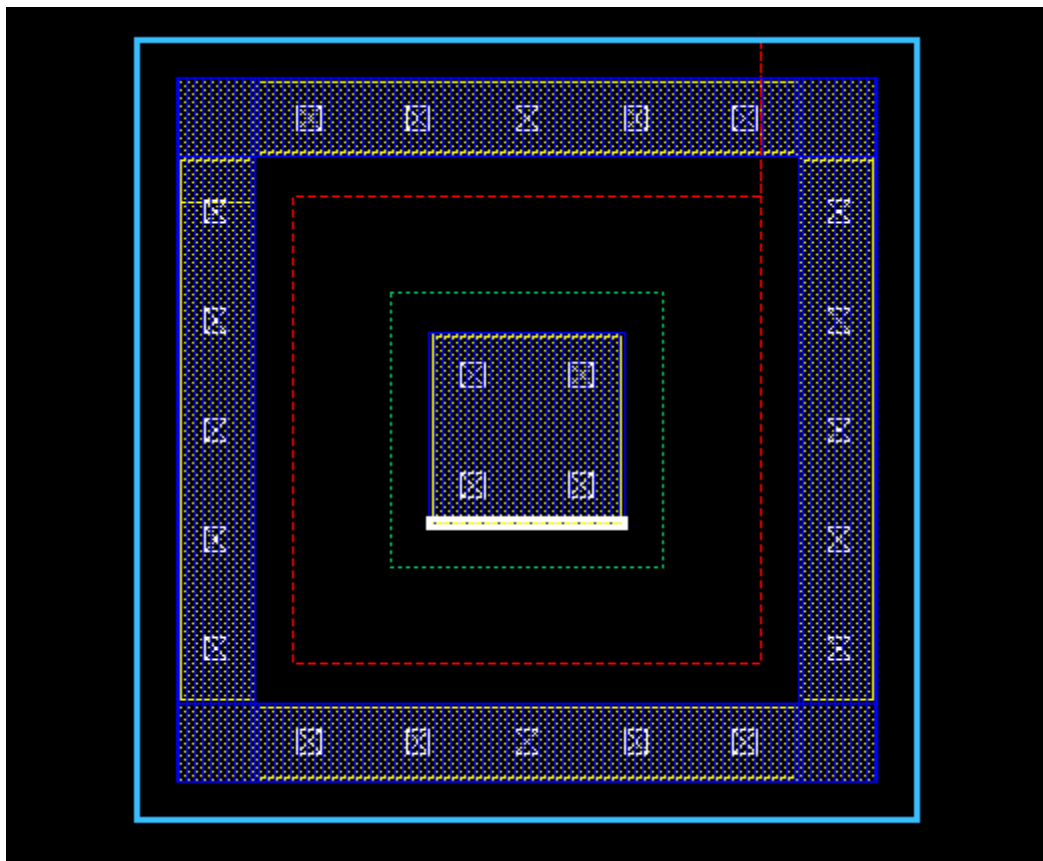


```
*****
* Library Name: ETRI_lay
* Cell Name:    ndiode5
* View Name:    schematic
*****
.SUBCKT ndiode5 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DDØ PLUS MINUS ndiode m=2 w=5u l=5u
.ENDS
```

<cdl output>

3.16 ndiode5 (Layout)

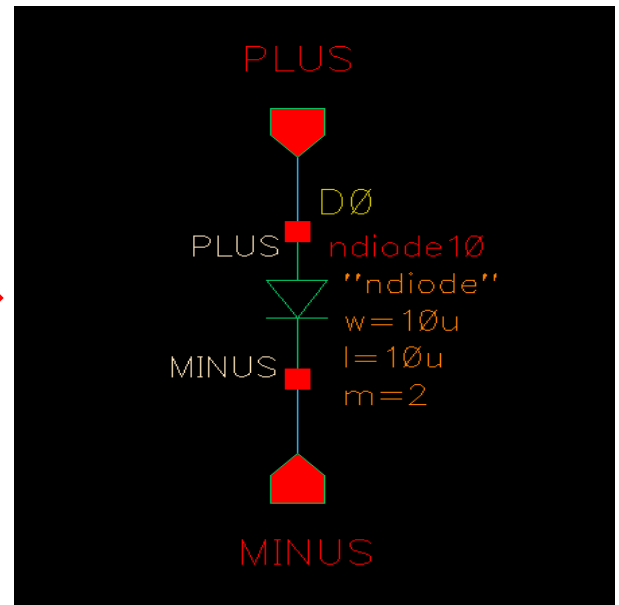
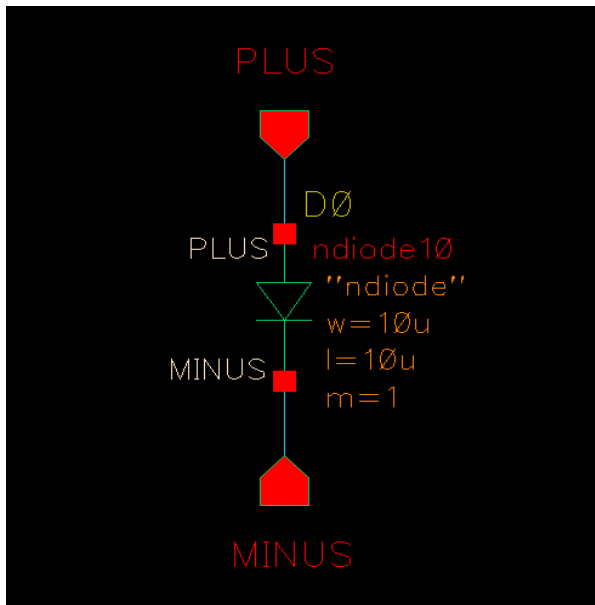
Parameter	Default	Change
-	-	-



NDIODE Fixed Size
→ doesn't provide parameter
modification

3.17 ndiode10 (Schematic)

Parameter					
Width	10.0u	Length	10.0u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI lay
* Cell Name:    ndiode10
* View Name:    schematic
*****
.SUBCKT ndiode10 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DDØ PLUS MINUS ndiode m=1 w=10u l=10u
.ENDS
```

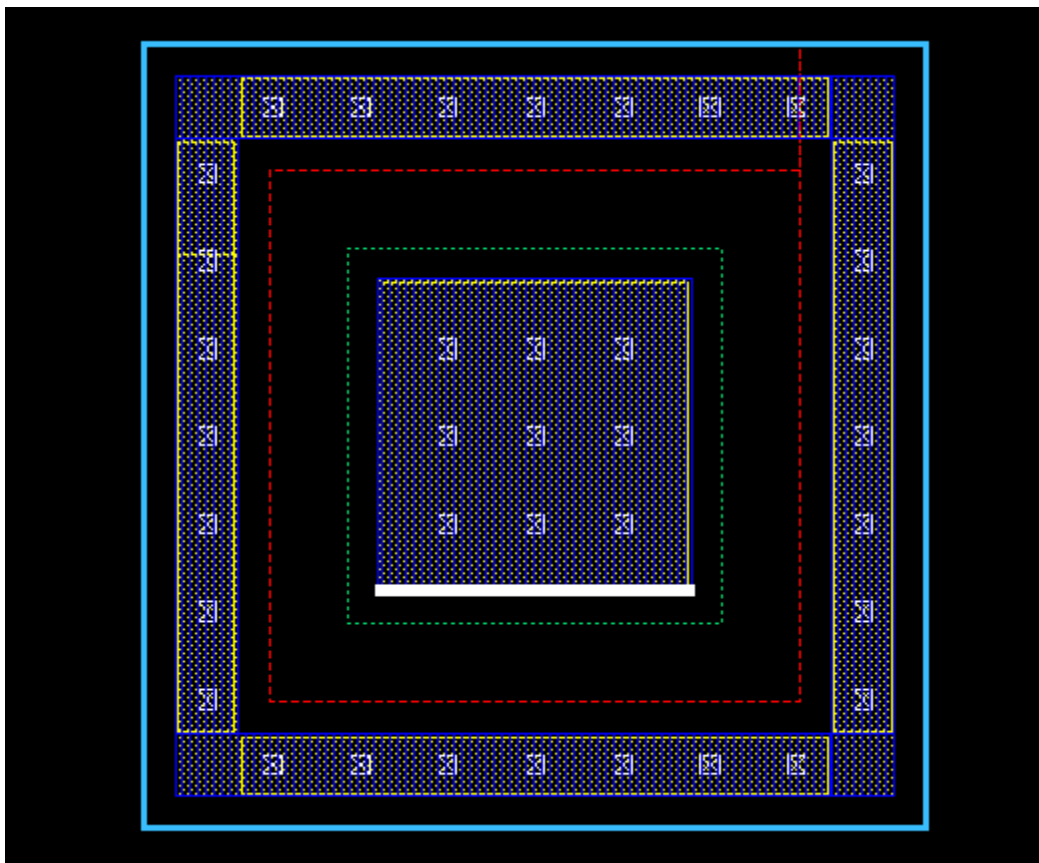


```
*****
* Library Name: ETRI ay
* Cell Name:    ndiode10
* View Name:    schematic
*****
.SUBCKT ndiode10 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DDØ PLUS MINUS ndiode m=2 w=10u l=10u
.ENDS
```

<cdl output>

3.17 ndiode10 (Layout)

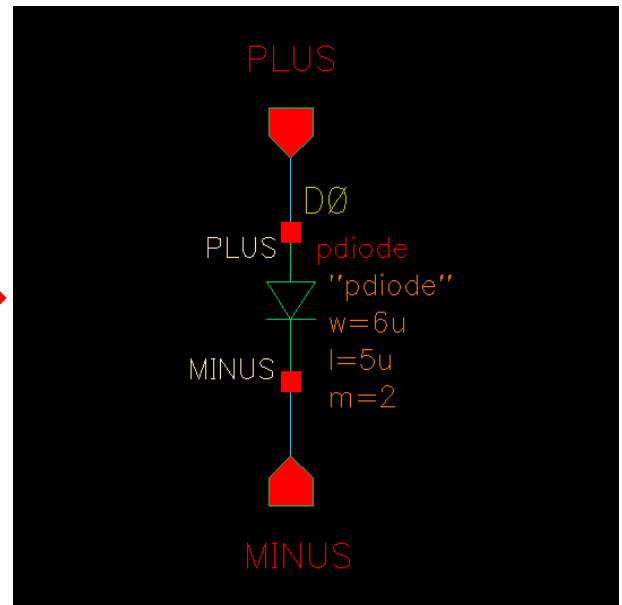
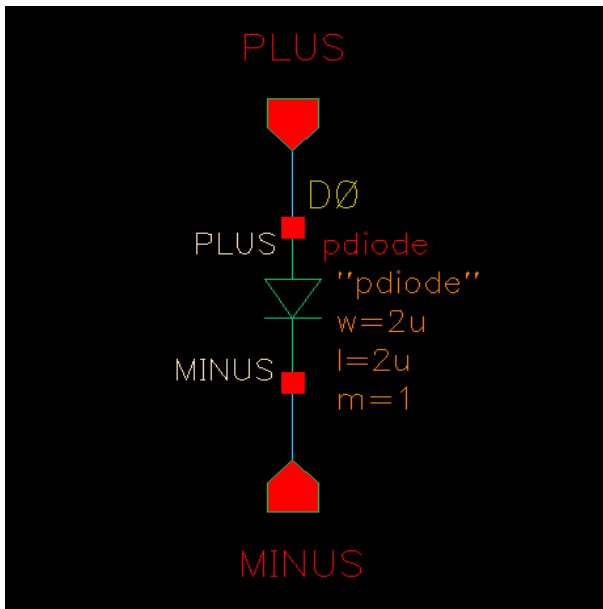
Parameter	Default	Change
-	-	-



NDIODE Fixed Size
→ doesn't provide parameter
modification

3.18 pdiode (Schematic)

Parameter					
Width	2.0u → 6u	Length	2.0u → 5u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI lay
* Cell Name:   pdiode
* View Name:   schematic
*****
.SUBCKT pdiode MINUS PLUS
*.PININFO MINUS:I PLUS:I
DDØ PLUS MINUS pdiode m=1 w=2u l=2u
.ENDS
```

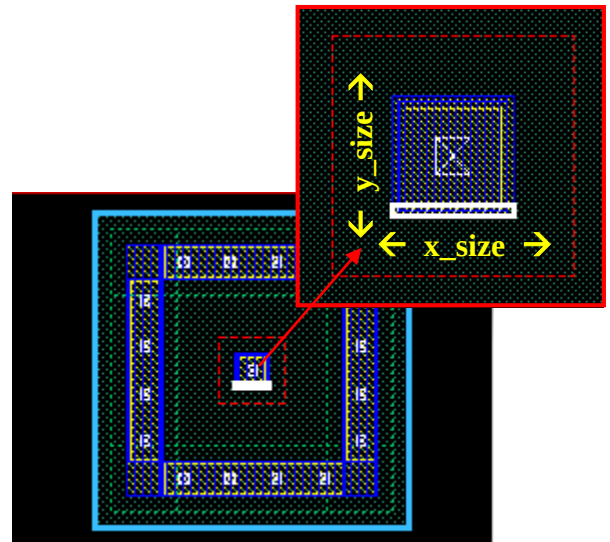


```
*****
* Library Name: ETRI lay
* Cell Name:   pdiode
* View Name:   schematic
*****
.SUBCKT pdiode MINUS PLUS
*.PININFO MINUS:I PLUS:I
DDØ PLUS MINUS pdiode m=2 w=6u l=5u
.ENDS
```

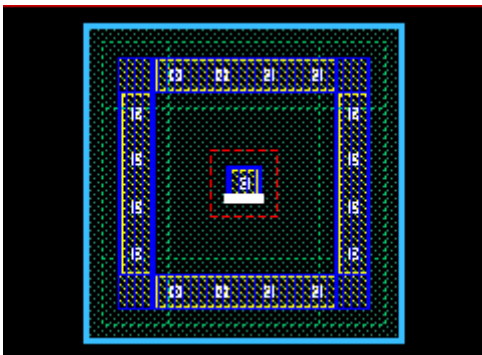
<cdl output>

3.18 pdiode (Layout)

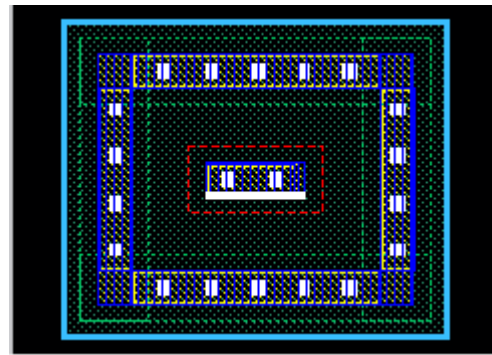
Parameter	Default	Change
x_size	2	value
y_size	2	value



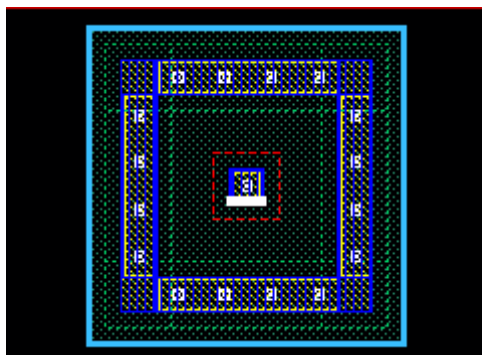
x_size	2
--------	---



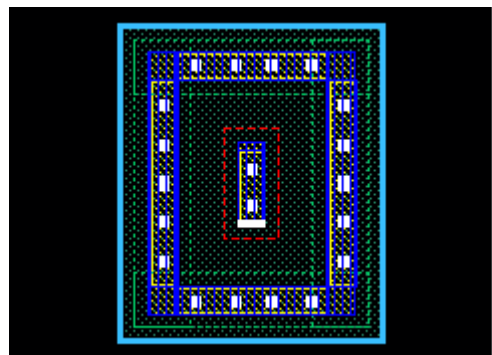
x_size	6
--------	---



y_size	2
--------	---

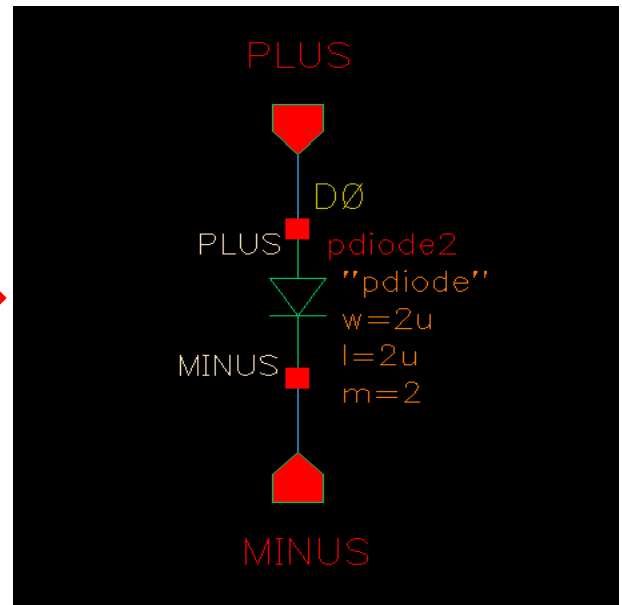
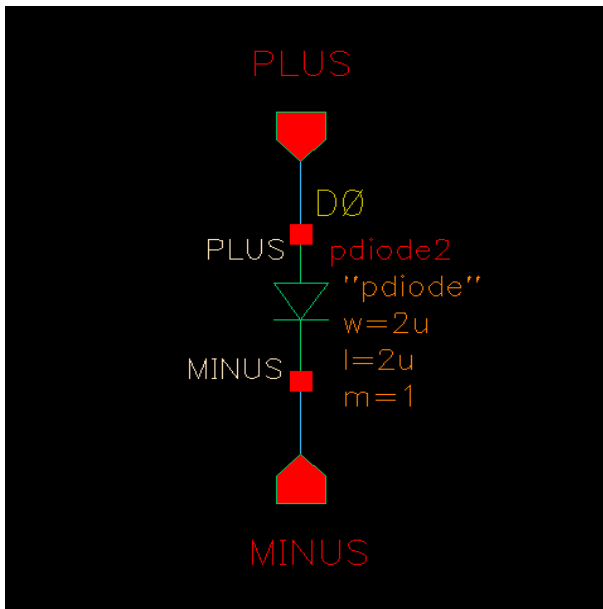


y_size	6
--------	---



3.19 pdiode2 (Schematic)

Parameter					
Width	2.0u	Length	2.0u	multiplier	1 → 2



<Schematic parameter>

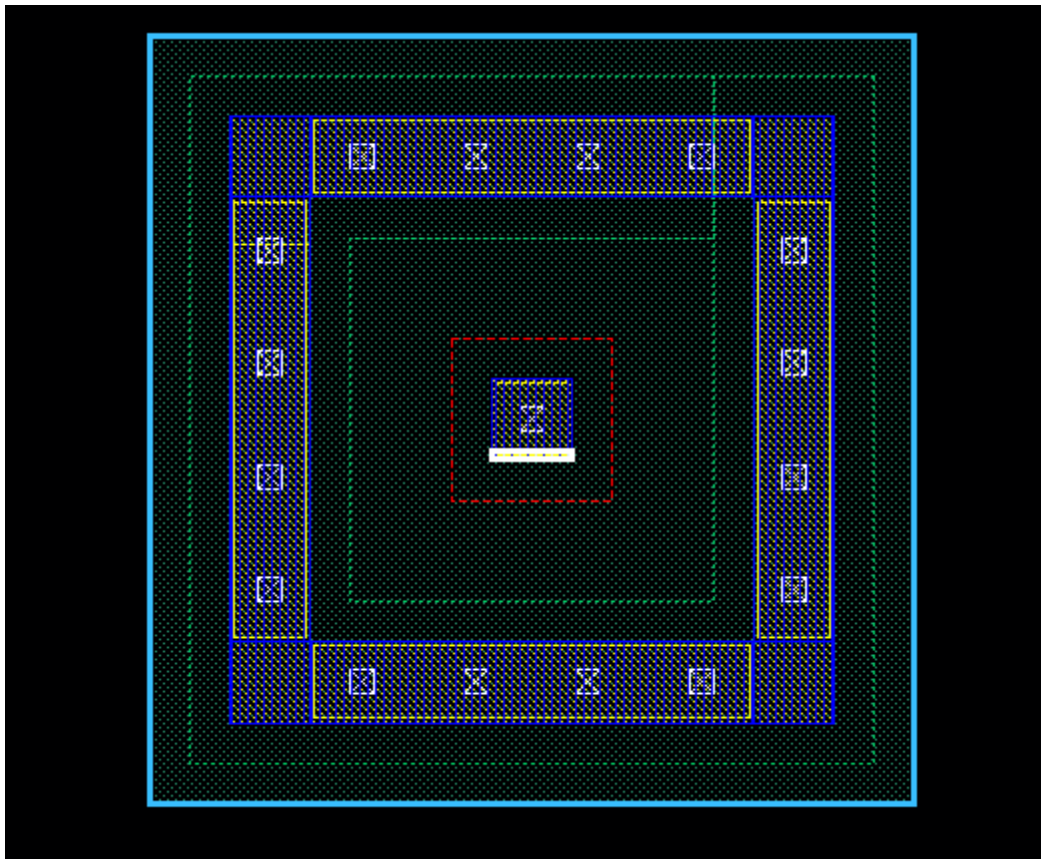
```
*****
* Library Name: ETRI lay
* Cell Name:   pdiode2
* View Name:   schematic
*****
.SUBCKT pdiode2 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS pdiode m=1 w=2u l=2u
.ENDS
```

```
*****
* Library Name: ETRI lay
* Cell Name:   pdiode2
* View Name:   schematic
*****
.SUBCKT pdiode2 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS pdiode m=2 w=2u l=2u
.ENDS
```

<cdl output>

3.19 pdiode2 (Layout)

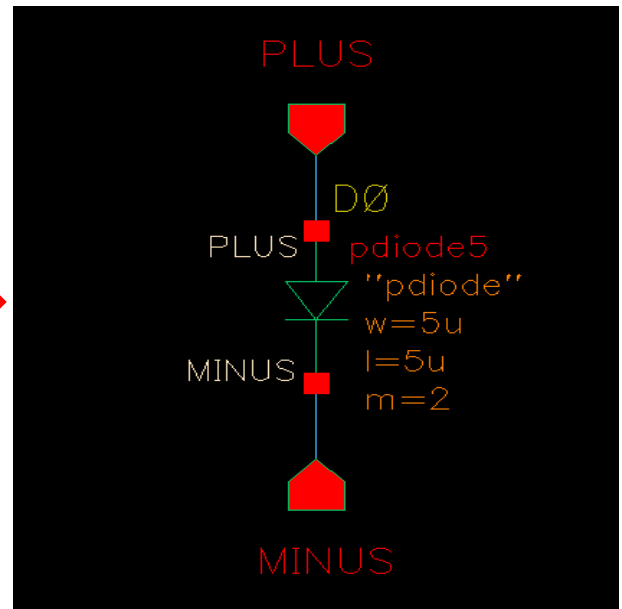
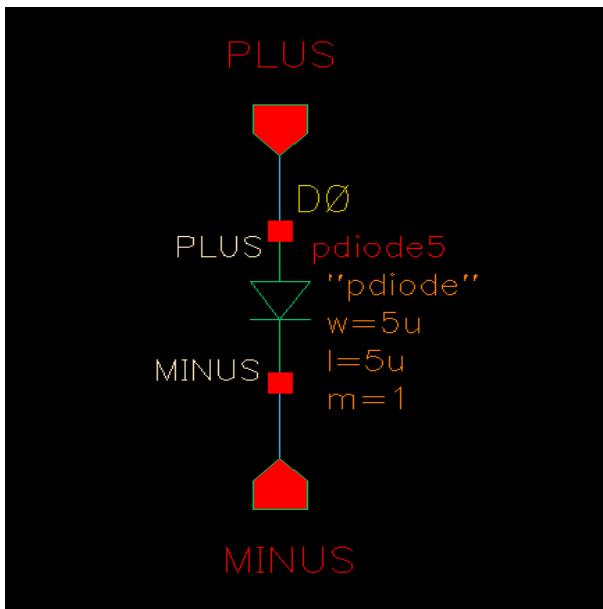
Parameter	Default	Change
-	-	-



PDIODE Fixed Size
→ doesn't provide parameter
modification

3.20 pdiode5 (Schematic)

Parameter					
Width	5.0u	Length	5.0u	multiplier	1 → 2



<Schematic parameter>

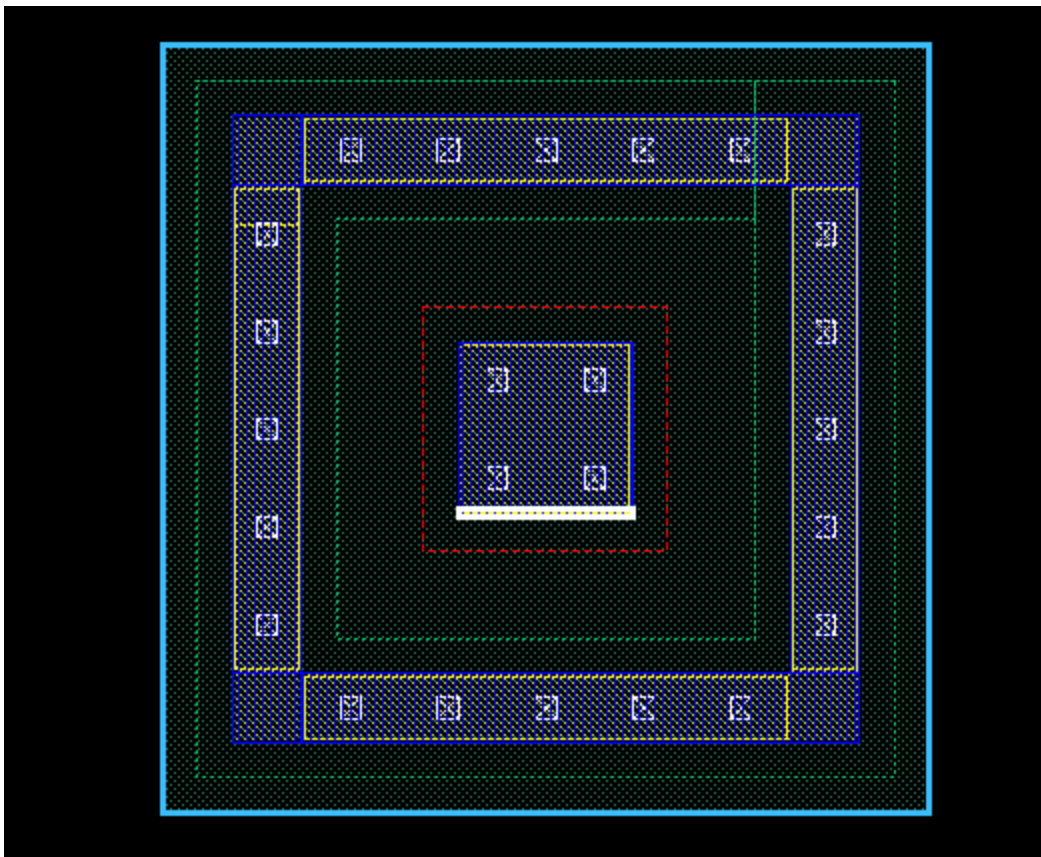
```
*****
* Library Name: ETRI_lay
* Cell Name:   pdiode5
* View Name:   schematic
*****
.SUBCKT pdiode5 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS pdiode m=1 w=5u l=5u
.ENDS
```

```
*****
* Library Name: ETRI_lay
* Cell Name:   pdiode5
* View Name:   schematic
*****
.SUBCKT pdiode5 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS pdiode m=2 w=5u l=5u
.ENDS
```

<cdl output>

3.20 pdiode5 (Layout)

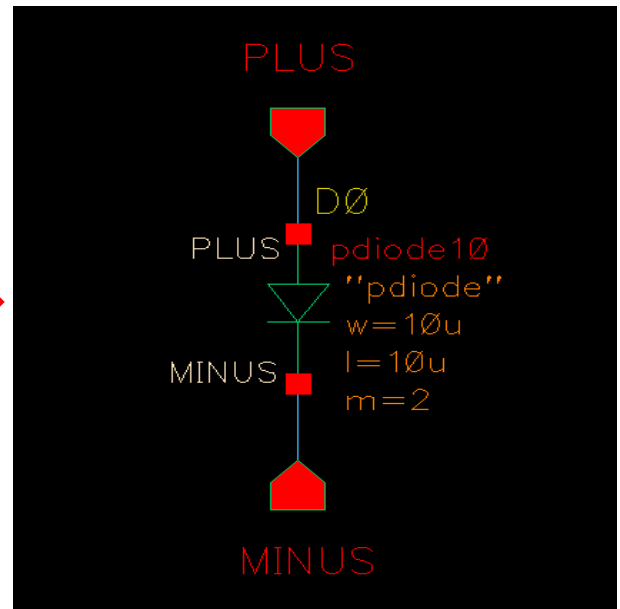
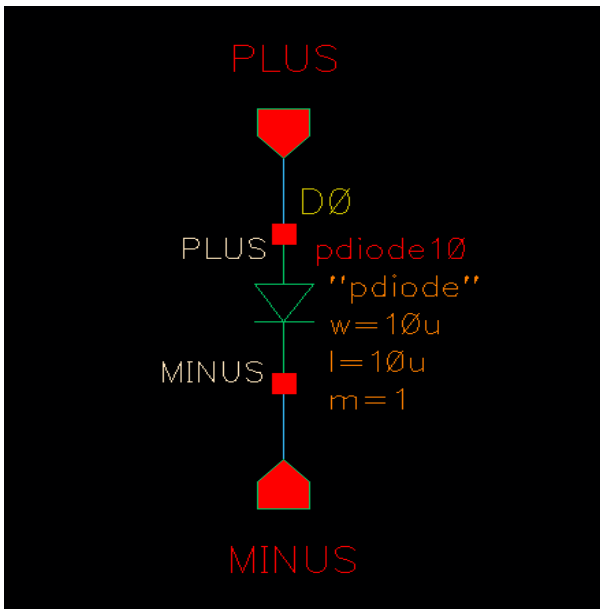
Parameter	Default	Change
-	-	-



PDIODE Fixed Size
→ doesn't provide parameter
modification

3.21 pdiode10 (Schematic)

Parameter					
Width	10.0u	Length	10.0u	multiplier	1 → 2



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:   pdiode10
* View Name:   schematic
*****
.SUBCKT pdiode10 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS pdiode m=1 w=10u l=10u
.ENDS
```

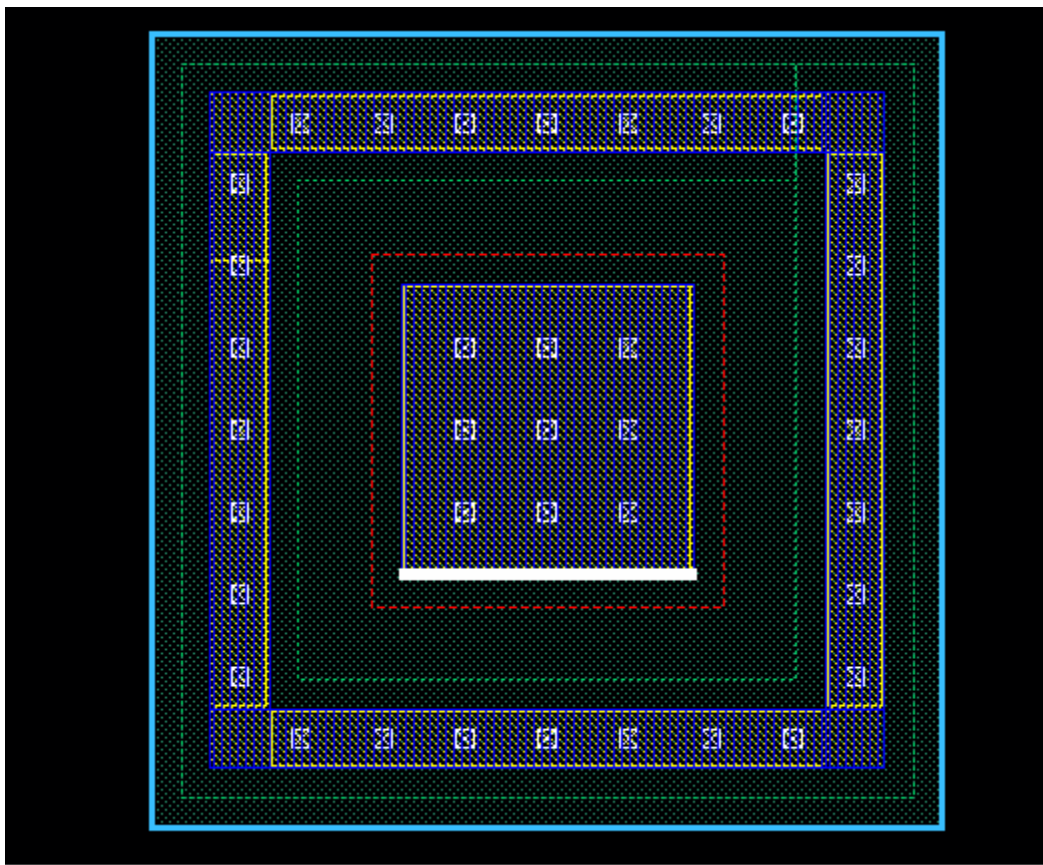


```
*****
* Library Name: ETRI_lay
* Cell Name:   pdiode10
* View Name:   schematic
*****
.SUBCKT pdiode10 MINUS PLUS
*.PININFO MINUS:I PLUS:I
DD0 PLUS MINUS pdiode m=2 w=10u l=10u
.ENDS
```

<cdl output>

3.21 pdiode10 (Layout)

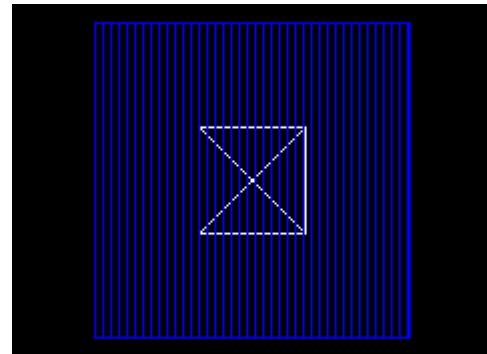
Parameter	Default	Change
-	-	-



PDIODE Fixed Size
→ doesn't provide parameter
modification

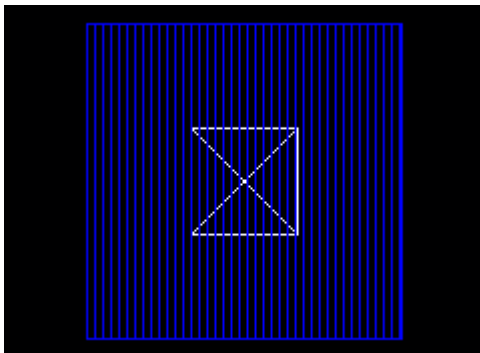
3.22 cont (Layout)

Parameter	Default	Change
x_size	1.8	<i>value</i>
y_size	1.8	<i>value</i>

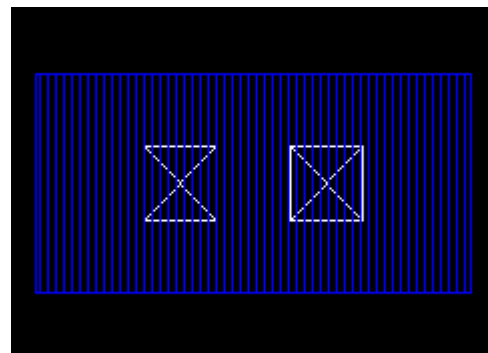


Size Criterion : MET1 Layer

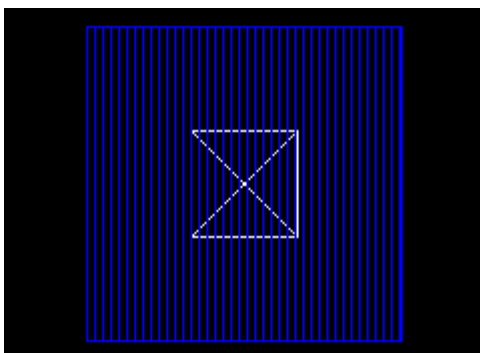
x_size	1.8
--------	-----



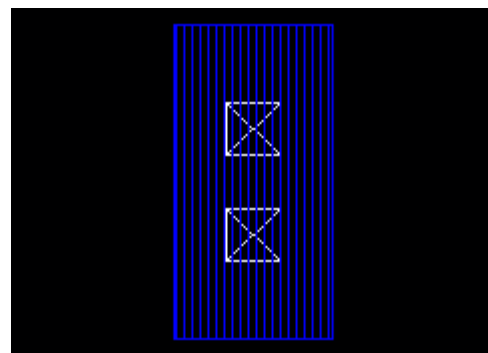
x_size	3.6
--------	-----



y_size	1.8
--------	-----

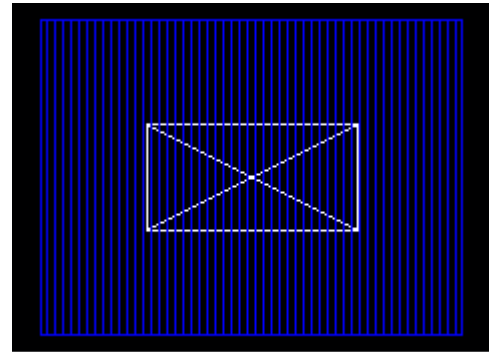


y_size	3.6
--------	-----



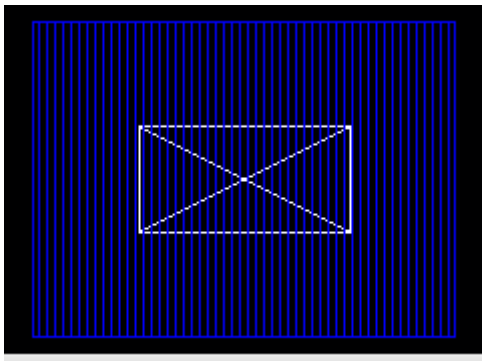
3.23 cont_bar (Layout)

Parameter	Default	Change
x_size	2.4	value
y_size	1.8	value

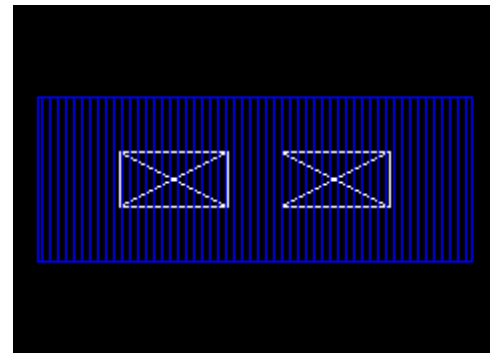


Size Criterion : MET1 Layer

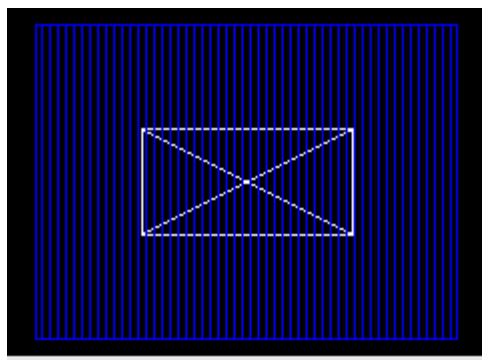
x_size	2.4
--------	-----



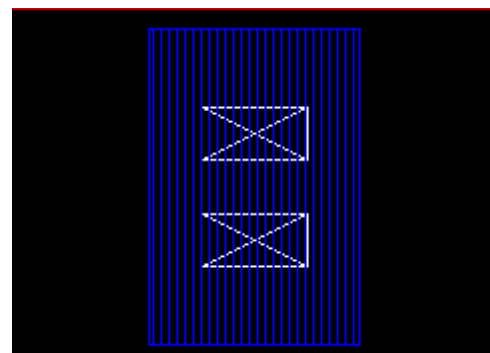
x_size	4.8
--------	-----



y_size	1.8
--------	-----

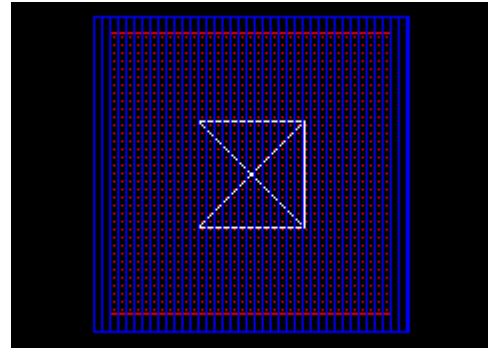


y_size	3.6
--------	-----



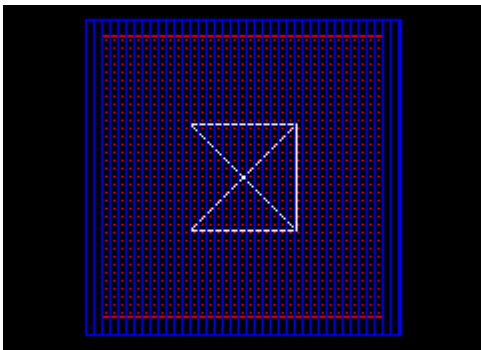
3.24 poly1cont (Layout)

Parameter	Default	Change
x_size	1.6	value
y_size	1.6	value

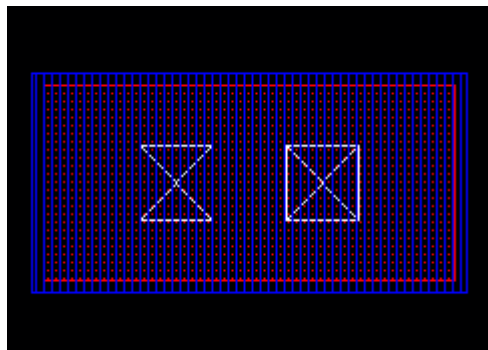


Size Criterion : POLY1 Layer

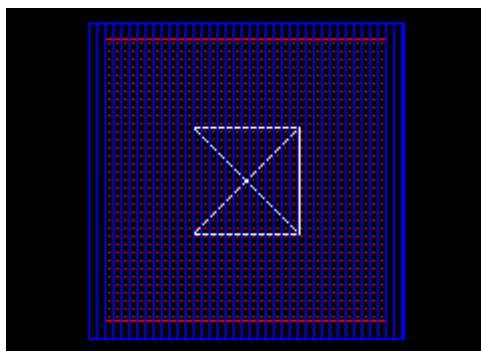
x_size	1.6
--------	-----



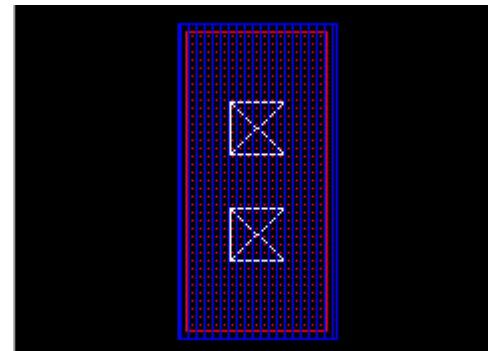
x_size	3.2
--------	-----



y_size	1.6
--------	-----

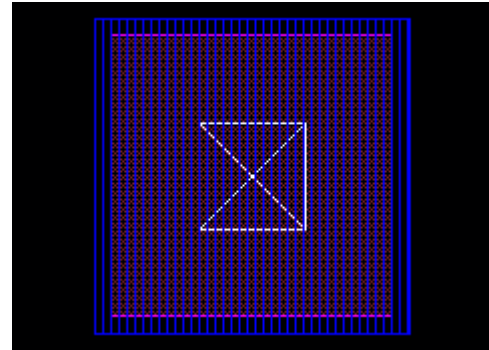


y_size	3.2
--------	-----



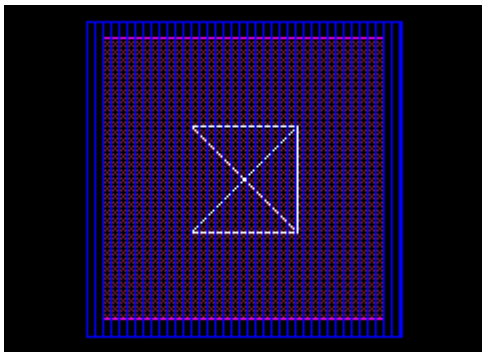
3.25 poly2cont (Layout)

Parameter	Default	Change
x_size	1.6	<i>value</i>
y_size	1.6	<i>value</i>

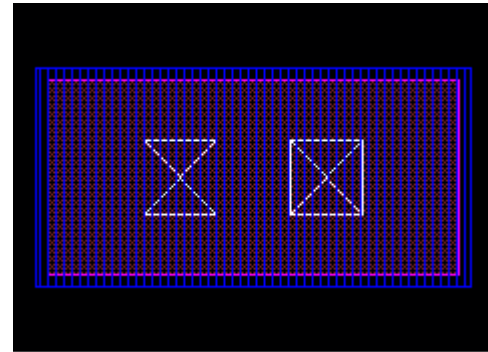


Size Criterion : POLY2 Layer

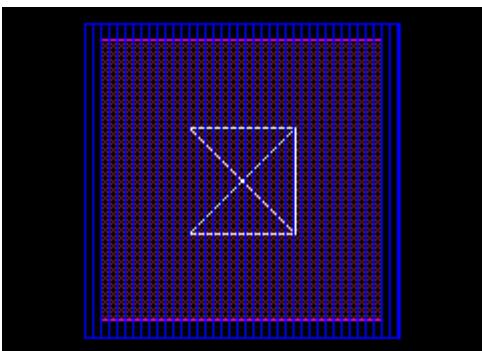
x_size	1.6
--------	-----



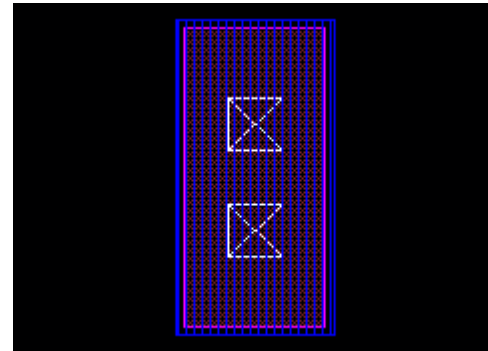
x_size	3.2
--------	-----



y_size	1.6
--------	-----

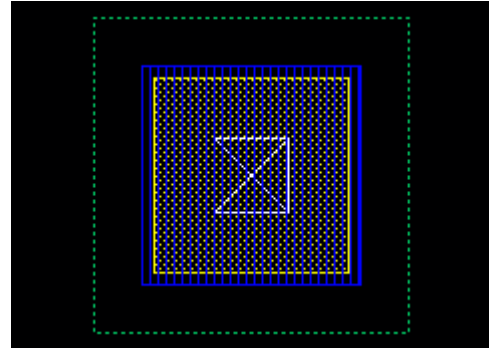


y_size	3.2
--------	-----



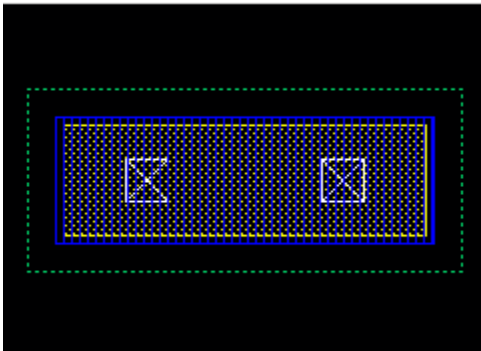
3.26 ntap (Layout)

Parameter	Default	Change
x_size	1.6	value
y_size	1.6	value
NSD	☒	☐

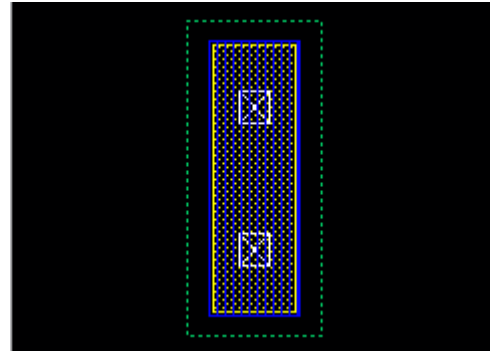


Size Criterion : ACT Layer

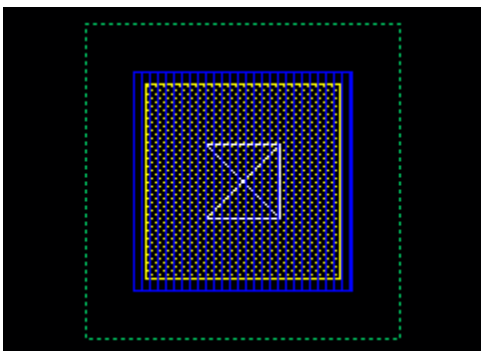
x_size	1.6	4.8
--------	-----	-----



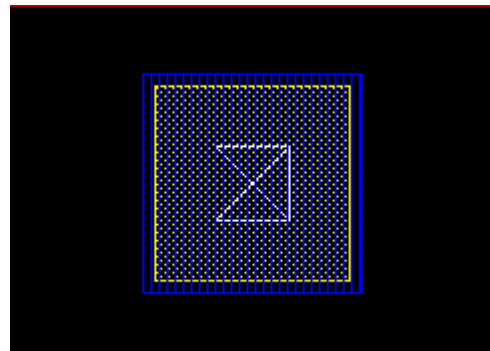
y_size	1.6	4.8
--------	-----	-----



NSD	☒
-----	-------------------------------------

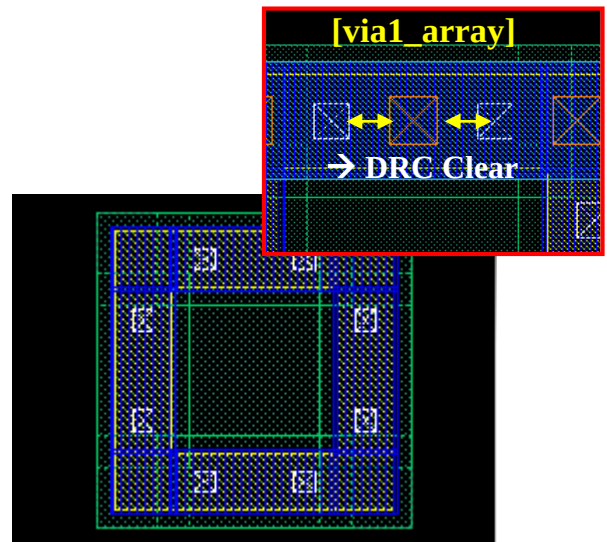


NSD	☐
-----	--------------------------



3.27 ntap_ring (Layout)

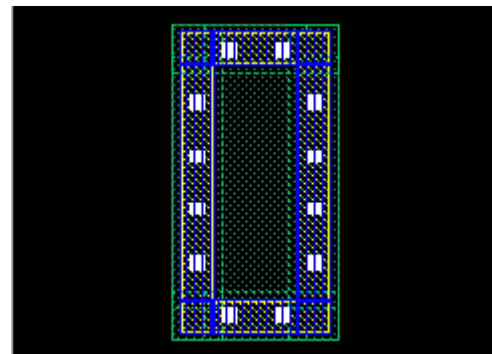
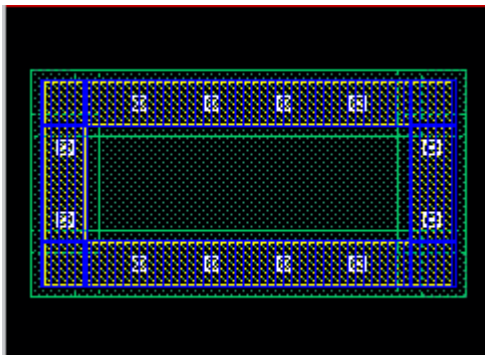
Parameter	Default	Change
x_size	7.8	value
y_size	7.8	value
TOP	☒	☐
BOTTOM	☒	☐
LEFT	☒	☐
RIGHT	☒	☐



Size Criterion : ACT Layer

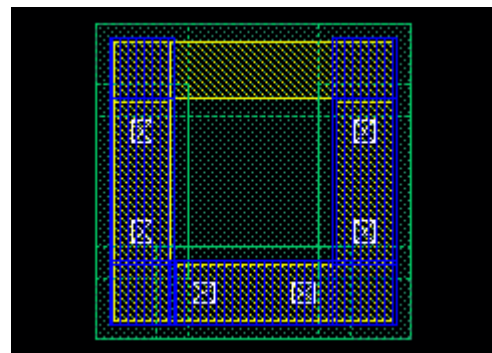
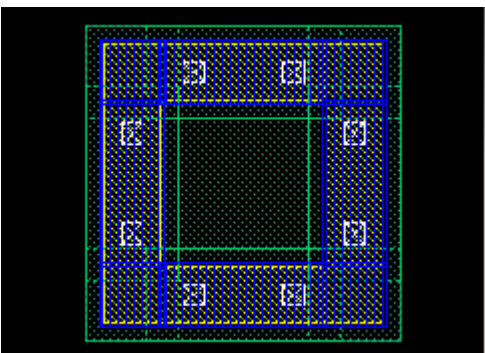
x_size	7.8	15.6
--------	-----	------

y_size	7.8	15.6
--------	-----	------



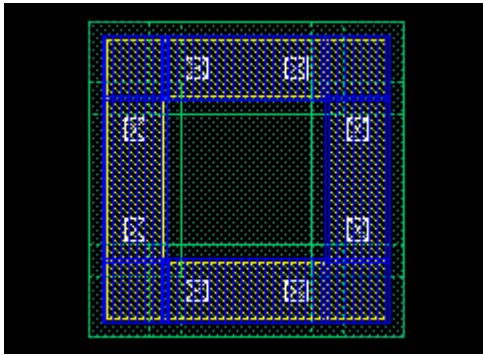
TOP	☒
-----	-------------------------------------

TOP	☐
-----	--------------------------

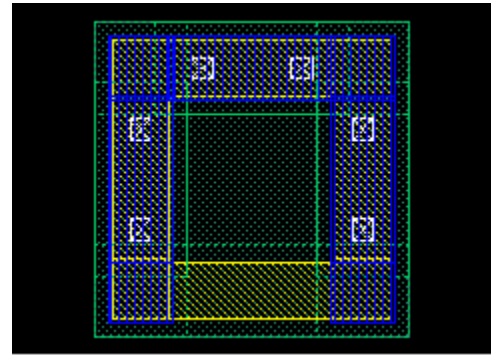


3.27 ntap_ring (Layout)

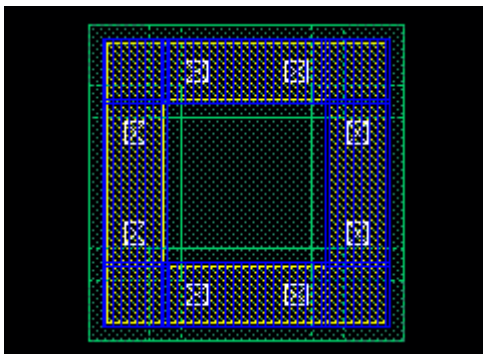
BOTTOM	☒
--------	-------------------------------------



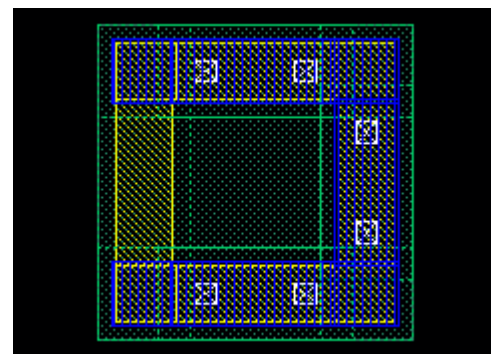
BOTTOM	☐
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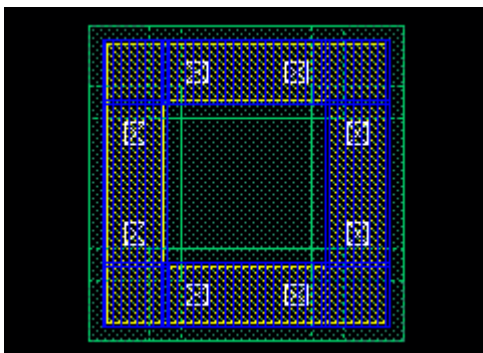
LEFT	☒
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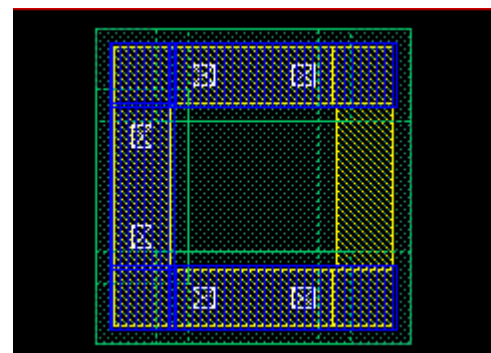
LEFT	☐
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RIGHT	☒
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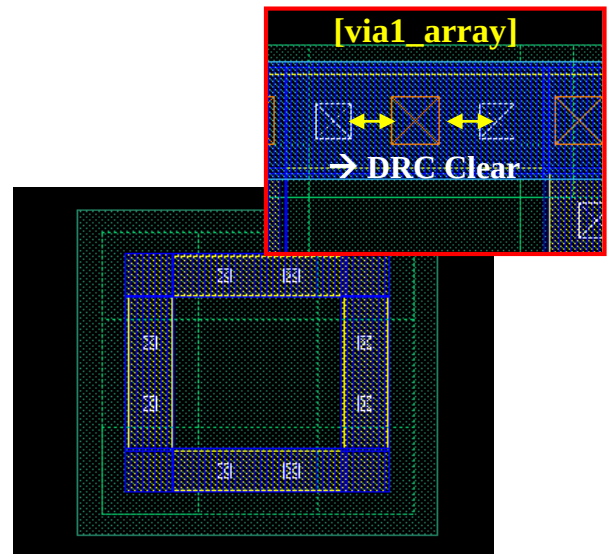


RIGHT	☐
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3.28 ntap_ring_pdiode (Layout)

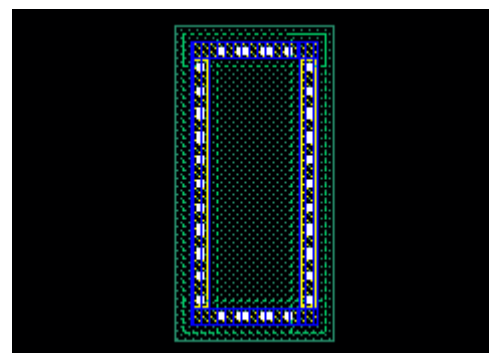
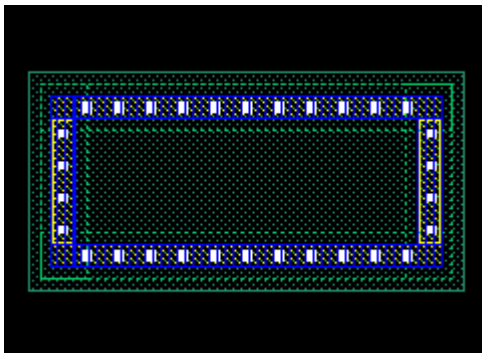
Parameter	Default	Change
x_size	15	value
y_size	15	value
TOP	☒	☐
BOTTOM	☒	☐
LEFT	☒	☐
RIGHT	☒	☐



Size Criterion : Cell Boundary

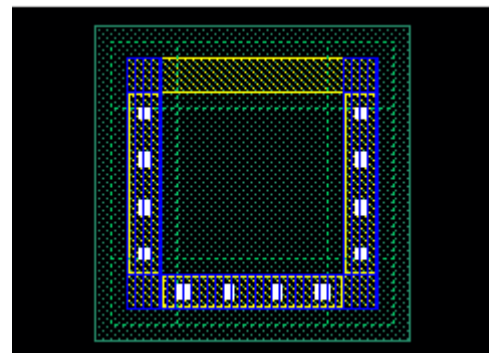
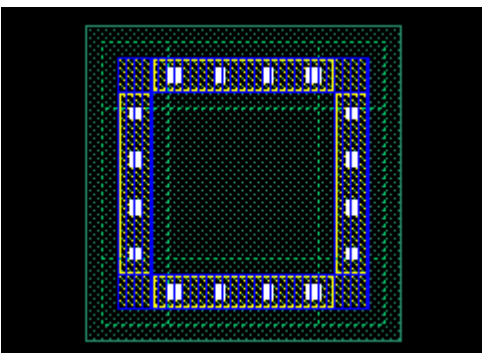
x_size	15	30
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y_size	15	30
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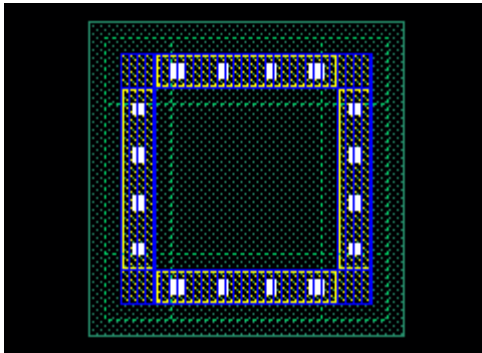
TOP	☒
-----	-------------------------------------

TOP	☐
-----	--------------------------

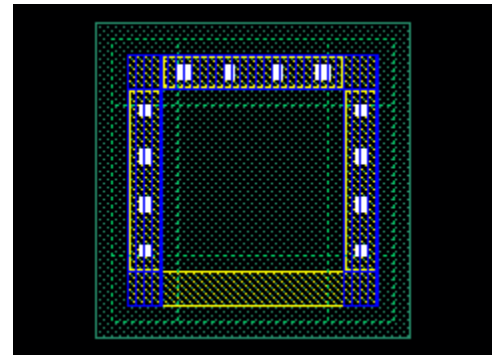


3.28 ntap_ring_pdiode (Layout)

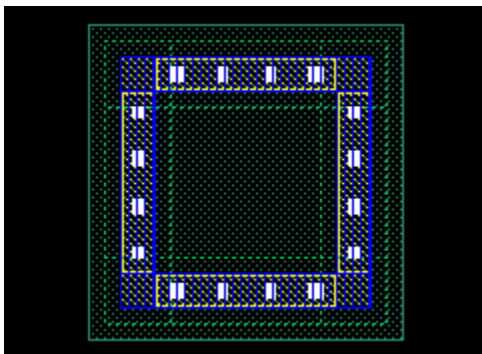
BOTTOM	☒
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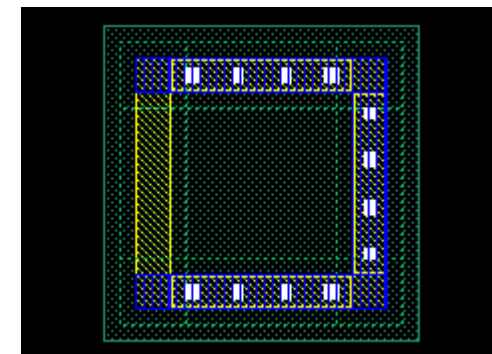
BOTTOM	☐
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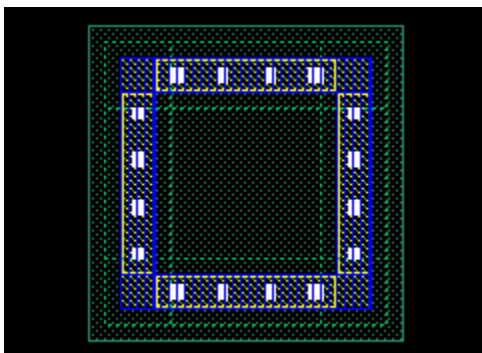
LEFT	☒
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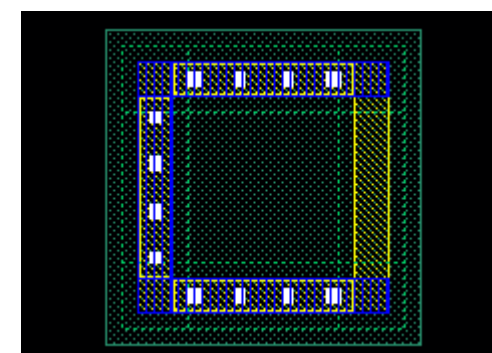
LEFT	☐
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RIGHT	☒
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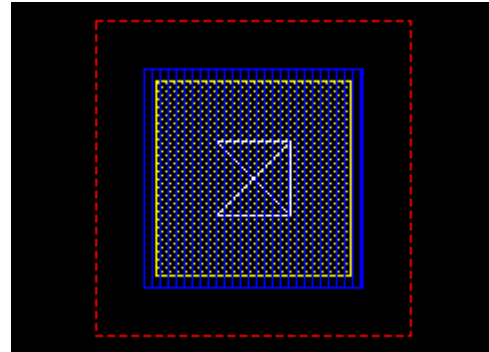


RIGHT	☐
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3.29 ptap (Layout)

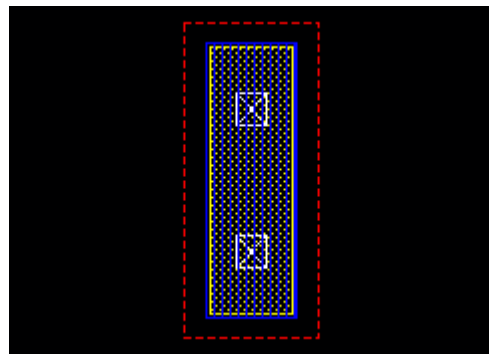
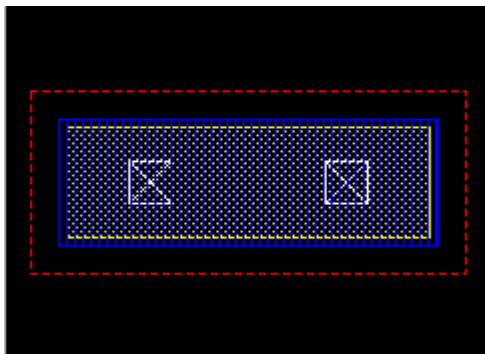
Parameter	Default	Change
x_size	1.6	value
y_size	1.6	value
PSD	☒	☐



Size Criterion : ACT Layer

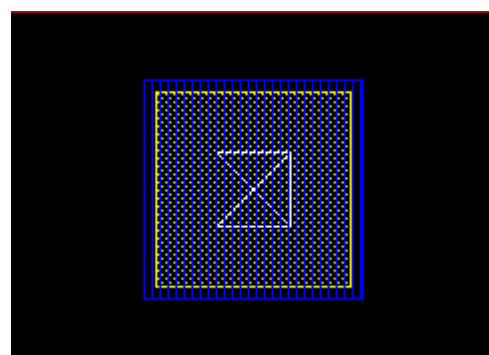
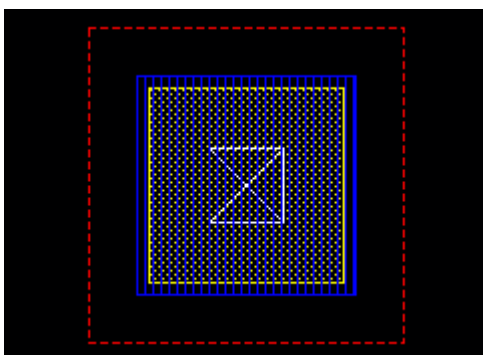
x_size	1.6	4.8
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y_size	1.6	4.8
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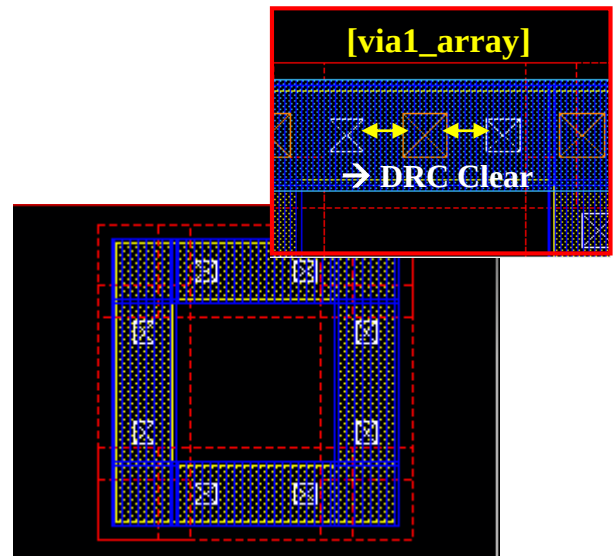
PSD	☒
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PSD	☐
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3.30 ptap_ring (Layout)

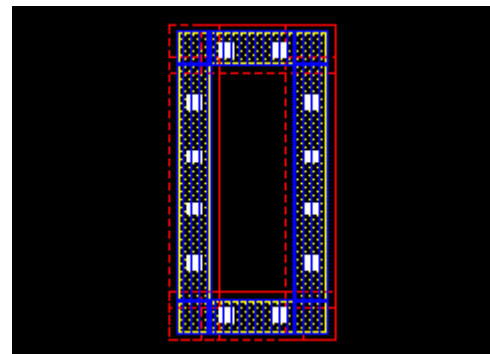
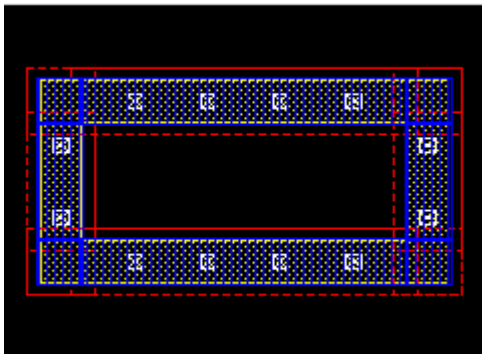
Parameter	Default	Change
x_size	7.8	value
y_size	7.8	value
TOP	☒	☐
BOTTOM	☒	☐
LEFT	☒	☐
RIGHT	☒	☐



Size Criterion : ACT Layer

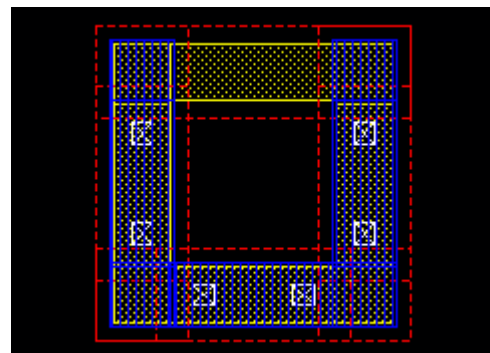
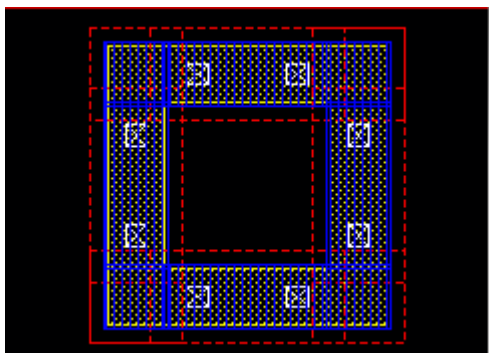
x_size	7.8	15.6
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y_size	7.8	15.6
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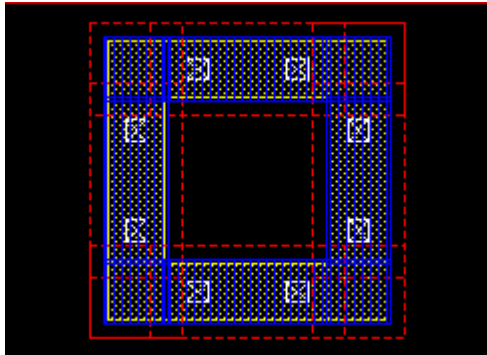
TOP	☒
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TOP	☐
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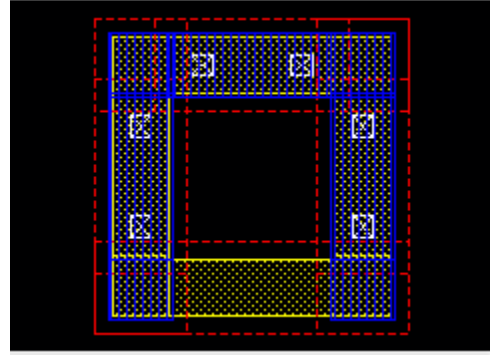


3.30 ptap_ring (Layout)

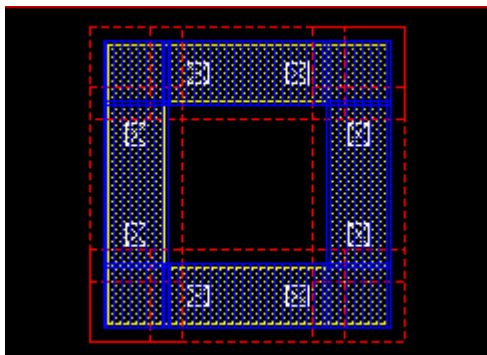
BOTTOM	☒
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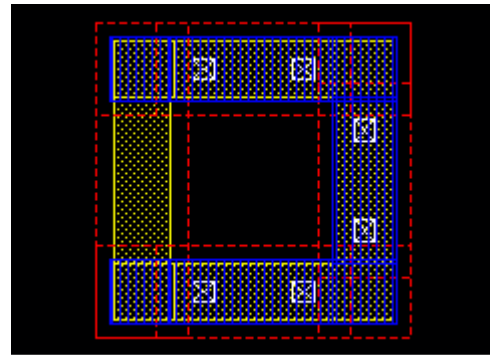
BOTTOM	☐
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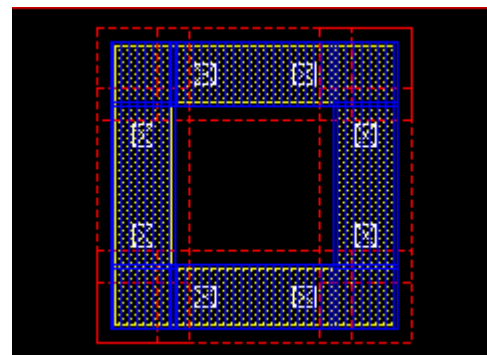
LEFT	☒
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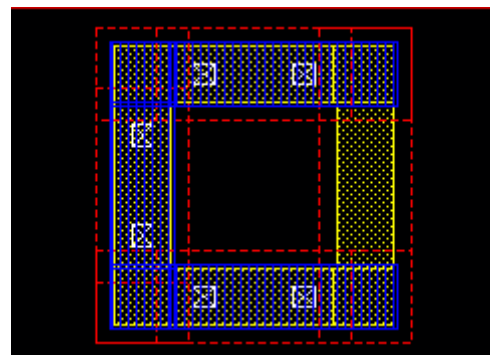
LEFT	☐
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RIGHT	☒
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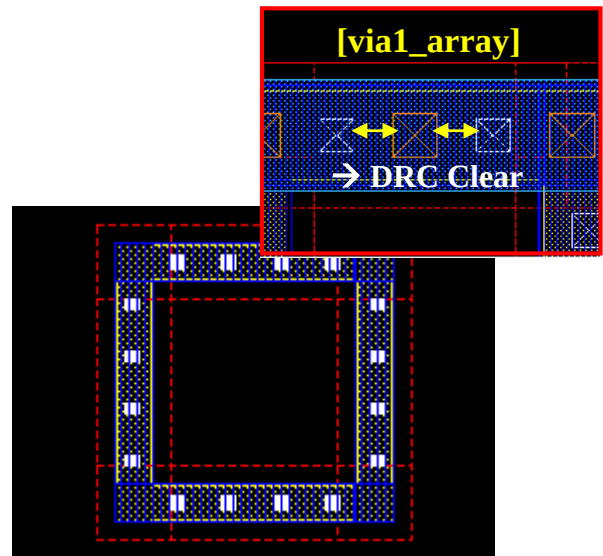


RIGHT	☐
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3.31 ptap_ring_ndiode (Layout)

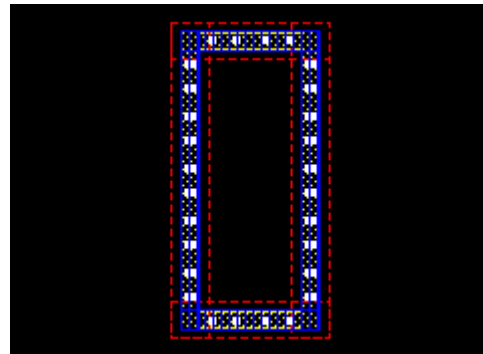
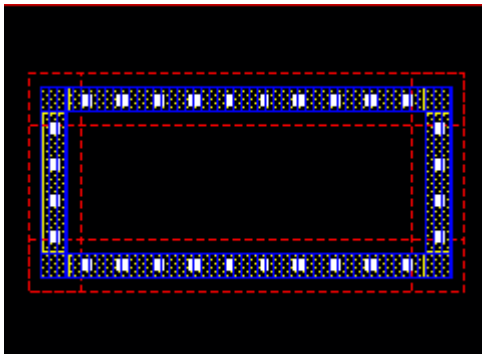
Parameter	Default	Change
x_size	17	value
y_size	17	value
TOP	☒	☐
BOTTOM	☒	☐
LEFT	☒	☐
RIGHT	☒	☐



Size Criterion : Cell Boundary

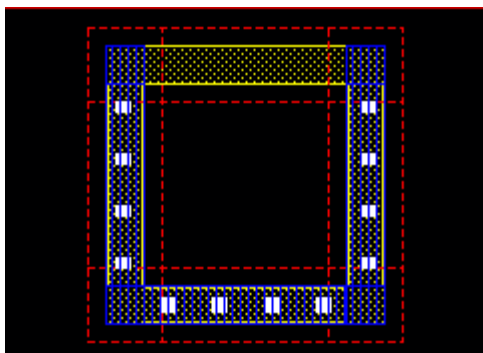
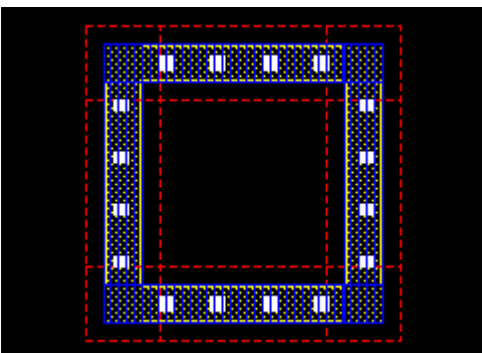
x_size	17	34
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y_size	17	34
--------	----	----



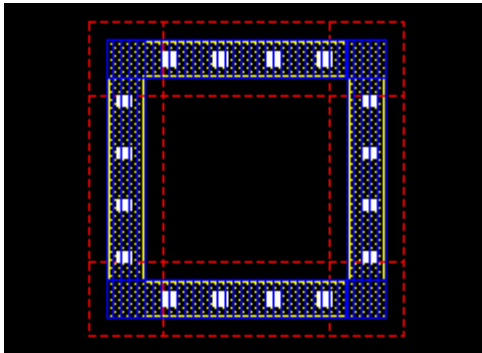
TOP	☒
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TOP	☐
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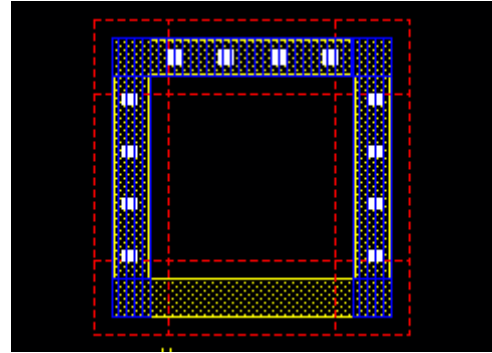


3.31 ptap_ring_ndiode (Layout)

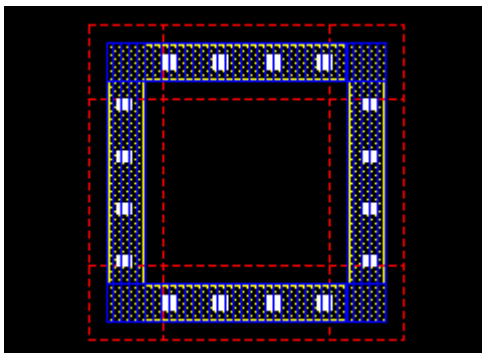
BOTTOM	☒
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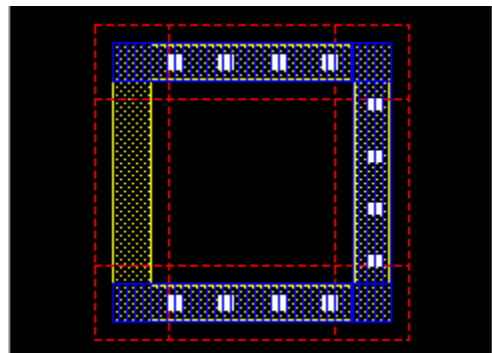
BOTTOM	☐
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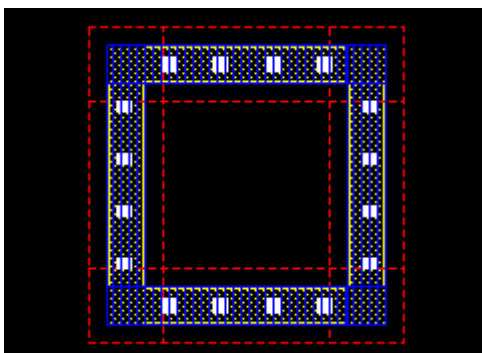
LEFT	☒
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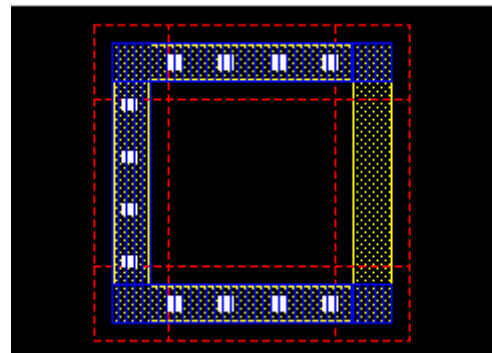
LEFT	☐
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RIGHT	☒
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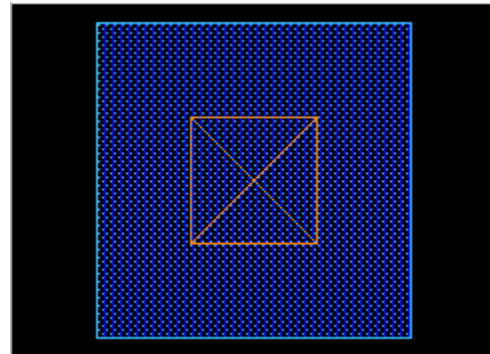


RIGHT	☐
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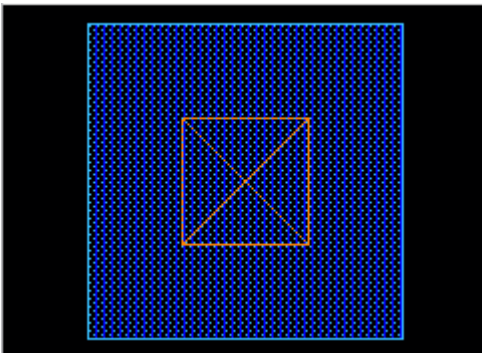


3.32 via1 (Layout)

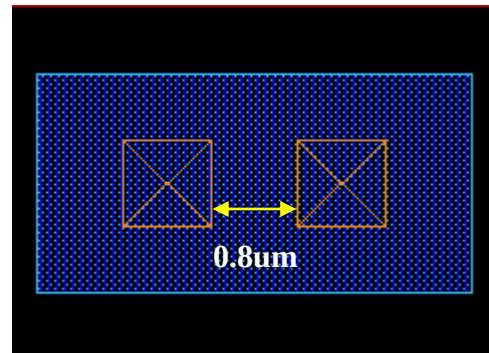
Parameter	Default	Change
x_size	2	value
y_size	2	value



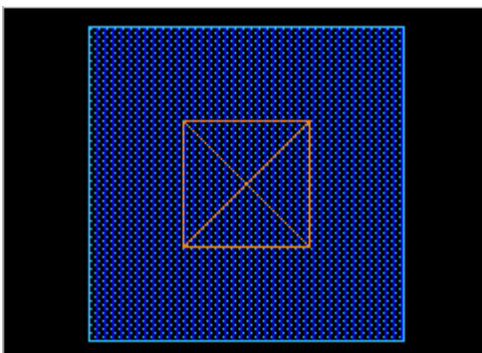
x_size	2
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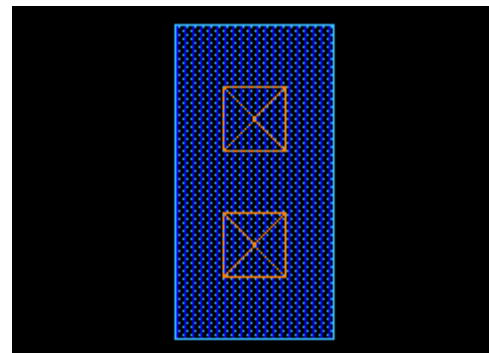
x_size	4
--------	---



y_size	2
--------	---

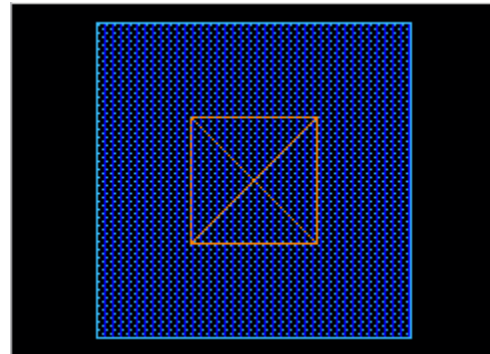


y_size	4
--------	---

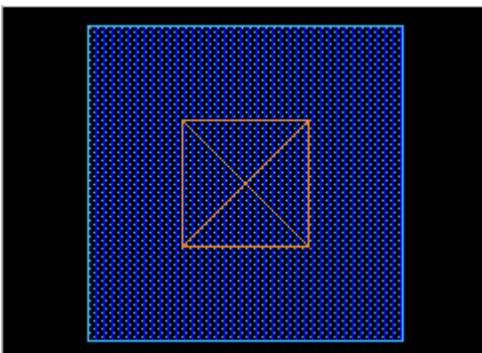


3.33 via1_array (Layout)

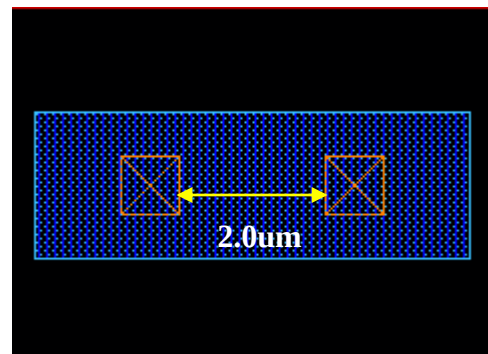
Parameter	Default	Change
x_size	2	value
y_size	2	value



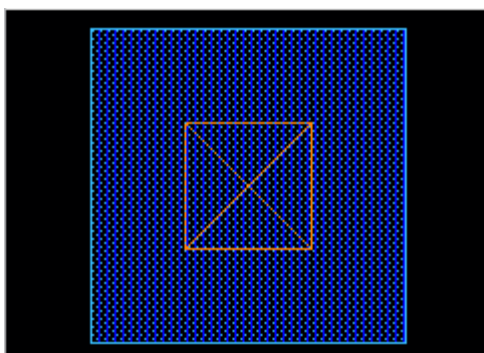
x_size	2
--------	---



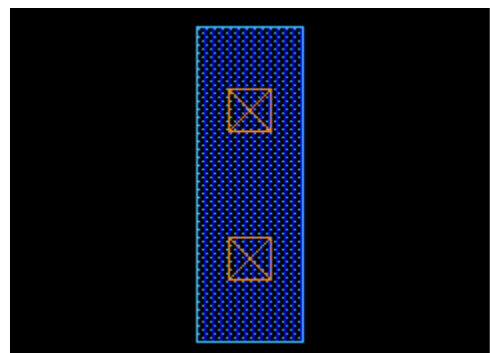
x_size	6
--------	---



y_size	2
--------	---

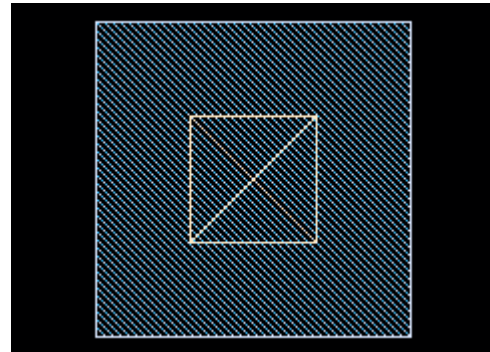


y_size	6
--------	---

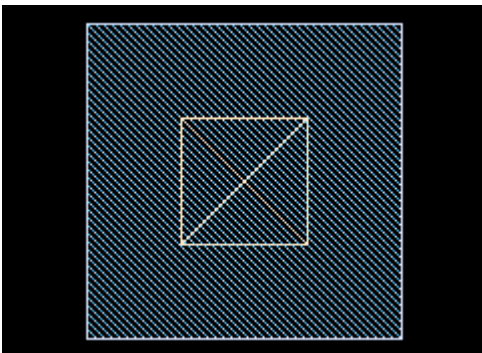


3.34 via2 (Layout)

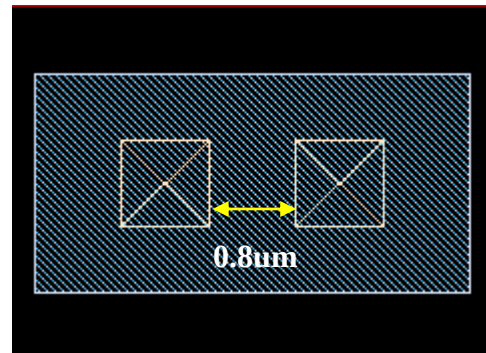
Parameter	Default	Change
x_size	2	value
y_size	2	value



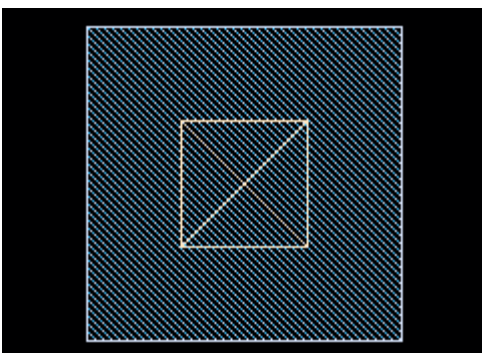
x_size	2
--------	---



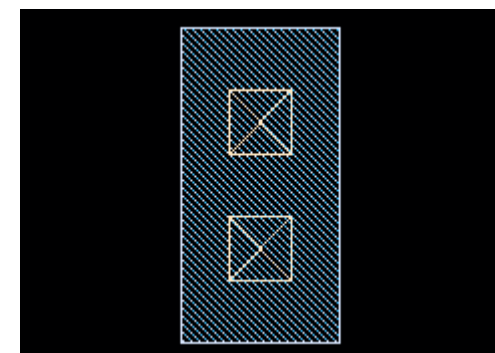
x_size	4
--------	---



y_size	2
--------	---

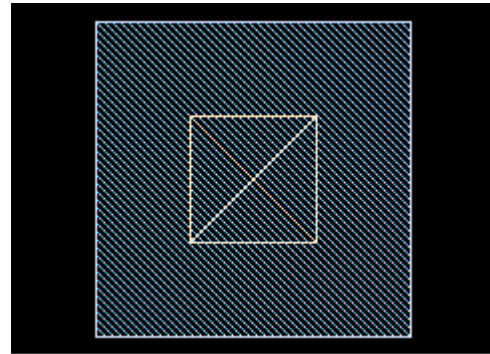


y_size	4
--------	---

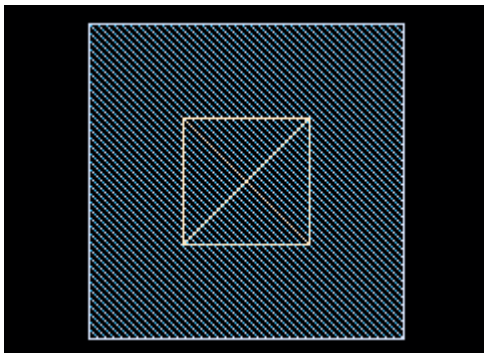


3.35 via2_array (Layout)

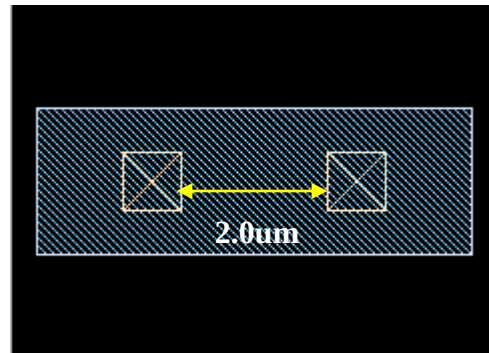
Parameter	Default	Change
x_size	2	value
y_size	2	value



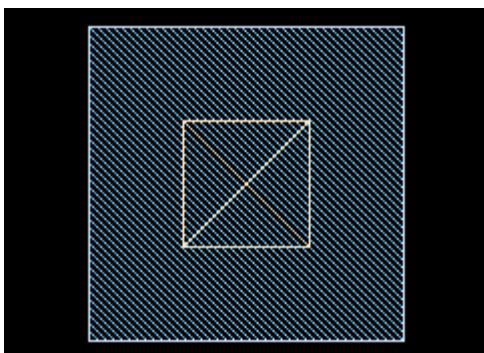
x_size	2
--------	---



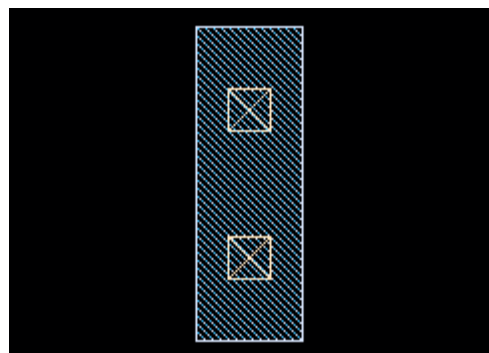
x_size	6
--------	---



y_size	2
--------	---



y_size	6
--------	---



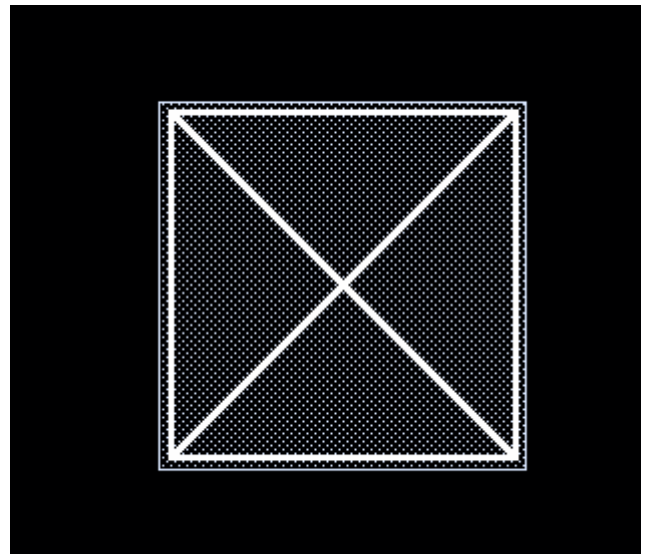
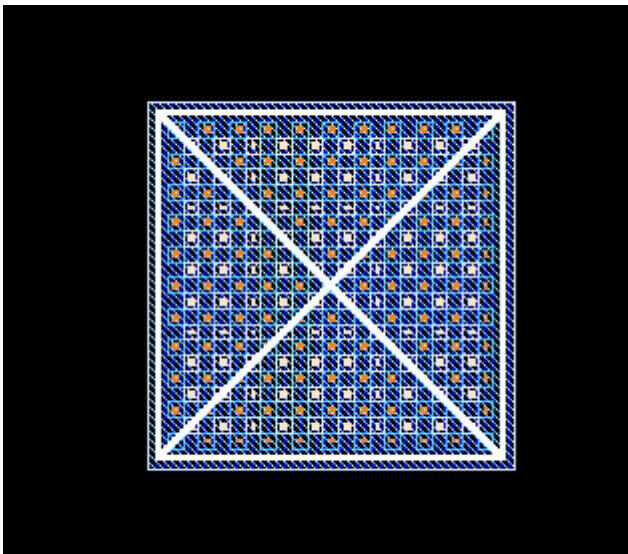
3.36 pad80 (Layout)

Parameter	Default	Change
-	-	-

PAD Fixed Size (80um * 80um)
→ doesn't provide parameter modification

PAD_EXT_M1M2	☒
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PAD_EXT_M1M2	☐
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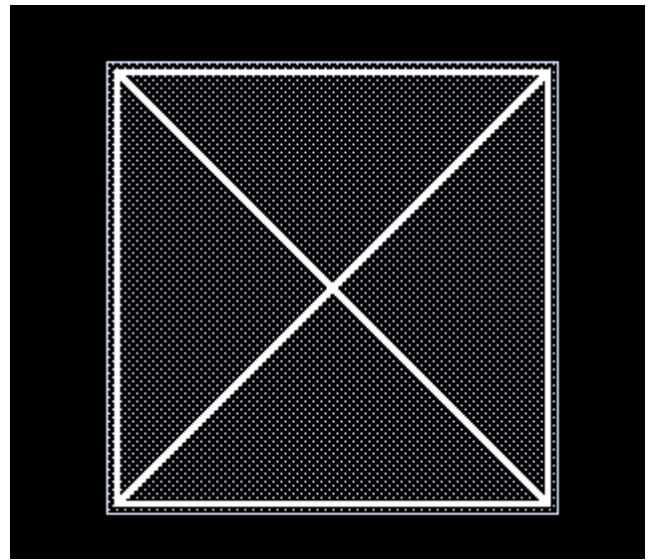
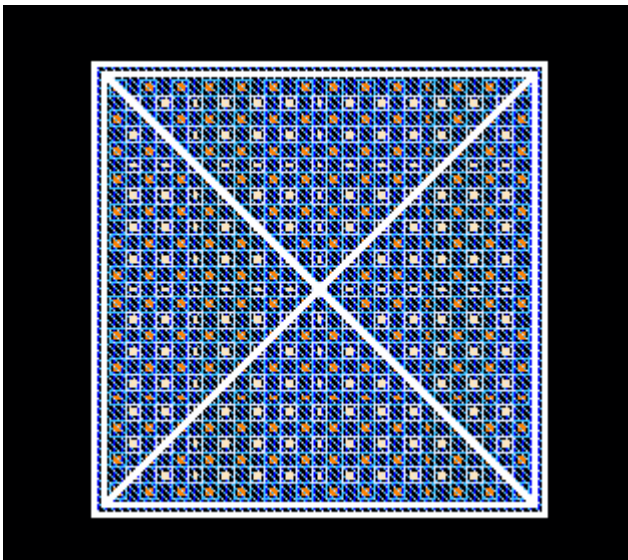
3.37 pad100 (Layout)

Parameter	Default	Change
-	-	-

PAD Fixed Size (100um * 100um)
→ doesn't provide parameter modification

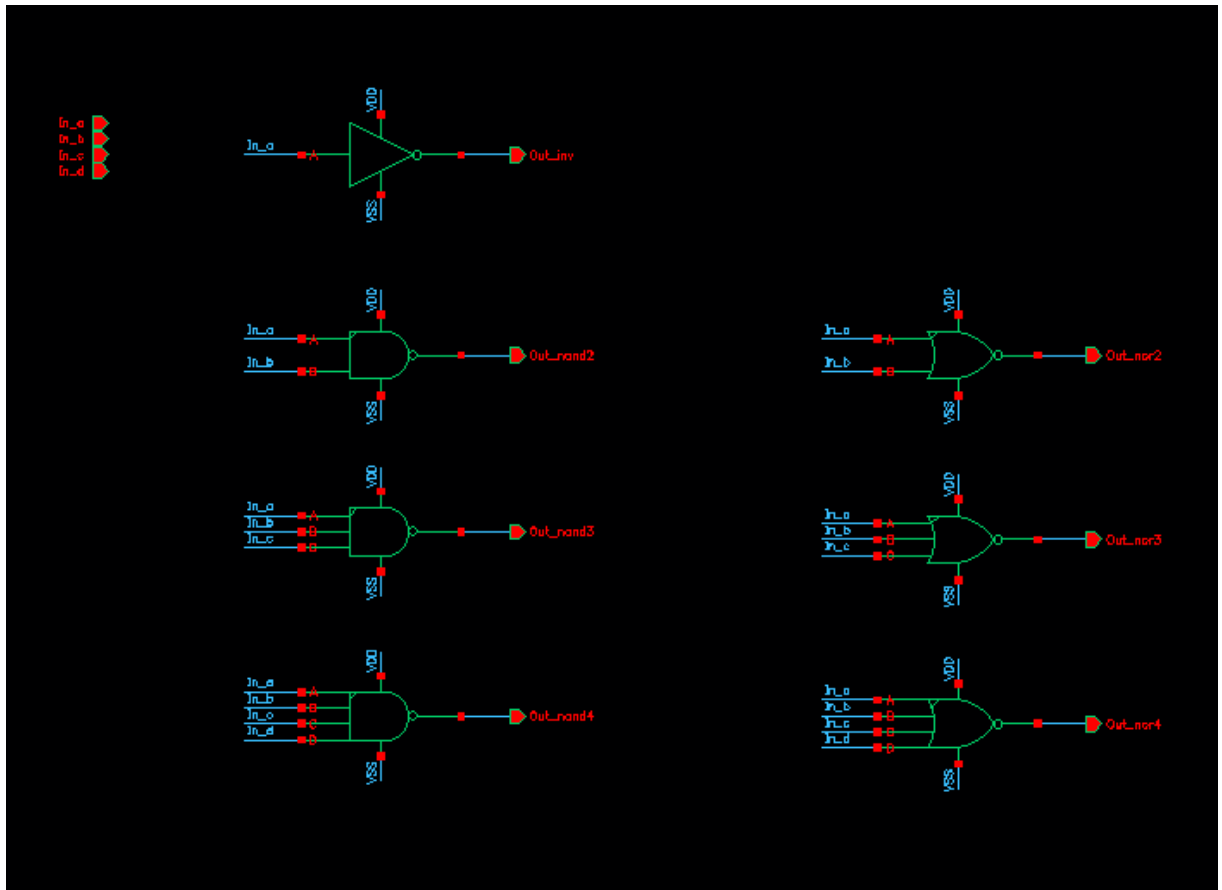
PAD_EXT_M1M2	☒
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PAD_EXT_M1M2	☐
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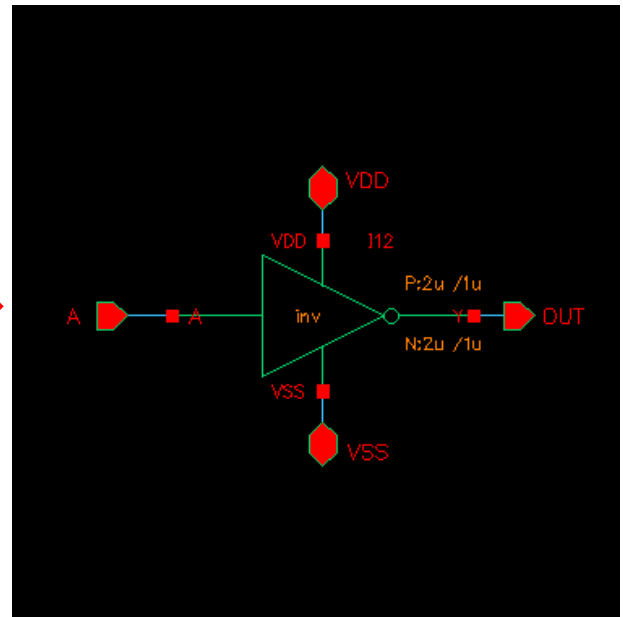
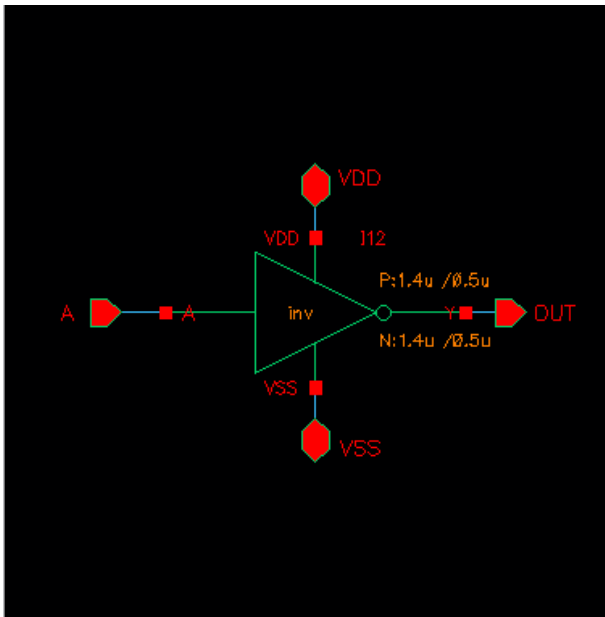
4. Standard Schematic Library

4.1 .SAMPLE



4.2 inv

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:   inv_schematic
* View Name:   schematic
*****

.SUBCKT inv_schematic A OUT VDD VSS
*.PININFO A:I OUT:O VDD:B VSS:B
XI12 A VDD VSS OUT / inv pl=0.5u pw=1.4u nl=0.5u nw=1.4u
.ENDS
```

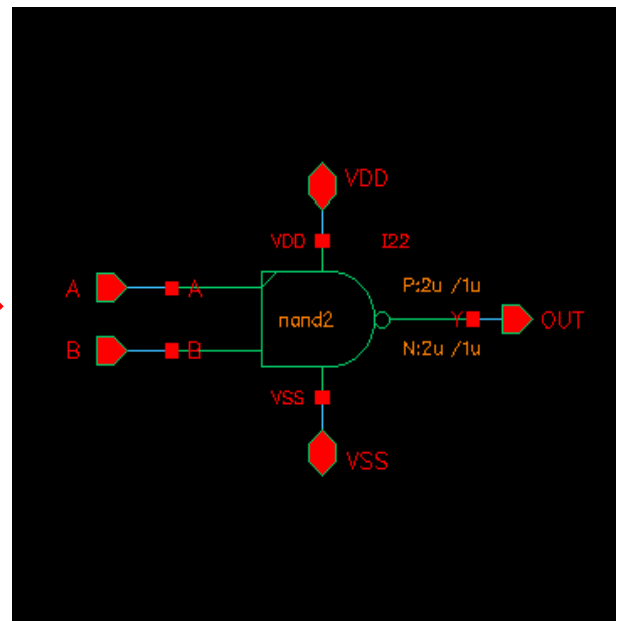
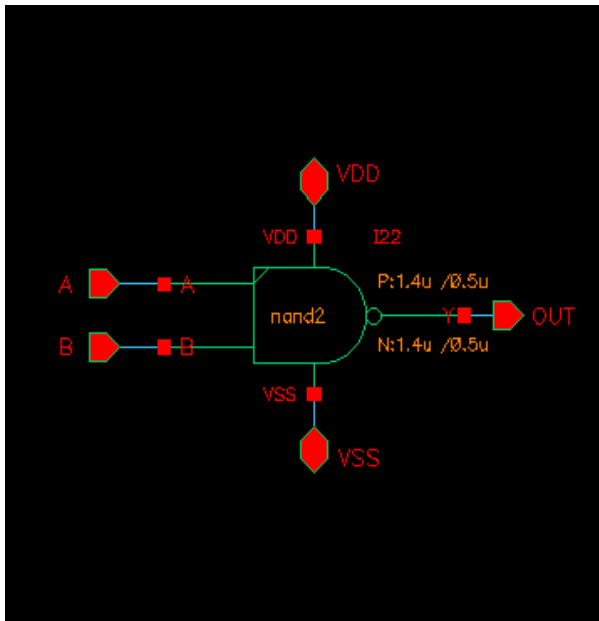
```
*****
* Library Name: ETRI_lay
* Cell Name:   inv_schematic
* View Name:   schematic
*****

.SUBCKT inv_schematic A OUT VDD VSS
*.PININFO A:I OUT:O VDD:B VSS:B
XI12 A VDD VSS OUT / inv pl=1u pw=2u nl=1u nw=2u
.ENDS
```

<cdl output>

4.3 nand2

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI lay
* Cell Name:   nand2_schematic
* View Name:   schematic
*****
.SUBCKT nand2_schematic A B OUT VDD VSS
*.PININFO A:I B:I OUT:O VDD:B VSS:B
XI22 A B VDD VSS OUT / nand2 nl=0.5u nw=1.4u pl=0.5u pw=1.4u
.ENDS
```

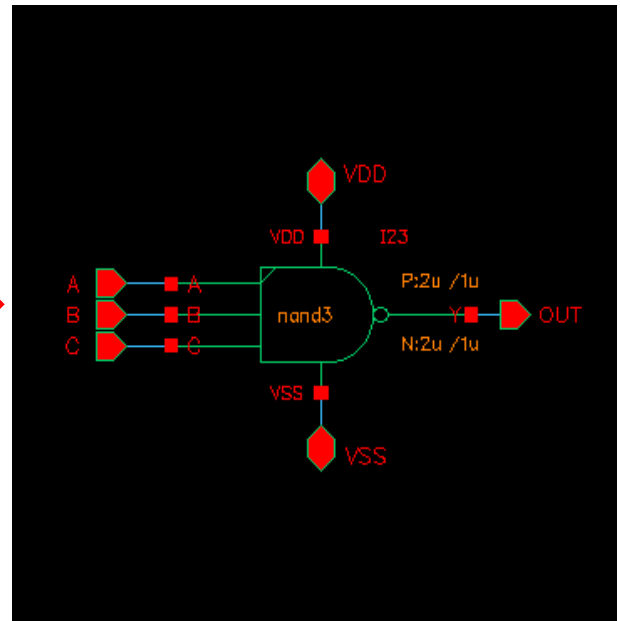
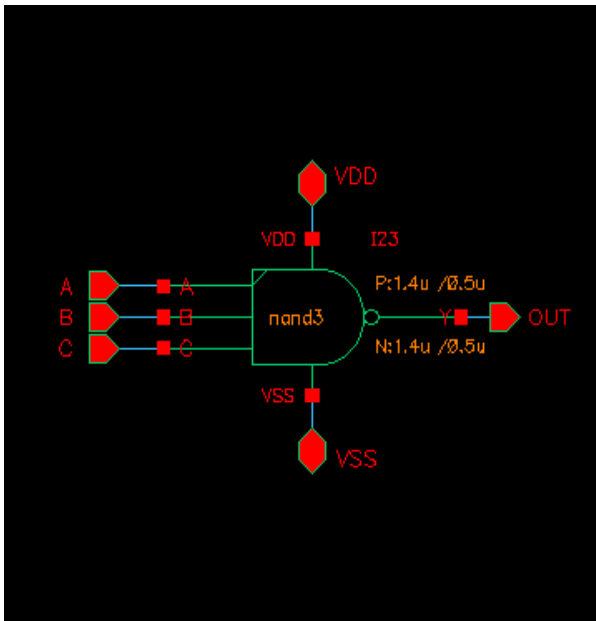


```
*****
* Library Name: ETRI lay
* Cell Name:   nand2_schematic
* View Name:   schematic
*****
.SUBCKT nand2_schematic A B OUT VDD VSS
*.PININFO A:I B:I OUT:O VDD:B VSS:B
XI22 A B VDD VSS OUT / nand2 nl=1u nw=2u pl=1u pw=2u
.ENDS
```

<cdl output>

4.4 nand3

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI lay
* Cell Name: nand3_schematic
* View Name: schematic
*****

.SUBCKT nand3_schematic A B C OUT VDD VSS
*.PININFO A:I B:I C:I OUT:O VDD:B VSS:B
XI23 A B C VDD VSS OUT / nand3 pl=0.5u pw=1.4u nl=0.5u nw=1.4u
.ENDS
```



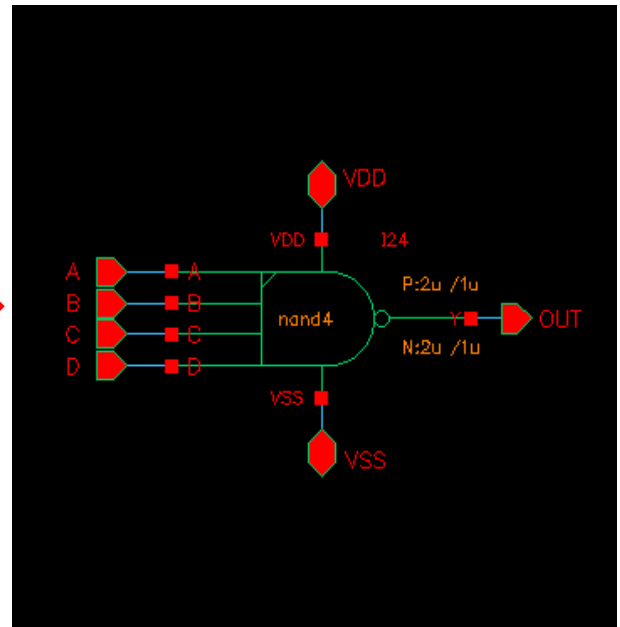
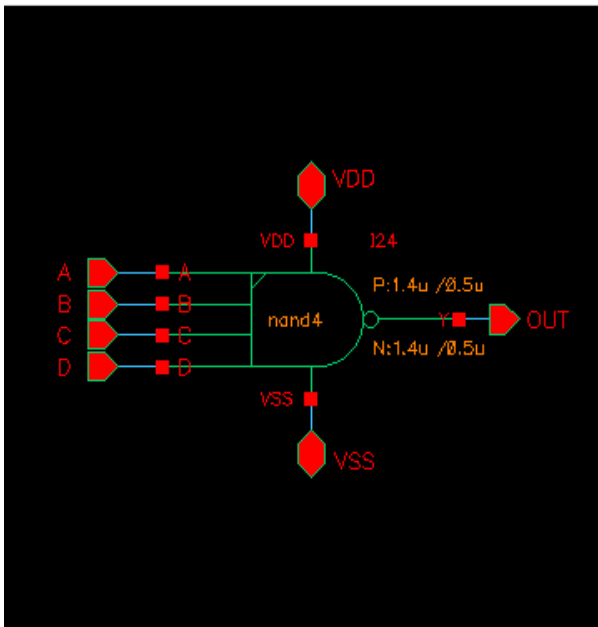
```
*****
* Library Name: ETRI lay
* Cell Name: nand3_schematic
* View Name: schematic
*****

.SUBCKT nand3_schematic A B C OUT VDD VSS
*.PININFO A:I B:I C:I OUT:O VDD:B VSS:B
XI23 A B C VDD VSS OUT / nand3 pl=1u pw=2u nl=1u nw=2u
.ENDS
```

<cdl output>

4.5 nand4

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI lay
* Cell Name:    nand4 schematic
* View Name:    schematic
*****
.SUBCKT nand4_schematic A B C D OUT VDD VSS
*.PININFO A:I B:I C:I D:I OUT:O VDD:B VSS:B
XI25 A B C D VDD VSS OUT / nand4 pl=0.5u pw=1.4u nl=0.5u nw=1.4u
.ENDS
```

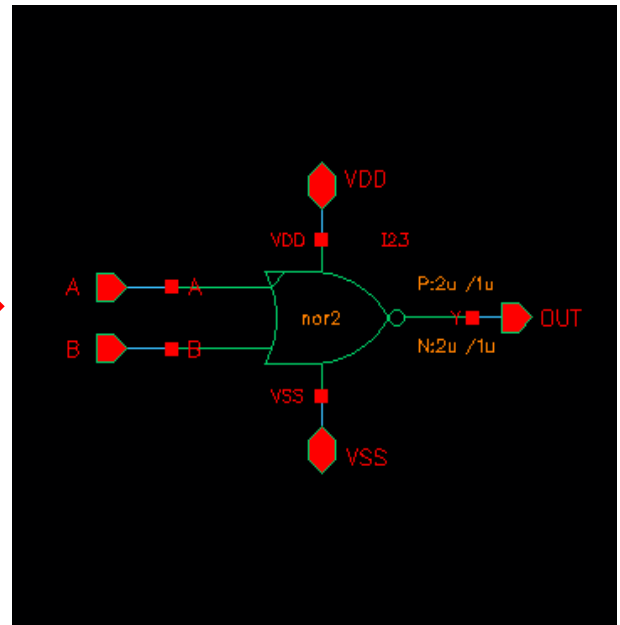
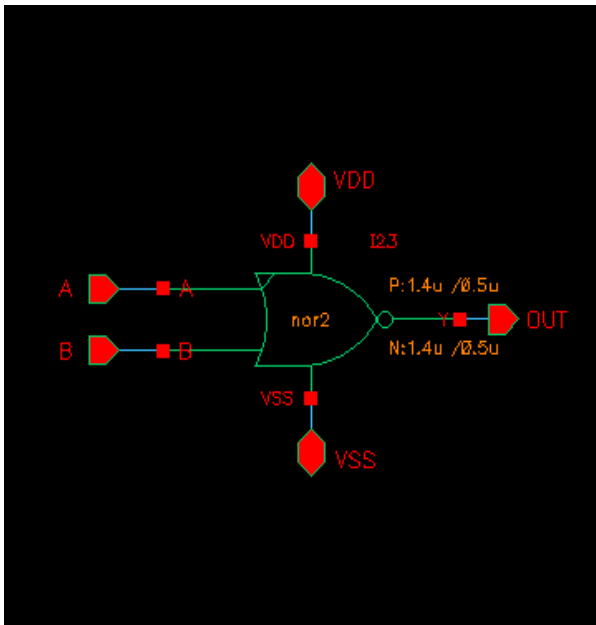


```
*****
* Library Name: ETRI lay
* Cell Name:    nand4 schematic
* View Name:    schematic
*****
.SUBCKT nand4_schematic A B C D OUT VDD VSS
*.PININFO A:I B:I C:I D:I OUT:O VDD:B VSS:B
XI24 A B C D VDD VSS OUT / nand4 pl=1u pw=2u nl=1u nw=2u
.ENDS
```

<cdl output>

4.6 nor2

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:   nor2_schematic
* View Name:   schematic
*****
.SUBCKT nor2_schematic A B OUT VDD VSS
*.PININFO A:I B:I OUT:O VDD:B VSS:B
XI23 A B VDD VSS OUT / nor2 pl=0.5u pw=1.4u nl=0.5u nw=1.4u
.ENDS
```

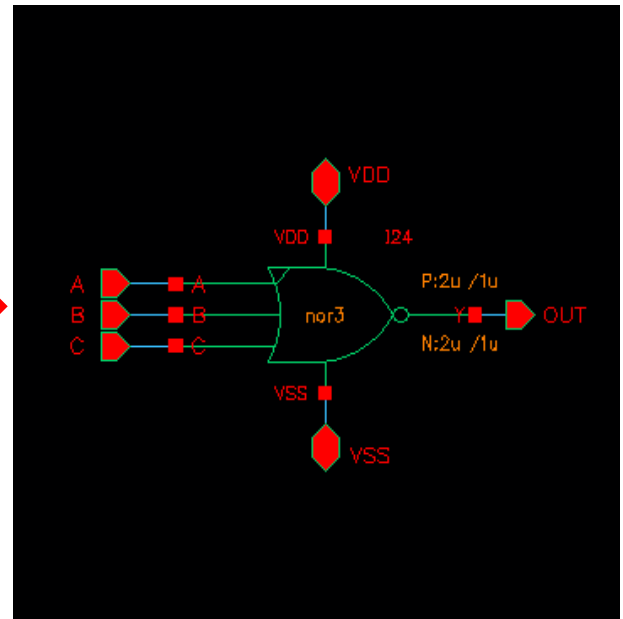
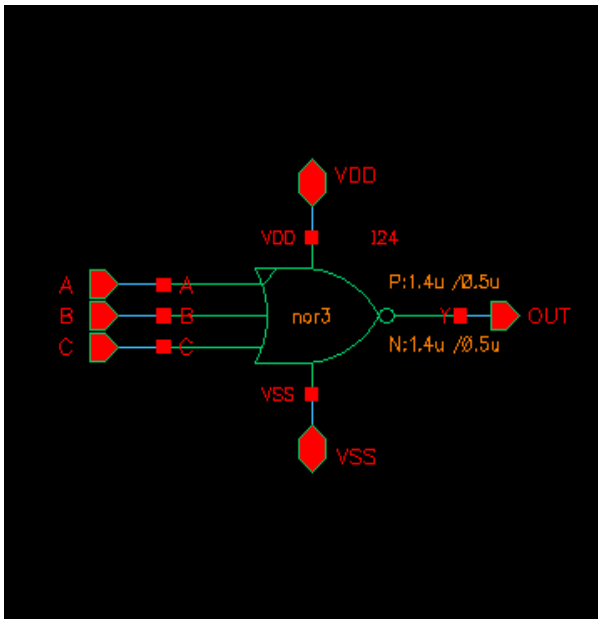


```
*****
* Library Name: ETRI_lay
* Cell Name:   nor2_schematic
* View Name:   schematic
*****
.SUBCKT nor2_schematic A B OUT VDD VSS
*.PININFO A:I B:I OUT:O VDD:B VSS:B
XI23 A B VDD VSS OUT / nor2 pl=1u pw=2u nl=1u nw=2u
.ENDS
```

<cdl output>

4.7 nor3

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:   nor3_schematic
* View Name:   schematic
*****
.SUBCKT nor3_schematic A B C OUT VDD VSS
*.PININFO A:I B:I C:I OUT:O VDD:B VSS:B
XI24 A B C VDD VSS OUT / nor3 pl=0.5u pw=1.4u nl=0.5u nw=1.4u
.ENDS
```

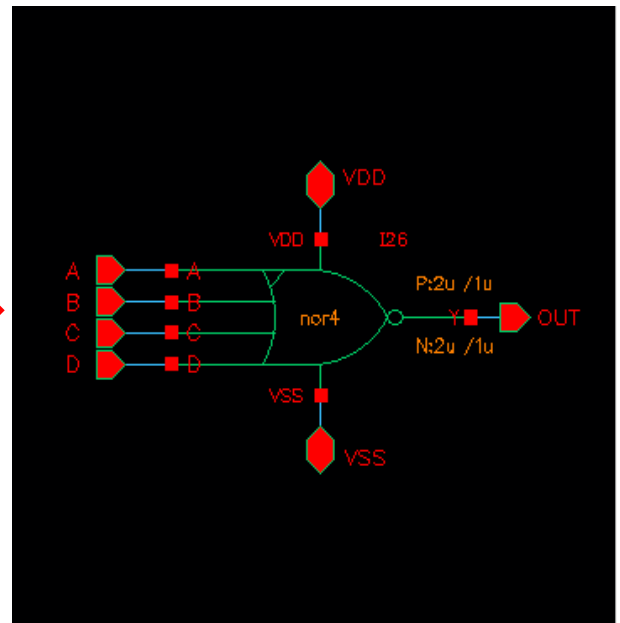
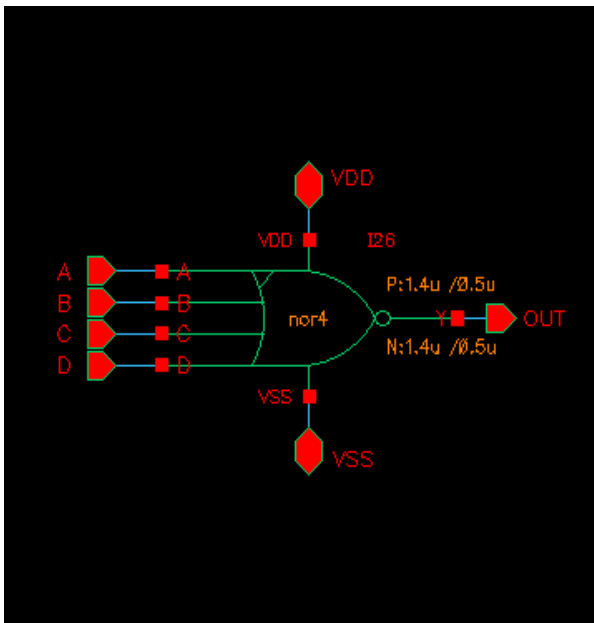


```
*****
* Library Name: ETRI_lay
* Cell Name:   nor3_schematic
* View Name:   schematic
*****
.SUBCKT nor3_schematic A B C OUT VDD VSS
*.PININFO A:I B:I C:I OUT:O VDD:B VSS:B
XI24 A B C VDD VSS OUT / nor3 pl=1u pw=2u nl=1u nw=2u
.ENDS
```

<cdl output>

4.8 nor4

Parameter							
pl		pw		nl		nw	
0.5u	1u	1.4u	2u	0.5u	1u	1.4u	2u



<Schematic parameter>

```
*****
* Library Name: ETRI_lay
* Cell Name:   nor4_schematic
* View Name:   schematic
*****
.SUBCKT nor4_schematic A B C D OUT VDD VSS
*.PININFO A:I B:I C:I D:I OUT:O VDD:B VSS:B
XI26 A B C D VDD VSS OUT / nor4 pl=0.5u pw=1.4u nl=0.5u nw=1.4u
.ENDS
```

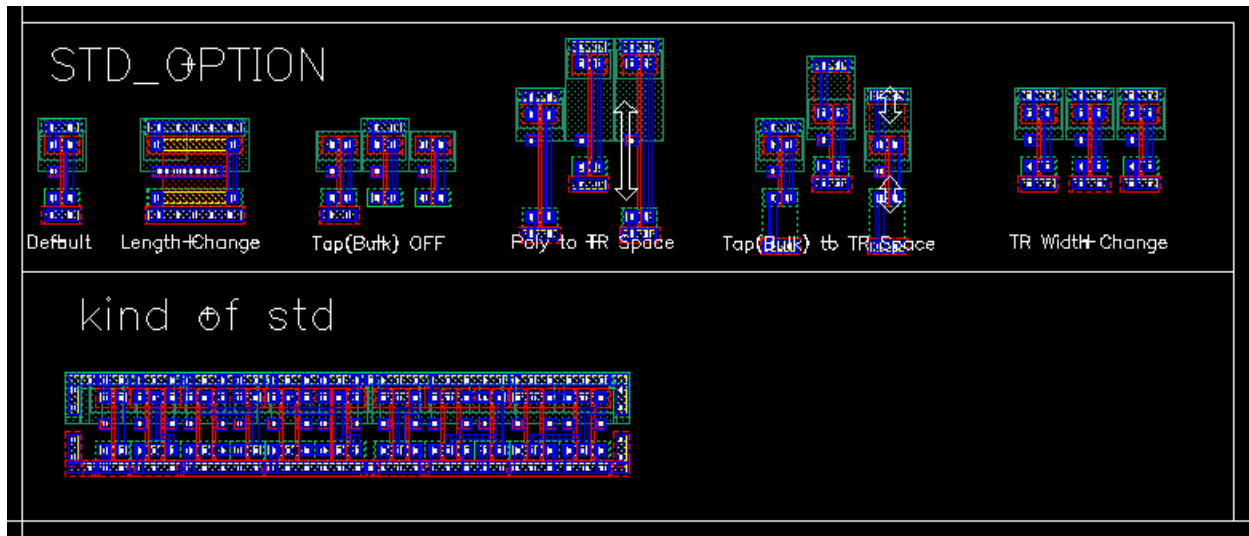


```
*****
* Library Name: ETRI_lay
* Cell Name:   nor4_schematic
* View Name:   schematic
*****
.SUBCKT nor4_schematic A B C D OUT VDD VSS
*.PININFO A:I B:I C:I D:I OUT:O VDD:B VSS:B
XI26 A B C D VDD VSS OUT / nor4 pl=1u pw=2u nl=1u nw=2u
.ENDS
```

<cdl output>

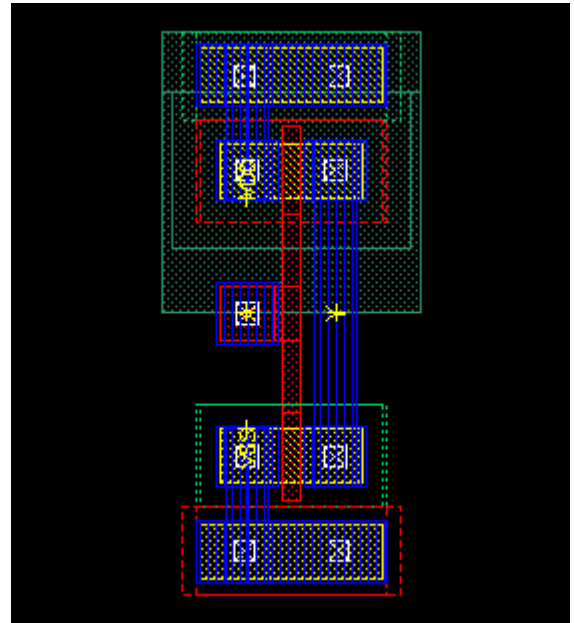
5. Standard Layout Library

5.1 .SAMPLE

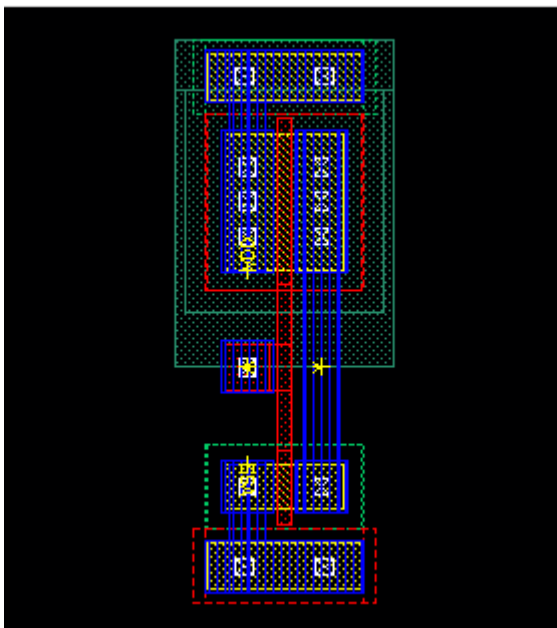


5.2 inv

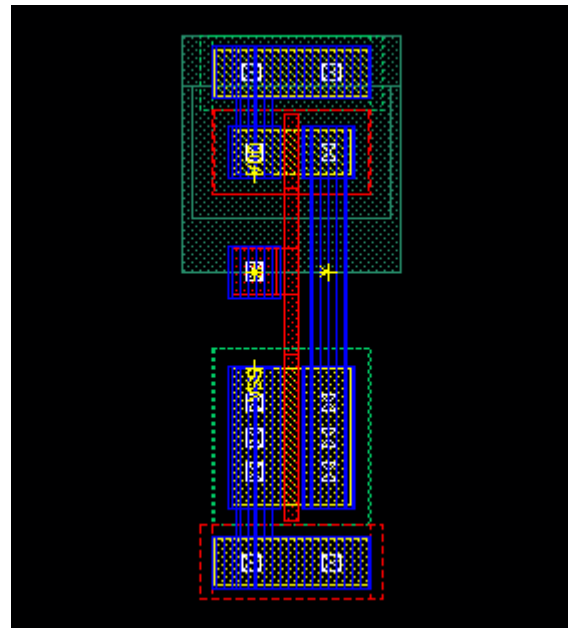
Parameter	Default	Change
P_WIDTH	1.6	<i>value</i>
N_WIDTH	1.6	<i>value</i>
LENGTH	0.5	<i>value</i>
POLY_TO_PTR_SPACE	2.6	<i>value</i>
POLY_TO_NTR_SPACE	2.6	<i>value</i>
PTAP_TO_TR_SPACE	0	<i>value</i>
NTAP_TO_TR_SPACE	0	<i>value</i>
PTAP	☒	☐
NTAP	☒	☐
POLY1CONT	☒	☐



P_WIDTH	1.6	4.8
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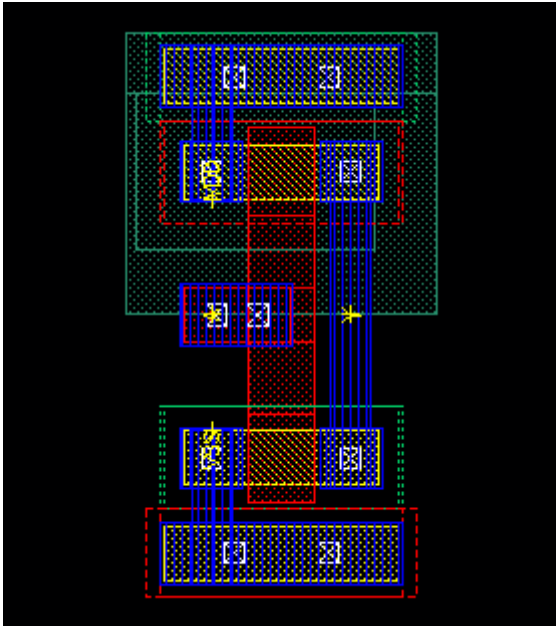


N_WIDTH	1.6	4.8
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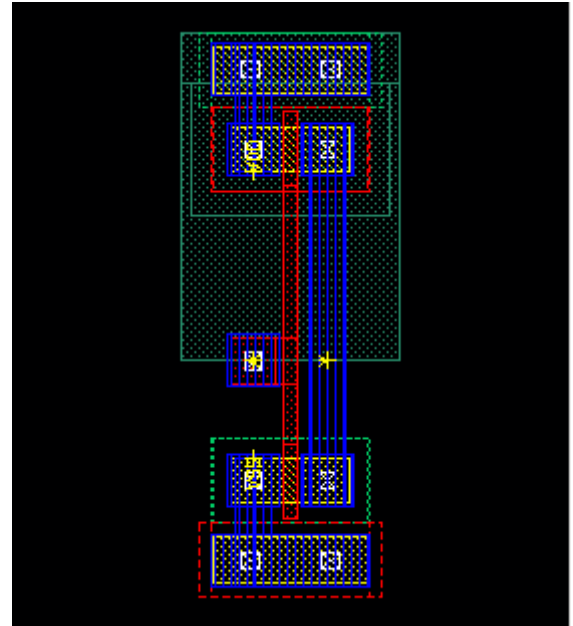


5.2 inv

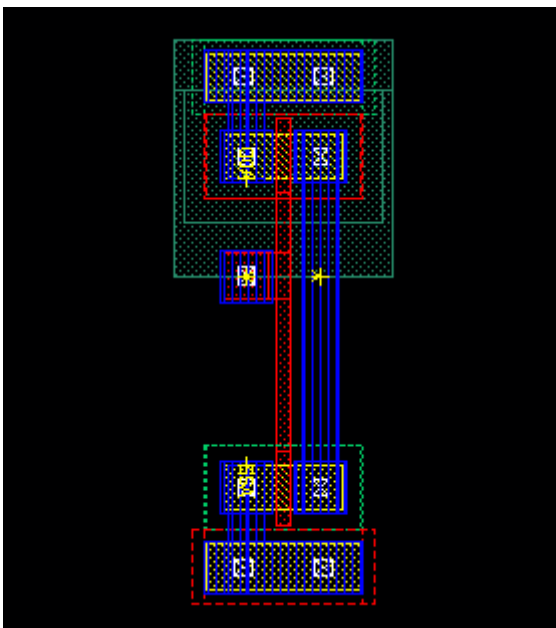
LENGTH	0.5	2
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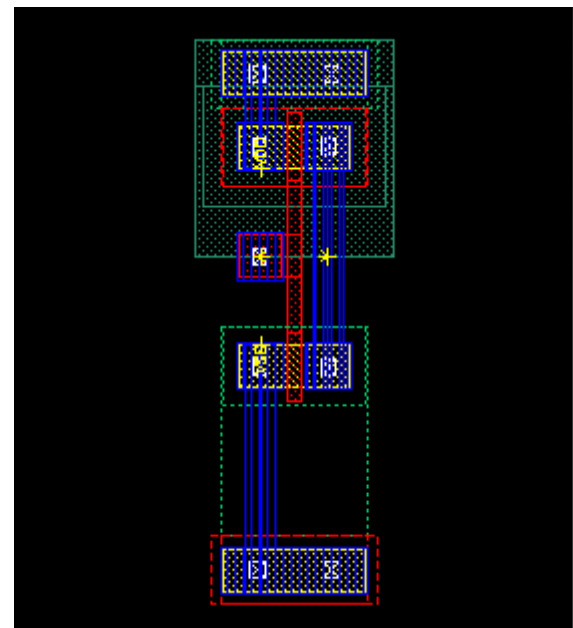
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
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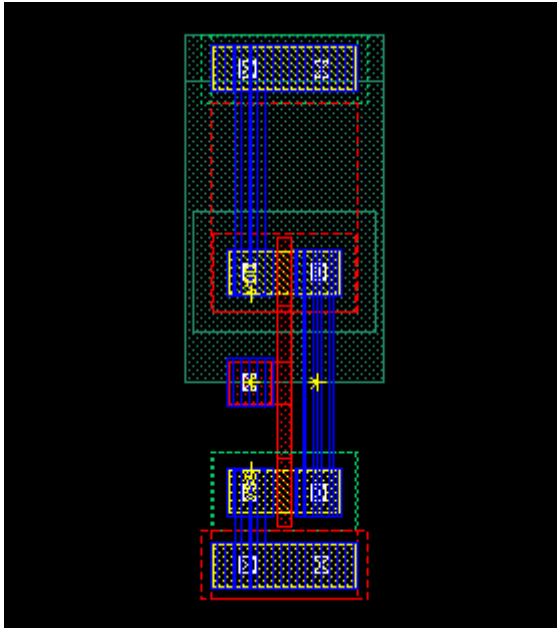


PTAP_TO_TR_SPACE	0	5.0
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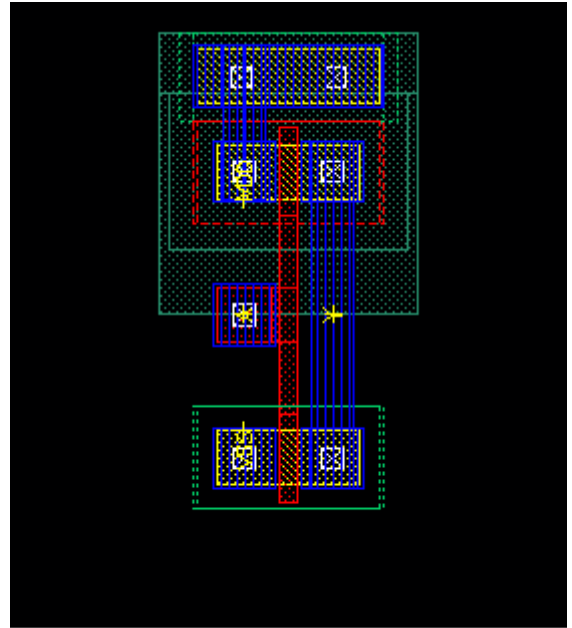


5.2 inv

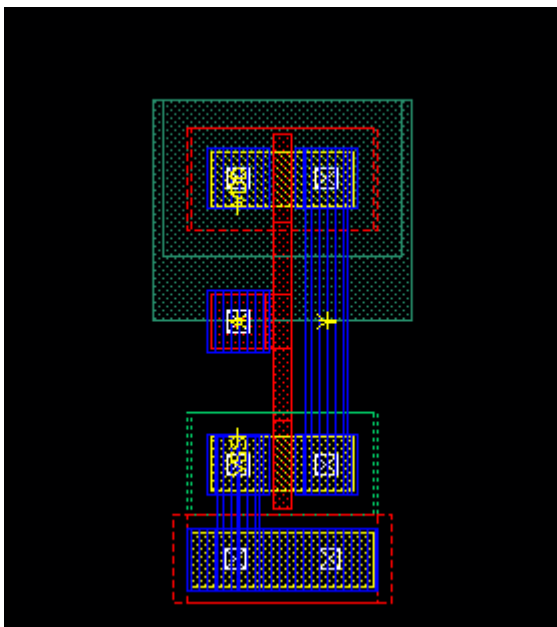
NTAP_TO_TR_SPACE	0	5
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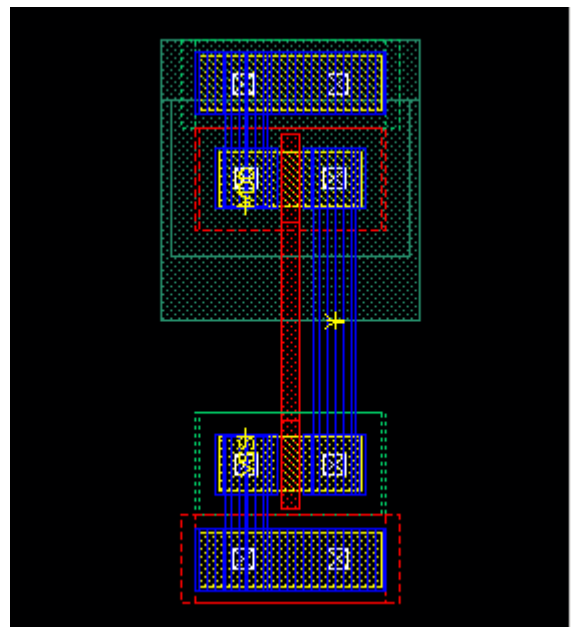
PTAP	☒	☐
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NTAP	☒	☐
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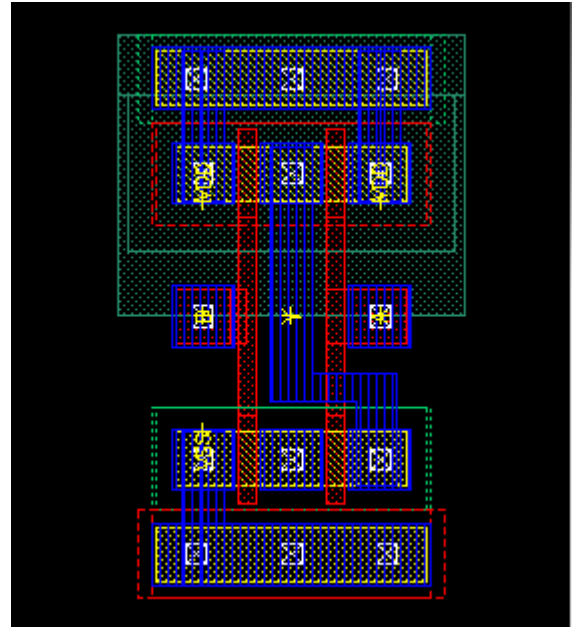


POLY1CONT	☒	☐
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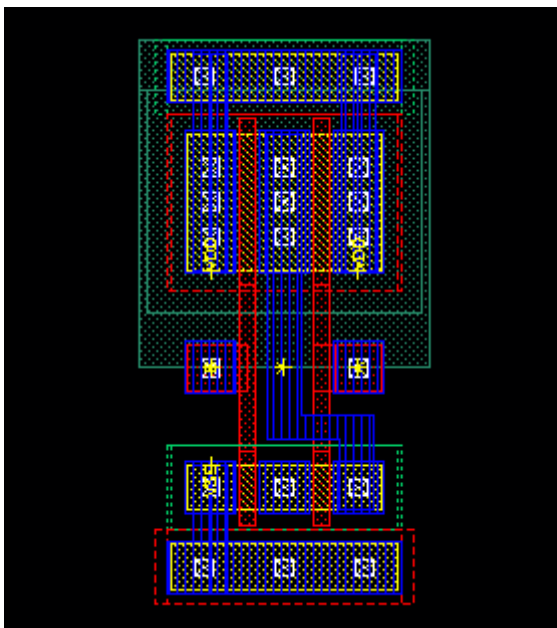


5.3 nand2

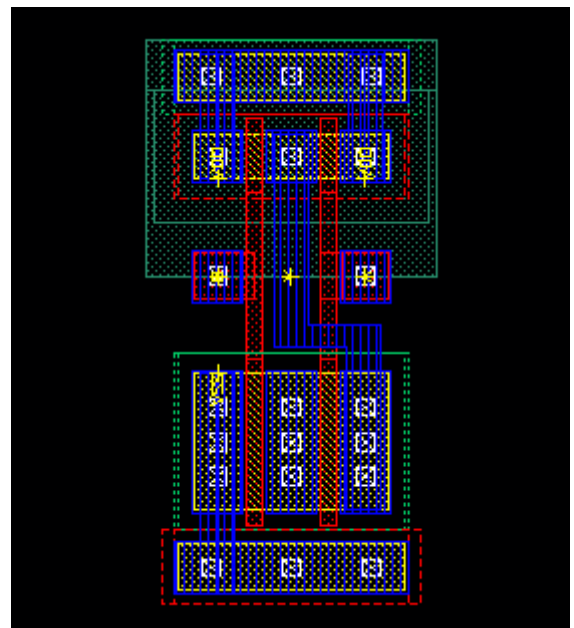
Parameter	Default	Change
P_WIDTH	1.6	value
N_WIDTH	1.6	value
LENGTH	0.5	value
POLY_TO_PTR_SPACE	2.6	value
POLY_TO_NTR_SPACE	2.6	value
PTAP_TO_TR_SPACE	0	value
NTAP_TO_TR_SPACE	0	value
PTAP	☒	☐
NTAP	☒	☐
POLY1CONTA	☒	☐
POLY1CONTB	☒	☐



P_WIDTH	1.6	4.8
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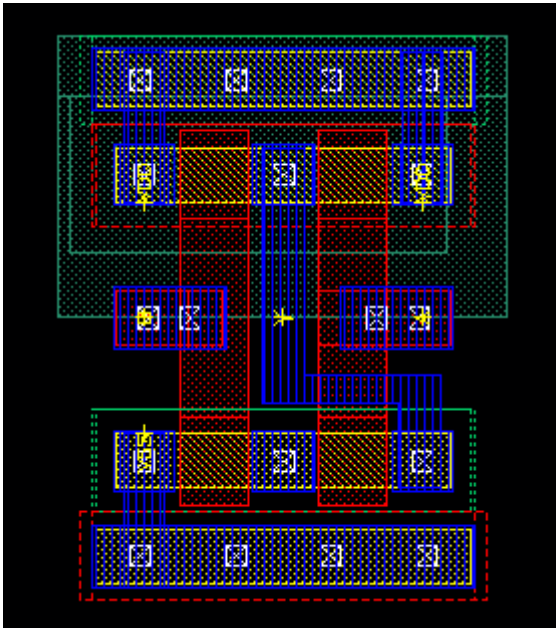


N_WIDTH	1.6	4.8
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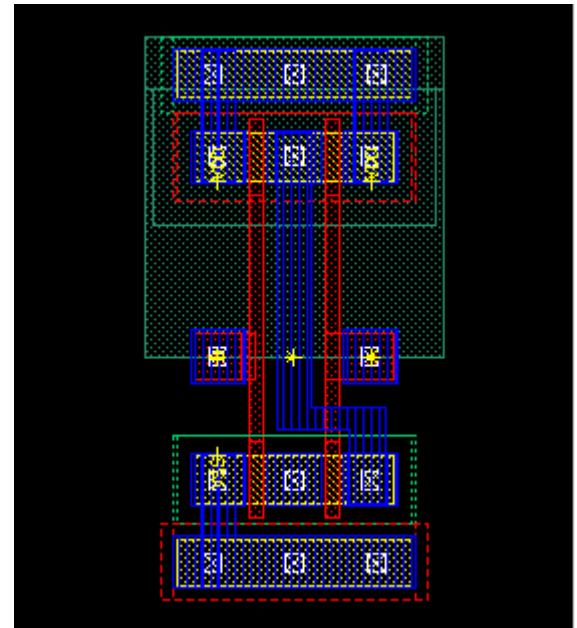


5.3 nand2

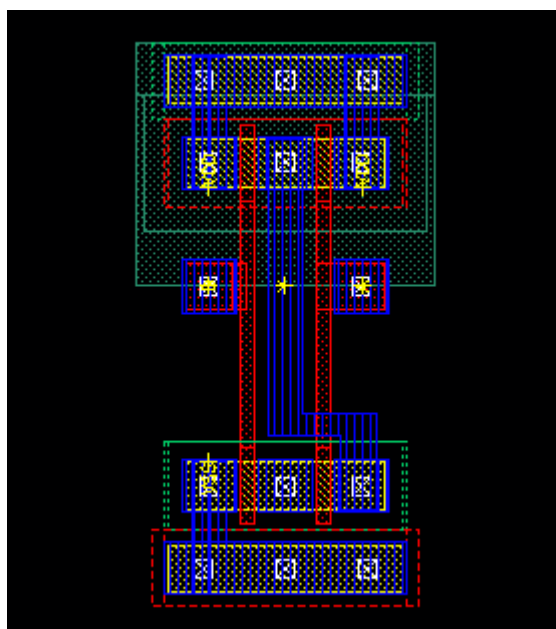
LENGTH	0.5	2
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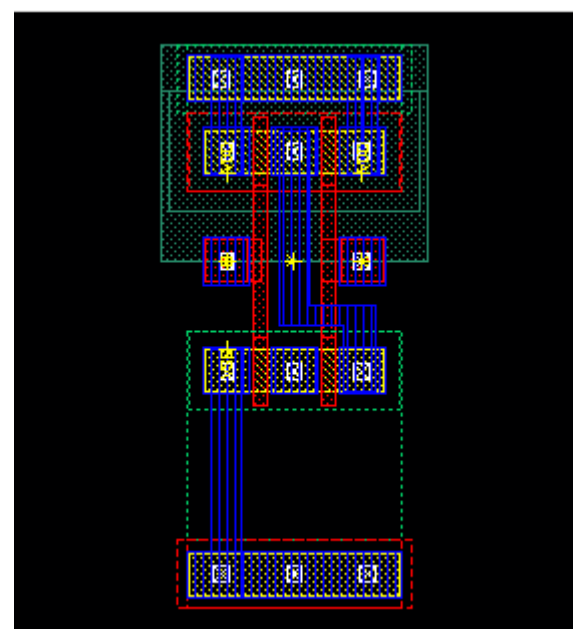
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
-------------------	-----	-----

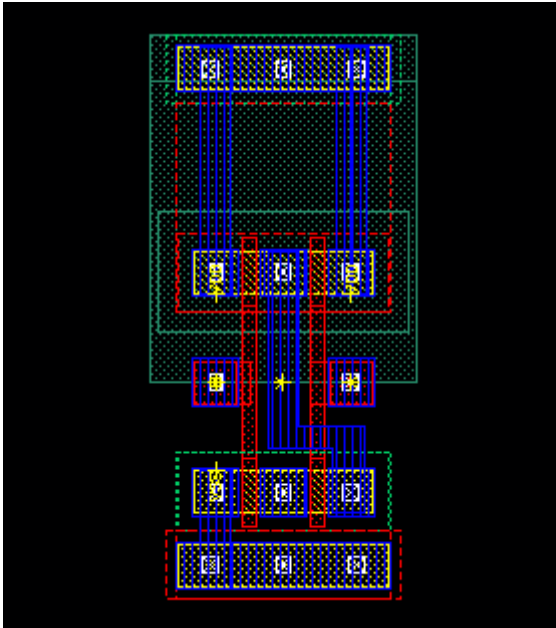


PTAP_TO_TR_SPACE	0	5.0
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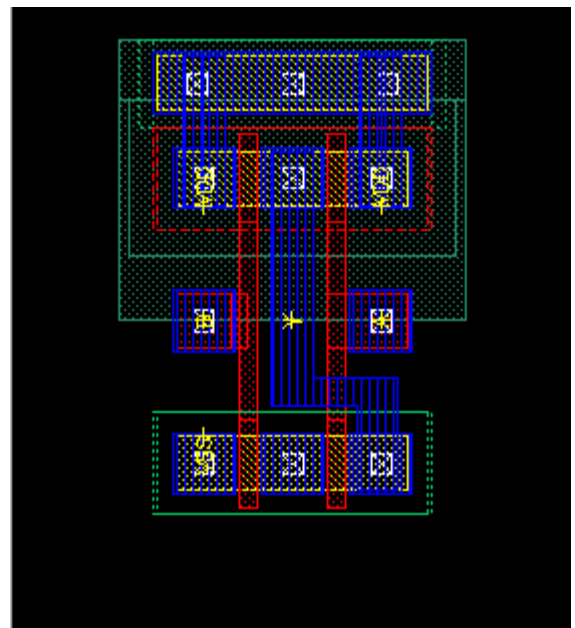


5.3 nand2

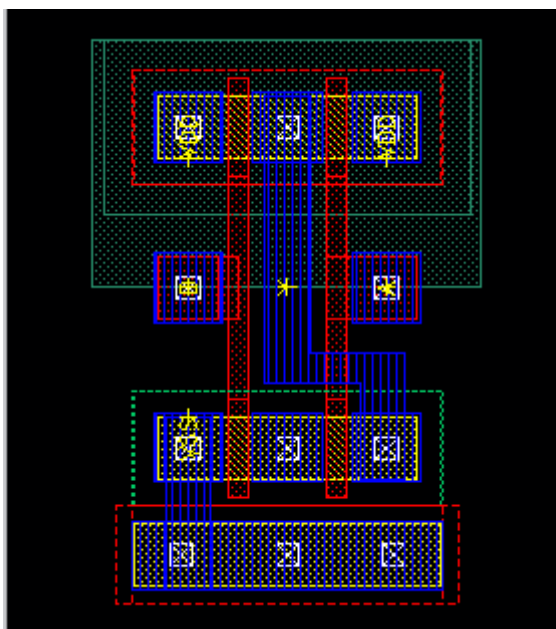
NTAP_TO_TR_SPACE	0	5
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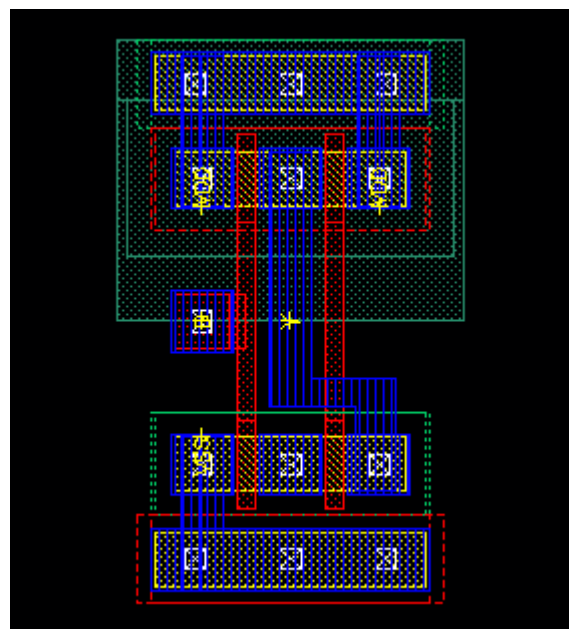
PTAP	☒	☐
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NTAP	☒	☐
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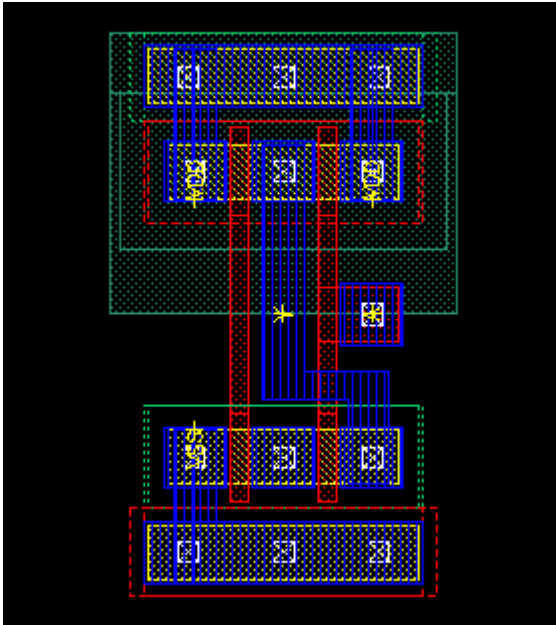


POLY1CONTA	☒	☐
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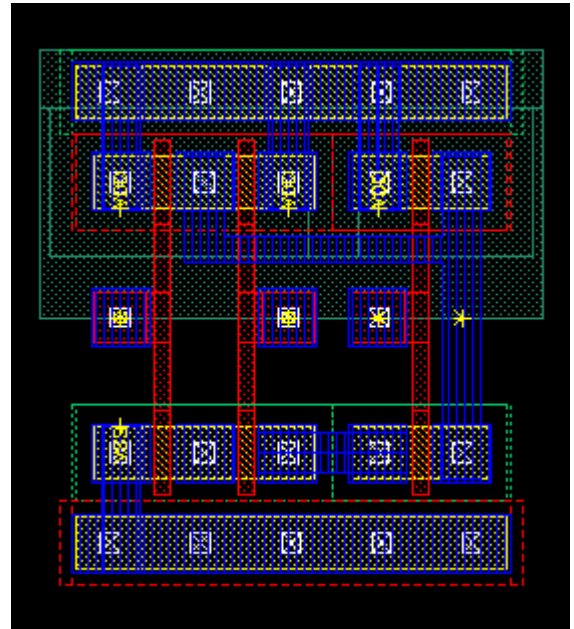
5.3 nand2

POLY1CONTB	☒	→	☐
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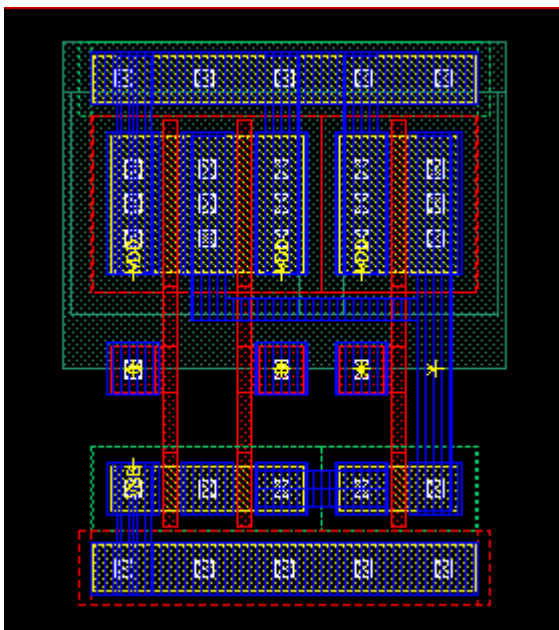


5.4 nand3

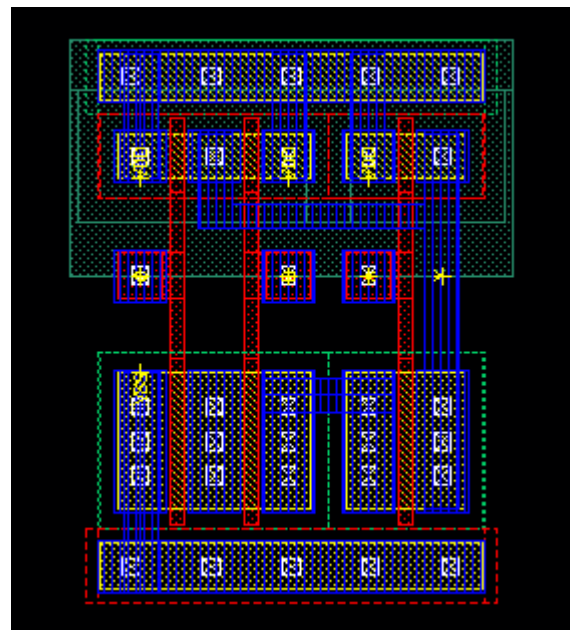
Parameter	Default	Change
P_WIDTH	1.6	value
N_WIDTH	1.6	value
LENGTH	0.5	value
POLY_TO_PTR_SPACE	2.6	value
POLY_TO_NTR_SPACE	2.6	value
PTAP_TO_TR_SPACE	0	value
NTAP_TO_TR_SPACE	0	value
PTAP	☒	☐
NTAP	☒	☐
POLY1CONTA	☒	☐
POLY1CONTB	☒	☐
POLY1CONTC	☒	☐



P_WIDTH	1.6	4.8
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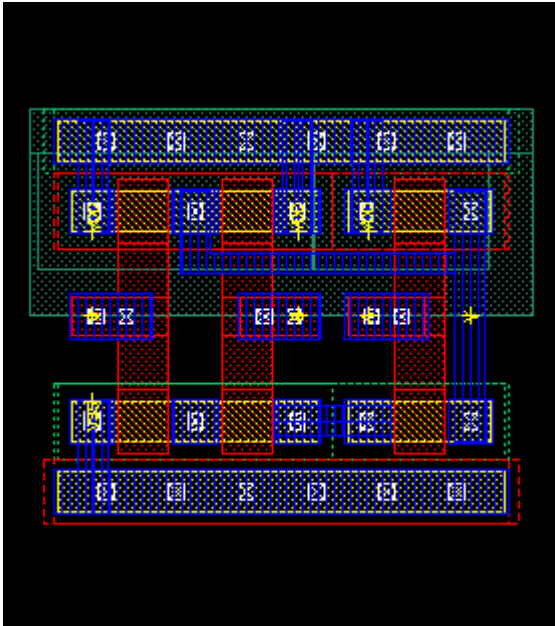


N_WIDTH	1.6	4.8
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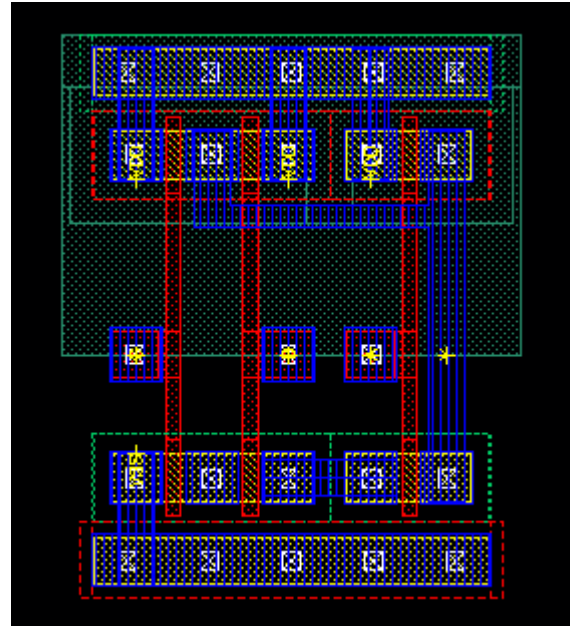


5.4 nand3

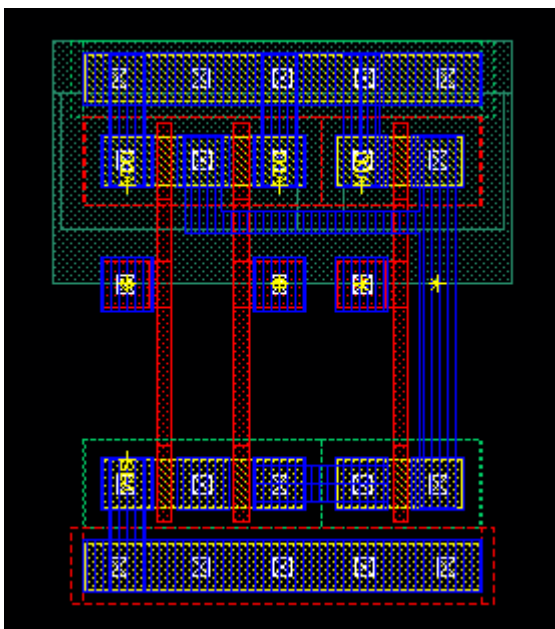
LENGTH	0.5	2
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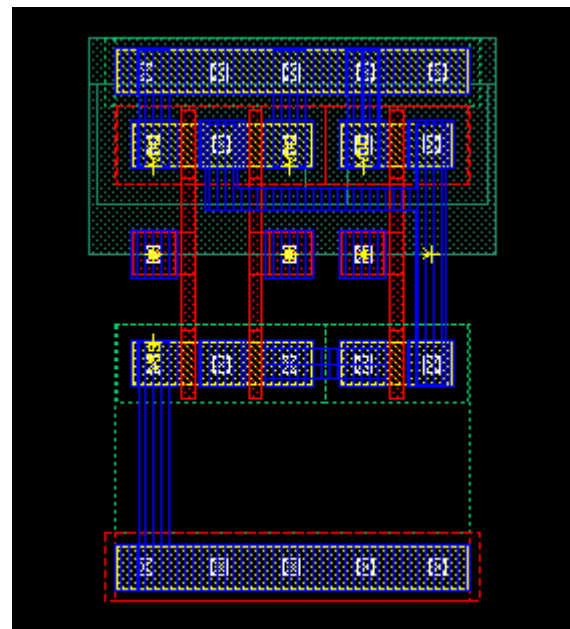
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
-------------------	-----	-----

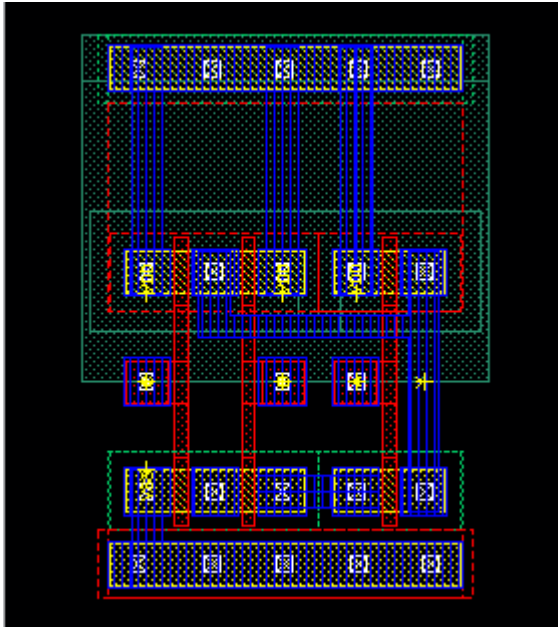


PTAP_TO_TR_SPACE	0	5.0
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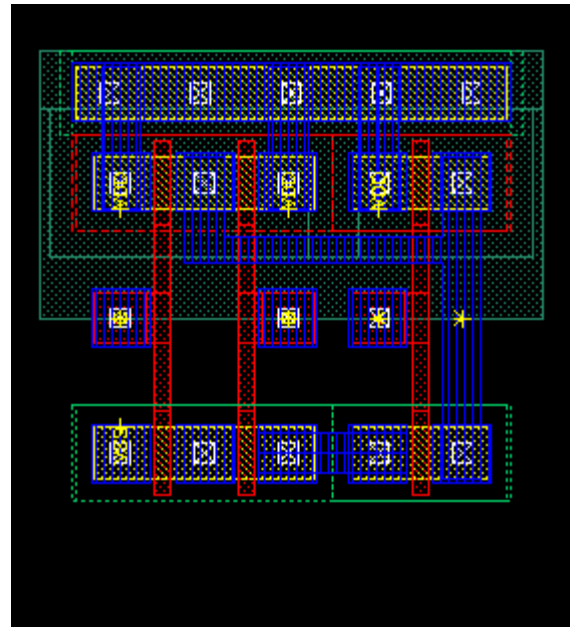


5.4 nand3

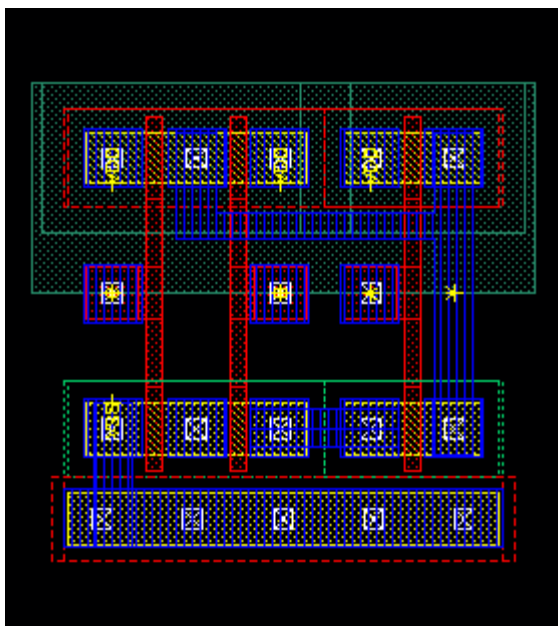
NTAP_TO_TR_SPACE	0	5
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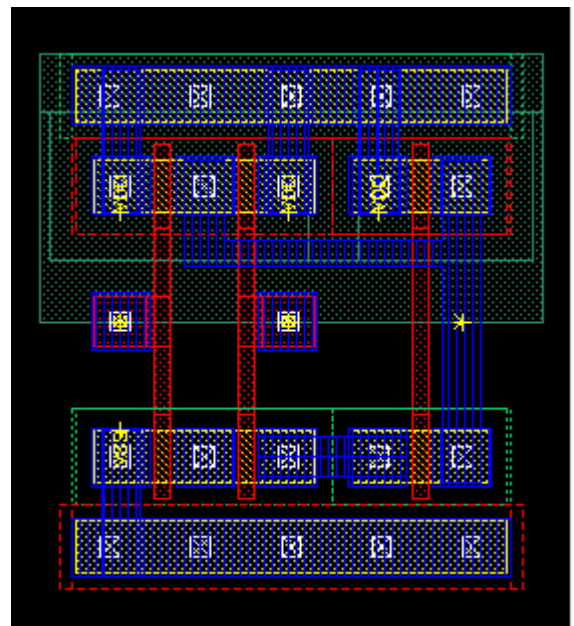
PTAP	☒	☐
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NTAP	☒	☐
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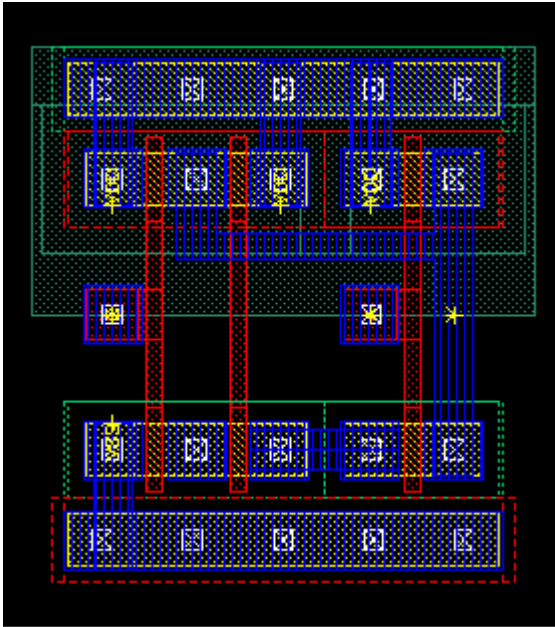


POLY1CONTA	☒	☐
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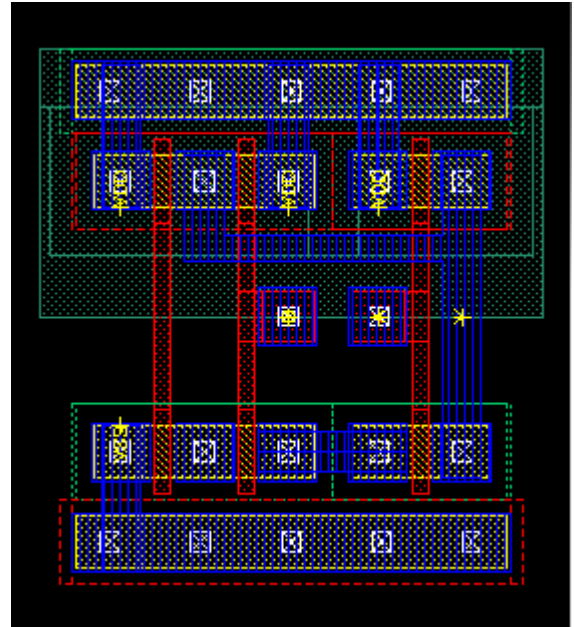


5.4 nand3

POLY1CONTB	☒	→	☐
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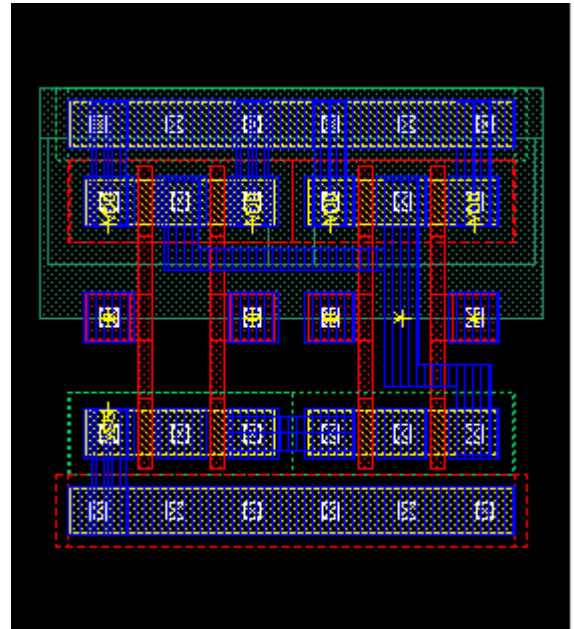


POLY1CONTC	☒	→	☐
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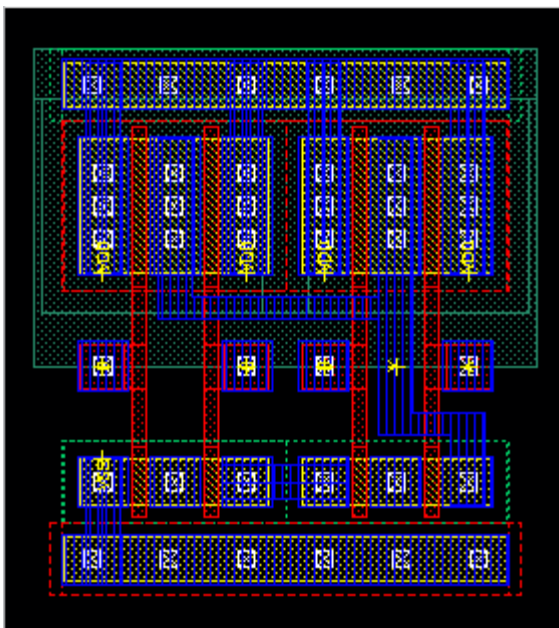


5.5 nand4

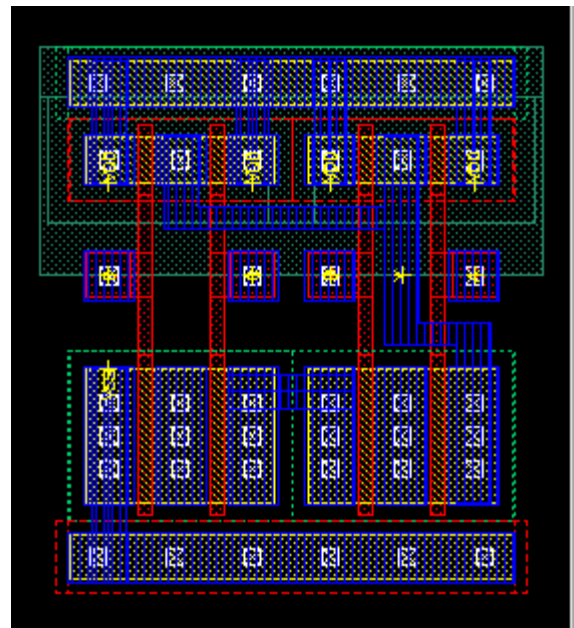
Parameter	Default	Change
P_WIDTH	1.6	<i>value</i>
N_WIDTH	1.6	<i>value</i>
LENGTH	0.5	<i>value</i>
POLY_TO_PTR_SPACE	2.6	<i>value</i>
POLY_TO_NTR_SPACE	2.6	<i>value</i>
PTAP_TO_TR_SPACE	0	<i>value</i>
N_TAP_TO_TR_SPACE	0	<i>value</i>
PTAP	☒	☐
N_TAP	☒	☐
POLY1CONTA	☒	☐
POLY1CONTB	☒	☐
POLY1CONTC	☒	☐
POLY1CONTD	☒	☐



P_WIDTH	1.6	4.8
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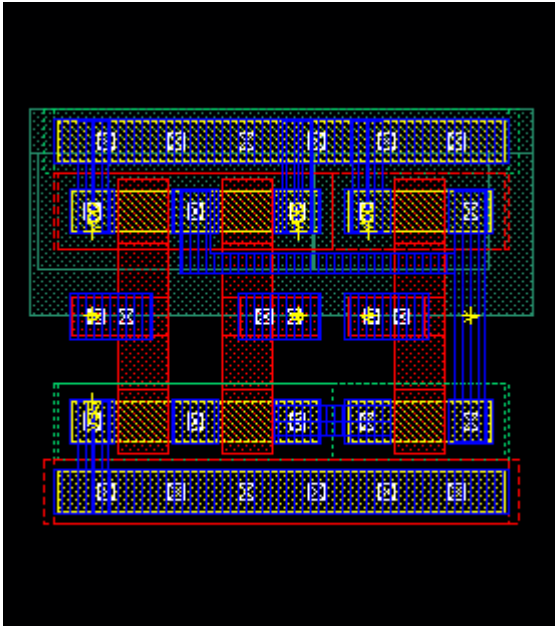


N_WIDTH	1.6	4.8
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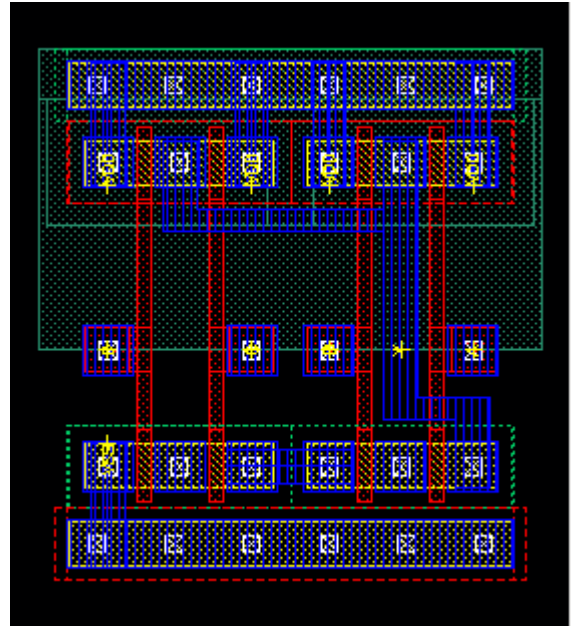


5.5 nand4

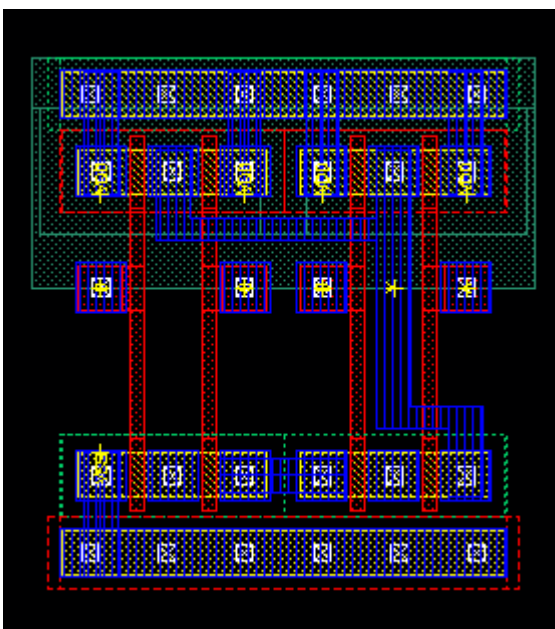
LENGTH	0.5	2
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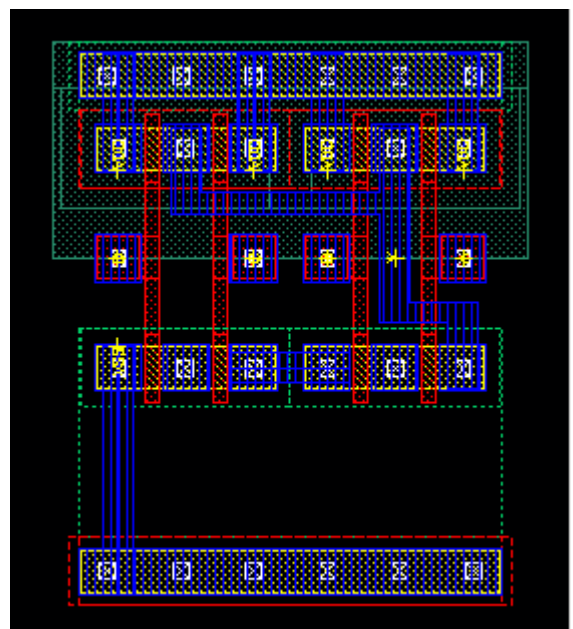
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
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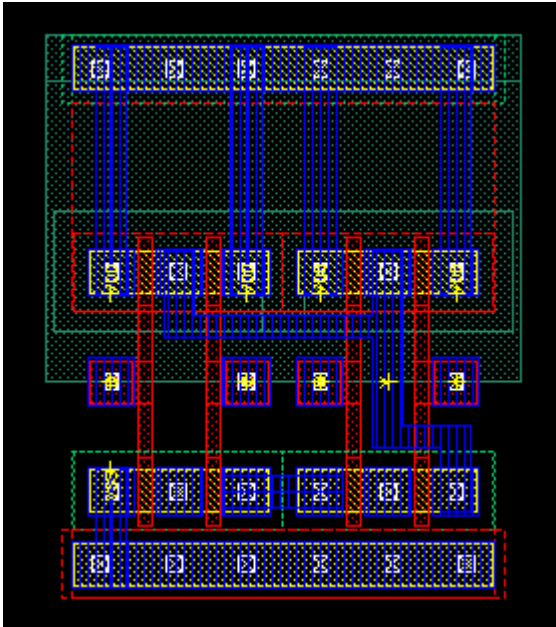


PTAP_TO_TR_SPACE	0	5.0
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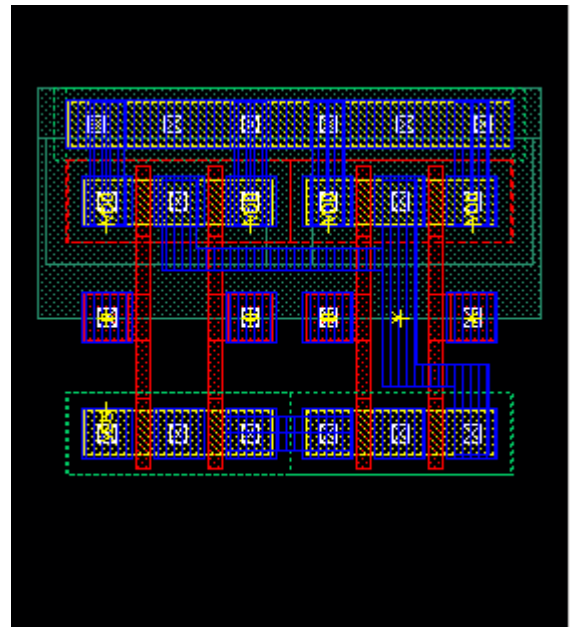


5.5 nand4

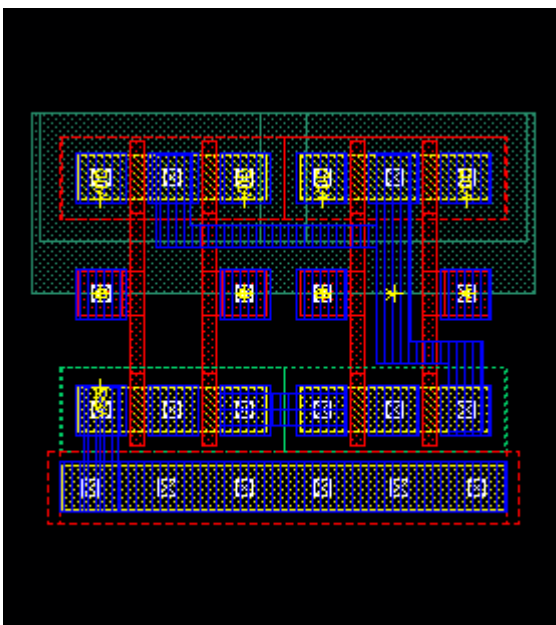
NTAP_TO_TR_SPACE	0	5
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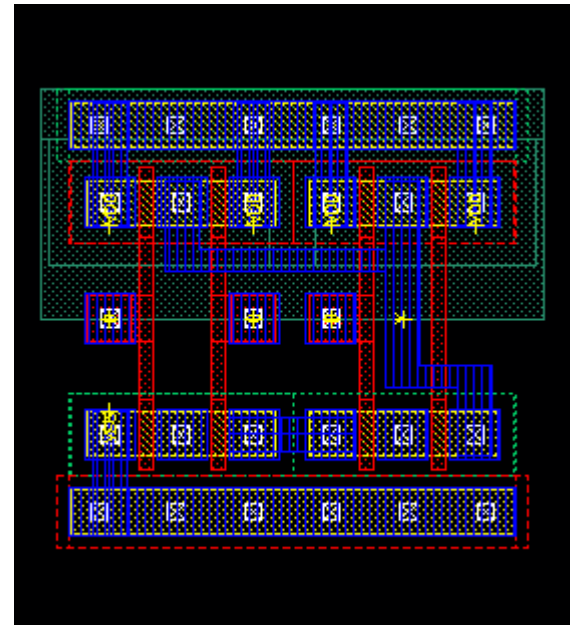
PTAP	☒	☐
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NTAP	☒	☐
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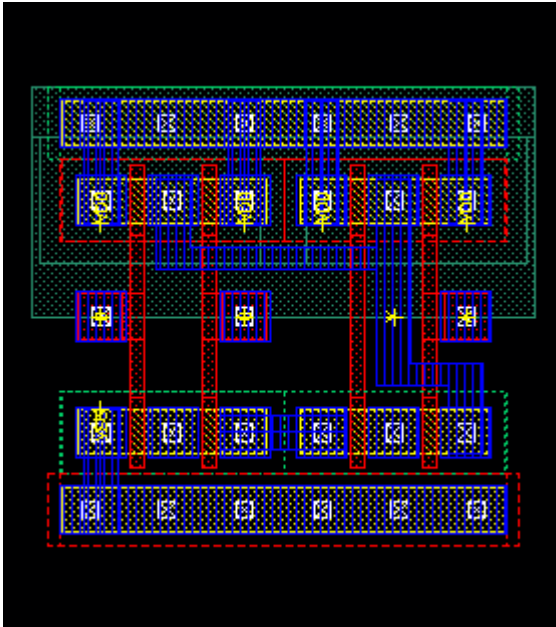


POLY1CONTA	☒	☐
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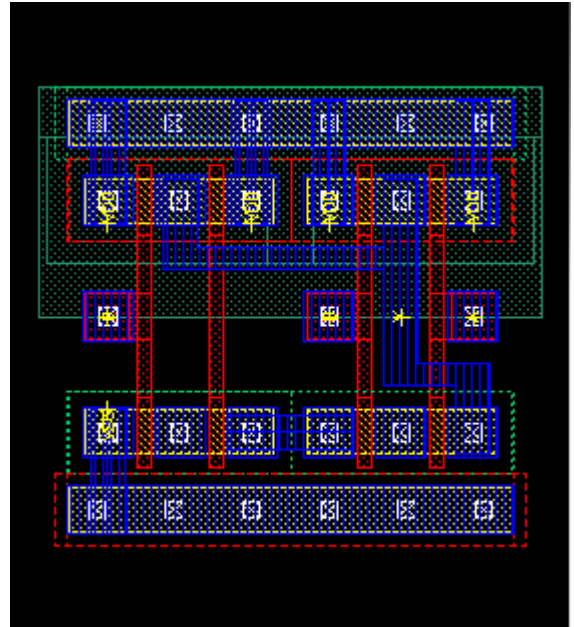


5.5 nand4

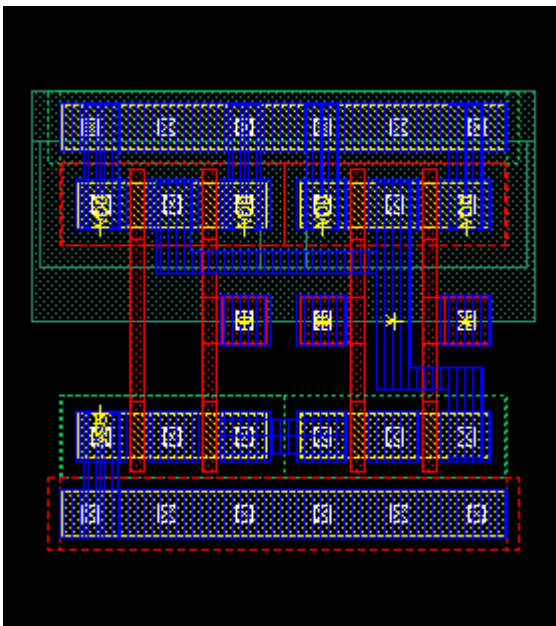
POLY1CONTB	☒	→	☐
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POLY1CONTC	☒	→	☐
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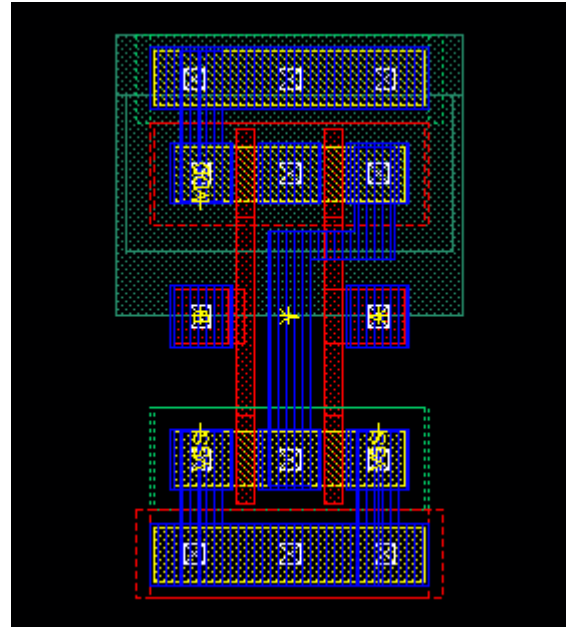


POLY1CONTD	☒	→	☐
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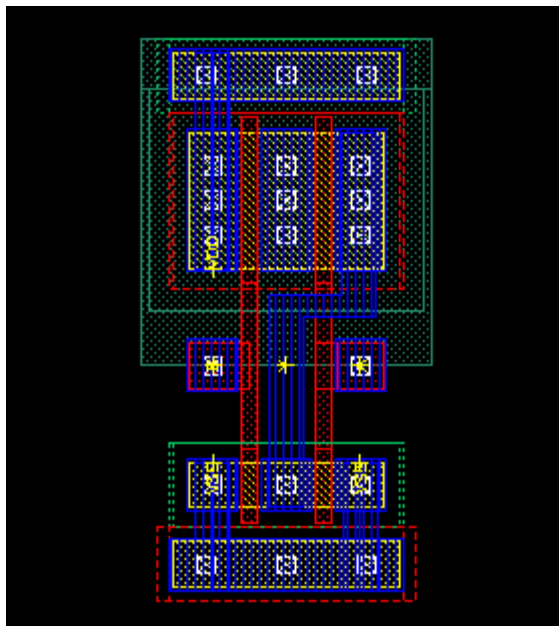


5.6 nor2

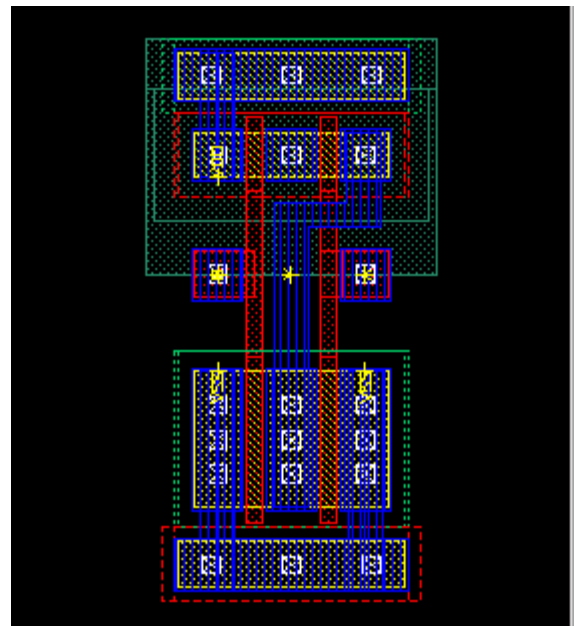
Parameter	Default	Change
P_WIDTH	1.6	value
N_WIDTH	1.6	value
LENGTH	0.5	value
POLY_TO_PTR_SPACE	2.6	value
POLY_TO_NTR_SPACE	2.6	value
PTAP_TO_TR_SPACE	0	value
NTAP_TO_TR_SPACE	0	value
PTAP	☒	☐
NTAP	☒	☐
POLY1CONTA	☒	☐
POLY1CONTB	☒	☐



P_WIDTH	1.6	4.8
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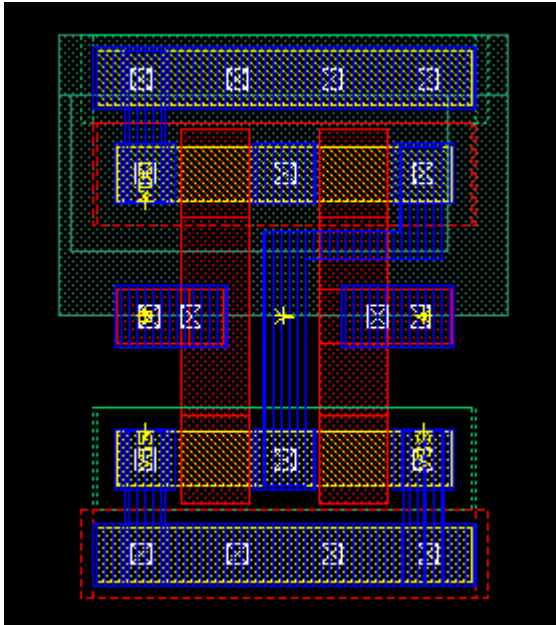


N_WIDTH	1.6	4.8
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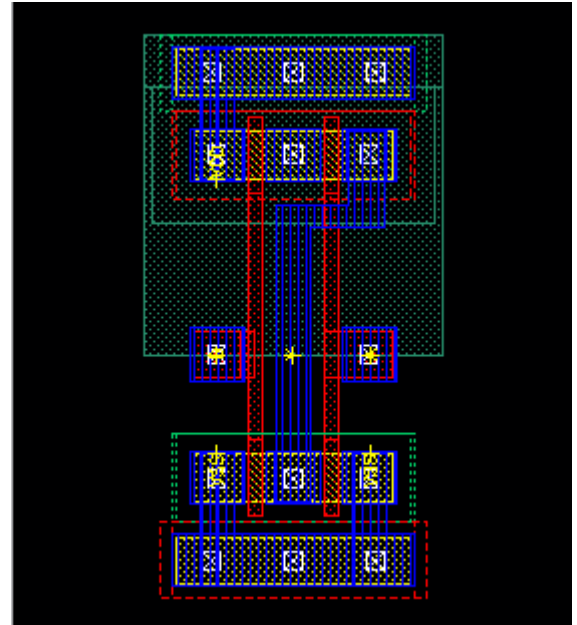


5.6 nor2

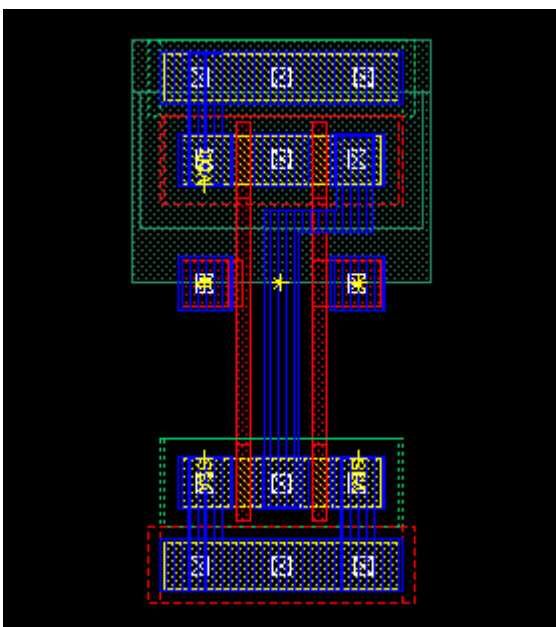
LENGTH	0.5	2
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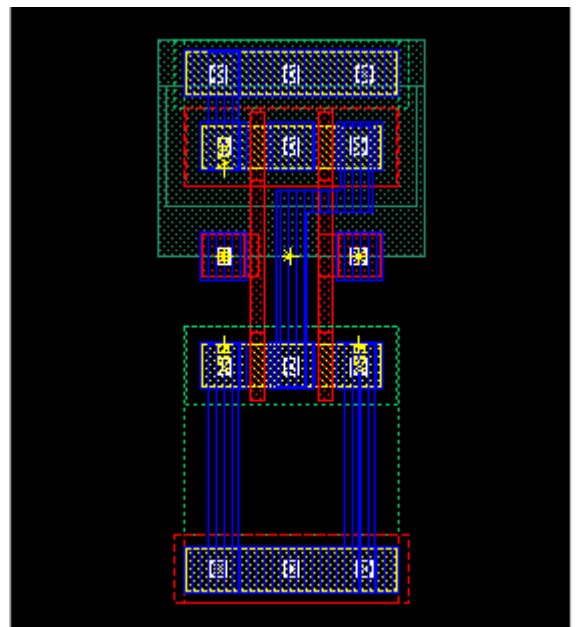
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
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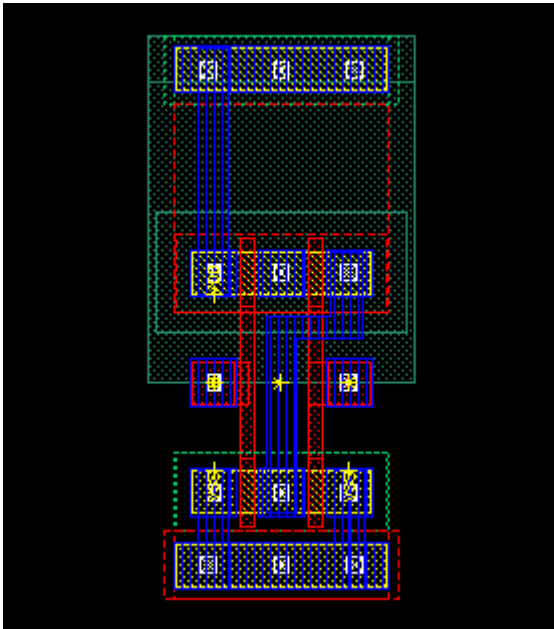


PTAP_TO_TR_SPACE	0	5.0
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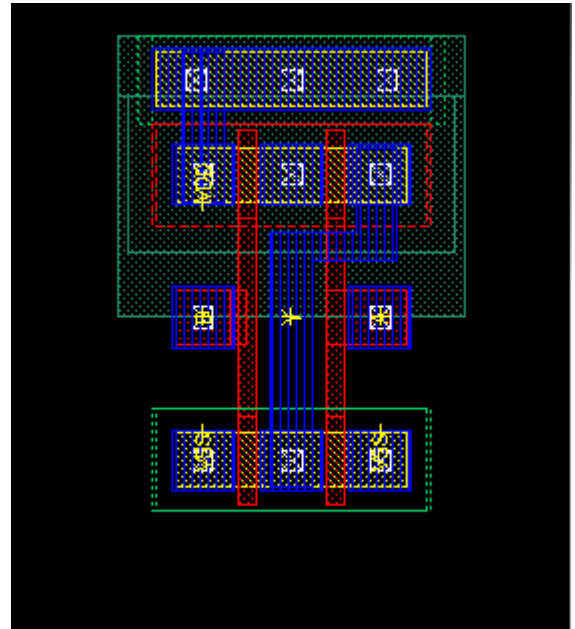


5.6 nor2

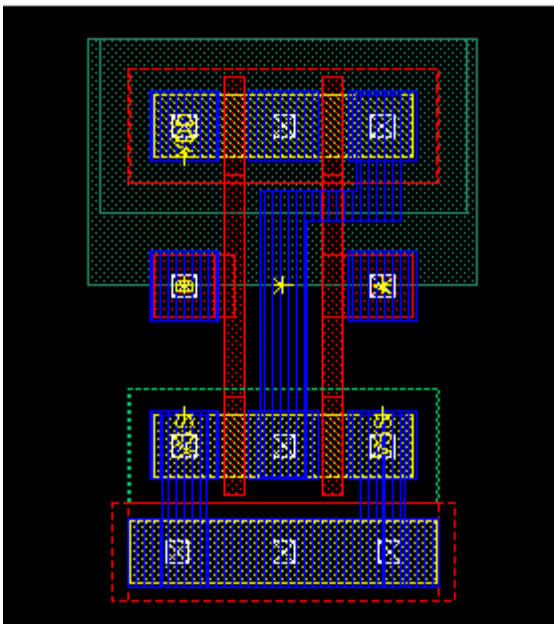
NTAP_TO_TR_SPACE	0	5
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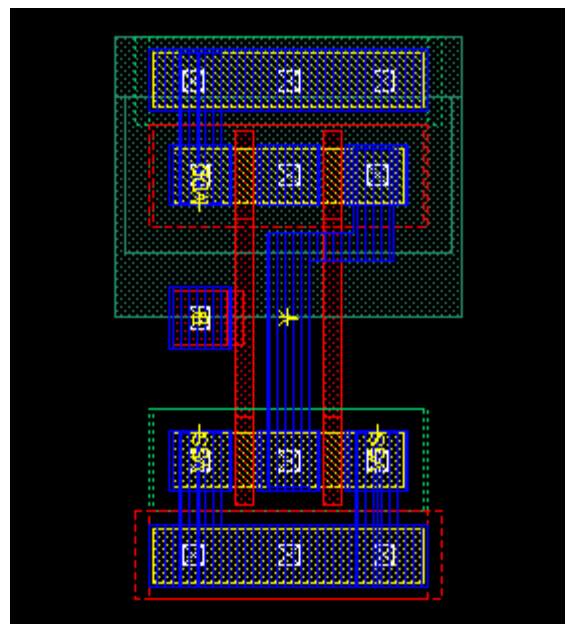
PTAP	☒	☐
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NTAP	☒	☐
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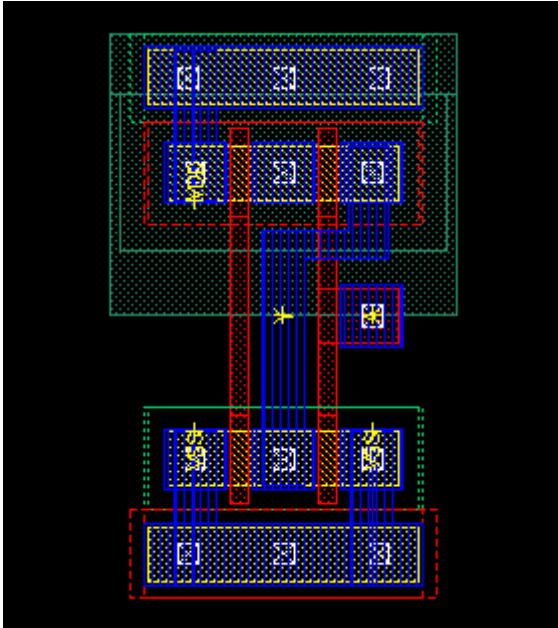


POLY1CONTA	☒	☐
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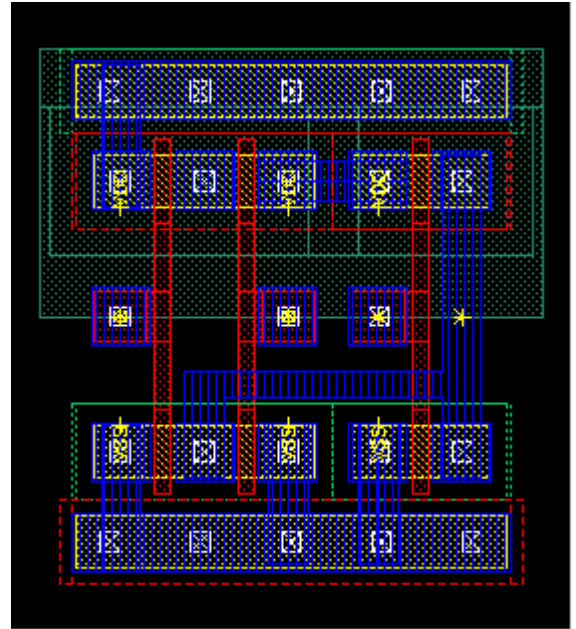
5.6 nor2

POLY1CONTB	☒	→	☐
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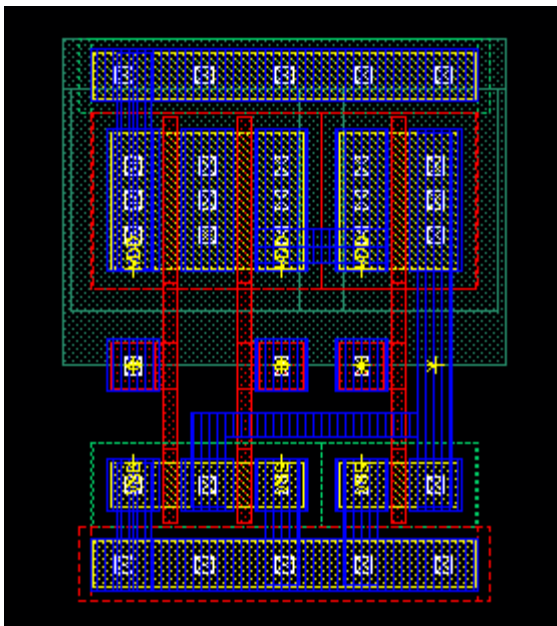


5.7 nor3

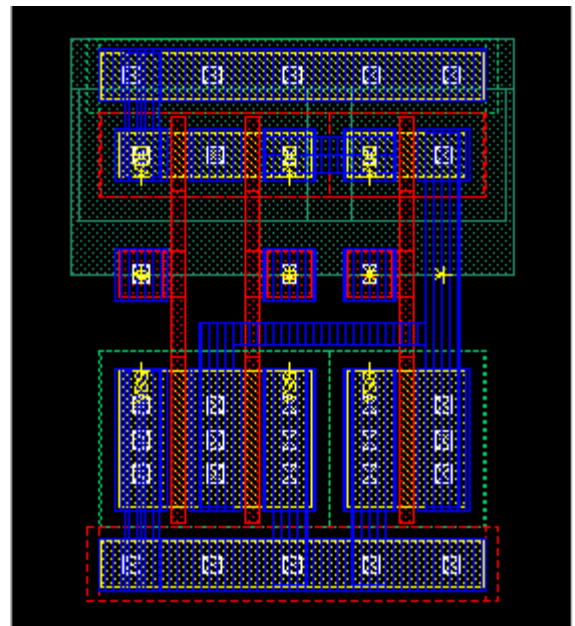
Parameter	Default	Change
P_WIDTH	1.6	<i>value</i>
N_WIDTH	1.6	<i>value</i>
LENGTH	0.5	<i>value</i>
POLY_TO_PTR_SPACE	2.6	<i>value</i>
POLY_TO_NTR_SPACE	2.6	<i>value</i>
PTAP_TO_TR_SPACE	0	<i>value</i>
NTAP_TO_TR_SPACE	0	<i>value</i>
PTAP	☒	☐
NTAP	☒	☐
POLY1CONTA	☒	☐
POLY1CONTB	☒	☐
POLY1CONTC	☒	☐



P_WIDTH	1.6	4.8
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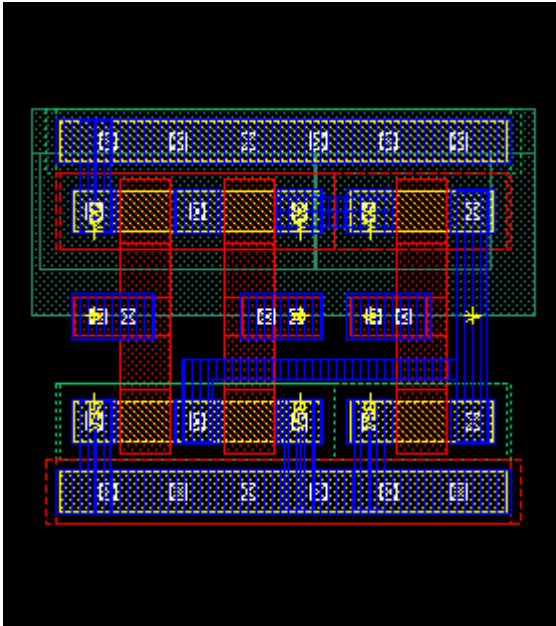


N_WIDTH	1.6	4.8
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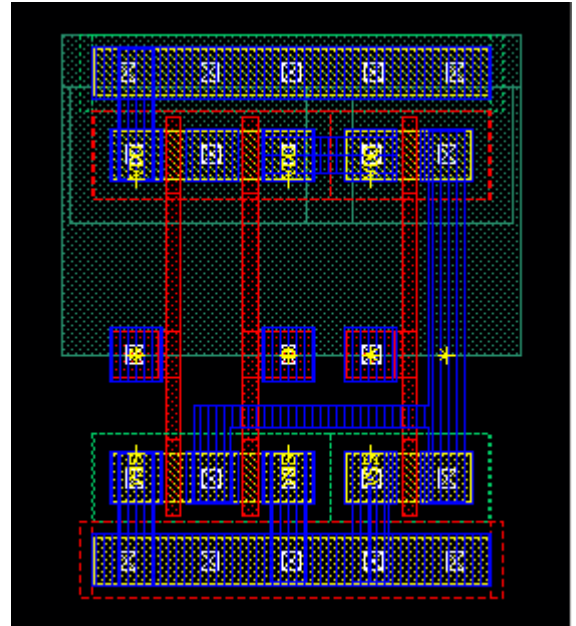


5.7 nor3

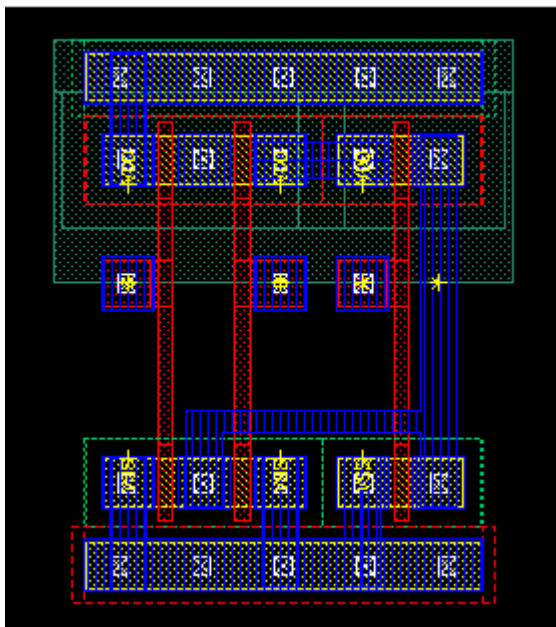
LENGTH	0.5	2
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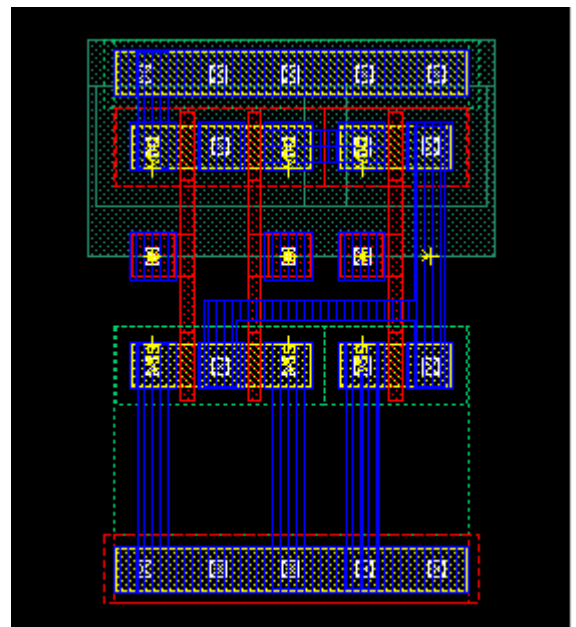
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
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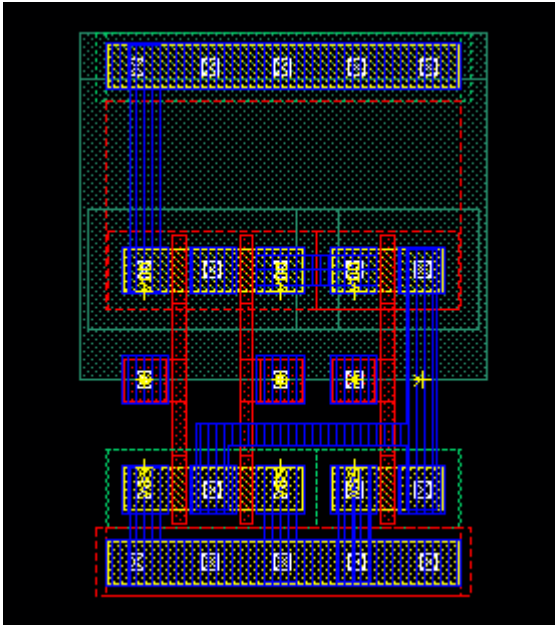


PTAP_TO_TR_SPACE	0	5.0
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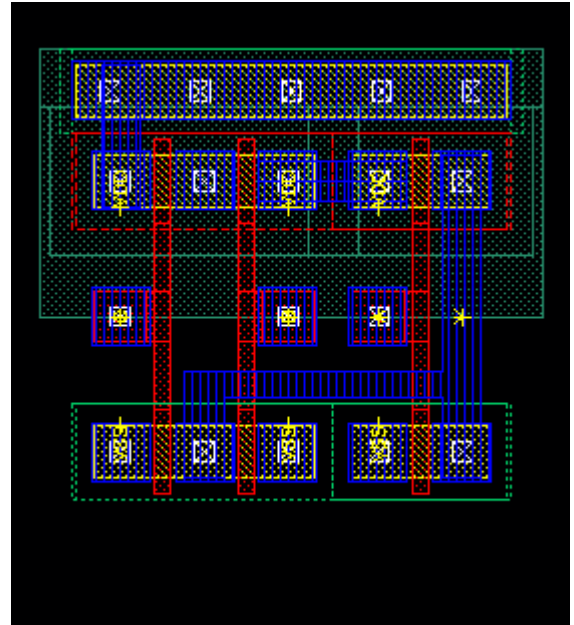


5.7 nor3

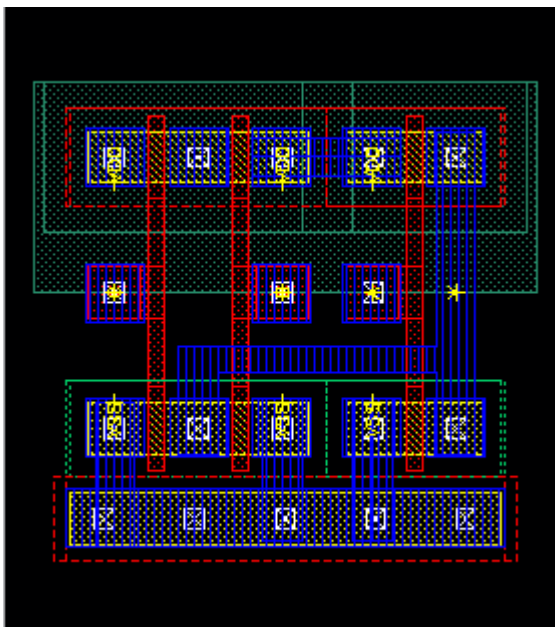
NTAP_TO_TR_SPACE	0	5
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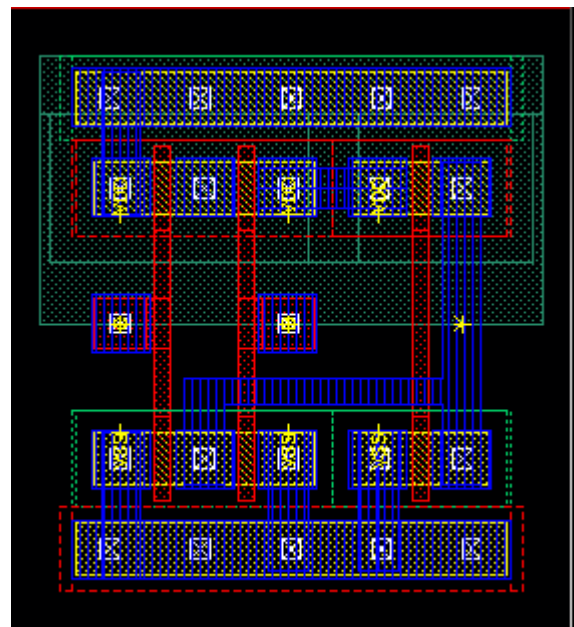
PTAP	☒	☐
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NTAP	☒	☐
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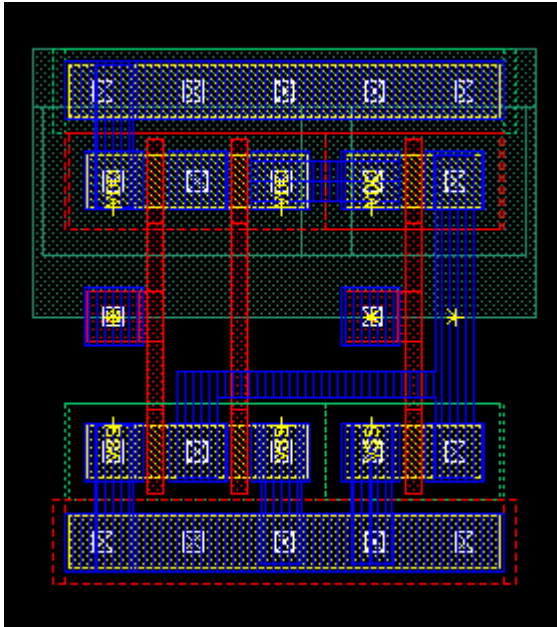


POLY1CONTA	☒	☐
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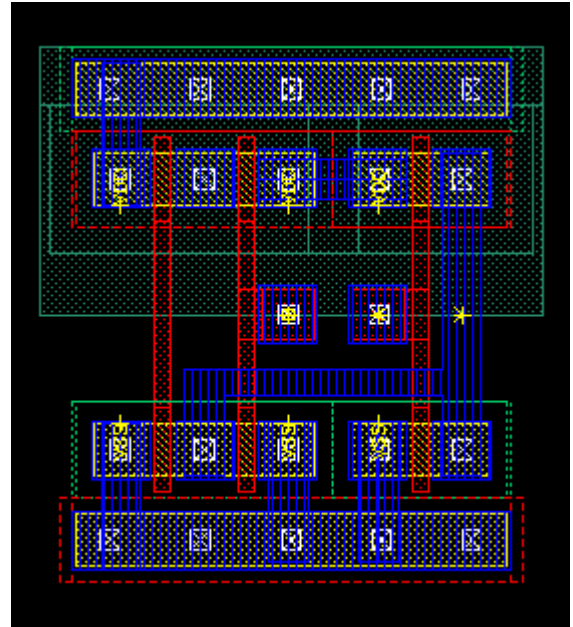


5.7 nor3

POLY1CONTB	☒	→	☐
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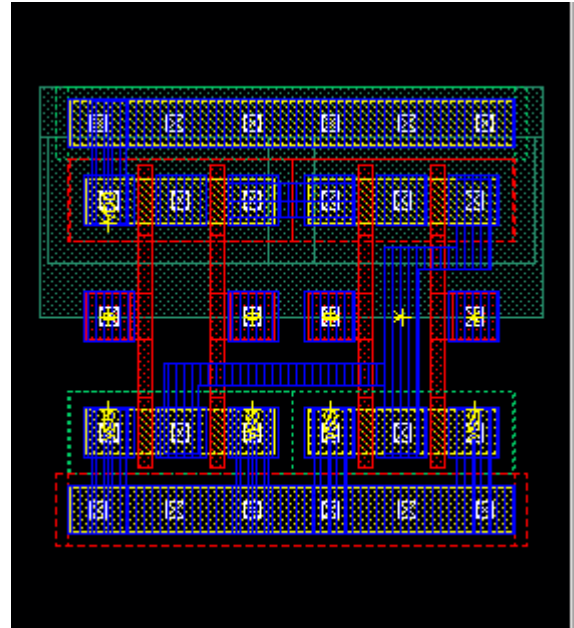


POLY1CONTC	☒	→	☐
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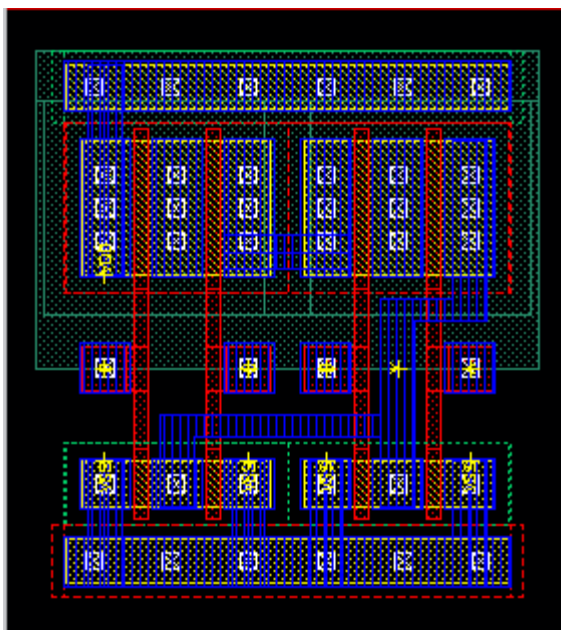


5.8 nor4

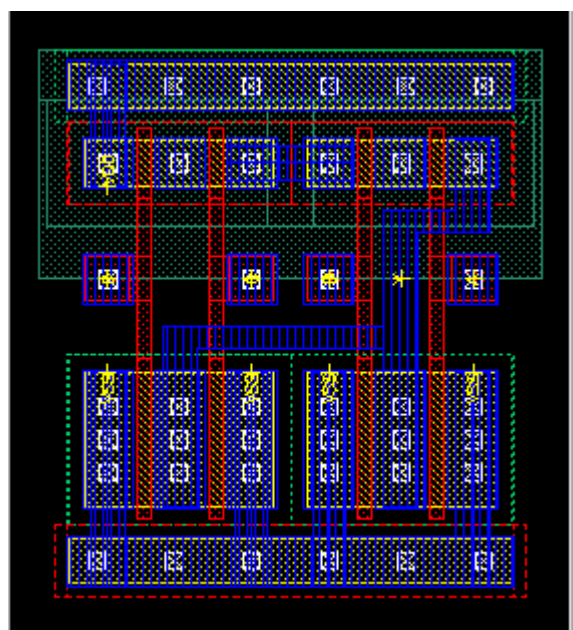
Parameter	Default	Change
P_WIDTH	1.6	<i>value</i>
N_WIDTH	1.6	<i>value</i>
LENGTH	0.5	<i>value</i>
POLY_TO_PTR_SPACE	2.6	<i>value</i>
POLY_TO_NTR_SPACE	2.6	<i>value</i>
PTAP_TO_TR_SPACE	0	<i>value</i>
NTAP_TO_TR_SPACE	0	<i>value</i>
PTAP	☒	☐
NTAP	☒	☐
POLY1CONTA	☒	☐
POLY1CONTB	☒	☐
POLY1CONTC	☒	☐
POLY1CONTD	☒	☐



P_WIDTH	1.6	4.8
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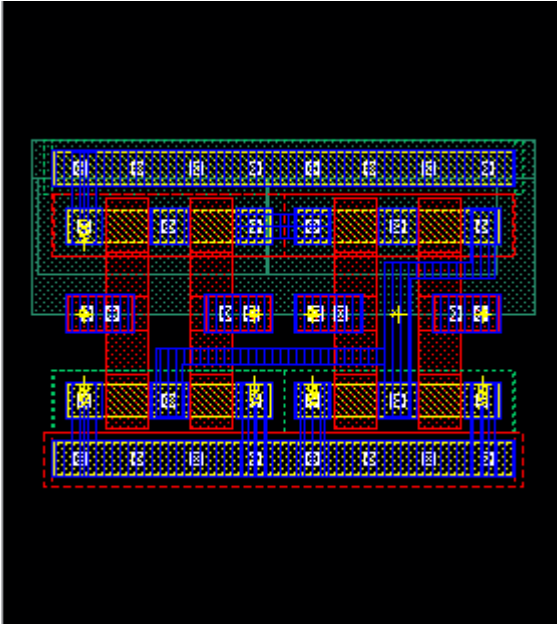


N_WIDTH	1.6	4.8
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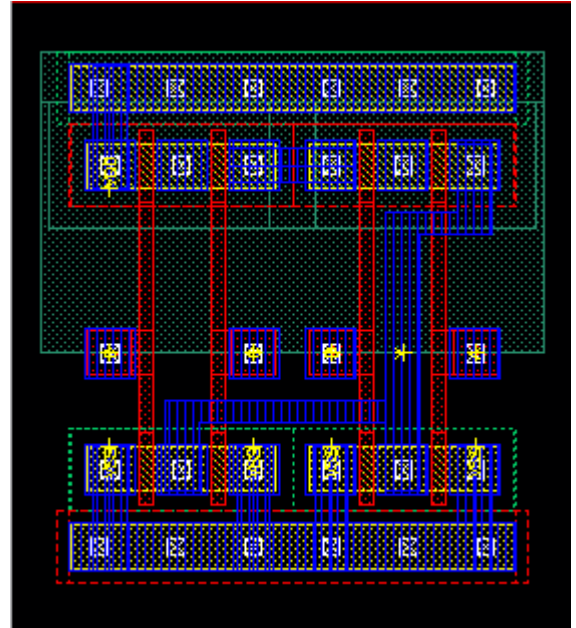


5.8 nor4

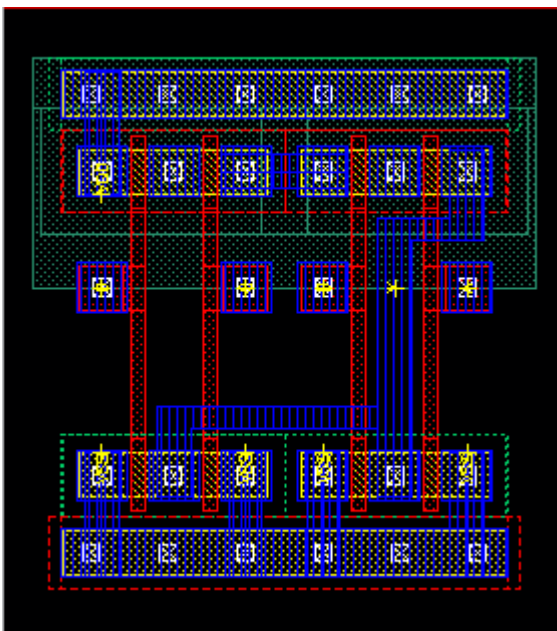
LENGTH	0.5	2
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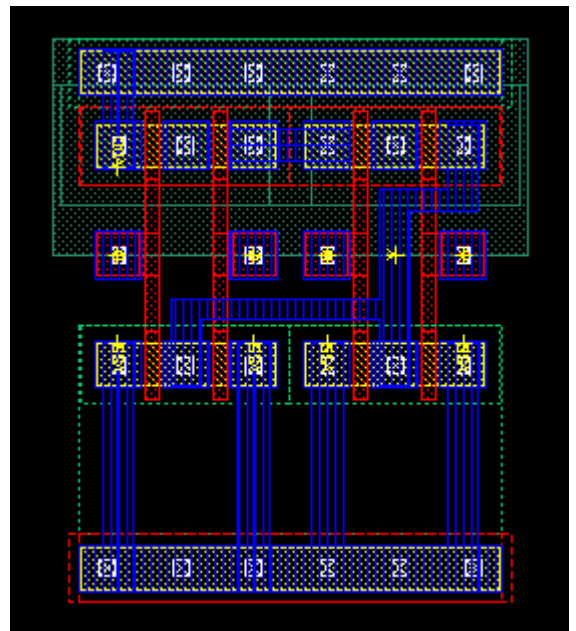
POLY_TO_PTR_SPACE	2.6	5.2
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POLY_TO_NTR_SPACE	2.6	5.2
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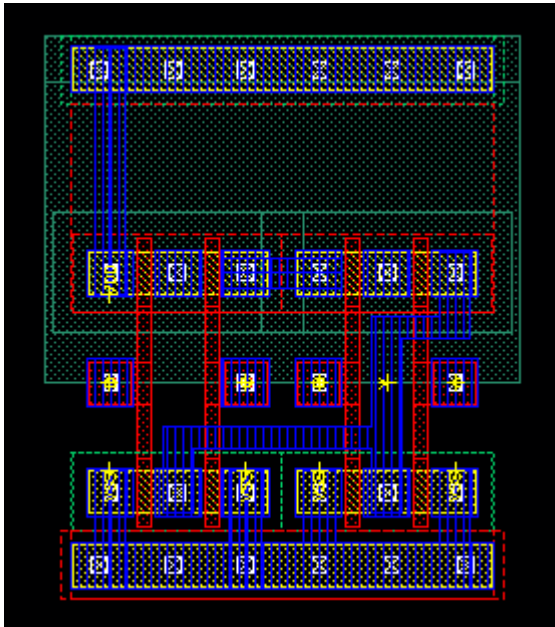


PTAP_TO_TR_SPACE	0	5.0
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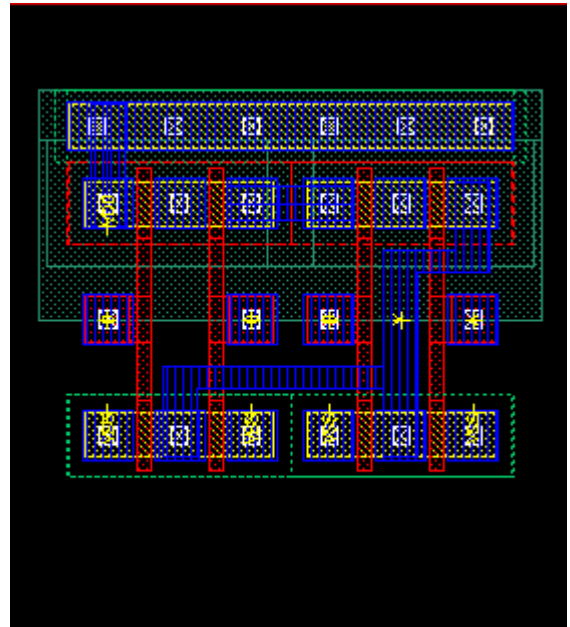


5.8 nor4

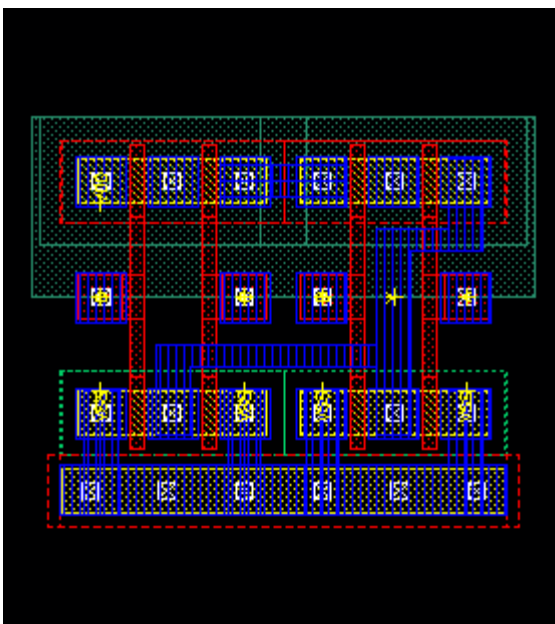
NTAP_TO_TR_SPACE	0	5
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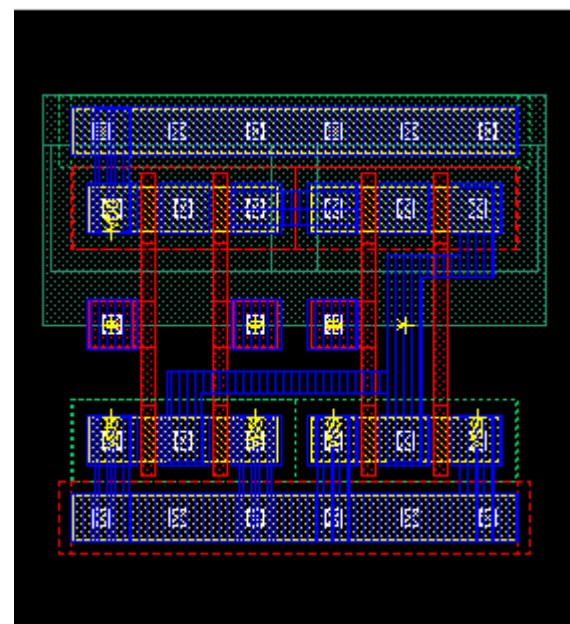
PTAP	☒	☐
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NTAP	☒	☐
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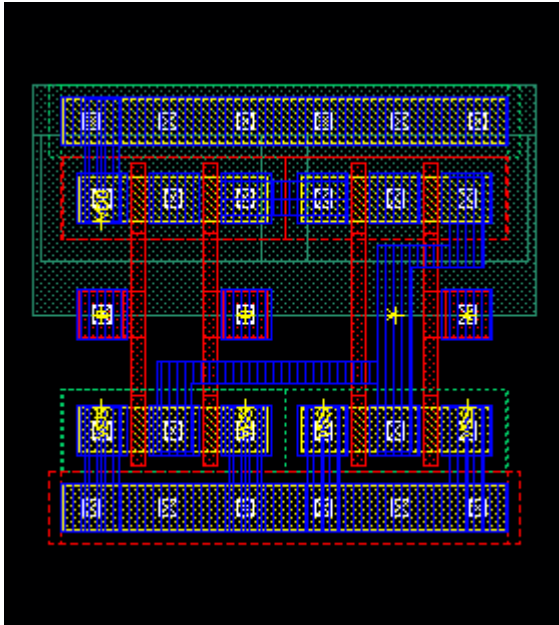


POLY1CONTA	☒	☐
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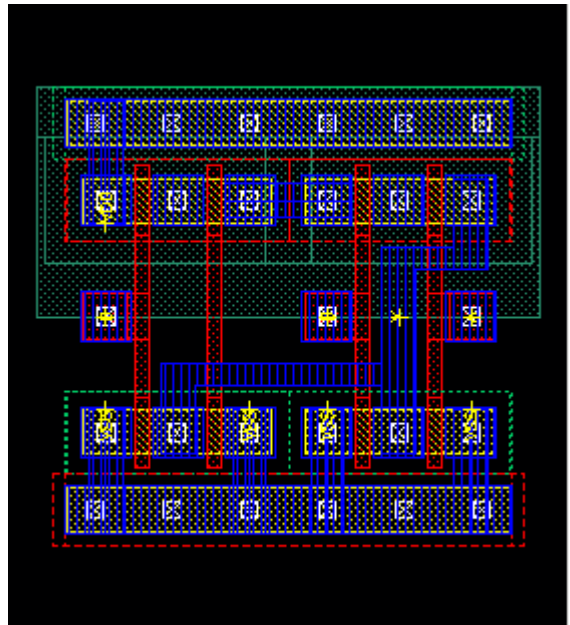


5.8 nor4

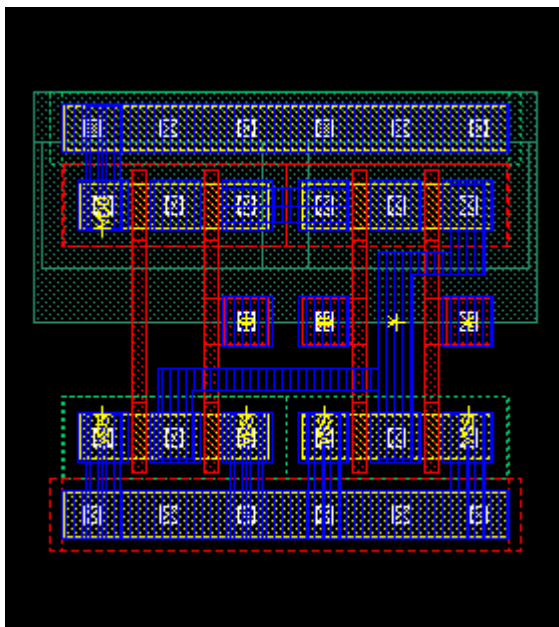
POLY1CONTB	☒	→	☐
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POLY1CONTC	☒	→	☐
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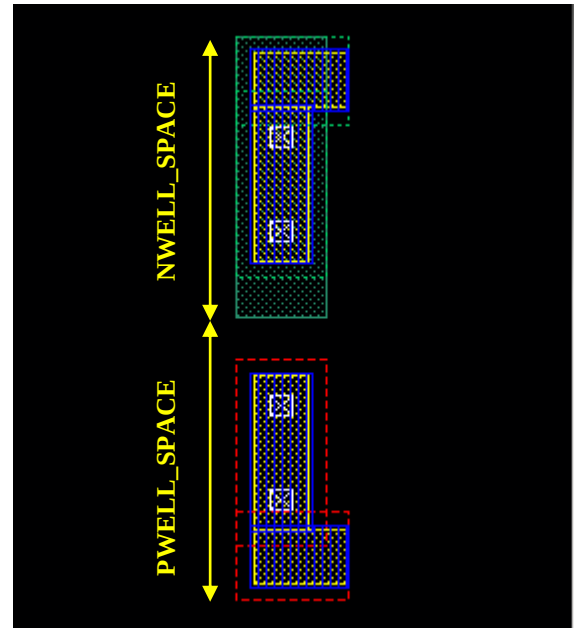


POLY1CONTD	☒	→	☐
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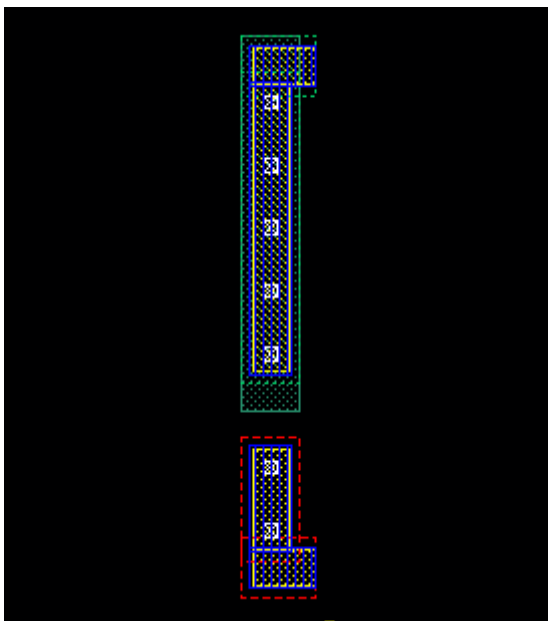


5.9 edge

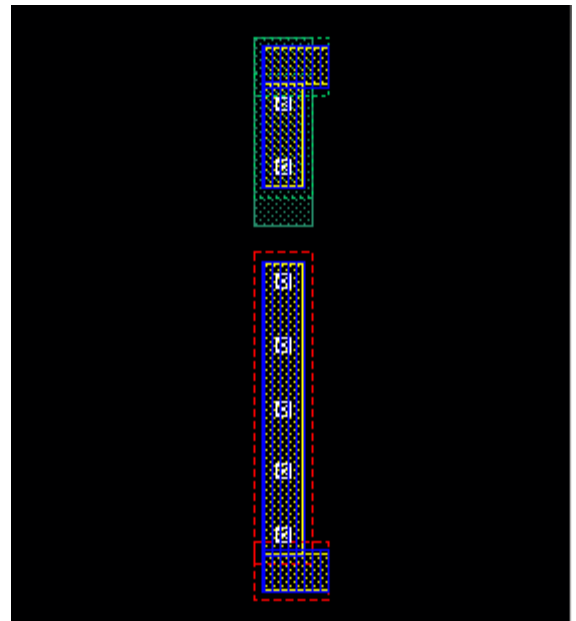
Parameter	Default	Change
NWELL_SPACE	8.3	<i>value</i>
PWELL_SPACE	8.3	<i>value</i>



NWELL_SPACE	8.3	16.6
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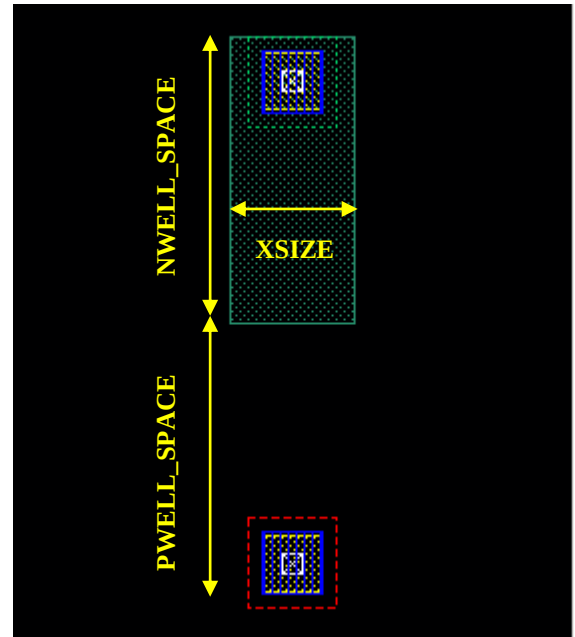


PWELL_SPACE	8.3	16.6
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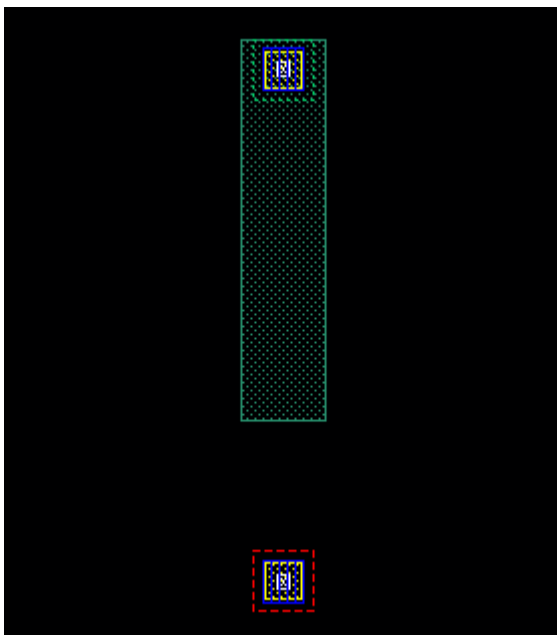


5.10 filler

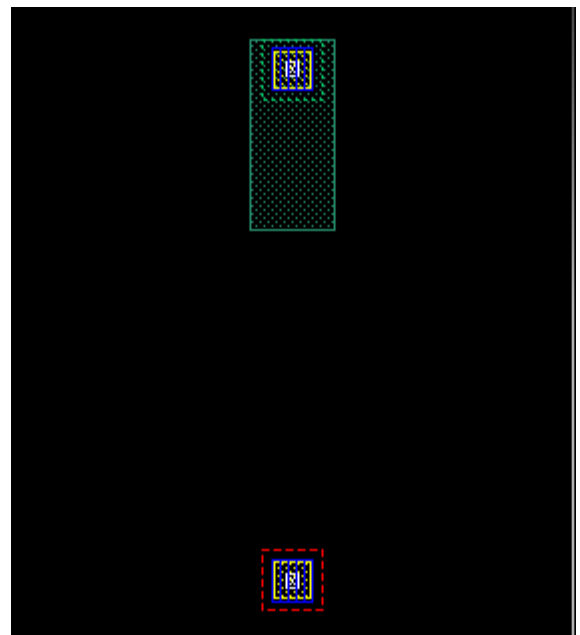
Parameter	Default	Change
NWELL_SPACE	8.3	<i>value</i>
PWELL_SPACE	8.3	<i>value</i>
XSIZE	1.8	<i>value</i>



NWELL_SPACE	8.3	16.6
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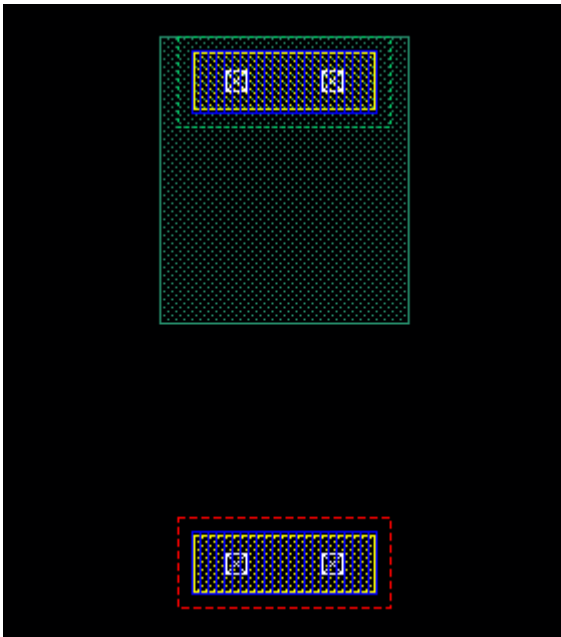


PWELL_SPACE	8.3	16.6
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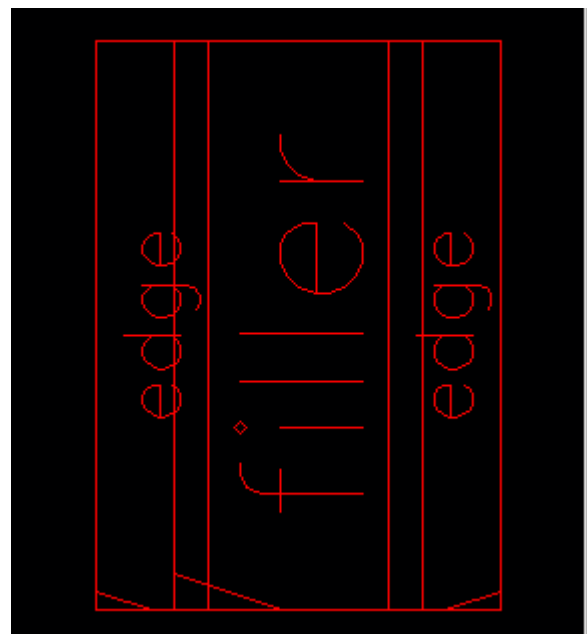
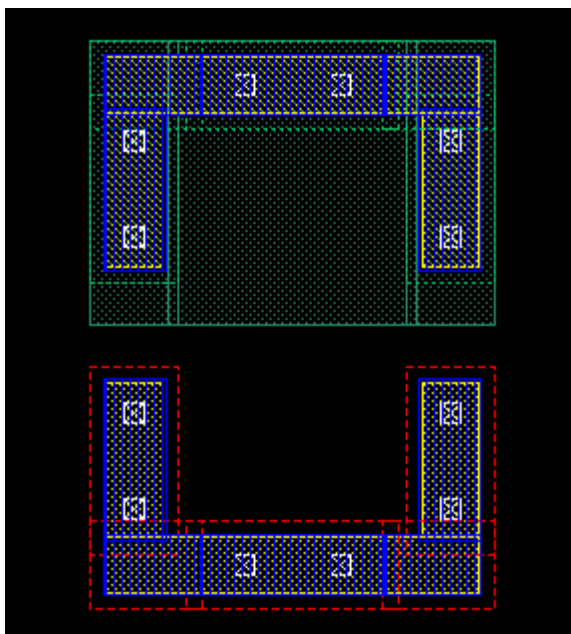


5.10 filler

XSIZE	1.8	5.4
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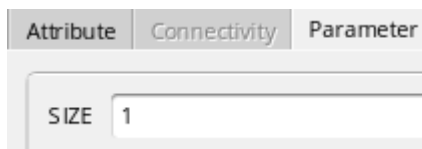
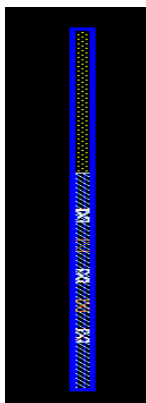
5.11 edge_filler



6. Sealing

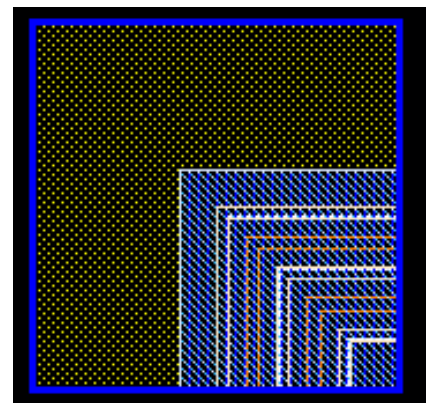
6.1 Sealing information [[ETRI_0p5um_Analog_CMOS_2P3M_5V_NSPL_pcell](#)]

- For protect layout block from DIE cutting, you can use sealing cell in ETRI_0p5um_Analog_CMOS_5p0V_pcell library.
(SEALRING_UNIT, SEALRING_CONER)



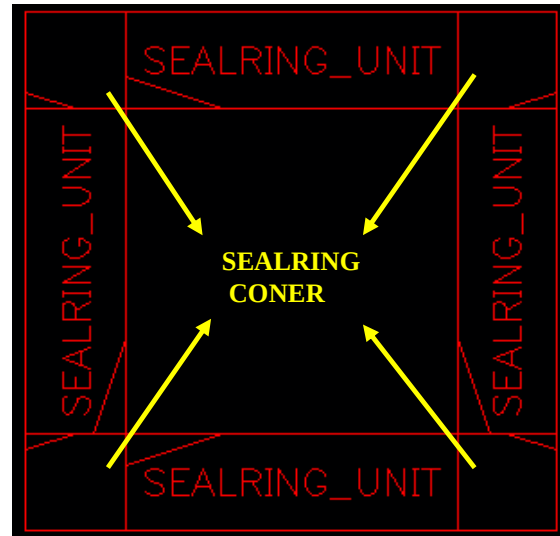
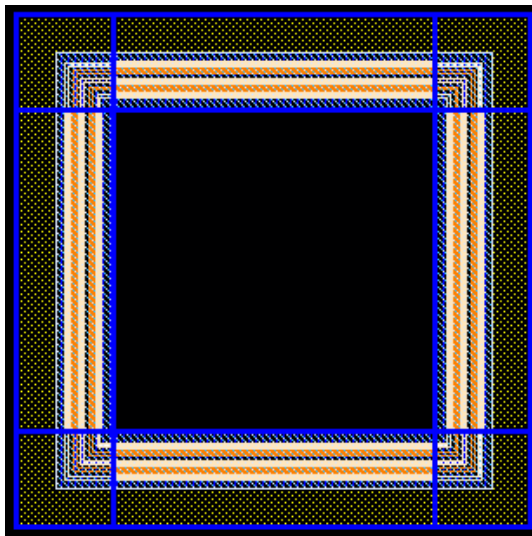
Can adjust size using SIZE parameter

SIZE Minimum value is 1



-SEALRING_UNIT

-SEALRING_CONER

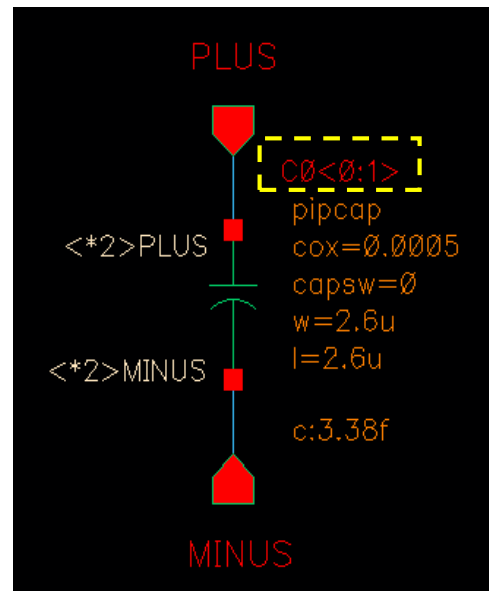
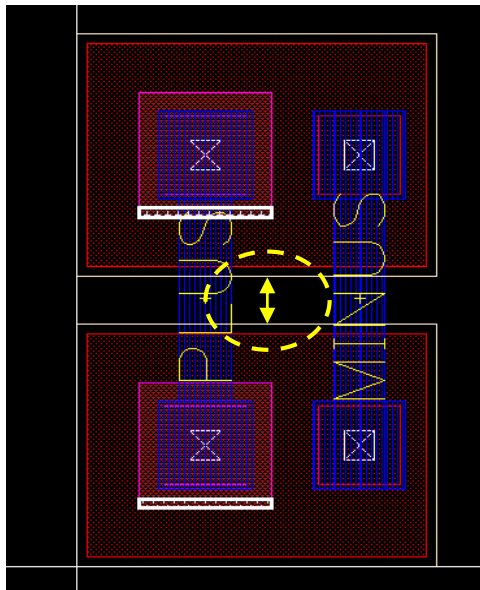


-Adjusted SEALRING_UNIT and SEALRING_CONER

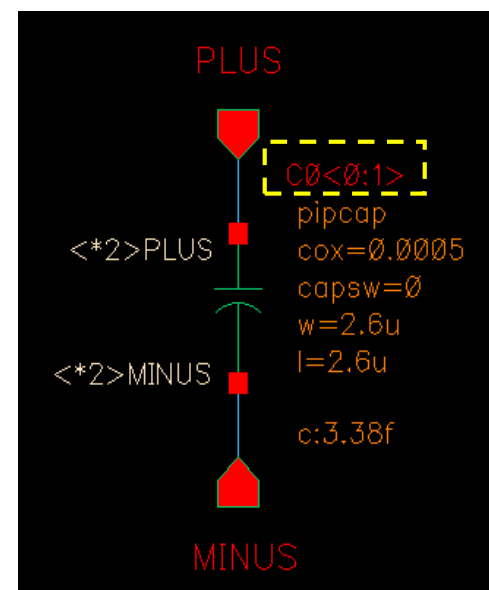
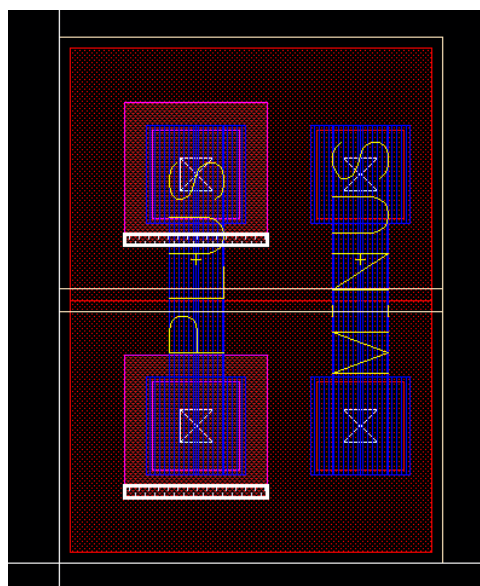
7. LVS Guide

7.1 PIPCAP Layout/Circuit Structure

- Correct Case : **Separated** PIPCAP

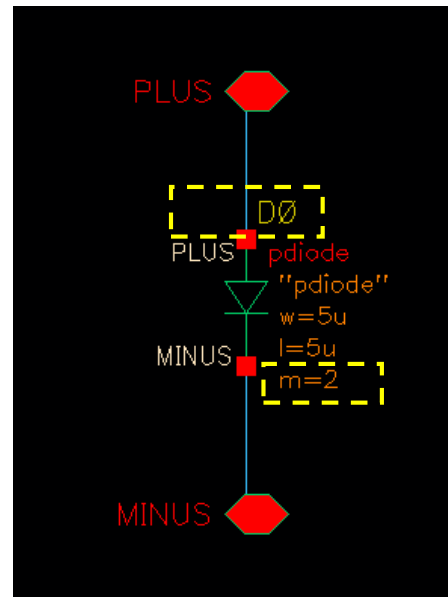
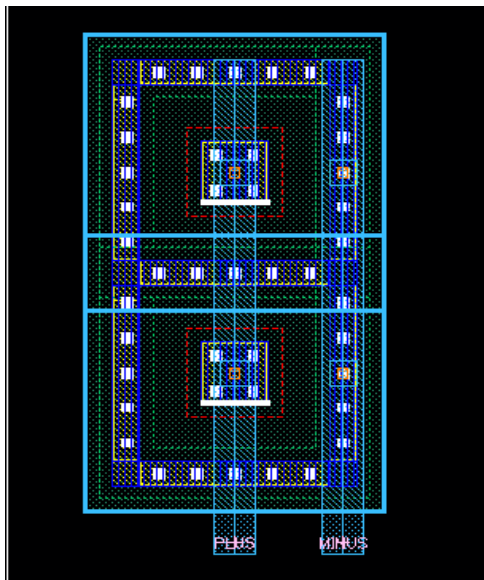


- Incorrect Case : **Merged** PDIODE

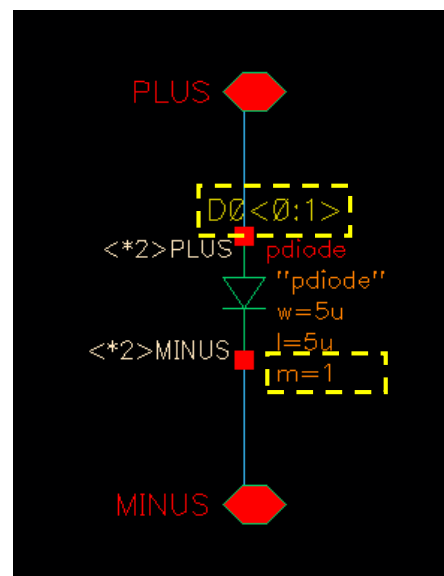
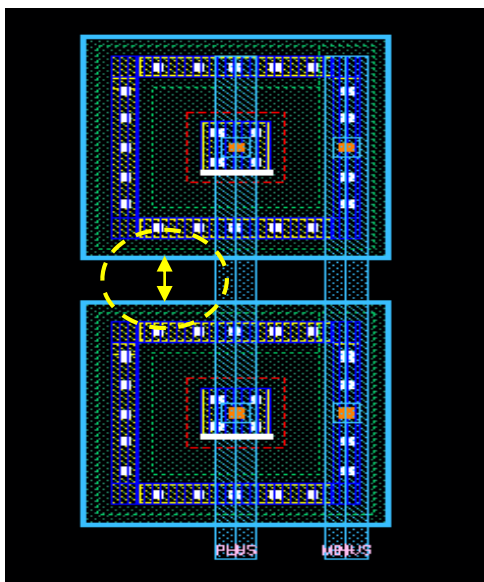


7.2 PDIODE Layout/Circuit Structure

- Case1 : **Merged** PDIODE

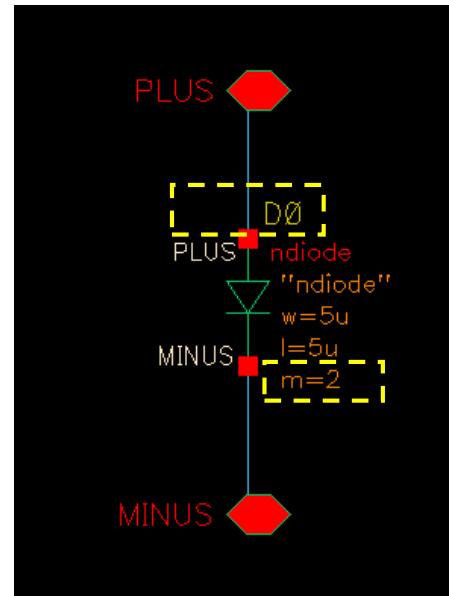
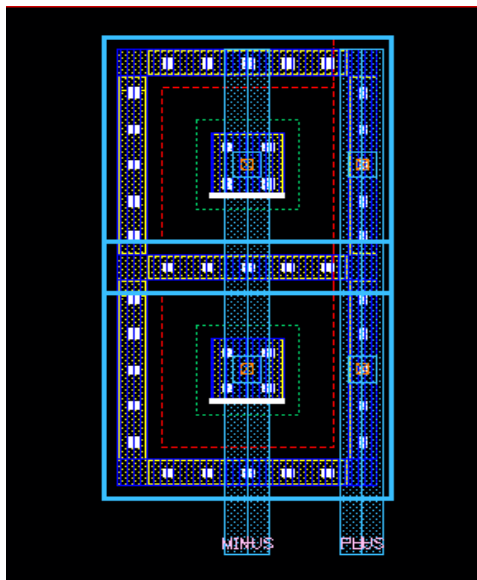


- Case2 : **Separated** PDIODE

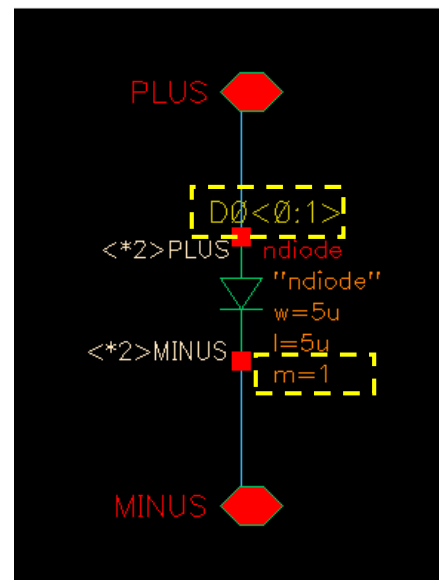
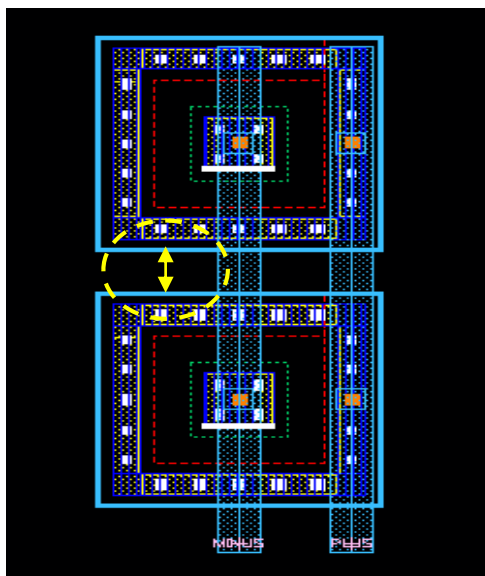


7.3 NDIODE Layout/Circuit Structure

- Case1 : **Merged** NDIODE



- Case2 : **Separated** NDIODE

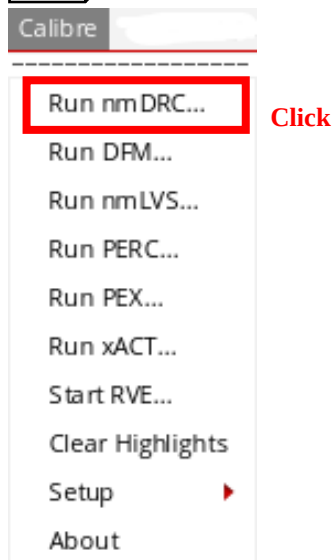


8. Calibre DRC

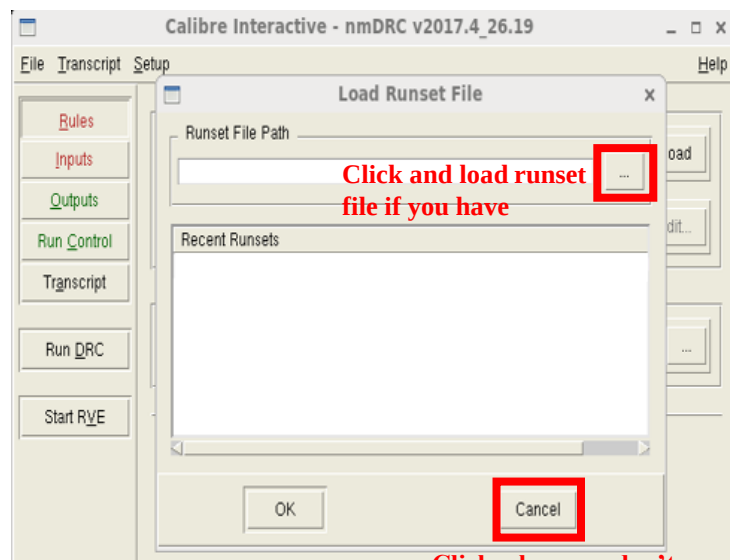
8.1 Run for DRC at GUI

- For running DRC(Design Rule Check) at GUI after layout, need to several steps like below pictures

1 Click Run nmDRC in Calibre menu

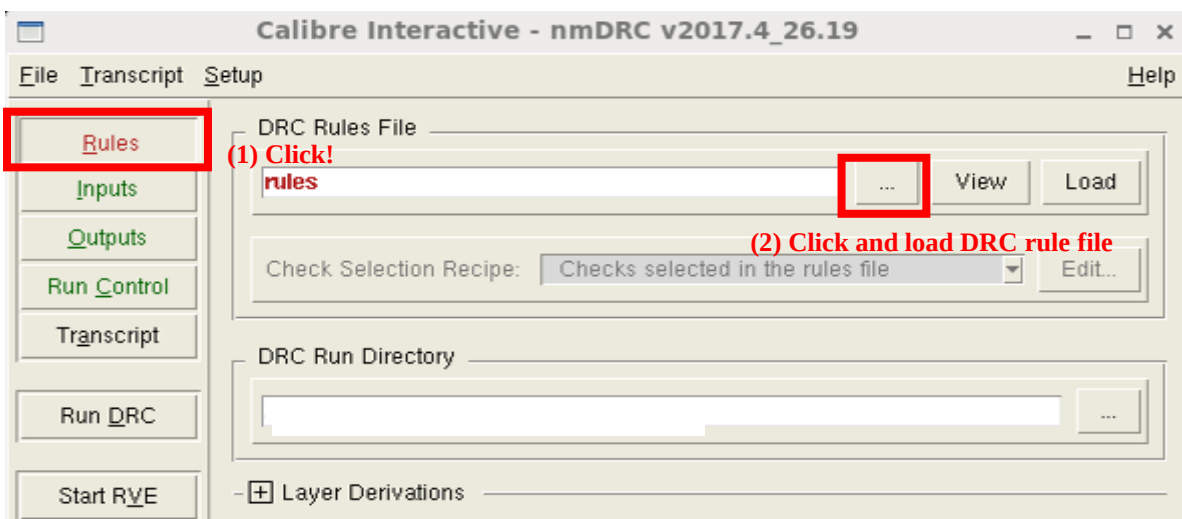


2 If runset file is exist, Load runset file. Runset file is not exist, click cancel

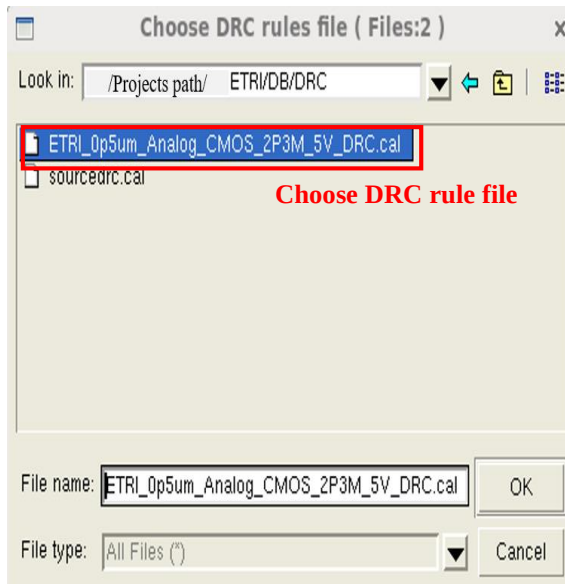


Click when you don't have any runset files

3 Load DRC rules file (ETRI_0p5um_Analog_CMOS_2P3M_5V_DRC.cal)

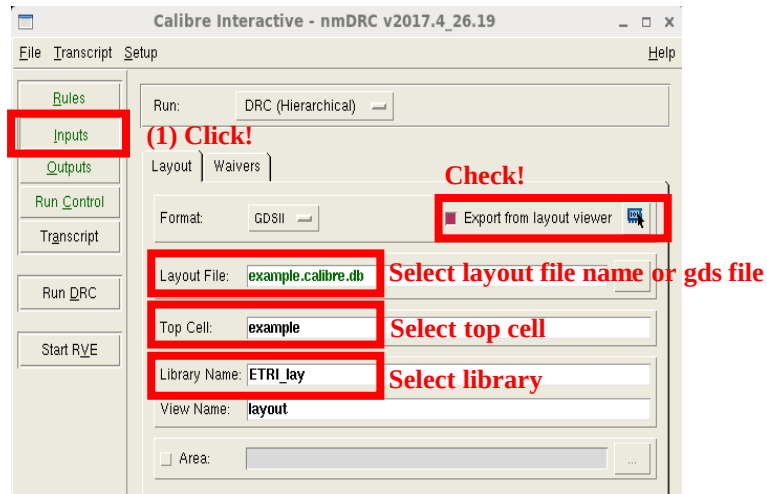


3 Load DRC rules file.

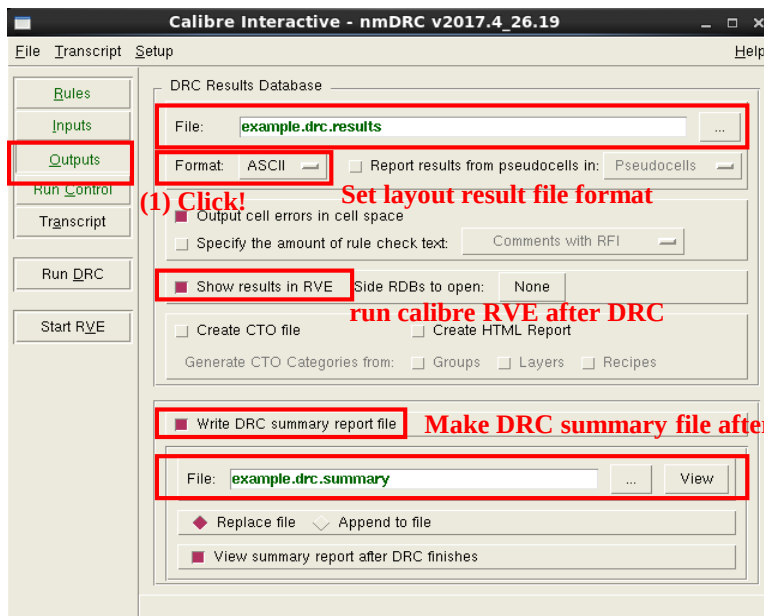


4

Set Input data. If you checked Export from layout viewer, insert layout file name to layout file. Not checked, Input your gds name want to run.

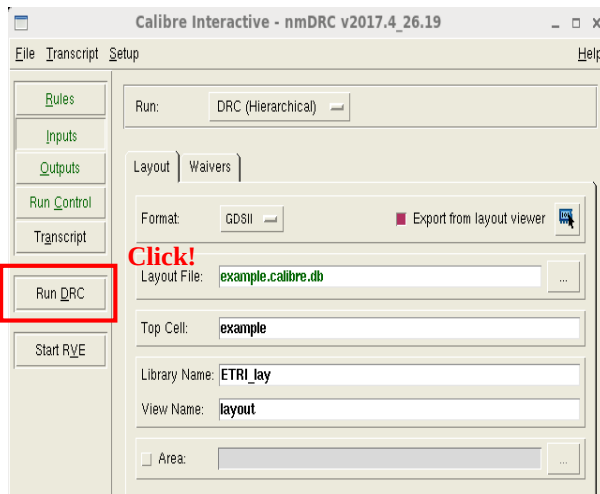


5 Set layout output data



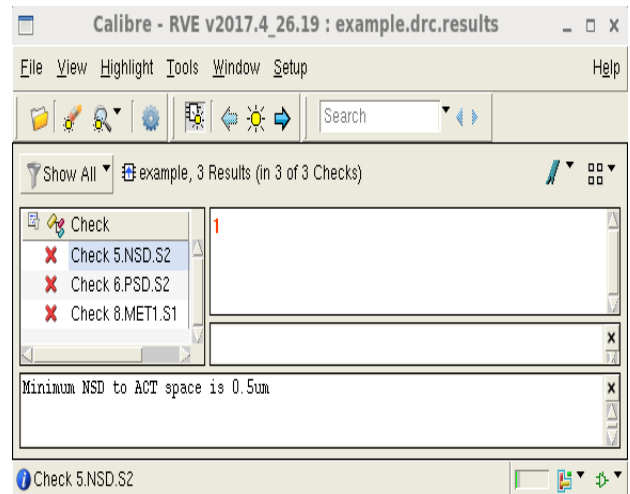
6

Run DRC



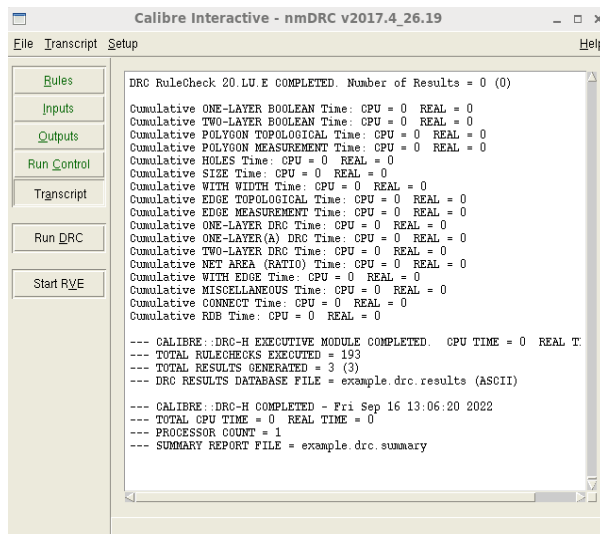
7

Get results (RVE)



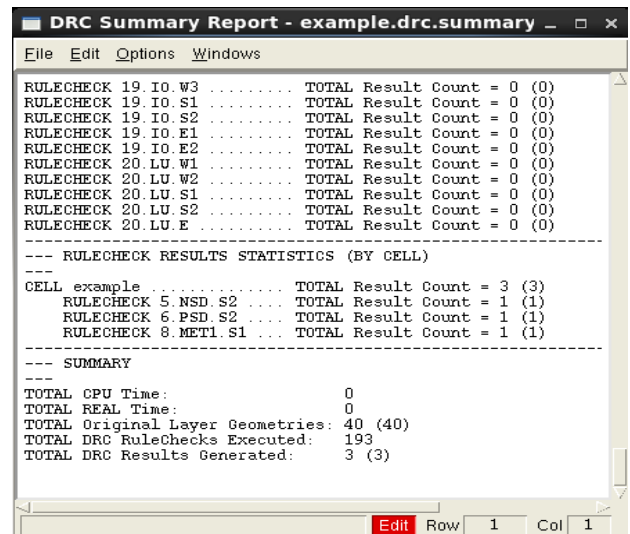
7

Get results (transcript)



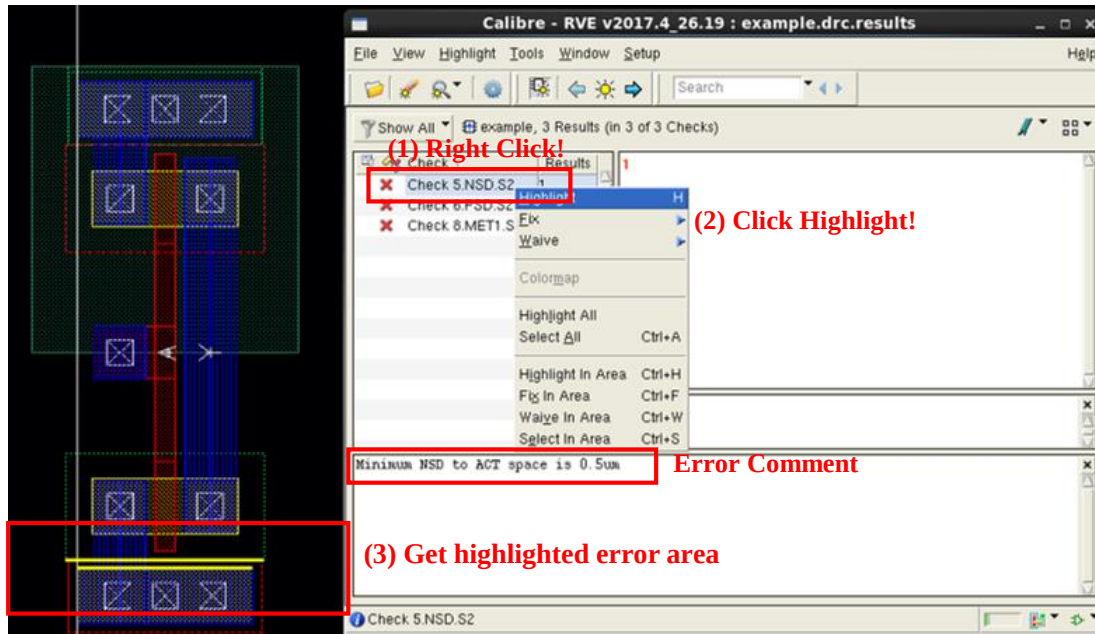
7

Get results (summary)



8

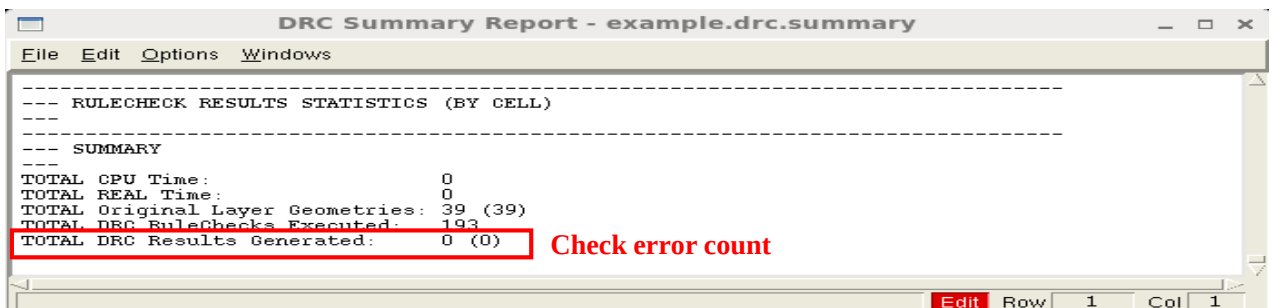
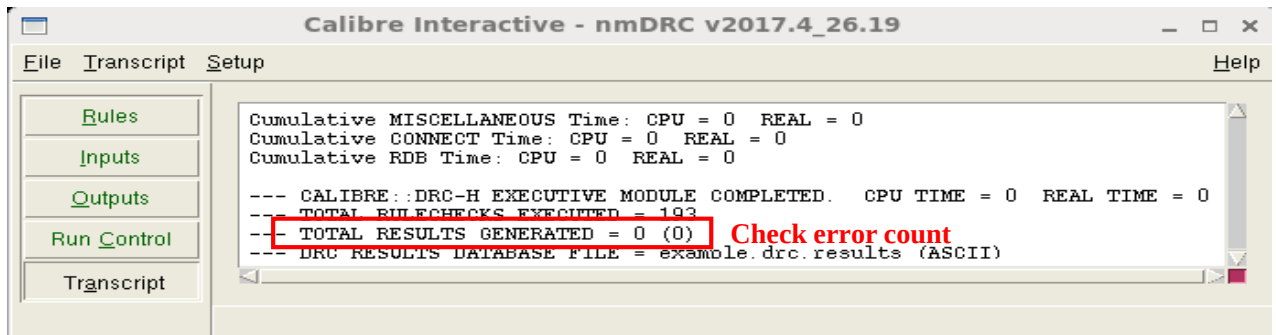
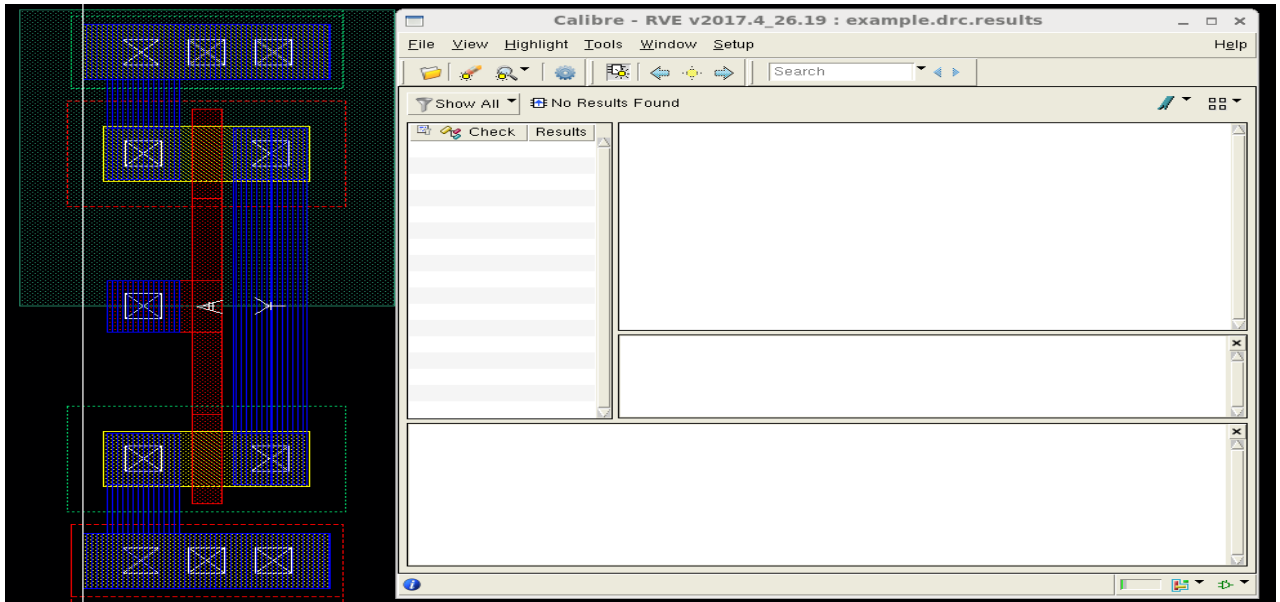
Run DRC



- After correct DRC errors, run again DRC.

9

Check Error Results



8.2 Script options

- In running DRC, you can use several options for convenience.
if you need, modify this comment in the drc_header.cal

option	value	comment
LAYOUT SYSTEM	GDSII OASIS	Select layout data format GDSII or OASIS
DRC MAXIMUM RESULTS	ALL [Maxresults]	Select maximum result of errors in running drc. Non-negative integer can be Maxresults. Default value is 1000
DRC SELECT CHECK	[rule name] [group name]	Only check selected rules You can add other rules or groups by writing name after values
DRC UNSELECT CHECK	[rule name] [group name]	Running drc except selected rules You can add other rules or groups by writing name after values
EXCLUDE CELL	[cell name]	Select cells not to be processed by drc verification. You can add other cells by writing name after values

- Before run drc, you have to modify script file(drc_header.cal) some parts below.

LAYOUT PATH “[gds_path]/[CELLNAME.gds]“

gds_path : the directory where layout gds file is exist

CELLNAME.gds : put in your gds file

LAYOUT PRIMARY “[CELLPRIMARY]“

Layout database topcell for running drc.

LAYOUT SYSTEM [GDSII or OASIS]

You can choose layout data type GDSII or OASIS

INCLUDE “[rule_path]/ETRI_0p5um_Analog_CMOS_2P3M_5V_DRC.cal “

path that rule file is exist.

- If you want to run drc in terminal, you can use command described below.

- Example drc_header.cal

```

LAYOUT PATH "temp.gds"   Layout gds file name for DRC
LAYOUT PRIMARY "temp"    Layout primary cell name
LAYOUT SYSTEM GDSII      Select layout system GDSII or OASIS
DRC RESULTS DATABASE "TOPCELL.db" ASCII Set DRC result file name
DRC MAXIMUM RESULTS ALL  Set DRC maximum error counts
DRC SUMMARY REPORT "TOPCELL.rep" Set DRC summary report name
//DRC SELECT CHECK rulename or groupname Choose rules or groups want to check
//DRC UNSELECT CHECK rulename or groupname Except rules or groups want to check
//Groups -> 1.NWL, 2.ACT, 3.PL1, 4.PL2, 5.NSD, 6.PSD, 7.CONT, 8.MET1, 9.VIA1, 10.MET2,
11.VIA2, 12.MET3, 13.PAD, 14.PLN2R, 15.CAP, 16.PNP, 17.NPN, 19.IO, 20.LU List of groups
//EXCLUDE CELL cellName1 cellName2 Set exclude cells in DRC

Remove "//" before using these options

INCLUDE "/PROJECT DIRECTORY/ETRI/DRC/ETRI_0p5um_Analog_CMOS_2P3M_5V_DRC.cal"

```

DRC rule path

- If you want to run drc in terminal, you can use command described below.

Command: calibre -drc -hier drc_header.cal

- If you want to see results at RVE in terminal, you can use command described below.

Command: calibre -rve -drc [DRC RESULTS DATABASE]

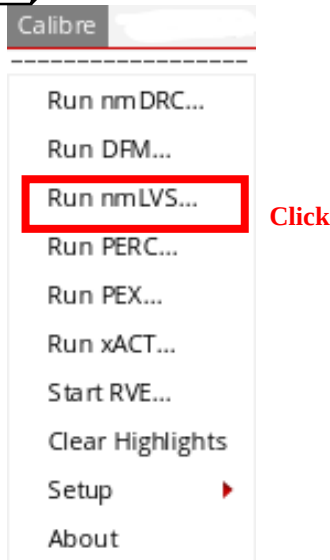
ex) calibre -rve -drc TOPCELL.db

9. Calibre LVS

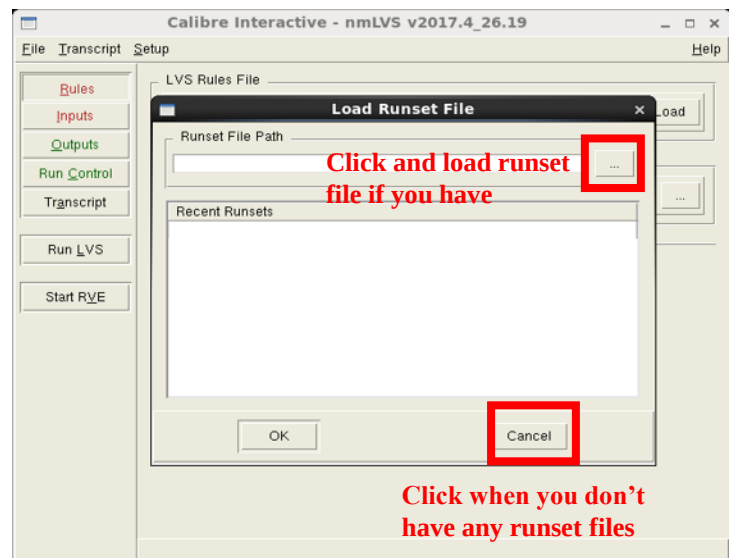
9.1 Run for LVS at GUI

- For running LVS(Layout VS Schematic) at GUI after layout, need to several steps like below pictures

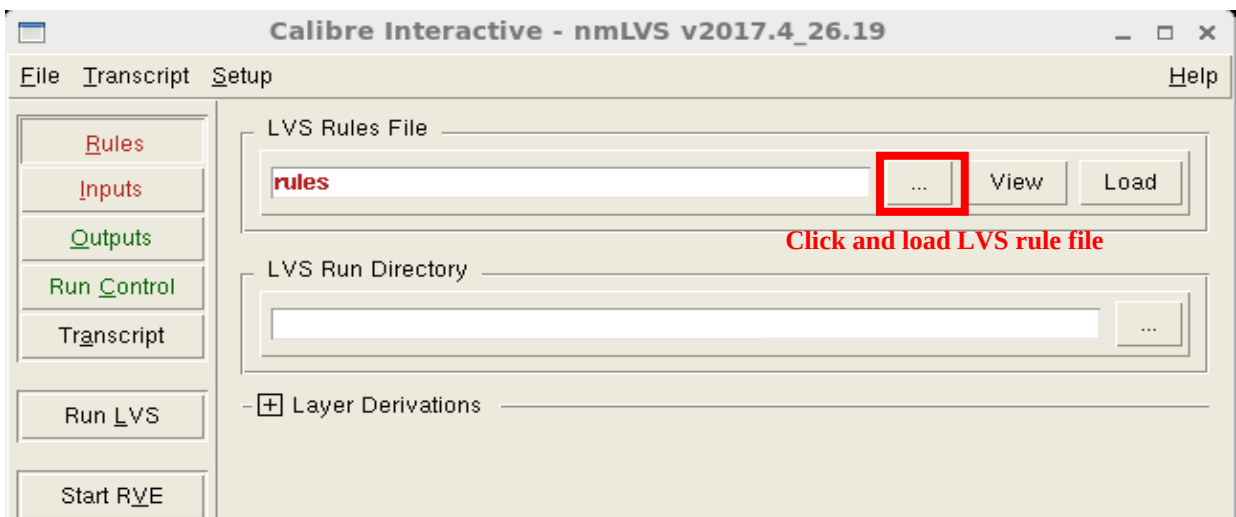
1 Click Run nmLVS in Calibre menu



2 If runset file is exist, Load runset file.
Runset file is not exist, click cancel



3 Load LVS rules file (ETRI_0p5um_Analog_CMOS_2P3M_5V_LVS.cal)

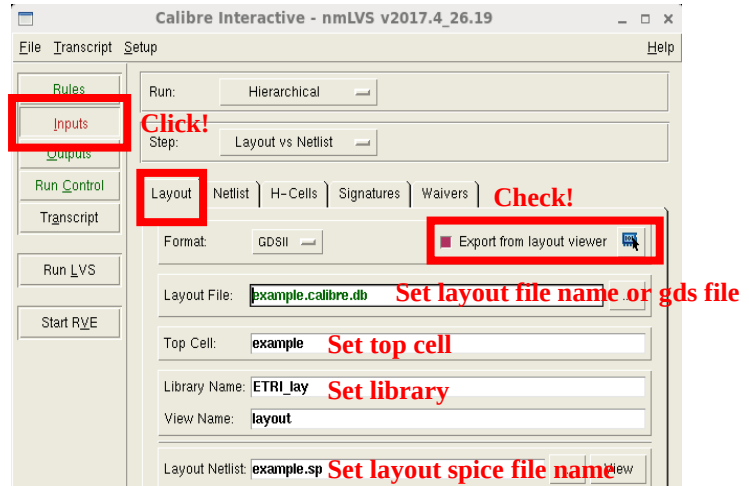
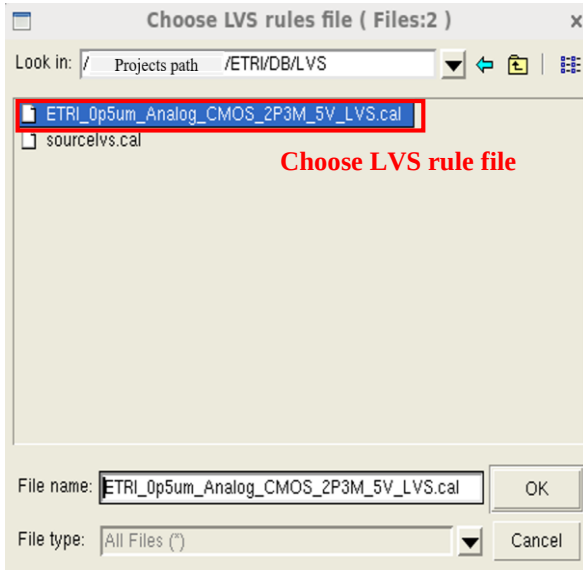


3

Load LVS rules file.

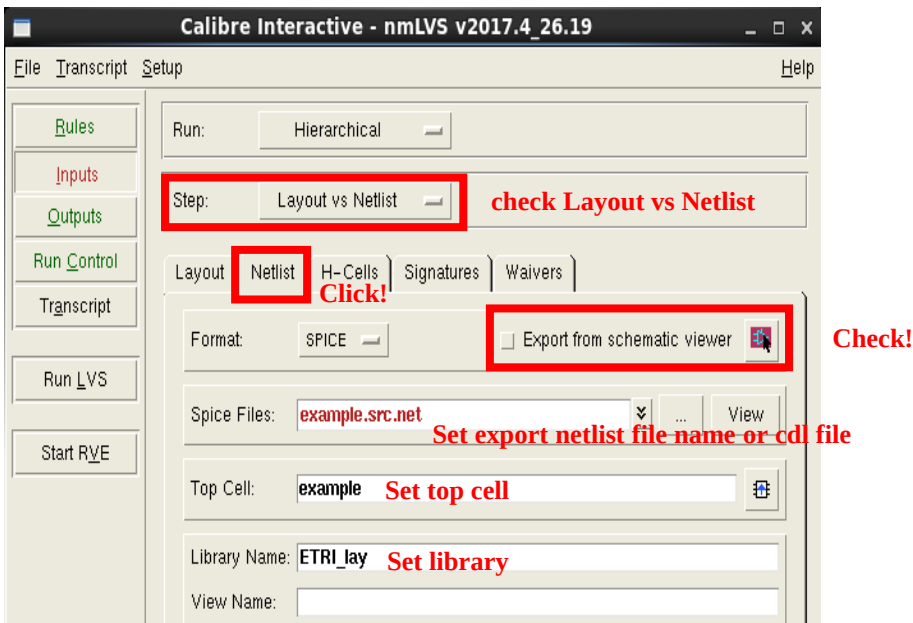
4

Set layout Input data. If you checked Export from layout viewer, insert layout file name to layout file. Not checked, Input your gds name want to run.

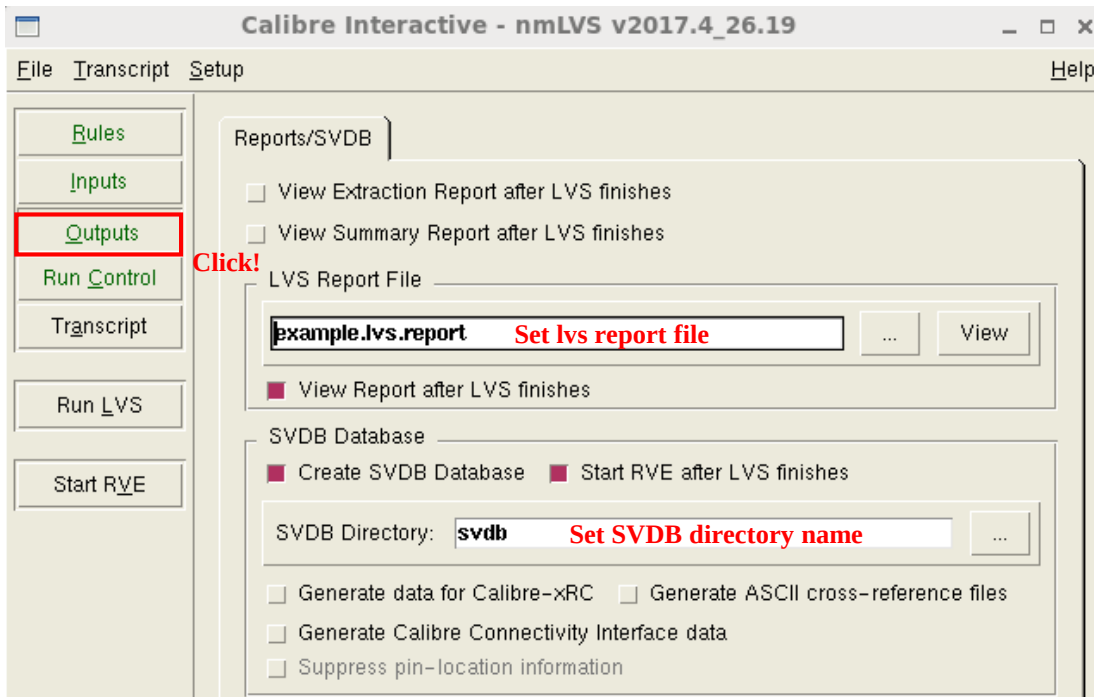


4

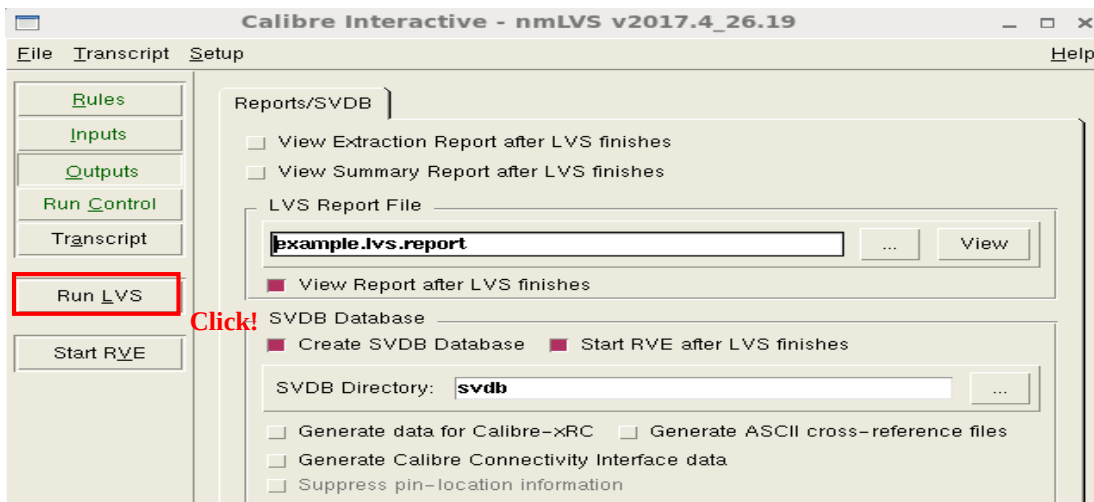
Set schematic Input data. If you checked Export from schematic viewer, insert layout file name to Sice files. Not checked, Input your netlist name want to run.



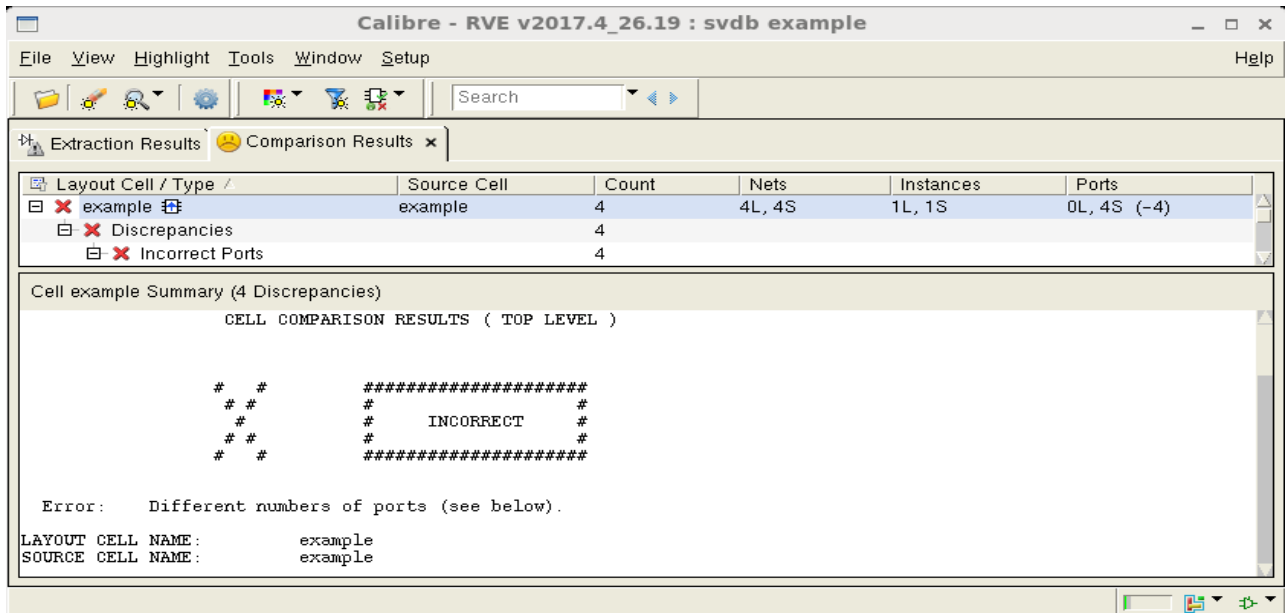
5 Set LVS outputs



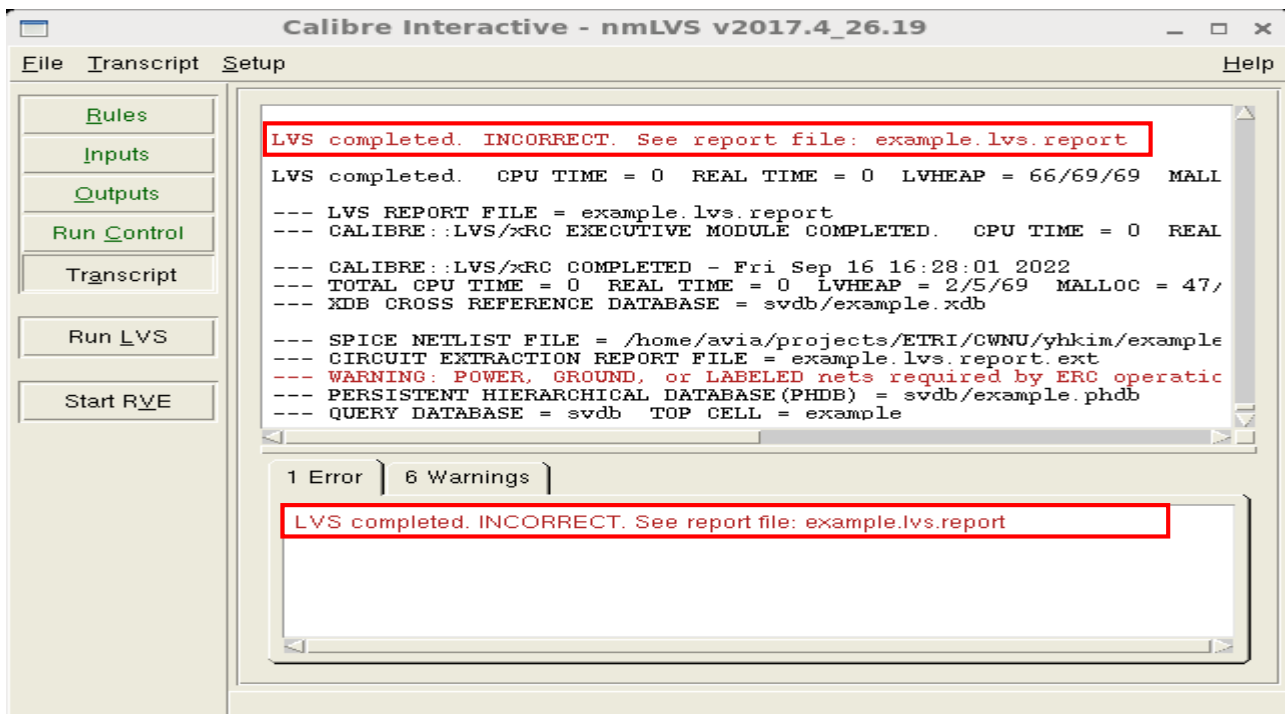
6 Run LVS



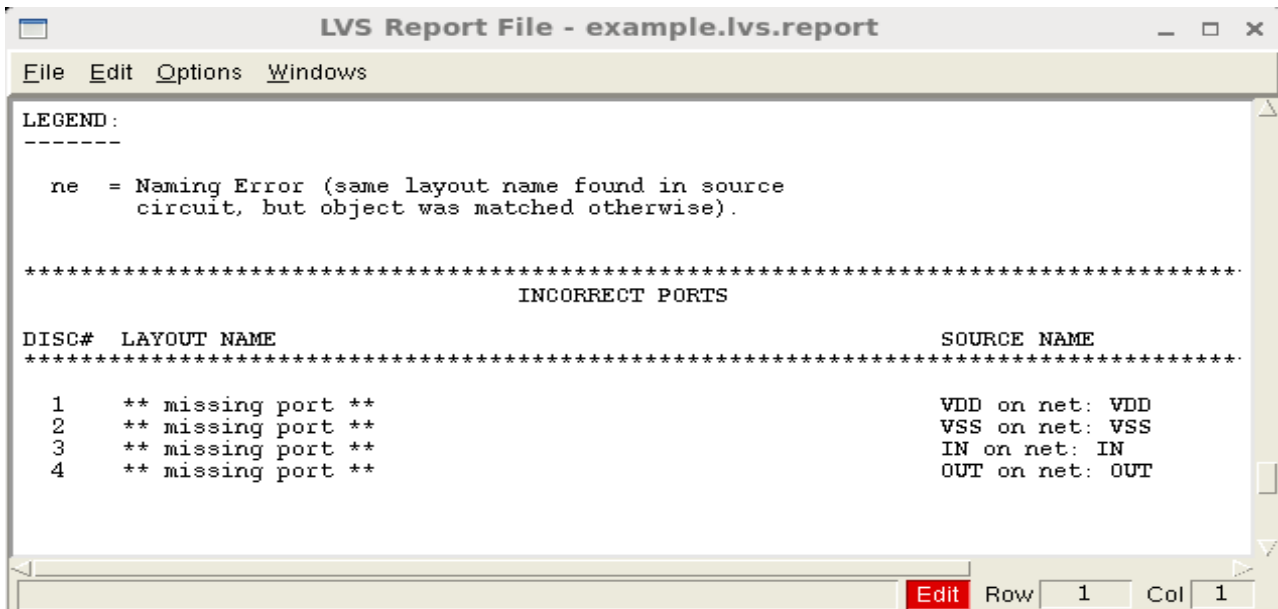
7 Get results(RVE)



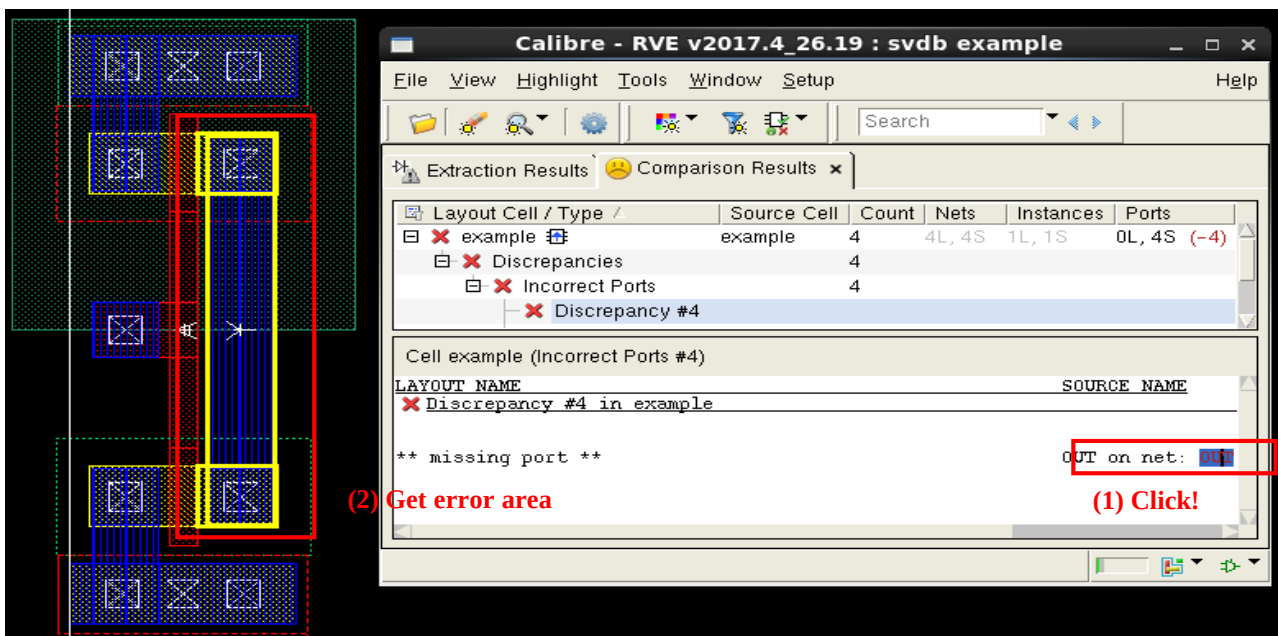
7 Get results(Transcript)



7 Get results(report)

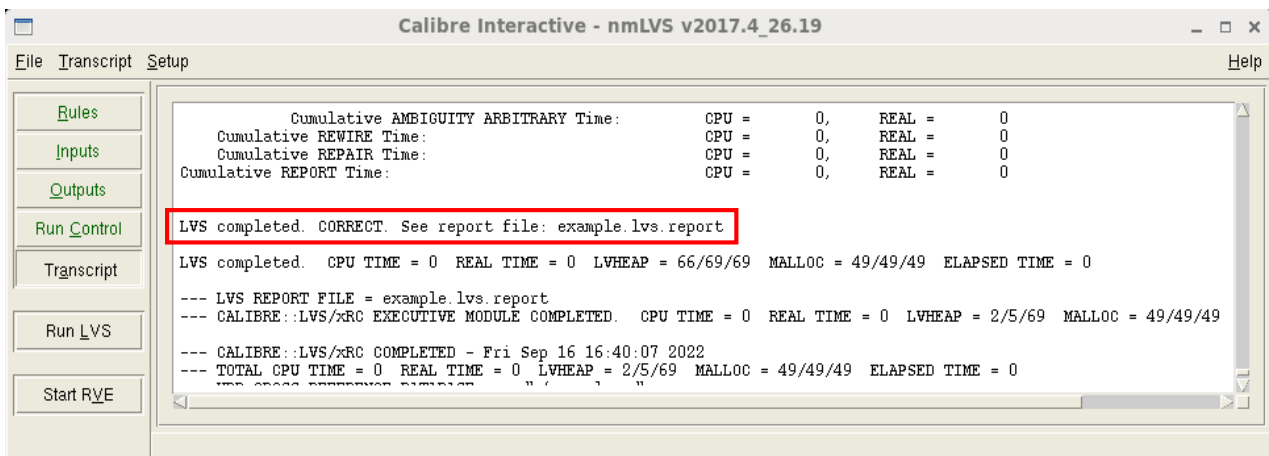
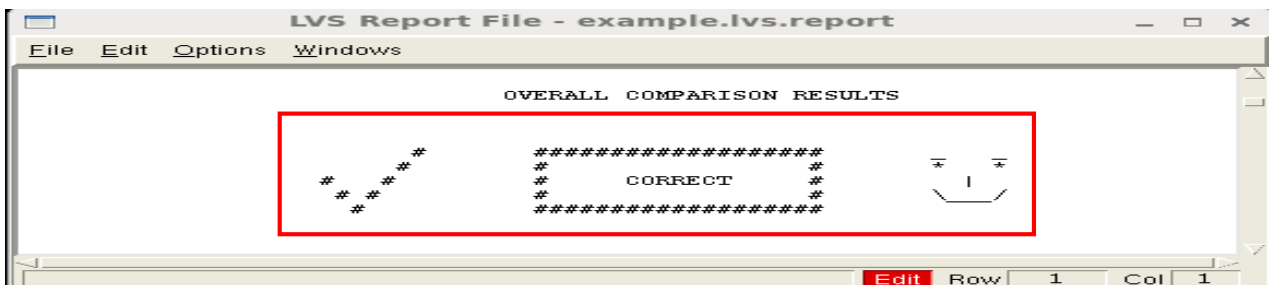
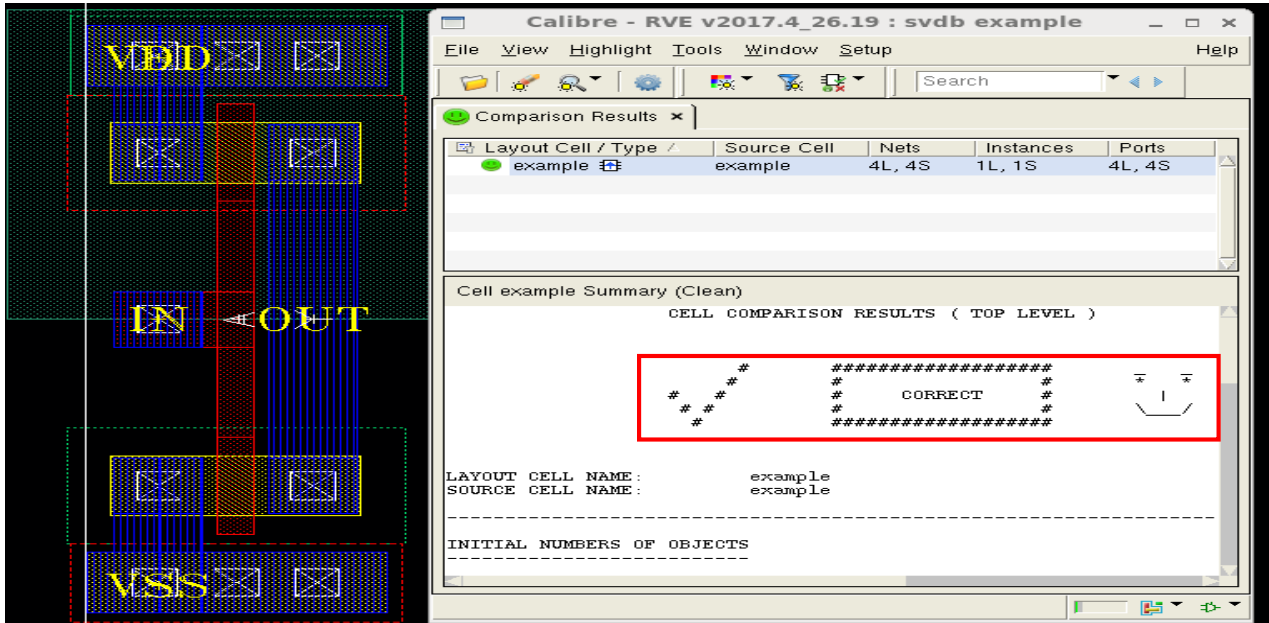


8 Run lvs



- After correct LVS errors, run again LVS.

9 Check error results



9.2 Script options

- In running LVS, you can use several options for convenience.
if you need, modify this comment in the lvs_header.cal.

- Before run lvs, you have to modify script file(lvs_header.cal) some parts below.

LAYOUT PATH “[gds_path]/[CELLNAME.gds]“

gds_path : the directory where layout gds file is existed.

CELLNAME.gds : put in your gds file

LAYOUT PRIMARY “[CELLPRIMARY]“

Layout database topcell for running lvs.

LAYOUT SYSTEM [GDSII or OASIS]

You can choose layout data type GDSII or OASIS

SOURCE PATH “[cdl_path]/[CELLNAME.cdl]“

cdl_path : the directory where schematic cdl file is exist

CELLNAME.cdl : put in your cdl file

SOURCE PRIMARY “[CELLPRIMARY]“

Schematic database topcell for running lvs.

SOURCE SYSTEM SPICE

ERC RESULTS DATABASE “[ERC_DB]“ ASCII

Set ERC DB name and format

ERC SUMMARY REPORT “[ERCSUMMARY]“

Set ERC summary report name

ERC MAXIMUM RESULTS 100

Set ERC maximum error counts

INCLUDE “[rule_path]/ETRI_0p5um_Analog_CMOS_2P3M_5V_LVS.cal “
path that rule file is exist.

- Example **lvs_header.cal**

```
LAYOUT PRIMARY "test" Layout primary cell name
LAYOUT PATH "test.gds" Layout gds file name for LVS
LAYOUT SYSTEM GDSII Select layout system GDSII or OASIS

SOURCE PRIMARY "test" Schematic primary cell name
SOURCE PATH "test.cdl" Schematic cdl file name for LVS
SOURCE SYSTEM SPICE Select source system SPICE

DRC RESULTS DATABASE "drc.db" ASCII // ASCII or GDSII Set DRC result file name
DRC SUMMARY REPORT "drc.sum" Set DRC summary report name

LVS REPORT "lvs.rep" List of groups
LVS REPORT MAXIMUM 1000 // ALL Set LVS maximum error counts
LVS REPORT OPTION A B C D S V Set LVS report option

ERC RESULTS DATABASE "erc.db" ASCII Set ERC result file name
ERC SUMMARY REPORT "erc.sum" Set ERC summary report name
ERC MAXIMUM RESULTS 100 Set ERC maximum error counts

INCLUDE "/ Project Directory /ETRI/LVS/ETRI_0p5um_Analog_CMOS_2P3M_5V_LVS.cal"
Include LVS rule path
```

- If you want to run lvs in terminal, you can use command described below.

Command: calibre -lvs -hier lvs_header.cal

- If you want to see results at RVE in terminal, you can use command described below.

Command: calibre -rve -lvs [SVDB directory]

ex) calibre -rve -lvs svdb

※ **LVS virtual connection option is set Yes. If you don't want virtual connect, Please set "VIRTUAL CONNECT COLON" to "NO" in LVS rule file.**

※ **LVS REDUCE PARALLEL/SERIES CAPACITORS option is set NO in LVS rule file.**

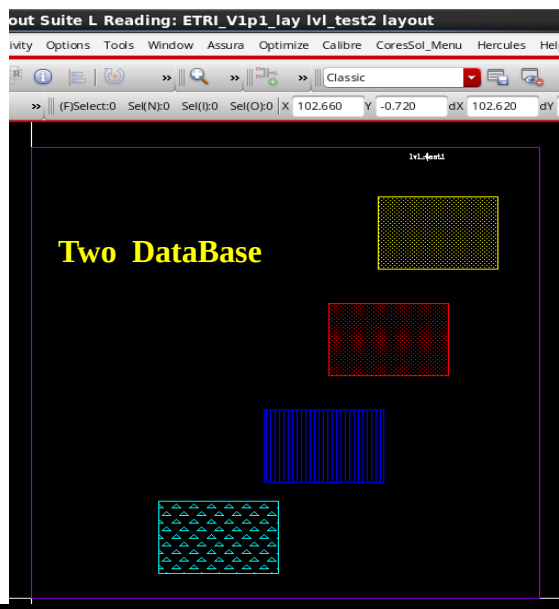
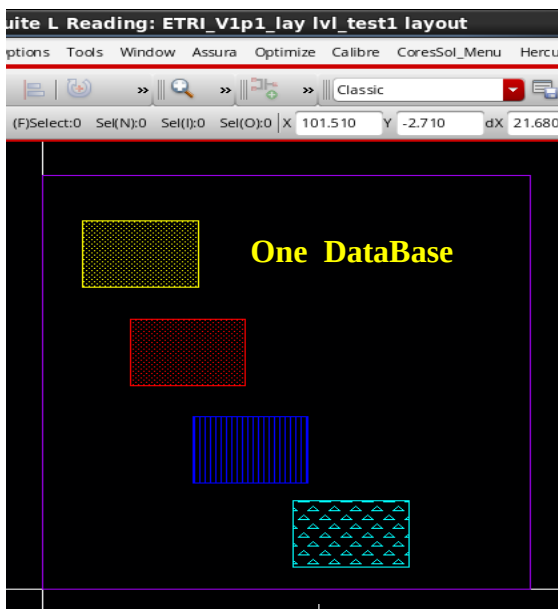
please do not change this option for exact LVS verification.

FILE NAME: ETRI_0p5um_Analog_CMOS_2P3M_5V_LVS.cal

10. Calibre LVL

10.1 LVL (Layout Versus Layout) Verification

- Distributes of the two input data that are not equal to each other as errors.
- Calibre has the Capability to compare two separate Layout Database
- When comparing two layout database :
 - * Specify One of the Database Using
 - LAYOUT SYSTEM
 - LAYOUT PATH
 - LAYOUT PRIMARY
 - * Specify the other Database Using
 - LAYOUT SYSTEM2
 - LAYOUT PATH2
 - LAYOUT PRIMARY2



LVL Error Results == XOR

Same Point = No Error

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

10.2 Script Option

- In running LVL, you can use several options for convenience.
if you need, modify this comment in the lvl_header.cal

option	value	comment
LAYOUT SYSTEM	GDSII OASIS	1 st Input : Select layout data format GDSII or OASIS
LAYOUT SYSTEM2	GDSII OASIS	2 nd Input : Select layout data format GDSII or OASIS
LAYOUT BUMP2	Number > 1 st DB	Increments 2 nd Layout Data Base Layer Number by Specified Value
DRC RESULTS DATABASE	<i>Filename[type]</i>	Specifies the filename and type of the results database
DRC SUMMARY REPORT	<i>Filename</i>	Specifies the DRC Summary report File

- Before run lvl, you have to modify script file(lvl_header.cal) some parts below.

LAYOUT PATH “[gds_path]/[CELLNAME.gds]“

gds_path : the directory where layout 1st gds file is exist

CELLNAME.gds : put in your gds file

LAYOUT PRIMARY “[CELLPRIMARY]“

Layout database 1st topcell for running drc.

LAYOUT PATH2 “[gds_path]/[CELLNAME.gds]“

gds_path : the directory where layout 2nd gds file is exist

CELLNAME.gds : put in your gds file

LAYOUT PRIMARY2 “[CELLPRIMARY]“

Layout database 2nd topcell for running drc

INCLUDE “[rule_path]/ETRI_0p5um_Analog_CMOS_2P3M_5V_LVL.cal “

path that rule file is exist.

- If you want to run lvl in terminal, you can use command described below.

10.2 Script Option

- Example lvl_header.cal

LAYOUT SYSTEM GDSII	1 st : Select layout system GDSII or OASIS
LAYOUT PATH "lvl_test1.gds"	1 st : Layout gds file name for LVL
LAYOUT PRIMARY "lvl_test1"	1 st : Layout primary cell name
LAYOUT SYSTEM2 GDSII	2 nd : Select layout system GDSII or OASIS
LAYOUT PATH2 "lvl_test2.gds"	2 nd : Layout gds file name for LVL
LAYOUT PRIMARY2 "lvl_test2"	2 nd : Layout primary cell name
LAYOUT BUMP2 2500	Set Number of 2 nd GDS Number
DRC RESULTS DATABASE "lvl.db" ASCII	Set LVL result file name
DRC SUMMARY REPORT "lvl.sum"	Set LVL summary report name
INCLUDE "./ETRI_0p5um_Analog_CMOS_2P3M_5V_LVL.cal"	LVL rule path

- If you want to run drc in terminal, you can use command described below.

Command: calibre -drc -hier lvl_header.cal

- If you want to see results at RVE in terminal, you can use command described below.

Command: calibre -rve -drc [DRC RESULTS DATABASE]

ex) calibre -rve -drc TOPCELL.db

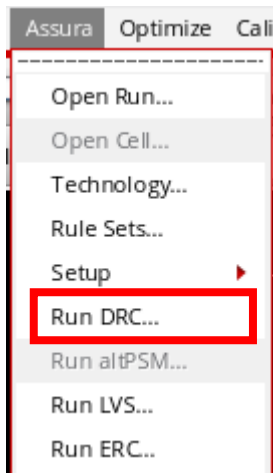
11. Assura DRC

11.1 Run for DRC at GUI

- For running DRC(Design Rule Check) at GUI after layout, need to several steps like below pictures

1

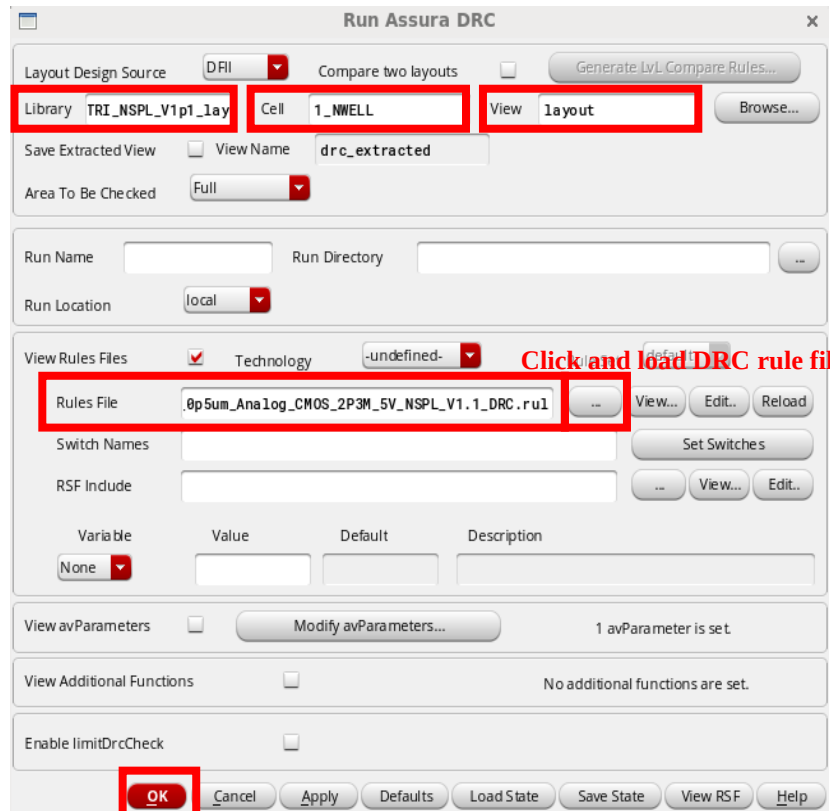
Click Run DRC in Assura menu.



Click

2

In the Run Assura DRC window, type Library, Cell, View, Rules File and click OK at the bottom.

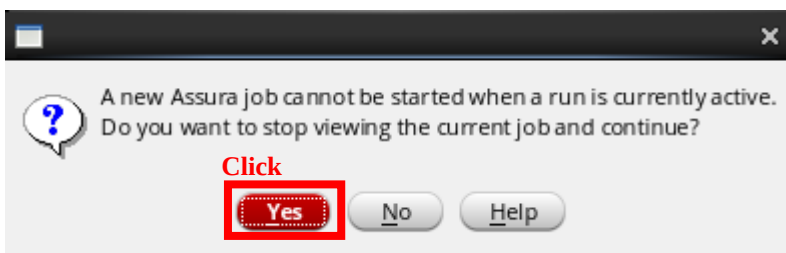


Click and load DRC rule file

Click

3

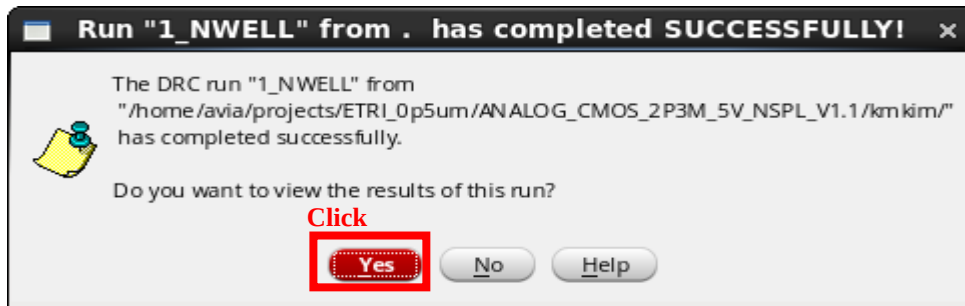
When this phrase comes up, press yes.



Click

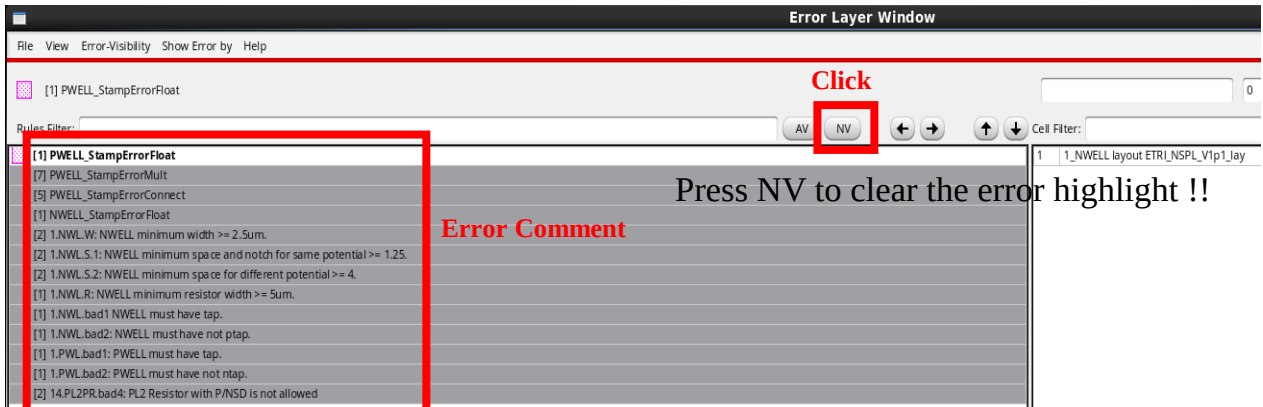
4

When this phrase comes up, press yes.



5

When the error layer window appears, press NV to select nothing.



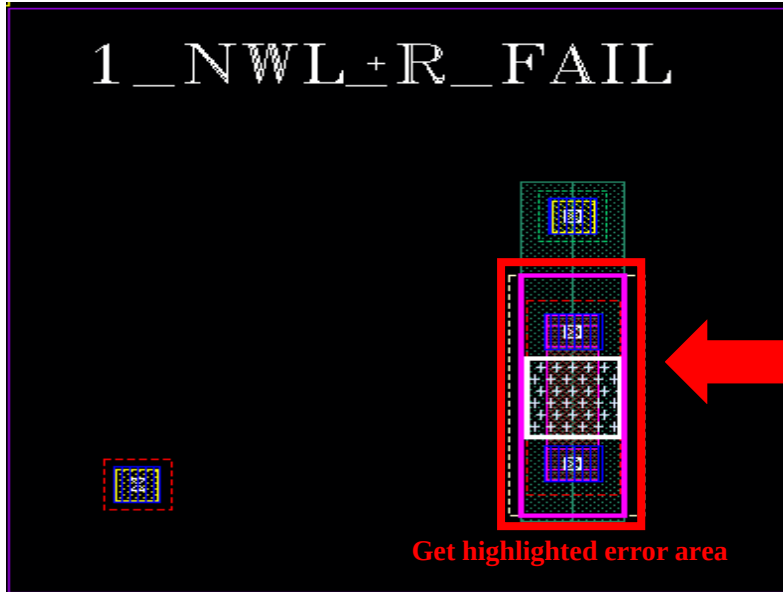
6

If you click on the error item and zoom in and out of the mouse on LAYOUT, you will see the error highlight.

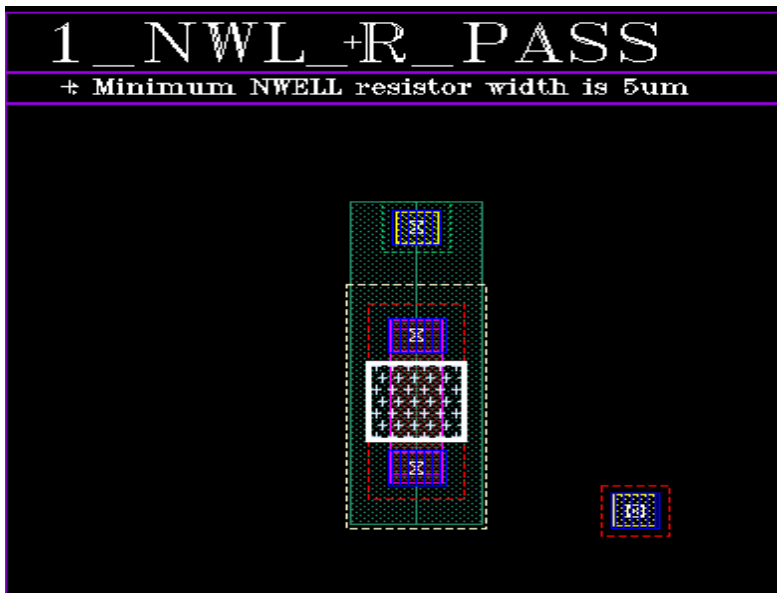


6

If you click on the error item and zoom in and out of the mouse on Layout, you will see the error highlight.



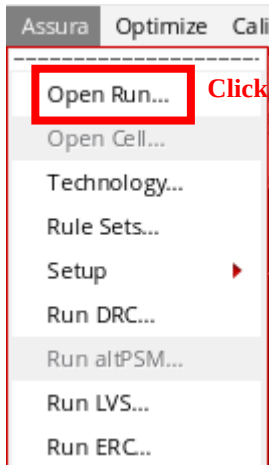
ERROR highlights are seen in the FAIL pattern.



ERROR highlights are not identified in the PASS pattern.

7

If you press open run, the most recent DRC-verified cell data remains. However, it does not apply to the layout modifications. (If the layout is modified, it should be verified as run drc.)



8

When this phrase comes up, press yes.



9

If you proceed with number 8, you can see the error layer window.

Click

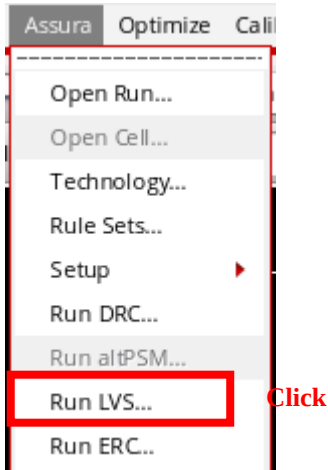


12. Assura LVS

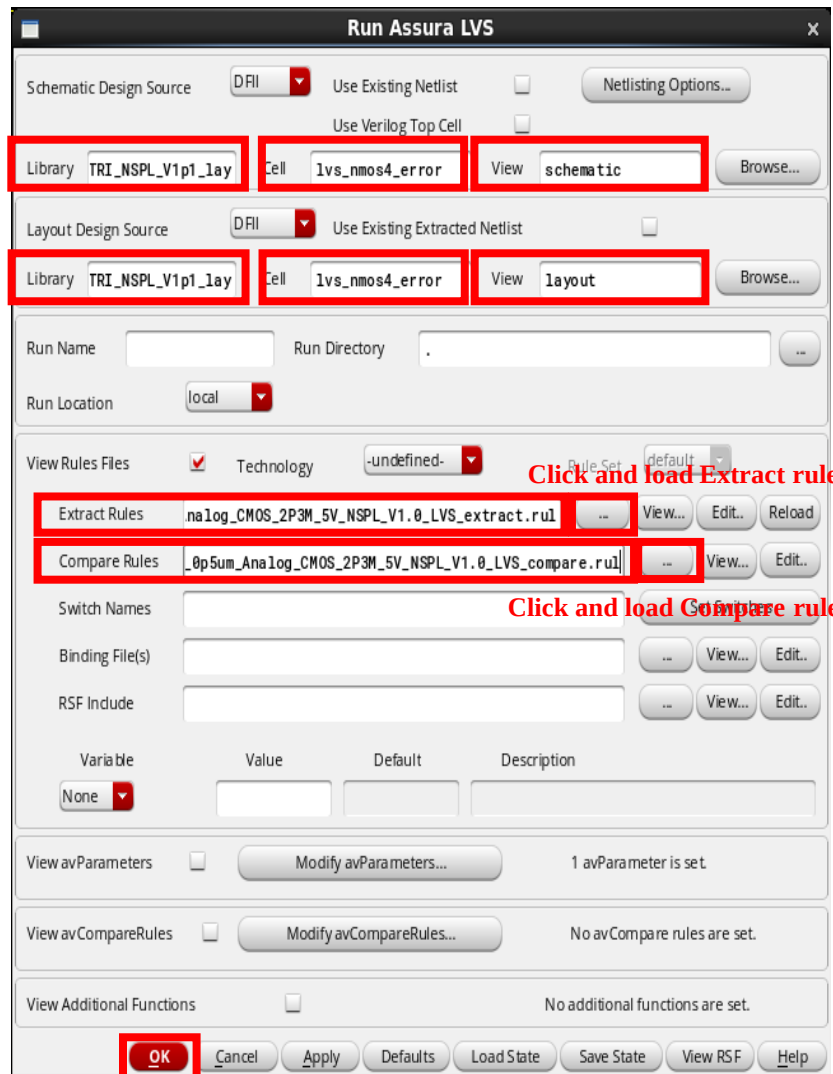
12.1 Run for LVS at GUI

- For running LVS(Layout VS Schematic) at GUI after layout, need to several steps like below pictures

1 Click Run LVS in Assura menu.



2 When the Run Assura LVS window opens, fill in Library, Cell, View, Extract rules, Compare rules and click OK at the bottom.



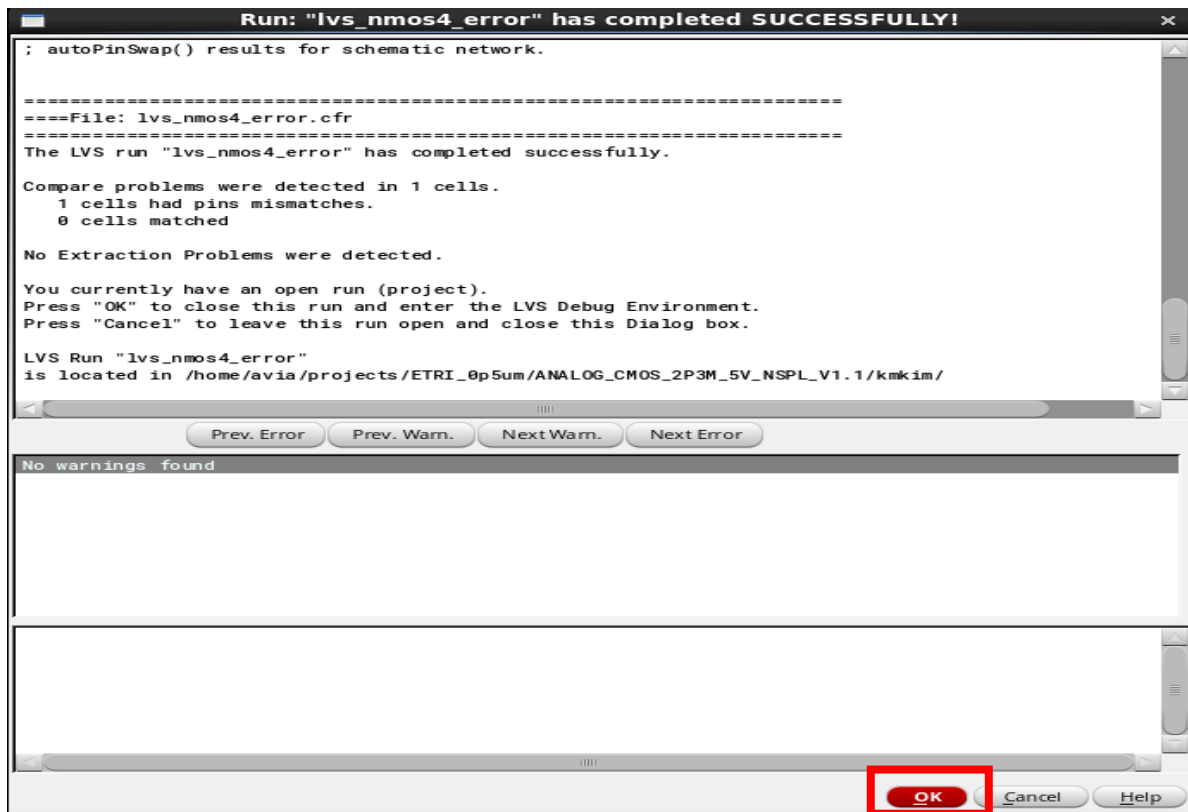
Click and load Extract rule file

Click and load Compare rule file

Click

3

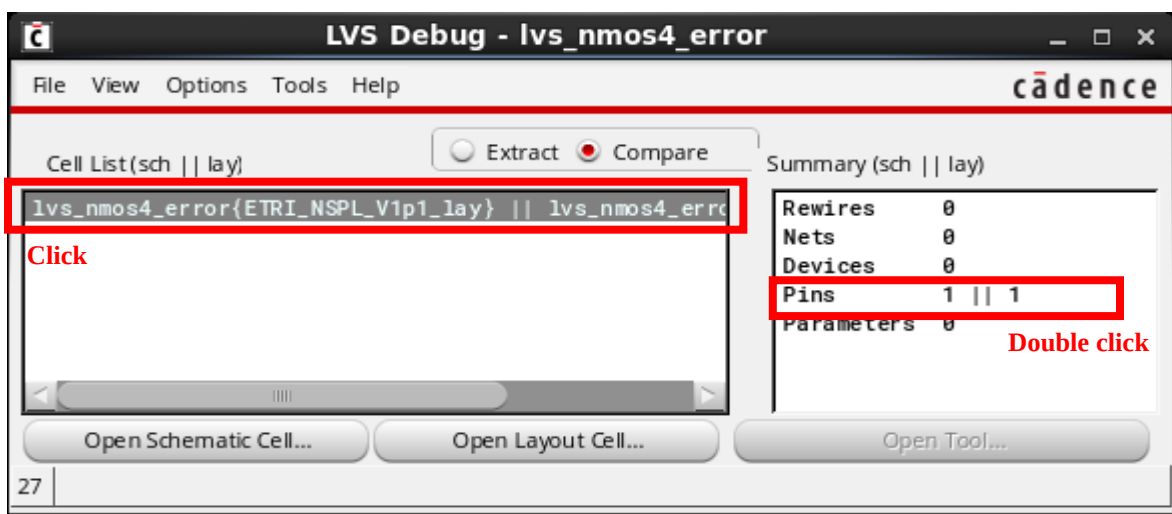
When this phrase comes up, press yes.



Click

4

Get error results

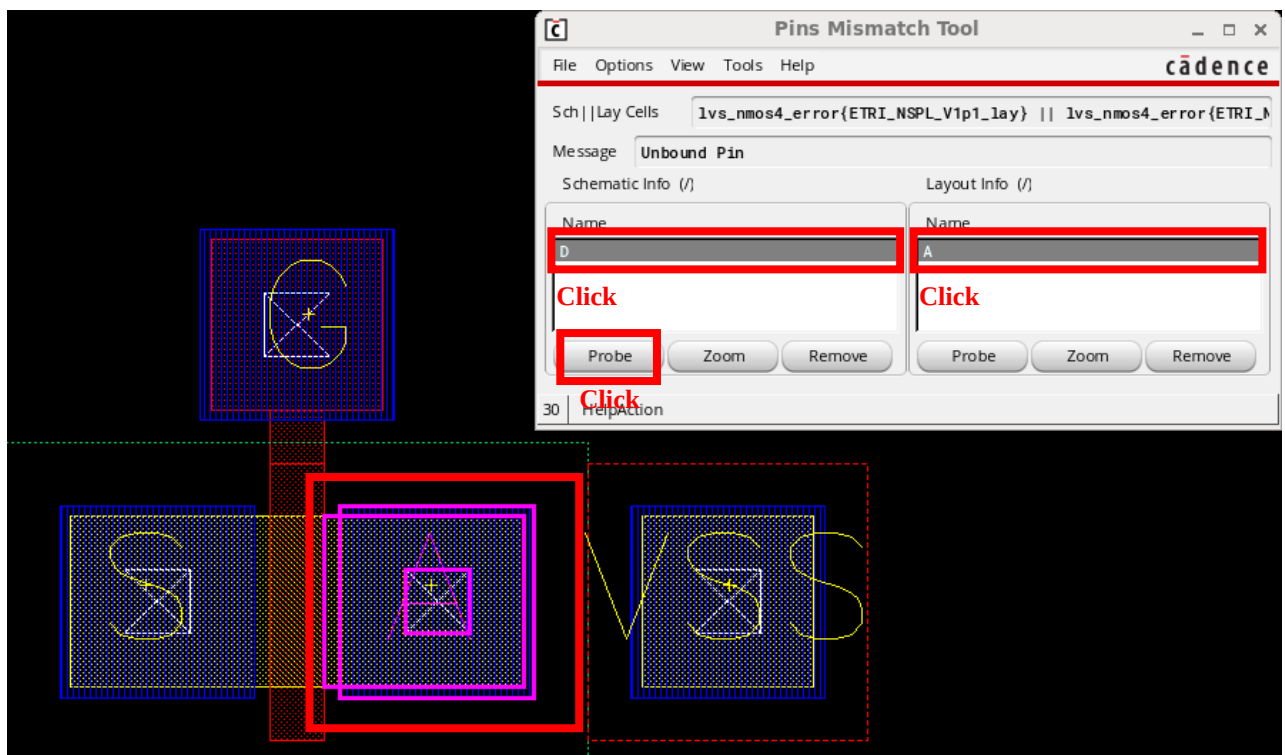
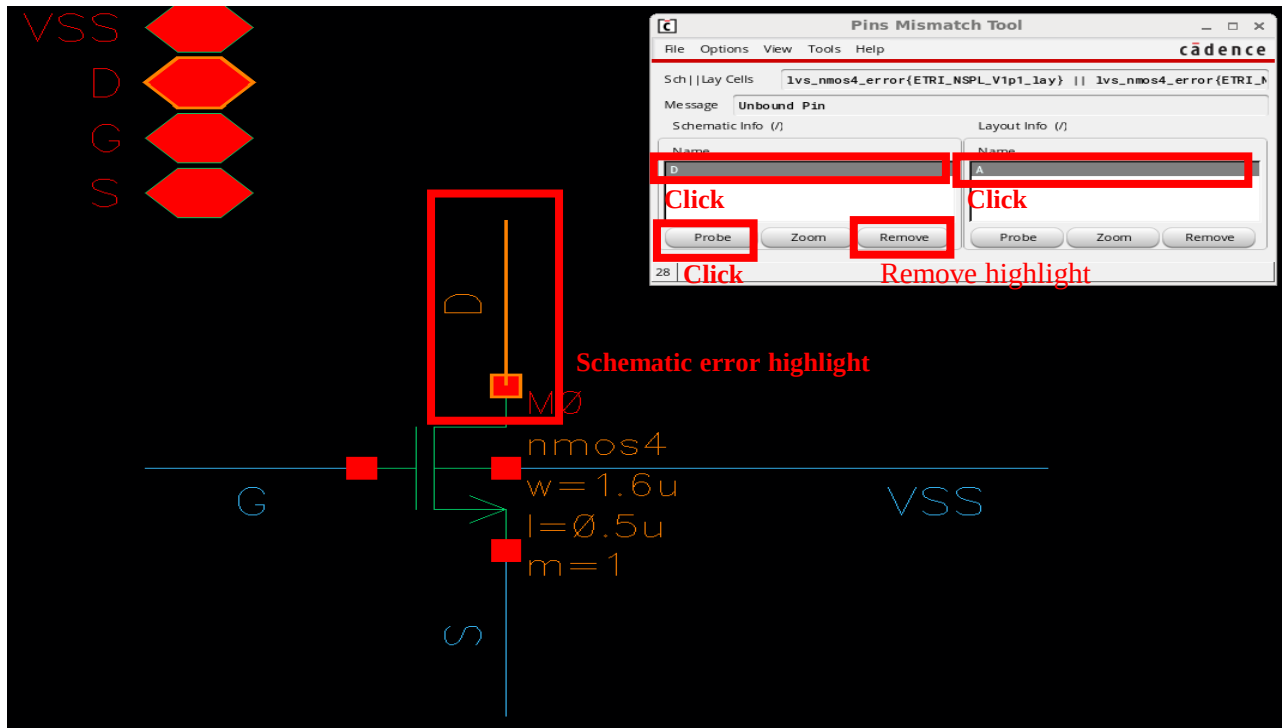


Click

Double click

5

If you click the part where the layout and schematic do not match, the part that does not match will be highlighted.



Layout error highlight

6

If the Layout and Schematic match, the LVS pass window comes out.

